Meeting of the Board
15 – 18 July 2024
Songdo, Incheon, Republic of Korea
Provisional agenda item 10

GCF/B.39/02/Add.15/Rev.01

15 July 2024

Consideration of funding proposals – Addendum XV

Funding proposal package for FP240

Summary

This addendum contains the following six parts:

a) A funding proposal summary titled "Collaborative R&DB Programme for Promoting the Innovation of Climate Technopreneurship" submitted by Korea Development Bank (KDB);
b) No-objection letter(s) issued by the national designated authority(ies) or focal point(s);
c) Environmental and Social report(s) disclosure;
d) Independent Technical Advisory Panel's assessment;
e) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
f) Gender documentation of the funding proposal.

These documents are presented as submitted by the accredited entity and the national designated authority(ies) or focal point(s), respectively. Pursuant to the Comprehensive Information Disclosure Policy of the Fund, the funding proposal titled "Collaborative R&DB Programme for Promoting the Innovation of Climate Technopreneurship" submitted by Korea Development Bank (KDB) is being circulated on a limited distribution basis only to Board Members and Alternate Board Members to ensure confidentiality of certain proprietary, legally privileged or commercially sensitive information of the entity.

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Funding Proposal

Project/Programme title:	*Collaborative R&DB Programme for Promoting the Innovation of Climate Technopreneurship*
Country(ies):	*Cambodia, Indonesia, Laos, Philippines, and Vietnam*
Accredited Entity:	*The Korea Development Bank (KDB)*
Date of first submission:	*[2022/02/28]*
Date of current submission	*[2024/06/17]*
Version number	*[V.11]*

GREEN CLIMATE FUND

GREEN CLIMATE FUND

Contents

Note to Accredited Entities on the use of the funding proposal template
• Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc. • Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents. • The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time. • The recommended font is Arial, size 11. • Under the GCF Information Disclosure Policy, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]"

A. PROJECT/PROGRAMME SUMMARY

A.1. Project or programme	Programme	**A.2. Public or private sector**	Private
A.3. Request for Proposals (RFP)	*If the funding proposal is being submitted in response to a specific GCF Request for Proposals, indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+.* Not applicable		

A.4. Result area(s)	*Check the applicable GCF result area(s) that the overall proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of **GCF and Co-financers' contribution** devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.*		
	**Figures (%) are all indicative based on investment scenario.*	**GCF Contribution**	**Co-financers' contribution**[1]
	Mitigation total	50 %	50 %
	☒ Energy generation and access	16 %	16 %
	☒ Low-emission transport	32 %	32 %
	☒ Buildings, cities, industries and appliances	2 %	2 %
	☐ Forestry and land use	Enter number %	Enter number %
	Adaptation total	50 %	50 %
	☒ Most vulnerable people and communities	10 %	10 %
	☒ Health and well-being, and food and water security	40 %	40 %
	☐ Infrastructure and built environment	Enter number %	Enter number %
	☐ Ecosystems and ecosystem services	Enter number %	Enter number %

A.5. Expected mitigation outcome *(Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)*	*Indicate greenhouse gas (GHG) emission reductions or removals in tCO2eq over total lifespan of the project/programme*[2] 1,639,681 *tCO2eq**	**A.6. Expected adaptation outcome** *(Core indicator 2: direct and indirect beneficiaries reached)*	*Indicate total number of direct and indirect beneficiaries*	
			Indicate number of direct beneficiaries 1,180,881 (women, 50%)	*Indicate number of indirect beneficiaries* 1,132,408 (women, 50%)
			Indicate % of direct beneficiaries vis-à-vis total population 2.4%	*Indicate % of indirect beneficiaries vis-à-vis total population* 4.0%
A.7. Total financing (GCF + co-finance[3]**)**	221.2149 million USD	**A.9. Project size**	Medium (Upto USD 250 million)	
A.8. Total GCF funding requested	104.471 million USD *For multi-country proposals, please fill out annex 17.*			

[1] Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.
[2] The total lifespan of the project/programme is defined as the maximum number of years over which the outcomes of the investment are expected to be effective. This is different from the project/programme implementation period.
[3] Refer to the Policy of Co-financing of the GCF.

A.10. Financial instrument(s) requested for the GCF funding	*Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.* ☒ Grant USD 20.721 million ☐ Loan Enter number ☐ Guarantee Enter number ☒ Equity USD 83.75 million ☐ Results-based payment Enter number		
A.11. Implementation period	11 years* * *Fund lifespan*	**A.12. Total lifespan**	30 years (5+25)* * *5 years of the fund investment period + a pipeline with the longest lifespan estimated*
A.13. Expected date of AE internal approval	*This is the date that the Accredited Entity obtained/will obtain its own approval to implement the project/ programme, if available.* *10/31/2024*	**A.14. ESS category**	*Refer to the AE's safeguard policy and GCF ESS Standards to assess your FP category.* I-2
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	**A.16. Has Readiness or PPF support been used to prepare this FP?**	Yes ☒ No ☐
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	**A.18. Is this FP included in the country programme?**	Yes ☒ No ☐
A.19. Complementarity and coherence	*Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.* Yes ☐ No ☒		

A.20. Executing Entity information	The programme has four Executing Entities (EEs) as below. (1) The Korea Development Bank (KDB): Accredited Entity, co-Executing Entity of Components 1 and 4 The programme will leverage the proven three-tier venture investment platforms (*KDB NextRound, NextOne,* and *NextRise*) and global VC and acceleration networks of KDB Headquarters and its Singapore/London Venture Desk and KDB Silicon Valley LLC (VC subsidiary). Above all, KDB will serve as the control tower that aims at seamless interlinked management amongst the four (4) different components across five (5) NOL countries in capacity of both Accredited Entity and Executing Entity. (2) Global Green Growth Institute (GGGI): co-Executing Entity of Components 1 and 4 GGGI is a treaty-based intergovernmental organisation that has been carrying out a number of climate actions and green growth promotion activities across the developing world. As a trusted advisor with a prominent in-country presence, GGGI has built trust with the five NOL countries' National Designated Authorities (NDAs). During the Project Preparation Facility (PPF) phase of the programme, GGGI had closely consulted the five NDAs for designing Components 1 and 4 to be properly aligned with the five governments' national policy direction and specific needs. GGGI will contribute to designing and operating the programme's core governance structure for strengthening country ownership – Sustainability Management Unit (SMU) – in consultation with NDAs and key local stakeholders in the five NOL countries. Its signature initiative, entitled *"Global Greenpreneurs Program"*, has supported high-potential local green entrepreneurs – ca. 800 youths in total over the last four years – while providing incubation and acceleration services for 57 teams from 20 countries. In addition, GGGI has extensive experience in working with the private sector particularly with start-ups and MSMEs in diverse sectors with various green investment interventions. (3) NH Investment & Securities (NHIS): Executing Entity of Component 2 NH Investment & Securities (NHIS) is the leading comprehensive financial investment company with overwhelming competitiveness in all business areas in Asia. Given that a call of the time for corporates to create ESG values and initiative is increasing, NHIS provides eco-friendly and green finance services, and has been committed to leading "Green Impact Finance" by fully leveraging its capabilities. NHIS has been striving to lay a solid foundation for practical execution of ESG by revamping ESG-related organizations and investing ca. USD 15 billion in green impact related investment banking & brokerage activities across green finance and CSR. In addition, its Carbon Finance Team has accumulated experiences in diverse GHG reduction projects in Korea and abroad, and has provided comprehensive carbon-offset consulting services, with a pursuit to lead the client-targeted carbon market through carbon credit brokerage and development of diverse financial products.

(4) NH Absolute Return Partners (NH ARP): Executing Entity of Component 3

NH ARP Pte. Ltd., a subsidiary 100% owed by NH Investment & Securities (NHIS) under the NH Financial Group (A+/S&P, A1/Moody's, A/Fitch), is a licensed fund management company in Singapore, with global investment expertise in private equity and debt deal opportunities. It has successfully driven performance with several leading investments in notable deals across Southeast Asian countries: in particular, specialised in growth stage investments in innovative technology-based companies within Southeast Asian.

NH ARP's capacity in leveraging core networks and securing high-quality investment opportunities will support the CTF to optimise investment portfolio and achieve returns. On the other hands, Carbon Finance Team of NH IS HQ, experienced with carbon financial services as the right solution, will support the fund's maximising climate outcomes. NH ARP will be the general partner and investment manager of the CTF, a limited partnership to be established in Singapore.

A.21. Executive summary (max. 750 words, approximately 1.5 pages)

"Climate Technology Acceleration" at the Core of a Paradigm Shift towards Low Carbon and Climate Resilient Innovation Pathways. This climate technology acceleration initiative has been inspired by the United Nations Framework Convention on Climate Change (UNFCCC, 2018)[4] that spotlights a need to support developing countries to "*facilitate access to foreign exchange for entrepreneurs to purchase technologies not available in local markets that they need for developing their solution on an economically viable scale* (p.50)." The programme, *'Collaborative R&DB Programme for Promoting the Innovation of Climate Technopreneurship,'* highlights the importance of supporting the Association of Southeast Asian Nations (ASEAN) to source and digest climate technology solutions through technology transfer in advancing mitigation and adaptation actions amid rapidly exacerbating climate crisis, with a strategic focus on five countries from Cambodia, Indonesia, Laos, and the Philippines to Vietnam (*in alphabetical order*). Pivoting from the five countries, the programme will support the regional innovation ecosystem conducive to replication and scaling of technology-enabled climate solutions in place, which will be instrumental in achieving a commonly forged vision of shared prosperity in the ASEAN community.

Programme Canvas

NOL Countries Shall Accelerate Climate Technology Leapfrogging as the Solution, while Climate Actions Are More Urgent than Ever. Many sources demonstrate that the five ASEAN economies are

[4] UNFCCC. 2018. Climate Technology Incubators and Accelerators. Bonn: United Nations Framework Convention on Climate Change.

severely exposed to climate vulnerability and its adverse impacts and associated risks. Alarming consequences include worsening natural disasters and extreme weather events such as hurricanes, tornadoes, heavy rainfall and floods, and droughts, in terms of frequency, intensity, and impacts of submersion of inhabited coastlines, deteriorating quality and quantity of water supply, declining agricultural productivity, increased outbreak and spread of water-borne diseases, and reduced effectiveness of natural coastal defences. The ravages of COVID-19 and the post-pandemic economic slowdown have disproportionately increased climate risks to the most vulnerable, compounded the negative impacts of past and current resource management practices, and magnified the sustainable development challenges.

The fastest climate solutions therefore need to be deployed throughout the five NOL countries given the climate urgency. Surrounded by worsening multi-dimensional ravages of global warming, developed economies have affordable and market-ready climate solutions. The programme shall blaze the trail for the five countries to kick their carbon intensive pathways through partnering with entrepreneurs armed with already existing technology-enabled climate solutions. It will offer a collaborative research, development and business (R&DB) platform on a quest for tailored business models that best fit the five countries' mitigation and adaptation technology needs, set up an investment fund at scale to encourage the market penetration of disruptive technology solutions, and empower the regional ecosystem eager for innovation uptake.

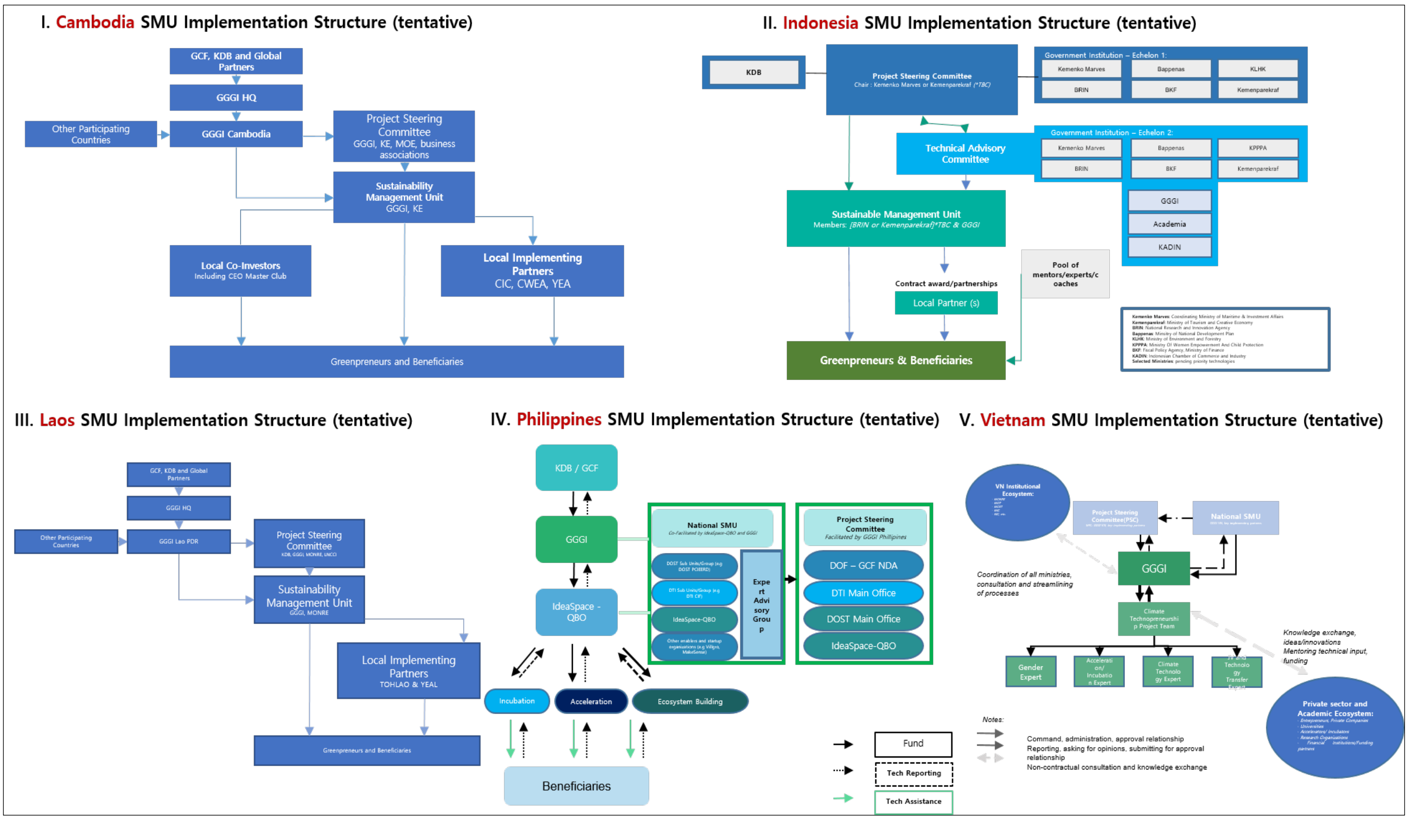
Local – Global Joint Venture (JV) Partnership as the Key Technology Transfer Mechanism that Promotes Climate Technology Leapfrogging in NOL Countries. The programme conceives *'technology transfer via a joint venture (JV) as a result of collaborative R&DB acceleration'* as a key profound concept, given that UNFCCC (2018)[5] spotlights the importance of supporting developing countries to source, digest, and deploy climate technology solutions through technology development and transfer in advancing mitigation and adaptation actions amid rapidly exacerbating climate crisis. JVs will eventually lead up to multi-faceted impacts that would curtail greenhouse gas (GHG) emissions, enhance climate resiliency, and, most importantly, strengthen local human capital capacity that is a crucial factor in internalising and accelerating technology innovation towards net zero in the long run, along with the engagement of diverse, cross-border innovation ecosystem stakeholders from academic institutions and start-up support intermediaries to large corporates under the common agenda of sourcing and disseminating tech-inspired climate solutions.

NOL Countries' Strong Ownership. The programme will build on existing local platforms and initiatives for the sake of the absorption of the programme into local governance. Every programme country has a national body for promoting entrepreneurship or relevant mechanism, though they demonstrate different forms, levels, and maturity. The programme will team up with such bodies to maximise the use of the contextually settled system and infrastructure, ensuring timely and efficient operationalisation, and staying on track even after the GCF-KDB exit. It shall minimise expenses for a new office set-up with newly purchased office supplies, electronics, and furniture. This satisfies a governmental demand to minimise the creation of new mechanisms that typically fragment, duplicate, or overlap the efforts of local governments. In addition, recognising that the public sector engagement on public policy and regulatory framework is key to a genuine success of the private sector intervention, the programme will invite NDAs and line ministries of the five NOL countries to be part of the Project Steering Committee that will give guidance for the overall programme execution by leading a high-level ministerial coordination and providing advice to each country-specific activities in the country-specific Sustainable Management Unit (SMU). The contextually fit SMU governance will ensure strong country ownership and participation of individual NOL countries.

[5] UNFCCC. 2018. Climate Technology Incubators and Accelerators. Bonn: United Nations Framework Convention on Climate Change.

A

Country Ownership-built SMU Implementation Structure (indicative)



"SDG 17. Partnerships for the Goals" – Cross-border Collaboration with Advanced Innovation Ecosystems to be Convened through Global Climate Technology Providers Joining the Initiative. In pursuit of the vitalised global partnership embodied in the Sustainable Development Goal (SDG 17), the programme shall invite and attract climate technology innovators from advanced ecosystems, including, but not limited to, Canada, Denmark, Finland, the Netherlands, Singapore, the U.S., and the UK, where leading climate technologies are being developed and accelerated. By leveraging the extensive network consisting of VCs, accelerators, and other startup support intermediaries (e.g., CATAPULT, 500 Global, Techstars, Plug and Play) with global presence, global innovators in climate space will be sourced to scale up their impactful technologies to flow into the ultimate destination – the five emerging ASEAN markets as the test-bed. Some concrete examples of networks and how they are to be leveraged for the programme are as follows:

- **[KDB, Accredited Entity]** Global presence of KDB, through its Silicon Valley LLC and London and Singapore Venture Desk, will add value to the programme in sourcing global innovators in climate space through regular Global NextRound events (KDB's signature IR events), alongside leveraging its Global Partnership Fund (KDB-led Fund of Fund scheme investing in global VCs, e.g., Kleiner Perkins). Besides, KDB global branch offices will support regular/ad-hoc venture meet-ups and sourcing events in the format of demo day to reach out to top-tier climate tech providers.
- **[VC]** The GP will leverage its connection to VCs based in advanced climate responsive ecosystems, namely, Collaborative Capital, DCVC (based in San Francisco, US), Mekong Capital (based in Vietnam), SkyRiver Ventures (based in Boston, US) and PITANGO and Cardumen

Capital (based in Tel Aviv, Israel), Innovision (based in Singapore and Japan), Pana LCE (based in Singapore), Vertex (based in Singapore) that are currently executing climate tech-related investments or operating global open innovation programmes and have potential to join investment rounds for the programme. Furthermore, the vast VC network into which the GP is plugged also spreads over the ASEAN region, having room for major VCs on the ground, e.g., Archipelago Capital Partners, Openspace Ventures, Wavemaker Impact, Tin Men Capital, Genesis Alternative Ventures, Indogen Capital (based in Singapore), East Ventures, AC Ventures, Alpha JWC Ventures, Indies Capital (based in Indonesia), Viet Capital Ventures, Do Ventures (based in Vietnam), Kickstart, Gentree Investment Management, and Tallwood VC (based in Philippines), to be connected to the programme and draw climate innovators at both global and regional levels. The programme plans to strengthen collaboration with the VCs and surrounding communities for climate theme-centred seminar/IR events. Venture ESG, an EU-born, global ESG community of VCs and financiers, is a solid example in which the programme can potentially engage on such occasions.

- **[Corporate]** The GP will leverage the pool of global technology providers, including but not limited to, previously reviewed by the its connections in the US (New York), China (Beijing, Shanghai), Indonesia (Jakarta), Vietnam (Hanoi, Hochimin), and UK (London), to support businesses with potential for scalability in the ASEAN market. This initiative is part of the GP's endeavour to aid the business development of NHIS's global subsidiaries, with a special focus on Southeast Asia.
- **[Academia]** The programme will leverage global academic-corporate cooperation activities (see below) hosted by EcoLabs Centre of Innovation for Energy under Nanyang Technology University of Singapore as well as Bandung Institute of Technology of Indonesia which engages and invites global corporates in their open innovation projects that can connect with the programme.
- **[Embassies]** The programme will seek to interact with the embassies of advanced climate ecosystems operating innovation and R&D programmes to proactively access the global climate technology providers. Innovation Centre Denmark, administered by the Danish Ministry of Science, Innovation and Higher Education, and the Danish Ministry of Foreign Affairs, and Swissnex, managed by the Swiss Federal Department of Foreign Affairs and under the initiative of the State Secretariat for Education, Research, and Innovation are some of the global launchpad examples that the programme could collaborate with, ultimately to enlarge innovation on a global scale.

The Programme's Thematic Goal of Technology Transfer Shall Pitch its Prominent Role at the UNFCCC COP29 Climate Summit. Building on meaningful insights on the 9th agenda of "Matters relating to development and transfer of technologies" at the 27th session of the Conference of the Parties of the UNFCCC (COP27), at the COP29 to be hosted by the Azerbaijan in 2024, the programme will therefore pitch its role as the very first of its kind technology transfer initiative that will accelerate developing countries' capacity to uptake and digest climate responsive technologies. The year of 2024 will be a monumental year for this programme that goes beyond the BAU climate business model and embraces an ambition to tackle root causes which have hampered developing countries from being equipped with innovative capacity to combat climate change. The programme will be officially showcased at the COP29 as a kernel initiative for a sustainable future that would contribute to addressing innovation ecosystem hurdles and technology uptake/digestion barriers, thereby enhancing the developing world's access to climate responsive technologies. This technology transfer initiative will be poised to support, accelerate, boost, and scale up sustainability achievements in various technology-themed result areas. In addition, the programme will create a vibrant space for entrepreneurial participation of the conventionally disadvantaged groups such as youth in poverty, women and girls in engineering and technician fields, and vulnerable local SMEs without capacity to carry out technology-driven climate actions. In particular, the targeted ASEAN countries are to seize the golden opportunities of being powered to experience a broad spectrum of technologies that ensure sustainability in energy security, healthcare protection, water resource management, resilient agriculture, food supply chains, and so on. The programme will be part of supporting COP 29 to pave the way for a brighter future ambition to effectively tackle the global challenge of climate change.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

Climate Context of the Targeted NOL Countries. The five target countries for the programme make up 21% of the total population for East Asia and the Pacific[6]. The total population of Southeast Asia grew at an annual average rate of 1.02% between 2015 and 2020, while the urban population grew at a rate of 2.21%.[7] Economic growth in the region has increase at an annual average rate of 5%, while fossil fuel generated energy demand has increased at an annual average rate of 6%.[8] Though all economies are classified as lower middle income, poverty rate and access to goods and services vary widely within the region; countries like Cambodia and Lao PDR have higher poverty rates and lower levels of access to goods and services.[9] A table below provides a snapshot of the countries' socioeconomic context.

Socioeconomic Context for the Programme Countries[10]

	Economic Context				Access to Goods & Services		
Country	GDP ($ Billions)	Urban Population	Rural Population	Poverty Rate (%)	Food Insecurity (%)	Water Security (%)*	Access to Healthcare
Cambodia	26.96	4,092	12,496,843	17.7	15.1	28.0	0.2
Indonesia	1186.09	156,833	116,919,988	9.8	0.7	..	0.5
Lao PDR	18.82	2,742	4,682,315	18.3	8.3	18.0	0.4
Philippines	394.08	54,302	59,577,632	16.7	4.8	47.0	0.6
Vietnam	366.13	37,088	60,379,495	6.7	0.6	..	0.8

**Cells displaying ".." indicate that data are not present within World Bank indicators database for select*

Climatic patterns in the region are somewhat consistent with defined dry and wet seasons, generally substantial annual rainfall amounts, consistently moderate to high temperatures with intermittent extreme heat conditions, and annual cyclonic activity. The countries have a collective 142,000 km of coastline and support several large transboundary rivers (i.e., Mekong River (Cambodia), Tonle Sap River (Cambodia), Red River (Vietnam)) whose floodplains provide delicate ecosystems to cultivate rice and other crops. In all countries, national plans and policy frameworks have emphasised the critical need for climate technology in response to the countries' high climate vulnerability and aspirations for low-carbon and climate-resilient sustainable economic development.

- **Need for Adaptation.** Southeast Asia is considered one of the most climate vulnerable regions in the world.[11] The climate hazard trends in the target countries show increases in rising atmospheric temperatures, changing precipitation patterns, sea level rise, ocean warming, and cyclones. Populations, natural systems, and built infrastructure within the region are experiencing increased exposure to these climate hazards. Combined with pre-existing vulnerabilities that communities experience—high rates of poverty and inequality, urbanisation and population density, insufficient healthcare systems, unsustainable agriculture, inadequate water resources and infrastructure, and

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[6] World Bank (2023). "Indicators." https://data.worldbank.org/indicator/SP.POP.TOTL?locations=Z4
[7] Asean (n.d.). "Chapter 5: Urbanisation Wave and ASEAN Regional Agenda." Global Megatrends: Implications for the ASEAN Economic Community. https://asean.org/wp-content/uploads/2017/09/Ch.5_Urbanisation-Wave-and-ASEAN-Regional-Agenda.pdf
[8] Center for Strategic and International Studies (2022). "Southeast Asia's Challenge of Decarbonizing While Growing Rapidly." https://www.csis.org/analysis/southeast-asias-challenge-decarbonizing-while-growing-rapidly
[9] World Bank (2022). "New World Bank country classifications by income level: 2022-2023." https://blogs.worldbank.org/opendata/new-world-bank-country-classifications-income-level-2022-2023#:~:text=The%20World%20Bank%20assigns%20the,the%20previous%20year%20(2021).
[10] World Bank (2023). "Indicators." https://data.worldbank.org/indicator
[11] USAID (2022). "Addressing the Climate Crisis in Southeast Asia: A Regional Approach." https://www.usaid.gov/asia-regional/fact-sheets/addressing-climate-crisis-southeast-asia-regional-approach#:~:text=Southeast%20Asia%20is%20globally%20considered,intense%20and%20unpredictable%20weather%20events.

gender inequality—significant impacts (i.e., realised risks) will lead to predominantly negative effects on human health, productivity, the economy, financial resources, and more.

A below table shows the key climate-related natural hazards and vulnerabilities based on reviewed data and evidence; per capita GHG emissions; and the Notre Dame Global Adaptation Initiative Index (ND-GAIN)[12] score, which is a summary of a country's vulnerability to climate change impacts and its capacity to improve its resilience and adapt to climate impact.[13] The Climate Rationale (Annex 24) reveals a clear resilience gap in the five countries—evidenced by recent climatic trends and some empirical evidence for the effect on people, economies, natural systems, and infrastructure—and an increasing adaptation need demonstrated by accelerating effects from climatic changes.

Climate Hazards and Vulnerability Indicators by Country

			Cambodia	Indonesia	Lao PDR	Philippines	Vietnam
Climate Hazard Category	Temperature-related hazards	Extreme heat	●	●	●	●	●
		Ocean warming		●			
	Water-related hazards	Changing precipitation	●	●	●	●	●
		Sea level rise	●	●		●	●
	Wind-related hazards	Tropical cyclone	●			●	●
Vulnerability Indicators	Poverty		●	●	●	●	●
	Urbanisation			●		●	
	Insufficient healthcare		●	●	●		
	Unsuitable agriculture		●		●		●
	Inadequate water resources		●	●	●	●	●
	Gender inequality		●	●	●	●	●
GHG-Emissions (per capita)			1.0	2.3	2.6	1.3	3.5
ND-GAIN ranking			133	107	145	115	126

- **Need for Mitigation.** High rates of urbanisation and economic development coupled with an increase in energy access have led to a greater dependence on fossil fuels and higher emissions in Southeast Asia. In Vietnam, for example, rapid economic growth has propelled a steep increase in vehicle ownership – car registration increased 15% per year from 2014 to 18 - with transport emissions representing 11% of the country's GHG emissions in total. With one of the world's highest deforestation rates – at 1.2% annually – Southeast Asia experienced the most rapid increases in carbon dioxide emissions in the world between 1990 and 2010.

  To meet the Paris climate pledges, ASEAN nations will have to reduce emissions by 11% by 2030 as compared to current trajectories. Present lagging progress for needed investment and emissions rate of change, the International Energy Agency (IEA) concludes that making improvements in regulatory and financing frameworks in Southeast Asia would reduce the costs of clean energy projects. For example, the levelised cost of energy (LCOE) of solar PV in Indonesia could be around 40% lower if its investment and financing risks were comparable to advanced economies.

[12] *ND-GAIN Country Index.* University of Notre Dame. July 21. https://gain.nd.edu/our-work/country-index/

[13] Countries are given a combined score of vulnerability and adaptive capacity and ranked amongst nearly all sovereign nations (the index includes 182 countries). The bottom row in the table below identifies the relative ND-GAIN score for the countries, with most countries ranked in the latter half of the Index (i.e., below at least 90 countries), signalling vulnerability and a lack of adaptive capacity in the face of climate-related risks.

NOL Countries' Prioritised Climate Responsive Technology Needs. To better define the range of optimal interventions against these adaptation and mitigation needs, the programme performed climate vulnerability situation analysis and technology needs assessment, with reference to Technology Needs Assessment Step by Step guidebook (UNEP ETU Partnership, 2019). Extensive review of country-specific climate vulnerabilities, national development plans including NDCs, NAPs, and other climate strategies, relevant policies, multi-disciplinary assessments, and multi-level stakeholder consultations, including potential technology and/or innovation suppliers, revealed a range of appropriate technologies for which there is deployment demand pull, which can be segmented into seven (7) technology groups. A Table below shows these prioritised climate technology groups mapped to each country and GCF results area. For more information on the methodological approach and country-specific and/or cross-country analysis, see Annex 2.

Prioritisation of Climate Responsive Technology Needs

[Step 1] Climate Vulnerable Sector Prioritisation
Identify the target sectors via climate vulnerability situation analysis

[Step 2] Climate Technology Prioritisation Criteria Settting
Develop unified criteria across all sectors for technology prioritisation

[Step 3] Criteria Weighting
Consult with key stakeholders to allocate weighting for each criteria

[Step 4] Climate Technology Rating and Scoring
Rate and score each criteria per technology with a pre-determined scale

[Step 5] Climate Technology Prioritisation
Identify top priority technologies in each sector with sensitivity analysis

Grouping of Seven (7) Prioritised Climate Technology Needs

Technology Group (TG)	Sub Technologies	Result Area(s)	C	I	L	P	V
TG 1: Renewable energy generation	Ocean energy	M1		○		○	○
	***Bioenergy (renewable biomass)	M1		○	○	○	○
	Small-scale hydropower	M1			○		
	Solar energy	M1	○	○	○		○
	Wind energy	M1		○			○
	Other**	M1	○	○	○	○	○
TG 2: Transmission, distribution, and electricity storage	Modern renewable off-grid energy systems	M1	○		○	○	
	Connection of isolated grid	M1			○		
	Battery storage	M1	○			○	
	Other**	M1	○	○	○	○	○
TG 3: Low-emission transport	Low-carbon/energy efficient mass transport	M2	○	○			○
	EVs (e.g., e-cars, e-motorbikes, e-bicycles)	M2	○	○	○	○	○
	EV infrastructure	M2	○		○		
	Alternative transport fuels (next generation advanced zero-emission biofuels, green hydrogen)	M2		○	○	○	○
	Other**	M3, M1	○	○	○	○	○
TG 4: Residential and industrial energy efficiency technologies	Building energy modelling	M3, M1					
	Lighting	M3	○		○	○	○
	HVAC	M3, M1	○		○	○	○
	Appliances	M3, M1	○	○	○	○	○

	Other**	M3, M1	○	○	○	○	○
TG 5: Agricultural technologies and practices	Sustainable agriculture	A1, A2	○	○	○	○	○
	Climate-resilient plant/animal genetics/breeding	A1, A2	○	○			○
	Water saving and efficient irrigation methods	A2	○	○		○	○
	Flood and drought management technology	A1		○	○		○
	RE-powered equipment	A2	○	○	○	○	○
	Other low carbon and climate-resilient farming methods and climate-linked insurance products	A1, A2		○	○	○	○
	Other**	A1, A2	○	○	○	○	○
TG 6: Water management and treatment	Nature based solutions	A2		○	○		○
	Management of water resources	A2	○	○	○	○	○
	Small-scale, decentralised wastewater plants	A2	○	○			
	Wastewater treatment	A2	○				○
	Water treatment processing	A2	○	○			
	Water desalination	A2				○	
	Other**	A2	○	○	○	○	○
TG 7: Waste management and treatment	Mechanical-biological treatment	A2, M1	○	○	○		○
	Aerobic/semi-aerobic digestion	A2, M1	○		○		
	Waste to energy	M1	○		○	○	○
	Composting	A2, M1	○	○			○
	Other**	A2, M1	○	○	○	○	○

** [Abbreviation by Country] Cambodia (C), Indonesia (I), Laos (L), Philippines (P), and Vietnam (V)*
** [Indicator] O = Country provided examples of the respective technology needs or generic area of the needs*
*** Other is included as a technology group category to indicate that technology deployment opportunities will evolve over time as brand new and/or unprecedented climate responsive technologies are brought to market over the life of the investment cycle (i.e., by earlier 2030s) and that not all possible solutions are presently identified. The UN Climate Technology Centre & Network (CTCN) Taxonomy[14] or equivalent can be referenced for potential technologies that may be considered over the life of the fund.*
**** In case of bioenergy (renewable biomass), relevant investments should not give rise to negative climate and/or sustainability concerns.*

Tailored and Focus Investment Plan by Country. The indicative priority sectors have been identified as below by the GP, while the investment is open for all technology needs identified by the NOL countries. See Annex 25 for more information on GP investment plan.

- **Cambodia –** The GP sees impactful investment opportunities in agricultural technologies including but not limited to water saving, irrigation, flood and drought management, and climate-resilient crops/livestock/fishery technologies. While Cambodia aims to incorporate renewables into its energy mix, the GP seeks to invest in technologies and businesses that increase the efficiency of renewable energy generation, in the middle of the availability of the technologies together with government alignment and support.
- **Indonesia –** The GP seeks to invest in technologies in the area of the EV and adjacent ecosystem for the sake of low-carbon transformation in mobility, along with the electrification of transportation pushed by the Presidential Degree. Another focus is the entire agro-food sector contributing to 35% of the populous country's GDP.
- **Laos –** The GP will prioritise agriculture tech practices and waste amid a specific regulatory framework not yet established. In addition, as for forest management, it is believed that public-private-community partnership-based investment would be needed in the future as below.

Public/Government	NH ARP	Community

[14] UN Environment Programme, Climate Technology Centre & Network Taxonomy (published December 15, 2017): https://www.ctc-n.org/resources/ctcn-taxonomy

- Sustainable wood and NTFP supply and ecosystem services (forest cover, carbon sinks, biodiversity soil, water) provision - Political, institutional, and financial risks	- Climate impactful investments in forestry relevant industries with a reasonable return - policy, technical, financial and reputational risks	- Increased food security, reduced poverty, sustainable livelihoods resilience to climate-impacted market fluctuations - Risks mainly related to land access and livelihoods

- **Philippines –** The GP will prioritise renewable energy generation and low-carbon transportation as it is the most in need of solution, along with the governmental adoption of a national energy policy and regulatory framework with incentives to increase the use of electric and hybrid vehicles.
- **Vietnam –** The GP will mainly see opportunities in industries of renewable energy generation and storage and low-carbon transportation, in partnership with other financing entities with the aligned mission for climate action. Relevant industrial ecosystems – e.g., establishment of a value chain for greener battery production beyond the traditional lead-acid battery production will be supported under the programme.

Most likely Scenario in the Absence of Intervention. In the absence of technological intervention for a shift from carbon-intensive to low carbon development and increased climate resilience, climate driven impacts will continue and worsen. By 2100, rising temperatures in the region will reduce agricultural yields by 50% compared to 1990 levels[15] and decrease agricultural labour productivity by 8.8% due to heat stress.[16] Water stress in the region is projected to affect nearly 185 million people by 2050.[17] Sea level rise is predicted to cause a loss of nearly 1% of land and associated capital assets by 2050[18]—much of this damage occurring in highly populous, coastal cities such as Jakarta (Indonesia), Manila (Philippines), and Ho Chi Minh (Vietnam).[19] Simultaneously, carbon-intensive growth trends will continue.[20] Energy demand is projected to rise at an annual rate of 3% per year in Southeast Asia through 2030; 75% of this growth will be met by fossil fuels causing a 35% increase in CO_2 emissions.[21] Countries will continue to be reliant on fossil fuel and vulnerable to energy market shocks, outdated infrastructure, and energy supply fluctuations. Rural and remote areas will remain underserved, urban buildings will remain inefficient, and clean transportation infrastructure underdeveloped.

Technological Leapfrogging is Necessary for Quick, Strong, and Enhanced Climate Actions. Urgent climate solutions, in short, immediate actions, need to be deployed throughout the five countries during this state of climate urgency. Many climate technologies available today, like renewables and lithium-ion batteries, have recently emerged to compete with emission-intensive technologies; it has taken decades, not years, for such technologies to mature to the point where markets can bring them to scale and replace outdated rivals. Developing countries cannot wait that long for the next generation of clean technologies to become competitive in their markets to reach current climate goals. This is even more urgent in the context of the five target countries where the mainstream investment focus is tilted towards IT, fintech, and mobile industries, calling for the needs to source adequate climate technology to fill in the gaps with effectiveness.

The programme, therefore, appeals to the global community to collaborate in order for the countries to leapfrog over emission-heavy, inefficient, and polluting technologies, moving straight towards low-carbon, efficient, and clean technologies; they should not revisit the past industrialisation trajectories and

[15] Asian Development Bank (2020). "Climate Change in Southeast Asia: Focused Actions on the Frontlines of Climate Change." https://think-asia.org/bitstream/handle/11540/720/climate-change-sea.pdf?sequence=1
[16] International Labour Organization (2019). "Working on a warmer planet: The impact of heat stress on labour productivity and decent work." https://www.ilo.org/wcmsp5/groups/public/---dgreports/---dcomm/---publ/documents/publication/wcms_711919.pdf
[17] Asian Development Bank (2015)."Southeast Asia and the Economics of Global Climate Stabilization." https://www.adb.org/sites/default/files/publication/178615/sea-economics-global-climate-stabilization.pdf
[18] Asian Development Bank. "Climate Change in Southeast Asia: Focused Actions on the Frontlines of Climate Change." https://think-asia.org/bitstream/handle/11540/720/climate-change-sea.pdf?sequence=1
[19] Asian Development Bank (2015). "Southeast Asia and the Economics of Global Climate Stabilization." https://www.adb.org/sites/default/files/publication/178615/sea-economics-global-climate-stabilization.pdf
[20] IEA (2022). "Southeast Asia Energy Outlook 2022." https://iea.blob.core.windows.net/assets/e5d9b7ff-559b-4dc3-8faa-42381f80ce2e/SoutheastAsiaEnergyOutlook2022.pdf
[21] IEA (2022). "Southeast Asia Energy Outlook 2022." https://iea.blob.core.windows.net/assets/e5d9b7ff-559b-4dc3-8faa-42381f80ce2e/SoutheastAsiaEnergyOutlook2022.pdf

undesirable legacies – especially, the chief culprits of climate change – of the Industrial Revolution that have swept through today's developed countries for the last several decades.

A Conclusion: Technology Leapfrogging via Technology Transfer. Technological leapfrogging is achievable with technology transfer from industrialised countries to late-comer economies. The low-carbon and climate-resilient capacity of the developing world can be promptly fostered by absorbing commercially proven frontier technologies, skipping the technological evolution process that can take decades of experiments and trials and errors to get to the maturity point. Therefore, the programme shall offer a window of leapfrogging opportunities for partner countries to opt for, customise, apply, deploy, and scale best-fit and innovative technologies under cross-border technology transfer mechanisms.

Significance of Technology Transfer Globally Recognised but Gaps between Knowing and Doing. Achieving the objectives and goals of the Paris Agreement requires technological innovation that inevitably entails a widespread transfer of mitigation and adaptation technologies addressing the needs of developing countries; innovation consumes significant time and energy so accelerated progress is necessary. Decades of global discourse have generated a consensus that technology transfer is a core enabling element to limit global average temperature rise to 1.5℃. Regardless of such global recognition and some trials, barriers have continued to affect a broad set of processes covering the flow of innovative climate technology solutions being transferred to emerging markets: surely those of the ASEAN region.

(1) Addressing Supply-sided Controversy over Intellectual Property (IP) Rights. In actuality, the programme theme *'climate technology transfer'* has raised a fundamental and controversial question on IP rights on whether developed economies have a commanding position over patents in climate technologies as apparent market trend reflects. Real cases have disclosed that patent holders are reluctant to engage or simply refuse to license their patented technologies to a third party, while expressing fears of unreasonable competition from the licensee that could weaken their competitive advantage. Several companies expressed concern as they have been reminded of real-world experiences of others in foreign countries whose bargaining power enforced technology transfer to local actors through mechanisms outside the patent based system. Such cases have become more prevalent and protective actions by innovators have hindered developing countries from gaining access to adequate technology solutions.

The programme shall therefore catalyse the appropriate regime shift throughout relevant ecosystems, leveraging the bargaining power of the UNFCCC under which 197 Parties have committed to climate technology transfers. It particularly proposes a compulsory licensing framework utilising standard terms as a patent protection and safety tool, under which global players will voluntarily engage in technology transfer activities and local recipients will enjoy timely access to patent protected technologies without abusive practices sometimes committed by patent holders; i.e., breaking the vicious circle of distrust, secrecy, and patent abuse, such as excessive pricing. "Technology transfer can rarely stand on its own; it gets its support from a framework of intellectual property rights and know-how protection (Erstling,1992, p.215).[22]"

(2) Addressing Demand-sided Barriers via Ecosystem Building. Bringing climate technologies into transformative and scalable solutions in the five economies has been delayed due to barriers on the ground – though at different levels of severity according to varying stages of development. They are associated with limited government initiatives and public policies in support of innovation; heavy bureaucracy that frustrates effective inter-ministerial coordination; institutional capacity on technology based R&D in terms of budget, physical facilities, and qualified staff; shortage of risk capital and dormant private sector engagement; insufficiency of market readiness; a small-scale

[22] Jay Erstling. 1992. International Technology Transfer and Intellectual Property Rights: Some Essentials and Options for Technology Transfer Partners.

technology business incubation community; and stagnant inequalities and insecure social groups, including disadvantaged women and girls with low levels of literacy in science, technology, engineering, and mathematics (STEM).

With barriers found across different segments in shaping societies and market dynamics of the five economies, it is remotely possible to single out definitive root causes of the continuing challenges that hinder climate technology leaps and innovation potentials. On that account, the programme shall proactively endorse the appropriate ecosystem which can cover multi-dimensional barriers that have long been locked in a vicious interlinked circle.

(3) Minimising Fragmentation and Inefficiency in Harmony with On-going Interventions. Since early stages of idea brainstorming and concept designing, KDB has taken a proactive approach to explore ways of harnessing complementarities amongst a scattered set of similar interventions at both national and regional levels across partner countries. Market assessments indicate support is available in small slices from several different channels. A notable initiative is the Global Cleantech Innovation Programme (GCIP), supported by the Global Environment Facility (GEF) and implemented by the United Nations Industrial Development Organisation (UNIDO). Knowledge exchange and conceptual discussions with the UNIDO team were conducted to avoid wasteful duplications of work and inefficient spending of funds so that the two initiatives can operate in parallel and generate synergies to the fullest capacity possible.

Besides, the programme plans to make potential synergy points with other global or regional initiatives such as UNDP Country Accelerator Labs, ADB Mekong Business Initiative (MBI), the Mekong Challenge Programme, and GGGI Global and Regional Greenpreneurs Program: e.g., sharing lists of key stakeholders and lessons learnt from completed consultations, introducing a pool of graduates from other incubation programmes, and discussing joint matchmaking or networking events. In a nutshell, this initiative aims to avoid fragmentation between on-going, existing initiatives. Above all, it will build on existing infrastructures and resources on the ground – e.g., the five countries' business acceleration entities – for a true ownership and capacity enhancement of the five countries, thereby maximising complementarity and synergism with existing initiatives for technology business entrepreneurship.

(4) Unprecedented Uncertainty and Unpredictability in the Time of COVID-19 and Beyond. As the coronavirus has taken hold of the world and introduced a new level of uncertainty, the ASEAN bloc has been hit hard as much of the rest of the globe. An obvious downside is the adversity most entrepreneurs face amid economic fluctuations; in addition, it takes even longer to turn technologies into scalable market solutions during times of extreme stress. The Global Innovation Index (GII) 2021 (World Intellectual Property Organisation, 2021) indicates "a severe cutback in innovation investments (p.9)" throughout COVID-19, scoring the lowest innovation performance of economies over the course of the pandemic crisis. The programme shall therefore provide ASEAN-centric acceleration services with USD 200 million in patient capital amid the crisis like no other in modern times

Complementarity and Coherence to Existing Projects/Interventions. The programme aims to proactively endorse and create the appropriate ecosystem which can overcome multi-dimensional barriers that have collectively slowed progress. The R&DB Programme is the only equity-based platform that invests in SMEs in Southeast Asia to accelerate climate mitigation and adaptation entrepreneurship. Its targeted support for capacity building with institutions across the region, with active partnership and acceleration activities to match technology providers and entrepreneurs and facilitate market entry, and venture capital investing to move early-stage companies toward more stable financial foundations make the programme unique.

Since early stages of idea brainstorming and concept designing, KDB has taken a proactive approach to explore partnerships amongst a scattered set of similar interventions at both national and regional levels across partner countries. For example, project partners engaged the Global Cleantech Innovation Programme (GCIP),[23] to exchange knowledge and avoid wasteful duplications of work and inefficient spending of funds. KDB plans to actively connect and find synergies with other global or regional initiatives such as UNDP Country Accelerator Labs, ADB Mekong Business Initiative (MBI), the Mekong Challenge Programme, and GGGI Global and Regional Greenpreneurs Program, and share lessons learned as well as synergies.

Moreover, it will leverage the bargaining power of the UNFCCC under which 197 Parties have committed to climate technology transfers. It particularly proposes a compulsory licensing framework utilising standard terms as a patent protection and safety tool, under which global players will voluntarily engage in technology transfer activities and local recipients will enjoy timely access to patent protected technologies without abusive practices sometimes committed by patent holders; i.e., breaking the vicious circle of distrust, secrecy, and patent abuse, such as excessive pricing.

In sum, the programme will (i) address supply-sided controversy over intellectual property (IP) rights, (ii) address demand-side barriers via ecosystem building, (iii) improve supply-side opportunity through joint venture nurturing and critical growth financing, and (iv) minimise fragmentation and inefficiency in harmony with on-going interventions.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

Key Concept beneath the Surface of a Paradigm Shifting Transition. The UNFCCC (2018)[24] spotlights the importance of supporting developing countries to source, digest, and deploy climate technology solutions through technology development and transfer in advancing mitigation and adaptation actions amid rapidly exacerbating climate crisis. Against this backdrop, the programme has conceived its key profound concept: '*technology transfer via a joint venture (JV) as a result of collaborative R&DB.*' Technology innovation is all about massive collaboration; the range and types of required resources are so extensive that it is unlikely that those wishing to innovate can achieve it alone, even if they are exceptionally strong in their own areas. Therefore, the programme suggests a collaborative R&DB ('B' stands for business) module – further advanced than the existing collaborative R&D or R&DD ('D' stands for demonstration) module, taking one step forward to embrace both business modelling and investing.

Addressing Double Sided Barriers on the Part of Both the 'Receiver' and the 'Provider.' The programme shall serve the five ASEAN member states to achieve a genuine paradigm shift towards low-emission and climate-resilient growth pathways, based on the premise that the right intervention addressing both demand and supply side barriers can ensure the realisation of technology transfer into emerging markets. In short, this transfer initiative pays special attention to the hurdles that global innovators of the relevant supply chain in industrialised economies face, in addition to those confronted by the five recipient countries. Recognising that motivation for technology transfer need exists on both sides – not only recipients but also providers, the programme will tackle the joint set of demand and supply barriers, which differentiates it from previous trials to move disruptive technologies into developing countries. Under the programme designed with four components, Components 1 and 4 address demand-side barriers while the remaining two components manage supply-side obstacles as follows:

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[23] Supported by the Global Environment Facility (GEF) and implemented by the United Nations Industrial Development Organisation (UNIDO)

[24] UNFCCC. 2018. Climate Technology Incubators and Accelerators. Bonn: United Nations Framework Convention on Climate Change.

(1) Components 1 and 4 shall address demand-side barriers of the five different countries with varying forms and depths of intervention as the severity, facets, and dynamics of such barriers differ between countries: e.g., from a complete impasse and near-total lack of an ecosystem to a massive surge of the market with new-found popularity. In a nutshell, under the same umbrella, all five partner countries will enjoy acceleration activities that best fit individual local contexts. In fact, the current, conventional model of acceleration is not suitable to the emerging market context as it was borne over the course of supporting ICT start-ups in Silicon Valley. The two components will devise and deploy customised acceleration modules for the recipients, and will do its best to integrate it into the local ecosystems once proven contextually workable.

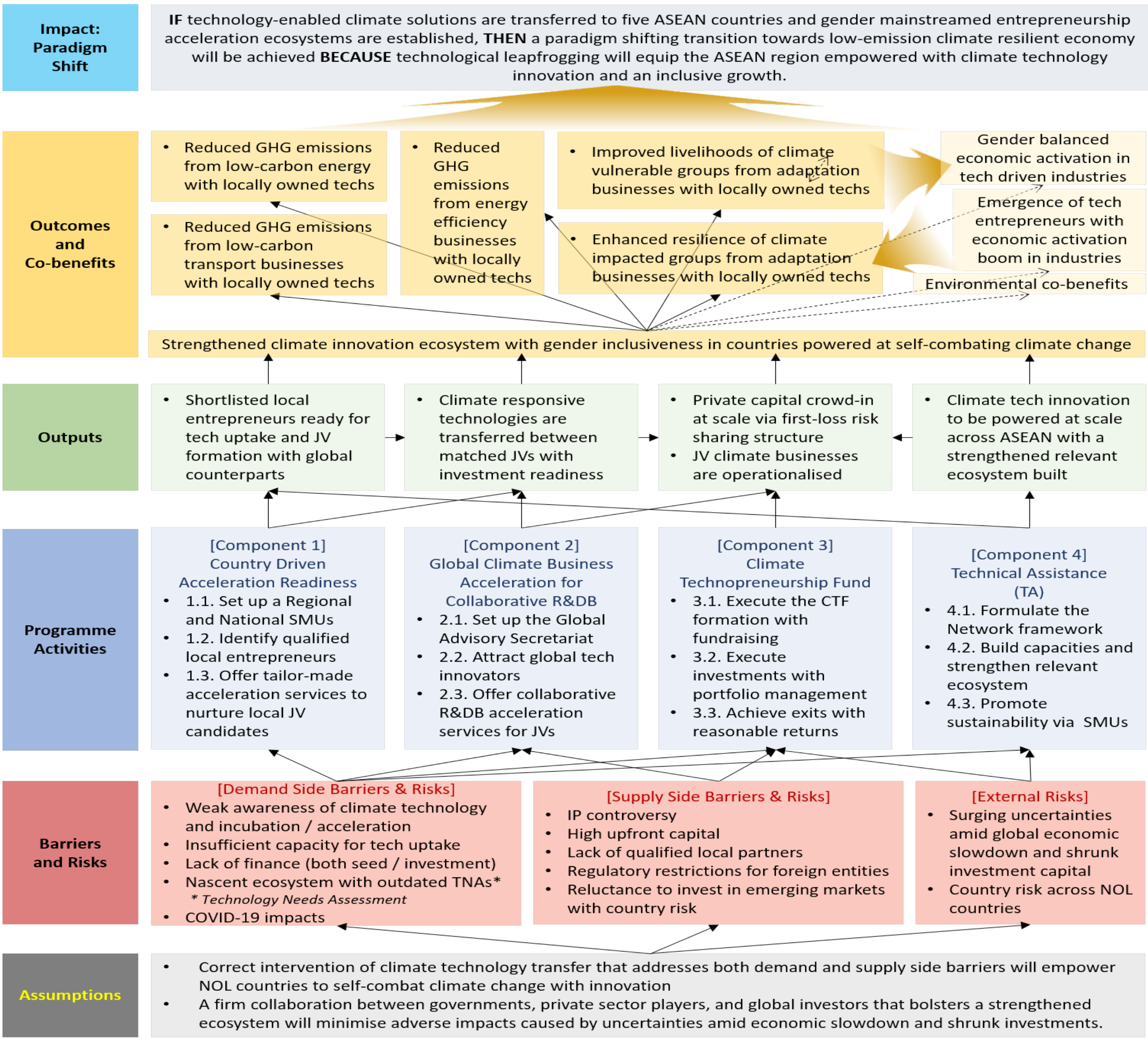
(2) Components 2 and 3 shall address supply-side barriers that have barred global technology providers, talented innovators, and private investors from carrying out technology-intensive transactions in emerging markets. Besides IP controversies prescribed in B1, there exist a plethora of barriers including, but not limited to, high upfront costs due to immature supply chains of a nascent market not integrated into the global trading system, difficulties in teaming up with the right local business partners like buyers, agents, distributors, and licensees, regulatory restrictions discriminating against foreign entities, and commercially supposed country risk factors.

Diversified Mitigation/Adaptation Potentials via Joint Venture (JV) Business Intervention that Leads Technological Leapfrogging in NOL Countries. Fitfully trained JVs will be supported throughout R&DB acceleration and patient capital injection to be able to deploy their business solutions equipped with climate responsive technologies. Over the course of the businesses being mature in market ecosystems to be nurtured with the TA intervention, their low-carbon and climate-resilient businesses will occupy a key space, be scaled, and eventually replace outdated rival businesses that feature emission-heavy, inefficient, and polluting technologies.

By immediately translating technology-based concepts and ideas into climate actions, JV businesses will accelerate a diffusion and stabilisation of the next generation of climate responsive technologies so that the transferred technologies will soon become competitive in individual markets, thereby moving straight towards reaching NDC and relevant climate goals. Meanwhile, the NOL countries will see a gender mainstreamed entrepreneurship ecosystem with a surging technologically skilled, capable, and passionate female entrepreneurs and women-led SMEs as the key innovation agency that will propel bright climate solutions after mainstreaming transferred technologies and applying them into locally customised businesses.

Case-specific ToC Scenarios on What Paradigm Shifting Changes to be Derived from JV Business Deployment and its Consequences. The programme will see newly created JVs with investment readiness to implement climate-technology enabled projects (Output of ToC) that will bring about mitigation and adaptation outcomes in a society with an emergence of female talents (Outcomes with Co-benefits of ToC), which will lead to a paradigm shifting innovation in the climate responsive entrepreneurial ecosystem of the five countries (Impacts of ToC). Case-specific ToC scenarios assume exemplary cases as below.

Theory of Change (ToC)



ToC Scenario 1: Post-harvest Processing Agro-Tech for Cambodia's Food Security

Impacts	Agro processing tech innovation will empower Cambodia to obtain climate resiliency in agriculture, leading the national leapfrogging in the global agricultural supply chain.	
▲		
Outcomes & Co-benefits	Enhanced capacity of food security in climate vulnerable agricultural areas with engagement of female talents throughout the processing supply chain	
▲		
Outputs	JV's agro-processing business deployment with local farmers' tech literacy enhancement	
▲		
Activities	Newly created agro-tech JV with investment readiness after acceleration services	
▲		
Barriers	**[Demand side barriers]** Weak technology capacity of post-harvest processing, systemic barriers in O&M, high	**[Supply side barriers]** High upfront capital burden and regulatory restrictions for foreign entities, lack of

	upfront costs for local farmers, COVID-19 impacts, etc.	qualified local partners, reluctance to invest in nascent agro-processing market
▲		
Rationale for Intervention	JV intervention that addresses demand and supply side barriers will empower Cambodia to self-combat low productivity and food insecurity caused by shifting climate and heat waves.	

ToC Scenario 2: AI-based Epidemic Warning Tech for the Philippines' Climate Resilient Healthcare

Impacts	AI-applied tech innovation will empower the Philippines to timely control epidemic disasters and obtain climate-resilient healthcare management system.	
▲		
Outcomes & Co-benefits	Enhanced capacity of tracking and managing infectious diseases with remedy actions in climate vulnerable areas with an emergence of female technicians and engineers.	
▲		
Outputs	JV's business deployment with localised big data analyses and diseases tracking software	
▲		
Activities	Newly created AI-tech JV with investment readiness after acceleration services	
▲		
Barriers	**[Demand side barriers]** Lack of access to relevant technology sources, high upfront costs, COVID-19 impacts intertwined with climate effects, etc.	**[Supply side barriers]** Regulatory ownership / investment restrictions for foreign entities, lack of qualified local partners, IP controversy, etc.
▲		
Rationale for Intervention	JV intervention that addresses demand and supply side barriers will empower the Philippines to self-combat uncontrollable spread of infectious diseases deteriorated by climate impacts.	

ToC Scenario 3: Clean/Economic Ocean Energy Technology for Vietnam's Energy Transition

Impacts	Clean ocean energy innovation will empower Vietnam to initiate low-carbon energy transition.	
▲		
Outcomes & Co-benefits	Reduced emissions (ca. 2.9 kt) with an increased access to electricity in isolated areas with engagement of female talents throughout the energy generation and transmission process	
▲		
Outputs	JV's clean and economic wave energy deployment in isolated rural areas	
▲		
Activities	Newly created wave energy tech JV with investment readiness after acceleration services	
▲		
Barriers	**[Demand side barriers]** Lack of awareness on technology amongst local stakeholders, high upfront costs	**[Supply side barriers]** High upfront capital and commercial funding gap, lack of qualified local partners
▲		
Rationale for Intervention	JV business intervention that addresses demand and supply side barriers will empower Vietnam to increase access to electricity in remote areas with energy transition.	

ToC Scenario 4: 4D Drone Control Platform for Laos' Climate Resilient Land Management

Impacts	Land management innovation will lead Laos into a leapfrogging move towards climate resiliency of the forestry sector and its vulnerable communities.
▲	
Outcomes & Co-benefits	Systemic land management and agricultural livelihoods improvement with active participation of women in vulnerable mountainous areas

▲		
Outputs	JV's cloud-based intelligent 4D ground control platform business that implements the mapping, real-time monitoring operation of drones in climate-damaged land areas	
▲		
Activities	Newly created 4D tech JV with investment readiness after acceleration services	
▲		
Barriers	**[Demand side barriers]** Lack of qualified local entities for technology uptake, high upfront costs for the vulnerable	**[Supply side barriers]** Lack of qualified local partners, nascent relevant innovation ecosystem
▲		
Rationale for Intervention	JV intervention that addresses demand and supply side barriers will empower Laos to self-combat against agricultural losses, caused by climate induced extreme weather events, on which over 75% of the total population relies for livelihoods.	

ToC Scenario 5: Industrial Energy Efficiency for Indonesia's Leap of Low-carbon Growth

Impacts	Energy efficiency innovation will empower Indonesia to make a leap of a low-carbon advancement, leading the national leapfrogging in the post-pandemic economic growth.	
▲		
Outcomes & Co-benefits	Reduced emissions across industries equipped with energy efficient equipment and technologies with engagement of female engineers and technicians in the market	
▲		
Outputs	JV's energy efficiency business deployment throughout the industrial manufacturing process	
▲		
Activities	Newly created agro-tech JV with investment readiness after acceleration services	
▲		
Barriers	**[Demand side barriers]** Lack of willingness for local stakeholders under no incentive measures, lack of energy efficiency finance, high upfront costs, etc.	**[Supply side barriers]** Reluctance to entre the nascent market, upfront capital burden and regulatory restrictions for foreign entities, etc.
▲		
Rationale for Intervention	JV intervention that addresses demand and supply side barriers will empower Indonesia to self-combat long delayed energy efficiency initiation without any successful precedents.	

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Outcome Mapping to GCF Result Areas with the Investment Portfolio Scenario Equipped with Robust Investment Criteria in terms of Eligibility and Selection. The CTF considers investment opportunities in five (5) result areas under a programmatic approach taken with a strong set of investment criteria at stages of eligibility screening and beneficiary selection. In that technologies, companies, and sub projects are not identified yet, all the five result areas are marked.

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	**MRA 2** Low-emission transport	**MRA 3** Building, cities, industries, appliances	**MRA 4** Forestry and land use	**ARA 1** Most vulnerable people and communities	**ARA 2** Health, well-being, food and water security	**ARA 3** Infrastructure and built environment	**ARA 4** Ecosystems and ecosystem services

High level Outcome (Strengthened climate innovation ecosystem with gender inclusiveness in countries powered at self-combating climate change)	☒	☒	☒	☐	☒	☒	☐	☐
Second level Outcome 1 (Mitigation from low-carbon energy with locally owned technologies)	☒	☐	☐	☐	☐	☐	☐	☐
Second level Outcome 2 (Mitigation from low-carbon transport businesses with locally owned technologies)	☐	☒	☐	☐	☐	☐	☐	☐
Second level Outcome 3 (Mitigation from energy efficiency with locally owned technologies)	☐	☐	☒	☐	☐	☐	☐	☐
Second level Outcome 4 (Improved livelihoods of climate vulnerable groups from adaptation actions with locally owned technologies)	☐	☐	☐	☐	☒	☐	☐	☐
Second level Outcome 5 (Enhanced resilience of climate impacted groups from adaptation businesses with locally owned technologies)	☐	☐	☐	☐	☐	☒	☐	☐

Ex-ante Prediction of Co-benefits with Scenario-based Evidence. The programme predicts that the five ASEAN members will enjoy variety of co-benefits, apart from adaptation and mitigation outcomes. For instance, energy efficient industrial processing technologies may align with co-benefits in environmental (less air pollution), gender sensitive (gender balance by employing female engineers and technicians) and economic and industrial (economic activation across climate and green growth sectors) areas.

Co-benefit Number	Environm't	Social	Economic	Gender	Adaptation	Mitigation
Gender balanced economic activation in tech driven industries	☐	☐	☐	☒	☐	☐
The emergence of tech entrepreneurs with economic activation boom in industries	☐	☐	☒	☐	☐	☐
Environmental co-benefits	☒	☐	☐	☐	☐	☐

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

Designing the Programme that Addresses Both Demand and Supply Side Barriers to Empower the ASEAN Community via Climate Technology Transfer and Innovation Acceleration. The programme shall address the systemic process of climate technology innovation that underpins a range of interacting actors with business accelerators anchored. It is structured with four different components using a two-tiered approach: locally driven (related to demand side barriers) and globally inspired (related to supply side barriers). Individual components and activities will closely function in order to reinforce synergies by bridging the gaps which have kept the five countries from a much-needed innovation leap.

<u>Designing Barriers-based Activities</u>

Pressing Barriers to be Addressed	Activities as Solutions
• Scattered small-scale international/national start-up support trials that are mostly grant based one-off events without an inter-ministerial level policy and regulatory engagement	• (Activity 1.1) Set up a National Sustainability Management Unit (SMU) that hold the Steering Committee with the government level engagement
• Limited awareness of climate technology, entrepreneurship and business acceleration with an unbalanced access to existing market opportunities by gender and others (geography, socio-economic status, language, etc.)	• (Activity 1.2) Launch contextually fit sourcing strategies to identify qualified local JV candidates with a gender focus (e.g., recruiting female students via partnership with academic institutions)
• Lack of qualified entrepreneurs with climate responsive technology uptake capacity (tech digestion capabilities, transparent IP trade concept, communication skills)	• (Activity 1.3) Offer tailor-made acceleration readiness services to nurture local JV candidates ready to partner with global tech entrepreneurs
• Lack of climate tech-supportive sophisticated acceleration infrastructure in a local space: esp. ASEAN-fit (emerging/developing market-fit) business accelerators versed in climate	• (Activity 2.1) Set up the Global Acceleration Advisory Secretariat in partnership with global accelerators of various business areas
• Few global tech companies willing to entre risky developing markets • Difficulties to identify qualified local partners under a fair standard competition format	• (Activity 2.2) Launch global sourcing strategies to attract global technology innovators and match them with trained/qualified local entrepreneurs
• Expensive foreign technologies • IP controversy • High upfront costs in a nascent market • Lack of locally customised tech-driven R&D and business acceleration opportunities	• (Activity 2.3) Support JV teams' technology transfer and business modelling in a legally permitted way through provision of gender-mainstreamed collaborative R&DB acceleration services
• Deficit of the government budget assigned for climate actions and tech entrepreneurship • Shrunk climate investments amid economic slowdown, market fluctuations	• (Activity 3.1) Set up and execute an investment fund with attracting and raising investment capital on a global scale
• Oversaturated, scattered small-scale grant based early-stage support schemes without continuing later stage investments • Lack of risk-taking equity capital in ASEAN	• (Activity 3.2) Execute equity investments with strategic portfolio management as per investment criteria incl. gender mainstreaming
• Few NOL countries have experienced exit achievements while their markets are nascent or almost non-existent.	• (Activity 3.3) Design detailed exit strategies and achieve exits with a reasonable rate of return
• Scattered similar programmes have been completed, without any follow-up actions amongst local ecosystem stakeholders	• (Activity 4.1) Formulate the best workable local ecosystem network framework

• Lack of enabling innovation ecosystem, including a weak regulatory framework and skilled climate business accelerators	• (Activity 4.2) Build capacities to enable and strengthen the relevant ecosystem stakeholders, including national climate tech-focused accelerators
• Disregarded/discontinued sustainability in one-off grant trials after international/national investment exits	• (Activity 4.3) Promote sustainability via the locally-owned SMU operation

Programme Framework by Output, Component, Activity, and Sub-activity. Maintaining a structural cohesion of the general programme framework (below table), country-specific programme details are developed to meet country-specific needs and gaps identified by feasibility studies and a series of consultations during the designing stage.

Output 1. Country Driven Shortlist of Local JV Candidates Ready for Climate Technology Uptake
Component 1. Country Driven Acceleration Readiness
• Activity 1.1. Set up a Regional and National Sustainability Management Unit (SMU)
- Sub-Activity 1.1.1. Set up a Regional SMU in GGGI HQ
- Sub-Activity 1.1.2. Set up a National SMU in each country
• Activity 1.2. Launch sourcing strategies to identify qualified local JV candidates
- Sub-Activity 1.2.1. Develop a robust sourcing strategy
- Sub-Activity 1.2.2. Prepare necessary platforms and materials for sourcing JV candidates
- Sub-Activity 1.2.3. Launch outreach activities
- Sub-Activity 1.2.4. Narrow down the pool of local JV candidates for acceleration readiness
• Activity 1.3. Offer tailor-made acceleration services to nurture local JV candidates
- Sub-Activity 1.3.1. Design the National Climate Entrepreneur Accelerator Programme (N-CEAPs)
- Sub-Activity 1.3.2. Train selected candidates for acceleration readiness
- Sub-Activity 1.3.3. Organise a competition to shortlist JV candidates for global acceleration
Output 2. Climate Technology Transfer between Global-ASEAN JVs with Investment Readiness
Component 2. Global Acceleration for Collaborative R&DB (R&D + Business)
• Activity 2.1. Set up the Global Acceleration Advisory Secretariat
- Sub-Activity 2.1.1. Implement the global component with necessary logistics
- Sub-Activity 2.1.2. Manage multi-dimensional stakeholder communication
- Sub-Activity 2.1.3. Conduct regular M&E and reporting to KDB
• Activity 2.2. Launch global sourcing strategies to attract global technology innovators
- Sub-Activity 2.2.1. Establish a Global-ASEAN pool of global technology innovators
- Sub-Activity 2.2.2. Assess the suitability and feasibility of global technology innovators
- Sub-Activity 2.2.3. Select the finalists for Global-ASEAN technology transfer
• Activity 2.3. Offer collaborative R&DB acceleration services for JVs
- Sub-Activity 2.3.1. Perform diagnostic analyses on participating JV candidates
- Sub-Activity 2.3.2. Offer tailor-made acceleration services for JV candidates
- Sub-Activity 2.3.3. Achieve Global-ASEAN technology transfer with best fit mechanisms
- Sub-Activity 2.3.4. Support investment attraction
Output 3. Climate Technopreneurship Fund (CTF) with Private Capital Crowd-in
Component 3. Climate Technopreneurship Fund (CTF)
• Activity 3.1. Execute the CTF formation with raising investment capital

- Sub-Activity 3.1.1. Establish the fund vehicle and related entities
- Sub-Activity 3.1.2. Close the GCF into the CTF
- Sub-Activity 3.1.3. Fundraise private investment capital
• Activity 3.2. Execute investments with portfolio management as per investment criteria
- Sub-Activity 3.2.1. Perform due diligence on proposed businesses
- Sub-Activity 3.2.2. Build, revise, and manage investment portfolios
- Sub-Activity 3.2.3. Monitor performance of portfolio companies with structured support
- Sub-Activity 3.2.4. Conduct regular reporting to KDB
• Activity 3.3. Achieve successful exits with a reasonable rate of return
- Sub-Activity 3.3.1. Realise exits after assessment on appropriate exit methods
- Sub-Activity 3.3.2. Distribute proceeds and returns
- Sub-Activity 3.3.3. Terminate the CTF
- Sub-Activity 3.3.4. Perform the fund's post-exit obligations
Output 4. Climate Technopreneurship to be Powered at Scale under Strengthened Ecosystem
Component 4. Technical Assistance (TA)
• Activity 4.1. Formulate the best workable network framework
- Sub-Activity 4.1.1. Promote national networks for climate technology innovation
- Sub-Activity 4.1.2. Develop a communication hub with knowledge sharing activities
• Activity 4.2. Build capacities to enable and strengthen the relevant ecosystem
- Sub-Activity 4.2.1. Build capacities of national accelerators to sustain N-CEAPs (1.3.1)
- Sub-Activity 4.2.2. Establish the national strategy on climate technopreneurship promotion
- Sub-Activity 4.2.3. Offer regulatory and policy recommendations
• Activity 4.3. Promote sustainability via the National SMUs
- Sub-Activity 4.3.1. Develop a detailed M&E implementation plan
- Sub-Activity 4.3.2. Conduct regular reporting to KDB
- Sub-Activity 4.3.3. Support ex-post impact assessments
- Sub-Activity 4.3.4. Coordinate with the Regional SMU and the Acceleration Secretariat

Component 1 – A Starting Point for NOL Countries to be the Climate Tech Capacitated Agent. The first component is this climate technology transfer initiative's starting point that sources promising local entrepreneurs and accelerate their capacity so that they could be equipped with the proper awareness and readiness to collaborate with global technology providers in Component 2. In particular, sourced and shortlisted local companies will be supported in order to satisfy the National Climate Entrepreneur Accelerator Programme (N-CEAP) Investment Criteria that requires qualifications NOL countries' local entities are not able to be equipped.

N-CEAP Investment Criteria for Local Entrepreneurs

	Criteria	Description
	Eligibility Criteria *(Local Entrepreneurs to be Screened Out if not Relevant or Satisfied)*	
1	Locally Owned Entities	• Entrepreneurs with valid business registration in NOL countries; or • Entrepreneurs with recommendation from the National SMU Project Steering Committee (composed by representatives from GCF NDAs and line ministries in NOL countries) ✓ *As a result of local consultation, in case that a vulnerable NOL country needs transfer of certain climate technologies but does not have qualified local entrepreneurs, the SMU Project Steering*

		Committee may consider a setup of a local subsidiary of global technology firms which must be registered as a local company with local employees in NOL countries.
2	Country Owned Tech Prioritisation	• Initial categorisation as per Seven (7) Technology Groups; or • Country fitness of a proposed technology to be confirmed by the SMU Project Steering Committee (composed by representatives from GCF NDAs and line ministries in NOL countries)
3	E&S Safeguards (ESS)	• [Initial Negative Screening on E&S Risk] Local entrepreneurs who consider a business defined in IFC Exclusion List and/or a business with adverse impacts on indigenous peoples are NOT eligible.
4	IP Rights Protection	• Applicant entrepreneurs' commitments to transparent technology uptake under the global level UNFCCC initiative that values IP rights shall be legally binding under confirmation of the National SMU Project Steering Committee (composed by representatives from GCF NDAs and line ministries in NOL countries)
	Selection Criteria *(Scoring-based Assessment for Admission to N-CEAP)*	
5	Technology Uptake Capacity	• Local entrepreneurs deemed to have capacity for foreign technology uptake and digestion in terms of staffing expertise – a minimum level of relevant technical/engineering expertise • (1st Round) To be assessed by the Technical Advisory • (2nd Round) To be assessed by the GP team via the Regional SMU
6	Gender Mainstreaming	• [Screening via Gender Profiling Form] Women-led SME candidates shall be prioritised for selection. ▪ SMEs with at least 26% of the total capital owned by women; ▪ SMEs with at least one woman participates in the Board of Directors and/or different types of boards or internal committees such as ESG committees, among others (CEO, COO, CTO, etc.); and/or ▪ SMEs with more than 30% of the total workforce are females. • To be consulted and assessed by gender experts of the SMU, based on Gender Profiling Form (Appendix 2 of Annex 8)
7	Funding Contribution	• [Co-investment Potential] Selection priority given to local entrepreneurs with investment capacity and/or willingness

NOL Country-specific Business Acceleration Provision in Component 1. This first component distinctively optimises specific contents of the National Climate Entrepreneur Accelerator Programme (N-CEAP) in each country while maintaining consistency in terms of an integrated goal and the overall structure. Under the general programme framework (above table), the detailed nature of sub-activities will vary across the five different countries: potentially even within a single country as per the needs and gaps of individual beneficiary firms. Consequently, all NOL countries will benefit from the N-CEAP, though the support structure may look like different across the different five countries. Assumed country-specific uniqueness is as follows (further detailed in Annex 2) and such uniqueness will be mainstreamed in N-CEAP elements.

- **Cambodia –** Given its crowded local incubation landscape, the Cambodian leg of the programme will focus on later-stage acceleration and JV readiness. The Cambodian SMU will partner with existing local associations and partners, including Cambodia Investment Club, Cambodia Women Entrepreneurs Association, Platform Impact, Young Entrepreneurs Association, UNIDO, and others to identify a pipeline of investable entrepreneurs and provide Cambodia's own N-CEAP courses to be climate tech-driven JV ready. In brief, Component 1 will pay particular attention to opportunities for technology transfer through subsidiaries and licensing, while Component 4 will strengthen the enabling environment for more local innovation.

- **Indonesia –** The Indonesian SMU will be set up in collaboration with the relevant Government of Indonesia's agency or ministry, consisting of the GGGI implementing team and representatives of the host ministry. The SMU will implement inclusive communication and outreach strategies that effectively attract qualified technology entrepreneurs with high changes to form JVs with global innovators. In particular, GGGI Indonesian team aims to shortlist at least five potential climate technopreneurs for global partnership.

- **Laos –** Given a very nascent nature of the relevant ecosystem, the Lao leg of the programme seeks for establishing the 'Lao National Climate Entrepreneurship Network' under the aegis of the Young Entrepreneurs Association of Laos. Strongly supported by the Ministry of Natural Resources and Environment, GCF NDA, in partnership with Lao National Chamber of Commerce and Industry (LNCCCI), the Lao SMU is expected to widely disseminate the calls for application and reach ca. 300 local firms in relevant industries.

- **Philippines –** Given that the Philippines has a wide pool of active local and international climate experts in the market, those potential mentors will be consulted for the N-CEAP design to ensure it necessary for addressing climate related issues on the ground. The Philippines' leg of the programme will mainly support both technology-based start-ups and social enterprises under Component 1, while strengthening local ecosystem stakeholders' capacity centred around the Philippine Businesses for Climate Impact Network under Component 4.

- **Vietnam –** Given quite a number of growing young entrepreneurs who recognise a desperate need of foreign technologies but cannot afford to buy due to realistic hurdles, which is different from situations of other participating countries in a severe vulnerability, the programme will proceed with competitive selection processes in order to identify qualified local players who could uptake transferred technologies from global peers after 3-6 months' N-CEAP provision and an advanced round of investor pitch.

Component 2 – Best Available Business Acceleration from a Global Team with Multi-Expertise. The second component aims to source and train global innovators and technology providers as per Acceleration Investment Criteria (see below), facilitate adequate partnership models to be established by matching between local entrepreneurs (N-CEAP graduates, i.e., domestically nominated as a result of N-CEAP under Component 1) and sourced global innovators, and to eventually achieve climate-responsive technology transfer by translating fit technology solutions into shovel-ready businesses of the matched teams.

Acceleration Investment Criteria for Global Entrepreneurs

	Criteria	Description
	Eligibility Criteria *(Global Entrepreneurs to be Screened Out if not Relevant or Satisfied)*	
1	Awareness of the Programme Mandate	• Global innovators must be aware of the fundamental programme mandate that climate (responsive) technologies are to be applied for business deployment across the five (5) NOL countries, in alignment with the five (5) GCF Result Areas (1) M1. Energy generation and access (2) M2. Low-emission transport (3) M3. Buildings, cities, industries and appliances (4) A1. Most vulnerable people and communities

		(5) A2. Health and well-being, and food and water security
2	Climate Technology	• A global entrepreneur must have an acknowledged qualification (e.g., intellectual property [IP] on a proposed technology) to have and trade a climate (responsive) technology.
3	Country Owned Tech Prioritisation	• To be screened by Seven (7) Prioritised Technology Groups; or • NOL countries' urgent need of a proposed technology should be proven, in case of a proposed technology not included in the Technology Groups – to be confirmed and approved by the National SMU Project Steering Committee (composed by representatives from GCF NDAs and line ministries in NOL countries)
4	Technology Transfer	• A proposal on technology transfer mechanism to be submitted to the Acceleration Secretariat for Global Acceleration, with acceptance on climate (responsive) technology to be transferred in legally permitted manners under compliance with a standard IP trade format
5	E&S Safeguards (ESS)	• [Secondary Negative Screening on E&S Risk] Global entrepreneurs who consider a business defined in IFC Exclusion List and/or a business with adverse impacts on indigenous peoples are NOT eligible. • Global entrepreneurs whose ESG policy and ESMS are existent but not aligned with the CTF E&S Safeguard Policy are recommended to adopt a common approach on the ESMS for their JV entities that will carry out the CTF activities. If candidates are equipped with their own (or stricter) strategies, theirs would be evaluated as per ESS/Gender compliance.
	Selection Criteria *(Scoring Based Assessment for Admission to Global Acceleration)*	
6	Local Partnership	• Partnership with N-CEAP graduates from Component 1; • Partnership with local entrepreneurs (non N-CEAP graduates) – need to be approved by the National SMU Project Steering Committee (composed by representatives from GCF NDAs and line ministries in NOL countries); or • Commitment to partnership with local entrepreneurs (if technologies are urgently needed amid lack of local peers) – need to be approved by the National SMU Project Steering Committee (composed by representatives from GCF NDAs and line ministries in NOL countries)
7	Gender Mainstreaming	• [Gender Mainstreaming Readiness Check] Scoring-based assessment on a global applicant's gender balanced appealing points, in addition to Gender Mainstreaming Commitment Plan (GMCP), upon request from international gender experts from Gaia Consult Inc. that includes, among others: ▪ Set specific employment and esp. high-value skill training targets of women and girls with concrete timeline ▪ Reflect gender inclusiveness in the design/business plans of the JV team (along through the process of climate technology R&DB acceleration): e.g., recruiting female talents intentionally with a certain target to be active agents of the enterprises and/or internationally targeting female population as major beneficiaries
8	Funding Contribution	• [Co-investment Potential]

		Selection priority given to global entrepreneurs with investment capacity and/or willingness

In order to support ASEAN-global teams to satisfy the Acceleration Investment Criteria above, the programme will deliver on collaborative R&DB activities overall as below.

(1) [Sourcing and Shortlisting of Global Technology Providers with Climate Responsive Solutions]

By leveraging the global innovation network consisting of VCs, accelerators, and other start-up support intermediaries with global presence, NHIS (as Executing Entity of Component 2) will launch a call for proposals to source tech-based businesses (start-ups, SMEs, etc.) in countries with advanced climate responsive ecosystems, including, but not limited to, Canada, Singapore, Denmark, Finland, the Netherlands, the US, and the UK. Global tech innovators in climate space will be sourced to engage in the programme to scale up their impactful technologies to flow into the five emerging ASEAN markets. The global JV candidates will be shortlisted upon a series of diagnostic assessment on the technological suitability, market viability and scalability potential (both in terms of technology fitness and climate outcomes) by the GP team.

(2) [Tailor-made Consultation and Guidance on Business Strategy]

With shortlisted global teams on board, a series of tailor-made consultation sessions will take place, to gauge and further assess the feasibility of technologies by closely examining risk factors, business projections and technology transfer capacity. Such baseline analysis will provide a solid ground for the GP to consistently keep track of back-to-back support provided to the companies.

Such consultation will also aim to help shape the go-to-market strategies of global tech providers by offering deep-dive sessions on respective local markets of the five NOL countries, including on-site, preliminary proof of concept (PoC) activities, local network building and setting up milestones on business expansion. Through this process, each team will have strategised measures to secure local business licenses and patents as well as plans on potential JV creation and entailed actions, while their business models with identified viable technologies will be confirmed, primed for proposing to local companies with technology needs subject to the JV establishment.

(3) [ASEAN Market Entry Readiness Training & PoC Support]

Based on previous consultation sessions validating core climate (responsive) technology as well as touching upon business strategies for scaling up, the global climate tech innovators will be provided with a series of activities enhancing their readiness for the ASEAN market. Training on prerequisites and associated procedural measures will be given, in conjunction with reviewing the general business ecosystem of the target NOL countries. Furthermore, similar practices that involve overseas market entry via technology transfer scheme will be analysed and reviewed as a benchmark, allowing the teams to examine potential issues in business expansion.

Upon formulating tailor-made market entry strategies, the programme aims to leverage local networks of extensive relevant stakeholders and partners, particularly state agencies, to facilitate substantial discussions on local business licenses and cooperation with local business ecosystems, in a close partnership and intensive discussions with the Regional SMU under KDB guidance.

(4) [Technology Transfer Modalities and Business Development]

With the global climate tech innovators identified, accelerated and ready to expand into the target NOL countries, a series of business matching consultations will take place, to undertake a genuine technology transfer through bespoke schemes: i.e., JV partnership (onshore target set-up as a locally registered JV between a local company and a global climate technology company contributing its technology transfer or asset including both cash and in-kind); and/or technology licensing agreement (late-stage, onshore target licensing climate technology from a global firm upon investment). In addition to aforementioned schemes, consulting sessions for enabling a genuine technology transfer through other likely recommended business models are to be offered as well in a close communication and intensive discussions with the Regional SMU under KDB guidance, following similar procedures involving feasibility studies on business and technology: e.g., local company under the concept of South-South Cooperation (onshore business with climate technology on its own, with no need for additional technology sourcing/licensing; rather, capable of enabling potential technology transfer as the transmitter amongst NOL countries).

In order to assess compatibility between the two partners, the matching arrangements between global and local entities will first entail feasibility studies covering business plan, profitability, financial projection and exit strategy. Furthermore, aside from a business perspective, technical aspects will be looked into, which an aim to review the potential commercialisation of respective climate tech-driven solutions.

(5) [Consultation on Climate Advisory and E&S/Gender Safeguards]

In order to align each business model with viable climate outcomes, a professional climate advisory firm will be engaged to provide businesses with adequate guidance and framework on capturing climate outcomes via Project Selection Tool based on GCF Additionality and Innovation Tool (AIT). Moreover, analysis on the environmental and social (E&S) impacts borne out of such technology transfer models will also be conducted to allow the businesses to better incorporate E&S/Gender related factors into their operation and monitoring of supply chains.

(6) [Investment Attraction and Facilitation Support]

Upon the set-up of technology transfer in local-global partnership formats, the programme will offer dedicated support to the selected businesses on attracting investment by not only sharpening their IR resources but also providing consultation to rebuild business strategies considering the investor-oriented approach. Moreover, aiming for a seamless integration into respective local markets, if necessary, the teams will further be mentored by global and local accelerators and venture capitals to strengthen the market linkage and build relationships for potential investment in the future.

In parallel with such support given to the businesses, the GP and local partner entities of Component 2 will host a periodic VC/investors' roundtable and networking session to highlight the accelerated teams' potential and expand the network of climate tech-driven investment space. As part of support, a regional demo day to a targeted audience of investors and industry stakeholders at both global and regional levels will take place to showcase the innovations and discuss the way forward.

Component 2 as the Channel of Global Innovators' Joining Climate Action. The kernel of the second component is to invite global companies that have not paid attention to the UNFCCC/GCF strategic goals that much, reframe their technology capacity as per above listed business acceleration services, and support them to be ready to contribute to climate action as per Acceleration and CTF Investment Criteria, including but not limited to strict E&S, gender, IP risk resolution requirements, thereby unlocking their climate responsive tech expertise fit to the target NOL countries. For this, the NHIS centred at the Team NH (see Section B for NH Financial Group's group-wide commitment for the programme success) will tap investment opportunities in partnership with more than 100 global VC/PE/AC partners in Singapore, the US, Switzerland, Japan, Australia, Thailand, and so on. Once proper global innovators identified, the second component will support them to be ready to satisfy the Acceleration Investment Criteria below and eventually CTF Investment Criteria.

ASEAN-fit Global Acceleration Teams to be Optimised with Kernel Professionals in Component 2. A global acceleration team with a diverse array of expertise, including IP and patent, accounting, tax, law, financial modelling, technology valuation, marketing, fundraising, investment, business mentoring, E&S risk management, and gender mainstreaming, will be organised to offer collaborative R&DB consulting services that are customised as per partnership and nature of technology. In a nutshell, agile and project-based acceleration teams will be optimised with kernel professionals who have the expertise each business needs – i.e., Collaborative R&DB acceleration services to be co-designed and/or operated with chartered patent agents, professional accelerators, university-based R&D centres, and consulting firms – to validate the sustainability of each team's core business model pertaining to climate outcomes, and to deliver on its mission.

In order to build its expansive relationship with global stakeholders nurturing and advancing innovation ecosystems, the programme, specifically NHIS, as Executing Entity of Component 2, shall make proactive collaborations with global acceleration partners with different expertise and focus areas that are proven capable of fostering acceleration in the climate technology space, for instance (see below), under engagement partnership. With the multi-disciplinary team on board, Component 2 shall proceed with sourcing talented global innovations, match-making with technology deployable local companies, training JVs to devise technology-enabled climate solutions, enabling selected forward-looking businesses to advance their scalable capabilities in five emerging markets, and finally reaching a commercialisation milestone, leading up to the investment phase of Component 3. This would allow the businesses to finalise their mid- to long-term business plans tailored to the local settings.

Component 3 – Climate Technopreneurship Fund (CTF) with GCF Catalytic First-loss Equity Capital under the Risk-Adjusted Investment Framework. Component 3 is the very core of this programme that establishes an investment fund of USD 200 million, *CTF*, with the GCF anchor investment, and finances graduates from Component 2 that have proven climate technology applied businesses most suitable to individual local contexts. Th CTF shall support JVs or other forms of entities that execute relevant technology transfer into NOL countries to grow as small and medium sized enterprises (SMEs) to expand climate responsive businesses while addressing a missing middle corporate structure caused by a wide gap between large and micro, mostly informal, enterprises in the five countries.

The CTF adopts a risk-adjusted investment framework to cover varying risks that arise from diverse portfolio companies in multiple NOL countries. Direct capital injection is the default mode of investment but, if needed, additional layer of a special purpose vehicle (SPV) may be structured to accommodate categorised risks such as tax/regulatory restrictions on foreign direct investment and entry/exist convenience. The SPV differentiates itself from a BAU Fund of Funds (FoF) that is an expensive vehicle investing in other hedge funds. Given that the programme pays attention to a challenging reality that levels of acceptable risk and capital configuration ratio differ country by country, even amongst industries within a single country, thereby necessitating deal-by-deal adjustments via navigating, calculating, and managing associated risks. Additionally, the configuration of debt and equity within a project can vary; setting full-equity investments as a default, a project with a higher risk may be structured with an 80/20 debt-equity configuration, while a lower-risk project may require a 50/50 debt-equity configuration.

It shall deal with the problematic finding of existing and/or similar initiatives that most graduates from initiatives with one-off seed grants have failed to attract further investment capital and progress to more advanced stages. In recognition of the finding, the fund shall invest in JVs or other forms of entities that execute relevant technology transfer into NOL countries as per CTF Investment Criteria (see below) in case barriers bar the JV model and prevent from achieving the programme objectives. Only applicants whose readiness and business fitness have been validated by the GP as per investment criteria below can benefit from the CTF's capital.

CTF Investment Criteria

	Criteria	Description
	Eligibility Criteria *(Local-Global Partnership to be Screened Out if not Relevant or Satisfied)*	
1	GCF Mandate	• [Alignment with the GCF Result Areas] An applicant must propose a business that is classified into (at least) one of the chosen GCF's five (5) result areas. (1) M1. Energy generation and access (2) M2. Low-emission transport (3) M3. Buildings, cities, industries and appliances (4) A1. Most vulnerable people and communities (5) A2. Health and well-being, and food and water security • Applicants must demonstrate their level of climate impact within the GCF result areas
2	NOL Countries' Technology Needs	• A proposed climate responsive tech-driven business must respond to NOL countries' technology needs in response to climate change. • To be screened by ***Priority Technologies by Result Areas with Specific Level of Needs*** in Section B1 • Other climate responsive technologies recommended by NOL countries' Project Steering Committee can be considered.
3	Technology Transfer	• An applicant must prove its technology transfer mechanism from global to NOL countries. • ***Joint Venture (JV) partnership*** is the default investment model with flexibility that is triggered by NOL countries' request. Such flexible cases will be confirmed by NOL countries' Project Steering Committee under Sustainability Management Unit (SMU). ✓ *Example 1: In case that NOL countries request 100% local ownership, technology licencing agreement can be considered as an alternative technology transfer case.* ✓ *Example 2: In case of a lack of local entrepreneurs who can digest a needed technology, NOL countries may consider a subsidiary of a foreign technology firm.* ✓ *Example 3: In case of a huge tech capacity gap between local and global side, NOL countries may consider South-South technology transfer modality, seeking for a local company with potential to execute South-South transfer/cooperation on climate responsive technology. In such case, the proposed local business should be able to demonstrate its pertinence to further establish potential South-South technology partnerships.* ✓ *Example 4: In case of little interest from global tech companies – e.g., due to lack of market opportunities – amid an urgent need of such technologies in NOL countries, NDAs via SMU's Project Steering Committee may suggest alternative investment options such as investments directed at such innovative companies with technologies in urgent need and/or debt/mezzanine for the NDA's proposed investments under a condition to achieve technology transfer via relevant tech*

		licensing agreement or an investment vehicle set-up in NOL countries for local job creation, among others.
4	Local Ownership of Innovation	• Clear evidence must prove that **a proposed technology-enabled climate solution is not owned by NOL countries**, being triggered by at least one of the followings, so that the innovation intervention is additional to NOL countries. ✓ A proposed technology-enabled climate transaction is a first of its kind in NOL countries; ✓ No technically feasible and more cost-effective alternatives to the solution exist; ✓ Similar technologies and/or viable alternatives are not available in the domestic market of a target NOL country; ✓ Similar technologies and/or alternatives should be necessarily supported by concessional finance and/or other public support mechanisms (e.g., tax incentives, subsidies) to be viable, while the support does not distort the market; ✓ The target market and available finance are insufficient to deploy a proposed technology-enabled climate solution so that scale-up and replication potential is highly low; or ✓ Similar technologies and/or viable alternatives are not locally owned, so that NOL countries cannot afford to be self-equipped with the solution to self-combat relevant adverse climate change impacts.
5	Additionality	• Clear evidence must be provided to demonstrate that a proposed solution overcomes existing barriers, contributes to market development and transformation and to strengthening knowledge and capacity, and applies best technology(ies). • To be screened by Project Selection Tool (Annex 21)
6	E&S Safeguards (ESS)	• Negative screening ✓ Businesses defined in IFC Exclusion List NOT eligible for investment ✓ Businesses with adverse impacts on indigenous peoples NOT eligible for investment, by falling under one of the categories below: (1) Business/enterprises with negative impacts on lands and natural resources subject to traditional ownership or customary use or occupation; (2) Business/enterprises that results in relocation of indigenous people from lands and natural resources subject to traditional ownership or under customary use or occupation, or; (3) Business/enterprises that may potentially impact cultural heritage[25], and where the commercial use of the cultural heritage of indigenous peoples would need FPIC). • E&S risk categorisation ✓ High risk businesses (Category A) are NOT eligible for investment

[25] "Cultural heritage includes but is not limited to natural areas with cultural and/or spiritual value, such as sacred groves, sacred bodies of water and waterways, sacred mountains, sacred trees, sacred rocks, burial grounds and sites, as well as the non-physical expression of culture, such as traditions, language, identity, ceremonial, or spiritual aspects of the affected indigenous peoples' lives." (GCF IPP, Para 63)

		✓ A proposed business must be classified as Category B or C • Post-investment commitment through Environmental and Social Management Plan (ESMP) for medium risk businesses (Category B) ✓ All the supported JVs with medium risk level (Category B) must comply with required duties and responsibilities regarding ESS compliance: e.g., E&S risk mitigation plan deployment, regular reporting.
	Selection Criteria – Scoring Priority *(Applicants are encouraged to strengthen their proposal to meet the selection criteria via acceleration)*	
7	Business Feasibility	• [JV dimension] Team capacity ✓ The founding team must be able to operate the business. (e.g., human resources in terms of size and expertise) • [Transaction dimension] Commercial viability ✓ Local market fit ✓ Technical feasibility with potential to scale ✓ Financial sustainability; bankability • CTF may finance a transaction that does not prove business feasibility, in case NOL countries need the business urgently.
8	Co-benefits & Sustainability	• Applicants are encouraged to fully deliver following co-benefits: ✓ Economic co-benefits, such as the creation of jobs, poverty alleviation and enhancement of income and financial inclusion, esp. amongst women; ✓ Social co-benefits, such as improvements in health and safety, access to education, cultural preservation, improved access to energy, social inclusion, improved sanitation facilities and improved quality of and access to other public utilities such as water supply; ✓ Environmental co-benefits, including increased air, water and soils quality, conservation and biodiversity ✓ Gender empowerment co-benefits outlining how the project will reduce gender inequalities (*Further gender component is also included in Criteria #9 below.) ✓ Promoting and empowering “indigenous-led/owned SMEs” to ensure indigenous climate technology and knowledge are fully tapped for the climate technology innovation in the five countries as shown in Criteria #10 below. • Where appropriate, proposals should present the ability of the projects to enable the achievement of one or more of the Sustainable Development Goals (SDGs).
9	Gender mainstreaming	• [JV formation dimension] ✓ At least 50% of the CTF-invested JVs are women-led JVs that are defined in Condition below; or alternatively, ✓ JVs are requested to present a gender mainstreaming commitment plan against the CTF. [Condition – “Women-led JV” Definition] It constitutes the following conditions[26]. Conditions below will be screened from

[26] Condition 1-1) and 1-2) constitutes the IFC’s definition of women-owned SMEs. It is reported that most of the Southeast Asian countries are using the IFC’s definitions, including Cambodia, Indonesia, Philippines and

			the selection process of entrepreneurs. The women-led SME candidates shall be prioritised for selection. ▪ SMEs with at least 26% of the total capital owned by women; ▪ SMEs with at least one woman participates in the Board of Directors and/or different types of boards or internal committees such as ESG committees, among others (CEO, COO, CTO, etc.); and/or ▪ SMEs with more than 30% of the total workforce are females. • [Enterprise dimension] ✓ All proposed business proposals must be gender mainstreamed as per its agreed minimum gender mainstreaming requirements that are, among others: ▪ gender-disaggregated impacts to be presented, monitored, and evaluated; and ▪ a simplified gender assessment (GA) and gender action plan (GAP) (see a standard format in Appendix 3 of Annex 8) to be developed and submitted as part of the CTF application package. • Overall gender performance shall be monitored and evaluated throughout the JV implementation (Component 3) against the submitted GMCP and GAP; in case that 50% target is not achieved due to any reasonable circumstances – e.g., failure to reach the target with a slight gap due to unexpected investments at the 5th year of the investment period, this will be consulted to KDB/GCF. • Prioritisation scoring system for a winning gender-balanced portfolio ✓ Higher gender scores will be weighted higher in selection and investment in cases where scores are identical in other criteria
	10	Indigenous-Led Entrepreneurship	• [Prioritisation of the indigenous-led enterprises/businesses] ✓ CTF will prioritise the applicant enterprises and activities that meet the following conditions: ▪ application/combination of indigenous technology for climate tech RD&B; and/or ▪ participatory and inclusive approach in the proposed enterprises/businesses in the design level • [Enterprise dimension] ✓ In case the nexus to indigenous people is confirmed, the JVs/activities may need to develop an indigenous peoples plan (IPP), as a stand-alone plan or as an integrated part of the environmental and social management plan (ESMP) of the proposed activities. ▪ The CTF will support, as appropriate, monitoring, evacuation, and any other due diligence including

Vietnam. See "Women-owned Small and Medium-sized Enterprises in Viet Nam: Situation Analysis and Policy Recommendations" (2017) A study requested by the Mekong Business Initiative (MBI) and conducted by a joint advisory facility of the Asian Development Bank (ADB) and the Government of Australia, and the Hanoi Women's Association of Small and Medium Enterprises (HAWASME)/ Full text at: http://www.mekongbiz.org/wp-content/uploads/2017/04/WBAs-Position-Paper_English.pdf

		response/remedial/corrective measures in case of occurrence of unintended adverse impacts as per Para 19 of the GCF IPP Guidelines (2019).
11	Balanced Portfolio	• CTF minimum investment threshold: at least 15% of the CTF investment proceeds to be allocated for Cambodia and/or Lao PDR ✓ If the NOL countries cannot be performed due to any force majeure events, it includes but not limited to (a) acts of God; (b) war, invasion, hostilities (whether war is declared or not); (c) action by any governmental authority; and (d) other similar events beyond the reasonable control of the executing entities, the threshold exemption in part or in whole shall be consulted with the GCF for the restructure approval. • At least 50% of the CTF investment proceeds to be allocated for the GCF adaptation result areas

Component 4 (Technical Assistance, TA) – Ecosystem Building in Pursuit of a Genuine Country Ownership and Sustainability. Component 4 will offer the country-specific TA activities to contribute to the relevant ecosystem development for the programme's smooth roll-out, while avoiding overlaps with similar grant funding policymaking or capacity-building projects of other international development agencies.

This TA component aims to create or develop each country's proper sourcing and acceleration training organism and ecosystem to nurture top talents on the ground, and build on existing local platforms and initiatives for the sake of the programme's harmony with the local governance. Almost every programme country has a national body or relevant mechanism for promoting entrepreneurship, though they demonstrate different forms, levels, governance, and maturity. The programme will team up with such bodies to maximise the use of the contextually settled system and infrastructure, ensuring timely and efficient operationalisation and staying on track even after the GCF-KDB exit; i.e., it shall minimise expenses for a new office set-up with newly purchased office supplies, electronics, and furniture. This satisfies a broader governmental demand to minimise the creation of new mechanisms that typically fragment, duplicate, or overlap the efforts of local governments.

In particular, each country has its own unique Intellectual Property (IP) related regulatory framework, though all of them are relatively vulnerable and require improvement with international support. Targeted five countries can be divided into two groups on the basis of what KDB has researched (see Annex 9 for country-specific IP relevant regulatory status).

- 1st group comprises Indonesia, the Philippines, Vietnam, all of which are at the initial stage of IP framework development, reflecting aspects such as technology license valuation and IP enforcement. Although there is a long way to go before the countries have sufficiently advanced IP framework to provide comfort for global tech providers, these governments' continued efforts to improve the IP environment deserve appreciation. It is expected that the countries developing IP regulatory framework will advance thanks to the governments' strong willingness, though their policy enforcement, such as penalties, are generally weak in the middle of existing barriers to licensing and commercialised IP assets: e.g., compulsory licenses to override IP rights have been issued by the governments. The programme shall support their direction of development, while addressing such local barriers on the ground.

- 2nd group consists of Cambodia and Laos, generally classified as the least developed countries. Due to a relatively smaller size of economies and trade volume, there seems comparatively less information and assessment references available on their IP frameworks. Therefore, the programme's PPF outcome will serve as a starting point for a long regulatory development journey towards a more advanced IP framework in this highly vulnerable group. While there are basic level of laws on IP and in-kind contributions like technology, copyrights, and patents can be registered for the issuance and consideration of shares for private companies, the evaluation method of value of each technology and enforcement process are not clear. The two countries are strongly recommended to refer to cases from other countries; i.e., TA activities are strongly needed.

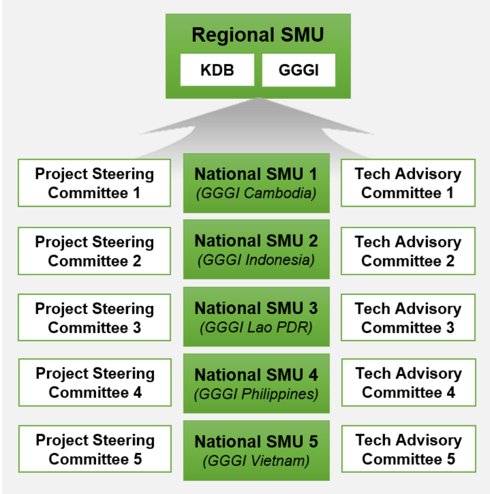
Based on the basic understanding of each country's IP regulatory framework ecosystem, the programme will develop a detailed and customised TA plan in pursuit of a more advanced IP protection ecosystem building. Meanwhile, lessons learnt by other target countries would be able to provide significant benefits amongst each other. It is anticipated that countries would go through a certain learning process via a case by case experience that they would perceive the concept of the IP rights and technical procedures in a practical fashion, and see the merits of how the IP protection scheme brings advanced technology providers and enhances their access to needed technologies. Most importantly, local JVs will deploy their own customised technologies with comfort that theirs could be protected under the newly framed or strengthened policies, laws, and regulations. This will lead countries to seriously consider the regulatory recommendations delivered on from Component 4, and to materialise TA deliverables on the ground – which is not common in a harsh reality of ODA or grant-funded TA areas. The programme admits a restraint that the whole legal framework of the five countries cannot be perfected in a few years under unlimited resources; but revision such as strengthening the existing references can be a practical and easier scenario with consideration of their individual country uniqueness.

Country Ownership Built Sustainability Management Unit (SMU), as the Core Decision-making Authority in Components 1 and 4. GGGI will organise in-house project teams named "Sustainability Management Unit (SMU) with an aim to strengthen country ownership in executing optimised and well-coordinated on-the-ground activities fit to individual NOL countries, and to concentrate allocated resources on the programme over the course of the implementation. Six (6) SMUs will be established as follows: (i) Regional SMU at the GGGI Seoul HQ; and (ii) National SMU at each NOL country's GGGI Country Office, thereby five National SMUs in total. In brief, the objectives of the SMU (GGGI's in-house team, not a separate legal entity) are as follow in brief: (i) country ownership; and (ii) effective programme management.

- The strengthening of country ownership: This creation of mechanism through which NDA and line ministries can make the decision-making intervention due to reasons (e.g., the technology needs evolve, considering the late implementation after board approval and FAA signing), which led to the SMU designing in different forms and composition by country. The National SMUs will be operational based on such feedback from the PSC under the principle of country ownership.

- The channel of the effective programme management: Given the multi-country targeting and the programme theme tech transfer per se leading to engagement of various stakeholders, the SMU will act as a core channel of the effective planning, resource management, communication, and coordination between various stakeholders in four components for the sake of the optimal investment outcomes and climate impacts.

Regional SMU – KDB Advisory Management & NH Engagement for Investment Linkages. The core decision-making of the two components will be advised, guided, and determined by the Regional SMU meetings on an as-needed basis. In capacity of Accredited Entity and Executing Entity, sitting at the Regional SMU meetings, KDB shall advise and approve the main agendas – i.e., National SMU team building, each country team's partner local entities (local bodies in the relevant acceleration and entrepreneurship realms in order to maximise the use of existing platforms and the eventual integration of the relevant resources into the ecosystem of the NOL countries in pursuit of a genuine post programme sustainability), procurement of properly qualified service providers – e.g., service provider with IP regulatory framework advisory capacity, budgeting, finalist company selection, and synergetic strategy amongst every component, as the control tower of the Regional SMU that gives guidance to the National SMUs.



The Regional SMU will invite NH affiliates in the decision-making process as for local private sector partnership, company sourcing, and acceleration activities (Activities 1.2 and 1.3) that require a close linkage with other subsequent components. At the Regional SMU meeting chaired by KDB, NH representatives will (i) recommend top-tier local VC/AC/PE partners that obtain abundant portfolio companies and local market expertise with established resources; (ii) will participate in the procurement and/or contracting process between GGGI and such recommended local top-tiers by supporting the ToR description with qualifications wanted from an investor's perspective, evaluating candidates' qualifications, along with the final selection. In addition to the selection of ASEAN-based local top-tiers, NH will directly engage discuss promising pipeline/portfolio company sharing with that selected local top-tiers, and approve the final selection of local JV candidates at demo days that eventually determine the finalists in Component 1. This NH engagement will support GGGI to identify and source qualified local entrepreneurs in Component 1, in addition to its own list of companies introduced by the NDAs and line ministries and its own private sector partners. All the procedures will be proceeded in accordance with the GGGI's established institutional system; but NH has the prior decision-making authority and veto rights in case such procedures and outcomes are not fit into investments planned afterwards in Components 2 and 3, thereby not repeating trials and errors having been found in most of one-off seed grant provision events without any follow-up investment at scale.

GGGI will set up different forms of the in-house National SMU fitting into each NOL Country' unique circumstances and the five National SMUs will lead operations of the locally driven activities of Components 1 and 4 in each country as a core implementation unit under the direction of the Regional SMU under KDB supervision and advisory guidance. The rationale of the five National SMUs is the different needs on the basis of the following general structure to enable effective delivery of the "National Climate Entrepreneur Accelerator Programme (N-CEAP)." As differentiated N-CEAPs should be designed and offered according to varying country needs and contexts, the GGGI will design and operate the locally customised N-CEAP provision with the support of both local governments and technical experts, thereby necessitating the Project Steering Committee (PSC) and Technical Advisory Committee (TAC) under the National SMU structure.

- <u>Project Steering Committee (PSC)</u>: As a high-level decision-making body, the PSC will be composed by representatives from the local government (e.g., NDA), GGGI, academia, and relevant public and private sectors; the composition varies as per NDA instructions as below. The duties of the PSC

include high-level ministerial coordination, overall policy guidance, and need-based institutional arrangements for timely and effective implementation.
- Technical Advisory Committee (TAC): An ad-hoc technical advisory body will be established with working-level government officials, relevant sectoral/thematic experts, technicians, and/or private sector representatives to address technical and operational issues that arise during project implementation in a timely manner, if necessary.

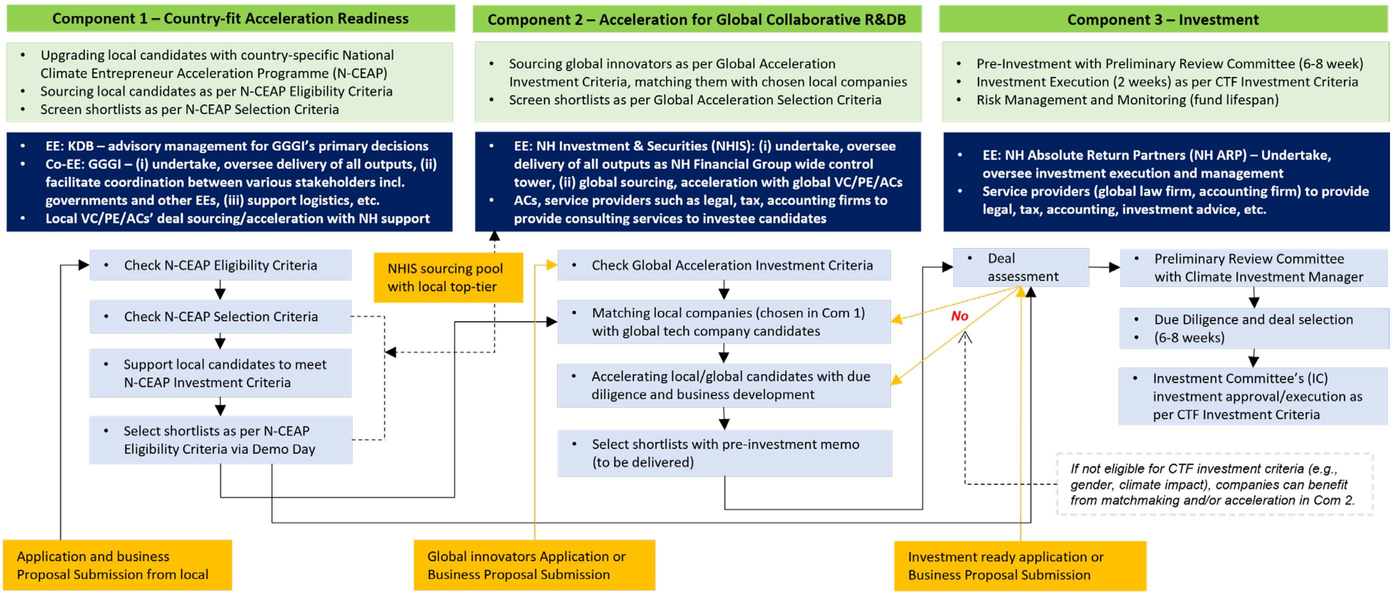
See below the tentative five National SMUs that have been formulated as a result of consultation with NDAs and their recommended local ecosystem players who consider country ownership significantly important for climate technology-enabled business acceleration activities. All the participating local partner entities were recommended or confirmed by the respective NDAs, line ministries, and their private sector partners as a result of extensive stakeholder consultation and feasibility studies, though it may be subject to change due to local ecosystem dynamics including the ministerial reshuffling.

In Components 1 & 4, the SMU aims at local companies' compliance with the N-CEAP investment criteria in Component 1 and readiness for collaboration with global companies in Component 2; only local graduates screened as per the investment criteria in Component 1 can enjoy collaboration with global companies in Component 2. Please see the N-CEAP investment criteria that include E&S safeguards screening and gender mainstreaming awareness - i.e., qualifications which local companies are rarely equipped with. The main roles and responsibilities of SMU are to: (i) strategise and develop operational work; (ii) undertake and oversee delivery of all outputs in a timely and cost-effective way; (iii) coordinate consultation with key governments, ministries, businesses, and vulnerable communities; (iv) communicate with NH in charge of Components 2 and 3 activities; among others. NH will approach and communicate with SMU on a regular and on-the-ground need-basis. Often, SMU will contribute its resources and in-kind supports such as venues and data sets) where needed.

Cross-Functional & Multi-Directional Collaboration between Components. The programme's success requires immense flexibility between components and agility in intuitive decision-making because the complete acceleration work-flow cycle is tremendously complex. In principle, the programme pursues the Joint Venture (JV) model as the mainstream technology transfer partnership modality, as prescribed in investment criteria: a default deal flow from Component 1 (locally driven readiness) and Component 2 (global collaboration) to Component 3 (risk-adjusted investment).

Nevertheless, a scenario may spin out of the lack of available local partner entities in certain industrial sectors and/or unpredictable barriers that bar investment approval and execution despite the strong needs. In such cases, the technology provider may be eligible for investment through setting a local subsidiary with ownership restriction and local employment condition. Another example is a local company that has already partnered with a foreign technology provider, in search of other methods of earning risk capital to address the second Death Valley. If deemed fit as per investment criteria, the local entrepreneur could be considered. Furthermore, if the investee's gender capacity is proven insufficient due to a lack of understanding of the GCF IRMF by the Investment Committee (Component 3), the company may be recommended to consult with related experts (Component 2) for impact assessment and gender mainstreaming trainings, a prerequisite for fund capital injections as shown below deal flow chart.

Deal Flow with Synergy between Components



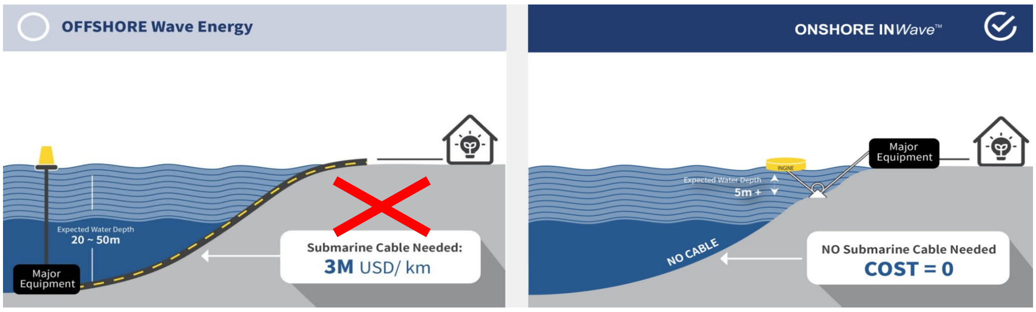
Below are case scenarios that illustrate and convey the spirit of the programme for ease of understanding.

Scenario 1. Climate-resilient Agriculture Strengthened with AI Technology

Global Tech Provider (Company A)	Climate-resilient Agriculture system for infectious diseases based on Artificial Intelligence (AI) technology, gathering data from over 100 different geographical dataset sources that can be utilised to quickly perform quantified risk assessments and identify potential high-risk areas and migration patterns
Local Entrepreneur (Company B)	Farming management company with needs on reducing the negative agricultural impacts primarily caused by climate-related disasters (e.g., recurring floods, typhoons, droughts) to improve harvesting management.
Programme	**Actions per Component**
Component 1	• Company B is selected for the programme by the national SMU in coordination with the regional SMU, based on the eligibility criteria • Company B undergoes a rapid needs assessment to then participate in a tailored N-CEAP programme primarily consisting of: a) the programme's pursuit and climate sensitisation; b) business and/or product development brainstorming; b) marketing and communications; c) significance of a transparent financial management; d) E&S/gender/IP requirements; and e) reporting and monitoring duties.
Component 2	• Matchmaking opportunities for Companies A and B in a hybrid format considering the ongoing pandemic context • Diagnostic analyses and training provided the NH Investment & Securities arranged team of accelerators and experts, to determine the best option for technology transfer between companies • Advisory, consulting, and due diligence services provided to better understand the legal, tax, E&S and practical implications of the different technology transfer options • Joint Venture model with in-kind technology contribution by Company A and local infrastructure by Company B → Company C created • Additional acceleration services for Company C focused on the technical training of personnel on utilising the AI-based predictive modelling tech • Legal resources to aid in negotiating the agreement terms and conditions between the two companies.

Component 3	• Company C goes under review by the CTF Investment Committee • Multi-faceted (reputational, legal, and financial) due diligence procedure • Shareholder's agreement is signed, and funds are provided along with the business development and execution support by the GP.
Component 4	• The SMU partners with the government agencies in the areas of e.g., public health and the local private sector on telecommunications to align with the Company C's actions, along with policy recommendation to be delivered on • Regular monitoring and evaluations are executed to ensure the climate impact created by the programme and Company C are being upheld.

Scenario 2. Bringing an Economic Wave Technology Solution to Remote and/or Small Islands

Global Tech Provider (Company A)	Onshore wave technology that collects wave power nearshore by multi-directional floating buoys with the power generation unit installed on land, enabling the provision of affordable yet clean and sustainable energy to remote islands, coastal areas. **Onshore Wave Technology** 
Local Entrepreneur (Company B)	Energy infrastructure company committed to responding to energy/water scarcity with valuable understanding of the local energy environment
Programme	**Actions per Component**
Component 1	• Company B is selected for the programme by the national SMU in coordination with the regional SMU, based on the eligibility criteria • Company B undergoes a rapid needs assessment to join a tailored N-CEAP
Component 2	• Matchmaking opportunities for Companies A and B in a hybrid format to discuss areas of potential collaboration on local sustainably energy resources advancement • Diagnostic analyses and training provided, to determine the best option for technology transfer between companies through collaborative R&DB activities for PoC (e.g., wave energy-backed desalination) with gap analyses and financial modelling • Advisory, consulting, and due diligence services provided to better understand the legal, tax, E&S and practical implications of the different technology transfer options • Joint Venture model with in-kind technology contribution by Company A and local infrastructure by Company B) → Company C created • Further extensive acceleration assistance on legal/business development/ technical capacity building provided to Company C, while legal resources to aid in negotiating the terms and conditions between the two companies
Component 3	• Company C goes under review by the Investment Committee for the CTF. • Multi-faceted (reputational, legal, and financial) due diligence procedure • Shareholder's agreement is signed, and funds are provided along with the business development and execution support by the GP.

Component 4	• The SMU offers country-specific TA activities considering the nascent nature of wave technology, namely knowledge-sharing events to deepen the understanding of policymakers, industry stakeholders, and end-users. • Policy-level intervention for incentivisation (e.g., Feed-in-Tariff) takes place backed by the national SMU in partnership with the local gov't • Regular monitoring and evaluations are executed to ensure the climate impact created by the programme and Company C are being upheld.

Scenario 3. Automatic Weather System for Disaster Risk Reduction

Global Player (Company A)	Meteorological technology with an Automatic Weather System enabling real-time, automatic measurement, recording and transmission of weather data
Local Entrepreneur (Company B)	Local company recommended by the local meteorological and hydrological administration with needs on strengthening human resources to operate AWS equipment on the ground
Programme	**Actions per Component**
Component 1	• Company B is recommended by the local meteorological and hydrological administration, and selected for the programme by the national SMU in coordination with the regional SMU upon assessment.
Component 2	• Matchmaking opportunities for Companies A and B in a hybrid format to discuss areas of potential collaboration on advancing weather forecasting and data management system • Diagnostic analyses and consultation provided the NH Investment & Securities arranged team of accelerators and experts, to determine the best option for technology transfer. As part of the global acceleration support, the Meteorological Administration and Meteorological Institute assess and verify the quality of meteorological instruments of Company A. • Technology licensing through a subscription-based model • Additional acceleration assistance is provided to Company B, including technical training on equipment installation, maintenance, and operation, as well as data processing and management, while legal resources to aid in negotiating the terms and conditions between the two companies.
Component 3	• Company B, now armed with the technology of Company A, goes under review by the Investment Committee for the CTF. • Multi-faceted (reputational, legal, and financial) due diligence procedure • Shareholder's agreement is signed, and funds are provided along with the business development and execution support by the GP.
Component 4	• With the SMU support, the relevant national meteorological agencies take part in knowledge exchanges to facilitate adequate information flow • Regular monitoring and evaluations are executed to ensure the climate impact created by the programme and Company B are being upheld.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

Partnership with Executing Entities with a Proven Capacity in Each Component. KDB partners with various entities to efficiently carry out activities in the five vulnerable NOL countries: as for Components 1 and 4, the Global Green Growth Institute (GGGI) which boasts a strong track record in climate and green growth initiatives across the five countries via professional government consultation, to broadly reach the best suited entrepreneurs and audiences on the ground; as for Component 2, NH Investment & Securities (NHIS) which is to execute collaborative R&DB acceleration activities on the basis of partnership with both local and global accelerators with different technology expertise and strategies; and as for Component 3, NH Absolute Return Partners Pte. Ltd. (NH ARP, General Partner, “GP”) which is undoubtedly a proven financier with impressive investment track record and performance globally.

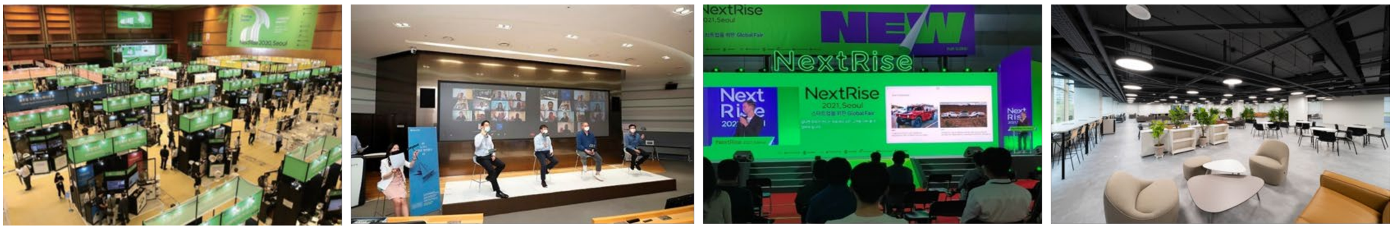
Component-wise Executing Entities and their Proven Capacity

Component	Activity	Executing Entity
Component 1 **Country Driven Acceleration Readiness**	1.1. Set up a Regional and National Sustainability Management Unit (SMU) 1.2. Launch sourcing strategies to identify qualified local JV candidates 1.3. Offer tailor-made acceleration services to nurture local JV candidates	**KDB** ▪ Start-up/VC investment platform systemised (in-house) with ecosystem building experience and know-how ▪ Global VC network incl. affiliates in Singapore, Silicon Valley, and London **GGGI** ▪ Local presence and network across the 5 NOL countries, thereby enabling outreach to key local stakeholders ▪ Track record on entrepreneurship by working with climate vulnerable start-ups and MSMEs as a *Greenpreneurs* initiator
Component 2 **Global Acceleration for Collaborative R&DB**	2.1. Set up the Global Acceleration Advisory Secretariat (Secretariat) 2.2. Launch global sourcing strategies to attract global technology innovators 2.3. Offer collaborative R&DB acceleration services for JVs	**NH Investment & Securities (NHIS)** ▪ A wide, strong business partnership with Southeast Asian top-tier VC/PEs ▪ Global business networks incl. NH affiliates in various geographies ▪ One of top-tiers, equipped with in-house VC investment wing and relevant resources and infrastructure to be leveraged for Team NH composition
Component 3 **Climate Technopreneurship Fund (CTF)**	3.1. Execute the CTF formation with raising investment capital 3.2. Execute investments with portfolio management as per investment criteria 3.3. Achieve successful exits with a reasonable rate of return	**NH Absolute Return Partners (NH ARP)** ▪ GP, composed with investment experts with track record in fund investment and portfolio management across Asia ▪ A strong, large-scale NH Financial Group-wide support as a subsidiary 100% owned by NHIS ▪ Singapore-based presence with a wide pool of ASEAN-oriented investors
Component 4 **Technical Assistance (TA)**	4.1. Formulate the best workable network framework 4.2. Build capacities to enable and strengthen the relevant ecosystem	**GGGI** ▪ Specialised in capacity building and knowledge sharing activities with a climate lens ▪ Key arranger, advisor for government consultations incl. NDAs, line ministries

	4.3. Promote sustainability via the National SMUs per country	▪ Obtained with a set of policies for executing and managing on-the-ground procurement, contracts, logistics, etc.

The Korea Development Bank (KDB), Accredited Entity. As an accredited entity, KDB shall run and supervise the programme overall during the implementation period – each component to be separately implemented but closely associated by working hand in hand with partner entities. KDB will lead, oversee, and supervise the programme by leveraging the bank's institutional capacity and resources that have fostered the economic growth and industrial advancement of Korea over the last seven decades. In particular, KDB has advanced the Asian venture capital market along with venture and technology banking services for promising start-ups and SMEs with technological prowess since the market began to take a shape during the late 1990s. Meanwhile, as a leading policy financing bank, KDB has set up an internal infrastructure to cultivate, sophisticate, and grow an enabling ecosystem that discovers small giants and supports them in growing into global unicorns for the country's sake of innovative growth through new industrial engines.

<u>**KDB Infrastructure for Start-up Ecosystem Building**</u>



Above all things, the bank's flagship venture investment platform called 'KDB Next Round' is an example of KDB's actions towards developing a start-up ecosystem. IR rounds are held three days a week in the auditorium on the ground floor of the bank's headquarters in Seoul, classified by the development stage of start-ups – (1) IR rounds for Seed and Pre-Series A on Tuesdays; (2) IR rounds for Series A and B on Wednesdays; and (3) IR rounds for Series C to Pre-IPO on Fridays – hence, covering the complete lifecycle of start-ups. The start-up and venture investment inducement platform has continued even in the midst of COVID-19 with live streaming services. It has so far arranged around USD 4 billion in total funding through more than 636 rounds since August 2018, as of 2020 4Q.

Such supervision and guidance as an accredited entity will be primarily provided by Climate Business Team of KDB HQ, while leveraging its in-house three-tier venture investment platform and global start-up/VC acceleration and investment networks: i.e., Singapore Venture Desk, London Venture Desk, and KDB Silicon Valley LLC (VC subsidiary in the US). Such KDB start-up/VC networks at the global financial centres will support the HQ to be equipped with information on timely trends about global acceleration and global technology company and/or climate transaction sourcing, which is critical for the bank's HQ to serve as the control tower that aims at seamless interlinked management amongst the four (4) different components across five (5) NOL countries.

<u>**Executing Entities of Components 1 & 4: KDB, Global Green Growth Institute (GGGI)**</u>

Garnering GGGI's Local Presence, a Noticeable Track Record as a "Greenpreneurs Program" Initiator, and Experience to Manage Non-reimbursable GCF Grant Proceeds. KDB developed a strategic partnership with GGGI amidst the wake of the pandemic crisis in appreciation of its in-country presence embedded in local governments and its prominent role as a trusted advisor for green growth promotion and NDC fulfilment. Garnering GGGI's established local presence, KDB-GGGI partnership has eventually resulted in a great success of obtaining NOLs from all the five NOL countries, while several NDAs had stopped their administrative function in the middle of COVID-19-caused national lockdown. In a nutshell, the programme has been able to minimise the impacts of COVID-19-imposed international travel restrictions, and build constructive stakeholder relationships with the NDAs and potential implementing partners, taking advantage of the five countries' GGGI membership. Such a strategic partnership with GGGI will strengthen the programme's alignment with the country ownership principle, through close communication with NDAs, line ministries, and influential market players across the five partner countries.

Above all things, GGGI Greenpreneurs Program is an incubator supporting young people to devise social enterprise solutions that address sustainability and climate change issues in the communities of GGGI member countries. It consists of web-based training modules, connecting with mentors and subject matter experts, business competition and provision of seed funding for winners. Since 2018, the initiative has engaged, on average, 200 youth green entrepreneurs every year, approximately 800 in total over four years, and has supported 57 teams from 20 different countries, including Indonesia and Cambodia. GGGI has extensive experience in working with the private sector particularly with start-ups and MSMEs in diverse sectors in GGGI member countries with various interventions.

In addition, GGGI has been closely working with various NDAs for the GCF Readiness Programmes as a Delivery Partner under "GCF-GGGI Framework Readiness and Preparatory Support Grant Agreement", which proves GGGI's requisite capacity to implement the GCF grant funds. With increased technical capacity in green and climate finance and strong relationship with public and private stakeholders in both donor countries as well as the developing world including the NOL countries, GGGI has mobilised over USD 8.6 billion in green investment commitments for its members and partners over the course of 2015–2022. In 2023, GGGI has contributed to green investment mobilisation valued at USD 1.8 billion, and has delivered 120 investment advisory outputs.

Executing Entity of Component 2: NH Investment & Securities (NHIS)

Strengthening a Linkage between Components 2 (Grant Component for Acceleration) and 3 (Equity Component for Investment). The programme pursues a close link between Components 2 and 3 because seamless operations between acceleration (Component 2) and investment (Component 3) are of great significance for the sake of achieving desirable climate impacts. Feasibility studies including situation assessment, gap analysis, and stakeholder consultations identified a key problematic issue that most of the graduates (local entrepreneurs) from scattered one-off seed grant initiatives have not obtained the next phase of investment capital needed for scale-up, market entry, and commercialisation. In consideration with such failed precedents learnt, the programme shall devise an implementation structure where later-stage accelerators and investment advisors are involved in the designing and programming of climate technology business acceleration services customised to individual JVs with different technology needs since the initial stage.

In order to achieve this pursuit, NHIS, to be invited into the CTF Investment Committee (IC) as an active advisor under Component 3, will function as Executing Entity of Component 2. As an Executing Entity of Component 2, NHIS will minimise undesirable cases that seed grant-supported entrepreneurs cannot approach the investment phase, are ineligible for component 3, and get stuck. This will enable a close linkage between multi-stakeholder engagement acceleration (Component 2) and investment portfolio management (Component 3), which will foster the programme's effectiveness in terms of resource management.

In order to allow accelerated high-potential JVs with measurable climate impacts to successfully grasp investment opportunities, NH Investment & Securities will serve as a control tower and focal point to take on the responsibilities of a smooth roll-out and the overall management of Component 2, including but not limited to creating acceleration work plans, facilitating partnership arrangements with local/global accelerators while coordinating the roles and responsibilities amongst business consulting advisors and accelerators under optimised communication and resource management, and tracking and reporting performance to KDB in accordance with the GCF principles.

Once the GCF Board Approval, existing service providers of NHIS and NHARP, composed by key experts in global level sourcing, business modelling, due diligence (legal, tax, tech valuation, IP trade, etc.), tech-enabled business localisation, E&S/gender/IP safeguards, and R&DB acceleration of climate tech ventures in partnership with various service providers.

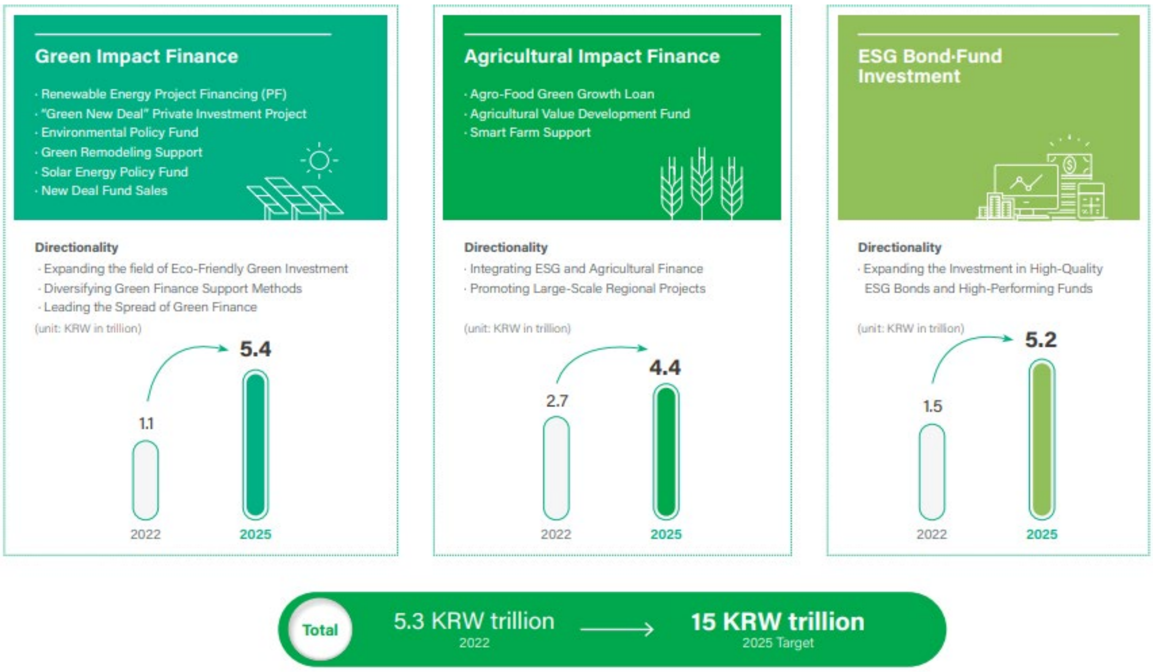
Executing Entity of Component 3: NH Absolute Return Partners (NH ARP)

Strong GP Team of NH Investment & Securities under the NH Financial Group. NH Absolute Return Partners (NH ARP) is the Executing Entity of Component 3 to be the General Partner (GP) entity for a Singapore-domiciled Climate Technopreneurship Fund (CTF).

NH ARP is a Singapore-based subsidiary 100% owned by NH Investment & Securities (NHIS) equipped with 3,136 investment experts, total assets of ca. USD 50 billion, and a stable credit rating of A3 (Moody's) and A-(S&P) (as of 2022), under the NH Financial Group with AUM-included total assets of ca. USD 600 billion. NH ARP is to establish and manage the CTF by fully leveraging the NH Financial Group and its NH Investment & Securities' competitiveness in diverse investment sectors, the Southeast Asian network with a strong local presence in Singapore, Indonesia, Vietnam, and Cambodia, and aggressive investment partnership with global peers (e.g., NH-Amundi Asset Management as the full-service asset management firm with more than ca. USD 47 billion in AUM).

NH targets ESG and green investments of ca. USD 13 billion (KRW15 trillion) in total by 2025, and boasts proactive sustainable business pursuits as a participant in various global initiatives such as the Partnership for Carbon Accounting Financials (PCAF), SBTi (Science Based Targets initiative), Net-Zero Banking Alliance (NZBA), Task Force on Climate-related Financial Disclosures (TCFD), UNEP FI, UN Global Compact, Carbon Disclosure Project (CDP), and Equator Principles (EP).

NH Ambition: ESG Investment Expansion by 2025



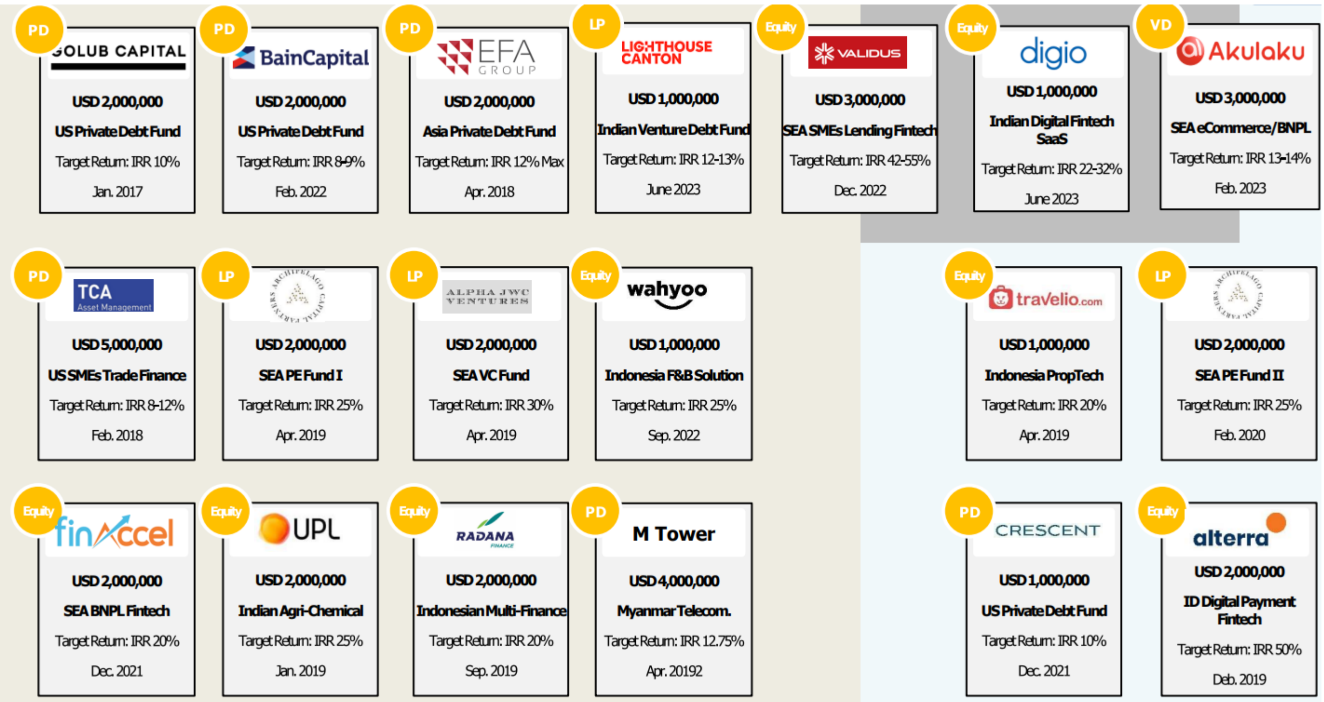
NH-wise Expertise in Financing the Climate Technology Needs of NOL Countries. GP team under the NH brand name that affiliates the National Agricultural Cooperative Federation (NACF) has shown a strong performance in agricultural finance. NH has financed the agricultural industry with diverse financial instruments and products – e.g., instalment financing products with an aim to expand the supply of renewable-fuelled agricultural machinery, farmers' insurance schemes, and agricultural and rural development fund set-up and management, while reinforcing internal and external collaboration (e.g., Foundation of Agricultural Technology Commercialisation & Transfer). Considering an obvious need in climate resilient and sustainable agricultural technologies from all the five countries, NH ARP's engagement as the GP entity will be able to garner the NH-wide stakeholders' demonstrated track record and accumulated experiences as the best-in-class player in financing the agricultural industry globally.

NH business strategy further reaches out to financing many other sectors in which the five NOL countries expressed their keen interest, including but not limited to renewable energy such as agricultural solar power deployment using reclaimed farmland and salt farms, floating solar power projects, and offshore wind power projects, which has contributed to increasing farmers' income as well as the expansion of renewable resources in climate vulnerable rural areas across countries. On the other hands, NH Venture Investment has created a virtuous circle that creates funds, invests in innovation-proven venture firms, increases their value, and finally maximise the returns. In addition, its investment in tech driven logistics companies via the ASEAN Technology Fund 2 in April 2022 proves NH's continued portfolio diversification efforts in innovative companies and promising industries with growth potential. Recently, it intends to expand green finance and investments in 100% carbon-neutral and green bio technologies and AI-based technologies.

It is also noticeable that climate finance professionals of NH Investment & Security (Carbon Finance Team-centred investors' team-up at NHIS Headquarters) are ready to contribute to enhancing climate investment speciality of the CTF. As a global climate financial service provider, NHIS has shown a brilliant track record in GHG reduction and removal projects globally, investment of climate technologies such as DAC and CCUS, investment in climate tech firms, and carbon investment platforms development. The programme anticipates effective delivery of climate outcomes, on the back of the GP's best-in-industry investment capabilities in terms of climate/carbon/green finance.

<u>NH Financial Group Organisation Chart: Garnering NH-wise Capacity</u>

The core person of the CTF key man group (Managing Director, CIO of NH ARP) has been leading all private equity and debt investments of NH Investment & Securities with a geographic focus on Singapore and its neighbouring Southeast Asian countries, in order to expand strategic business partnership opportunities with the ASEAN region's growing innovators, and secure reasonable investment returns in underexplored sectors across emerging markets. Under his leadership, NH ARP currently obtains a diversified investment portfolio as shown below with noticeable returns and performance. Before a move to NH ARP, the core key person had served as an investment professional and/or analyst in various global investment firms such as Merrill Lynch International, Macquarie Securities, Royal Bank of Scotland Asian Securities, and National Pension Service. The person's broad networking power and deep expertise in fund operations, fundraising, and investor relations, among others, will be fully leveraged to successfully operate and manage the CTF in Component 3 in pursuit of the intended programme objectives. In addition to the core person, the CTF GP, NHARP, plans to hire an additional key fund manager within the first anniversary of the FAA signing date to support and supplement the expertise of the core person and NHARP as a whole. The additional key fund manager will bring related expertise in sourcing and investing into climate tech companies to supplement the existing investment expertise of the core person and NHARP, in order to accelerate growth in climate tech investments and achieve both impact and fund returns.



As the CTF GP (fund manager), NHARP bears all authorities and responsibilities pertaining to the CTF general operations. The authorities related to investment is managed by NH ARP, and the CTF Investment Committee (IC) will initially consist of two (2) members appointed by the GP. The GP will actively search for appropriate advisors or consulting firms, and may from time to time appoint additional key professionals to serve as relevant personnel to ensure the perfection of the CTF IC decision-making process.

The authorities related to investment is managed by NHARP under which NH ARP appoints and invites members of the CTF Investment Committee (IC). The CTF IC, to be composed by two (2) seats along with Climate Impact and Compliance Managers as IC Observers with veto rights, is to be called. The CTF IC's investment decision is to be made by affirmative votes from the two representative professionals, along with climate impact and compliance managers with veto rights. While the role of the Climate Impact Managers is mainly to review, advise, monitor, and sustain (report) the climate impact of the investment deals, the managers will be given veto right during both preliminary review committees and investment committees, if the proposed investment deal severely lacks the representation of climate impact viability.

In addition, KDB is to sit in the CTF Advisory Committee (AC) in order to ensure the CTF's investment activities properly aligned with the GCF/KDB objectives which have been aimed at since the programme conceptualisation, approving matters including but not limited to any departures from or changes to pre-agreed investment objectives, policy, and/or restrictions.

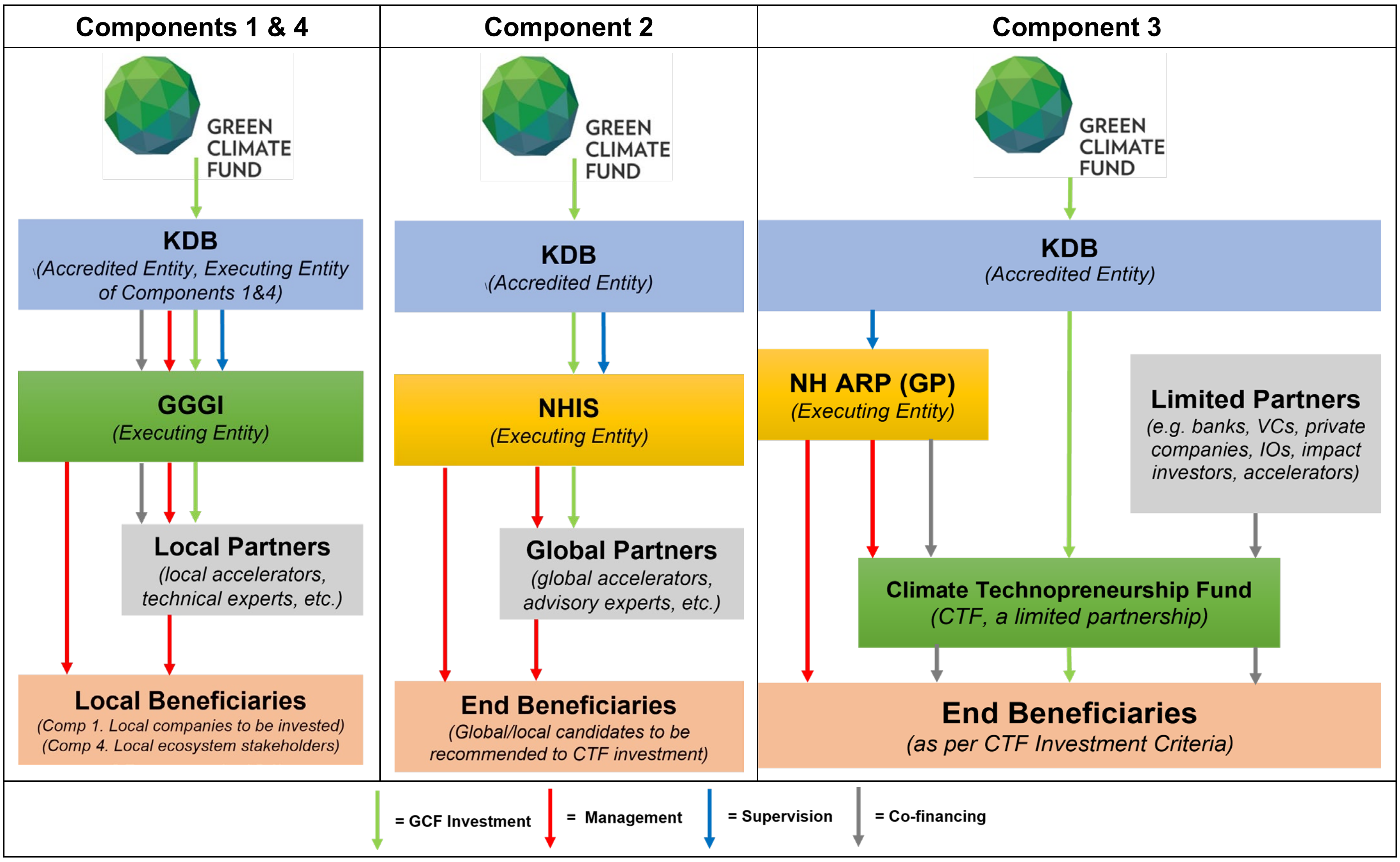
<u>CTF Investment Decision Making Process</u>

Pre-Investment ***(6-8 weeks)***	*Deal Assessment* *(2 weeks)*	• All potential deal pipeline (from inbound/outbound) to be reviewed by investment managers • Deal log to be presented in weekly investment review meetings • Pre-due diligence: macro, industry and company specific attributes to be reviewed
	Preliminary Review Committee (IC)	• Preliminary Review Committee to be called, composed of compliance and a respective climate impact manager (mitigation or adaptation) in addition to investment professionals
	Due Diligence *(6-8 weeks)*	• Deal selection key metrics: business model, growth potential, sector, competition, founders, existing shareholders, etc. • Rigorous modelling of projected financials to be added • On-site visit or virtual meeting with top management • NH-wise research capability in Southeast Asian countries • Review terms and conditions that relate to KDB/GCF standards – e.g., E&S risk assessment, gender mainstreaming, and climate impact measurement
Execution ***(2 weeks)***	*Investment Approval – Investment Committee (IC)* *(2 weeks)*	• Each IC member receives a pack of reports on the deal, prepared by investment managers. • IC called when investment managers come to a firm decision • IC responsible for investment decision making outcomes (performance)
Risk management	*Monitoring*	• Quarterly review on financial, commercial, and operational (including safeguards such as E&S, gender, and climate impact outcomes) performance • In cases of underperforming portfolio companies, investment managers to consider disposal in secondary market, among others

To strengthen the organic connectivity of the programme's four components and ensure more effective execution, NH Investment & Securities along with NHARP plan to establish "Team NH". This team will form an execution Task Force for Components 2 and 3, clearly defining the roles and responsibilities of participating personnel. Subsequently, Team NH aims to further leverage NH Financial Group's internal network and capitalise on professional resources for the programme success including a success of the fund close.

Implementation Structure with Capital Flow and Legal and Contractual Arrangements. Under KDB oversight and supervision in compliance with two bilateral agreements with the GCF – i.e., the Accreditation Master Agreement (AMA) and the Funded Activity Agreement (FAA), the programme's four components will be operational with different legal and contractual arrangements. Due to such varying engagement structures with different executing entities and local and global partner entities, the capital flow varies by component accordingly.

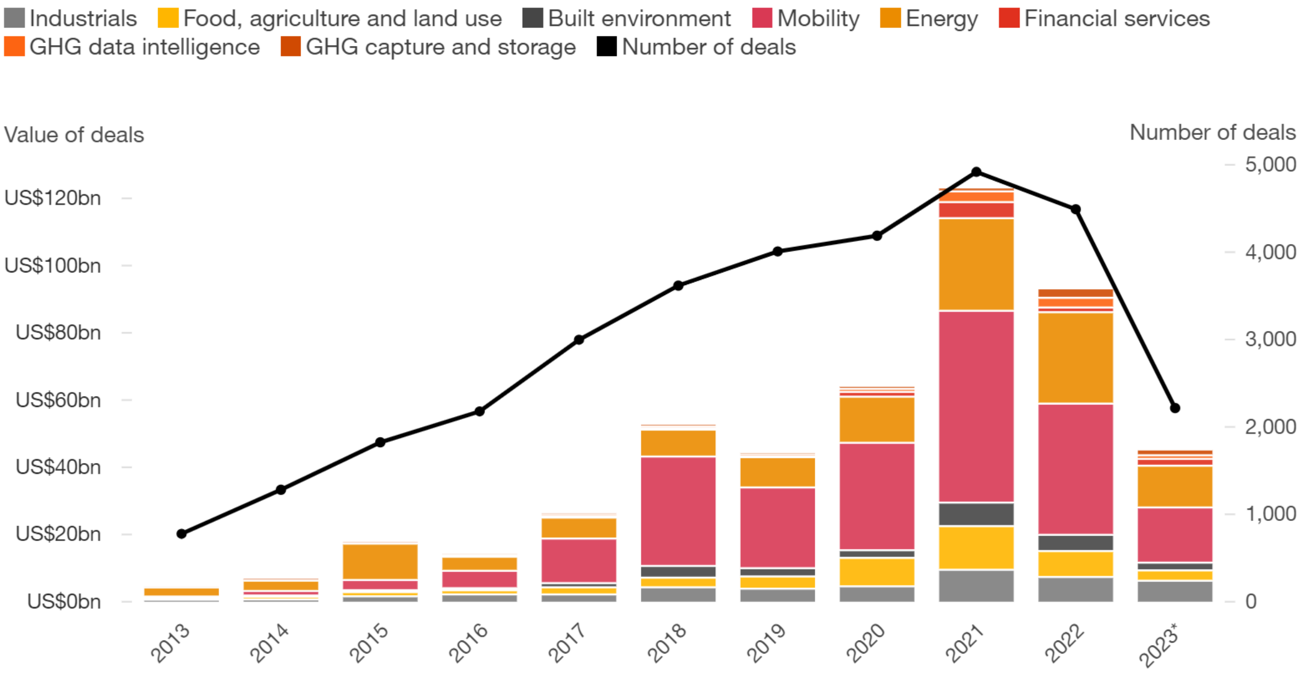
Component-wise Implementation Structure



B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

Make it Work for NOL Countries across ASEAN in Dire Need of Investment in Climate Responsive Technology and a Funding Shortfall. PwC's fourth *State of Climate Tech* report (PwC, 2023[27]) reveals a drop of investments in climate technology start-ups and ventures back to the level of five (5) years ago. The report emphasises that much larger amount of innovation funding is critical for the needed technologies with abatement and adaptation challenges to reach scale in constrained surrounding circumstances.

Climate Technology Investment Trends (Yearly)



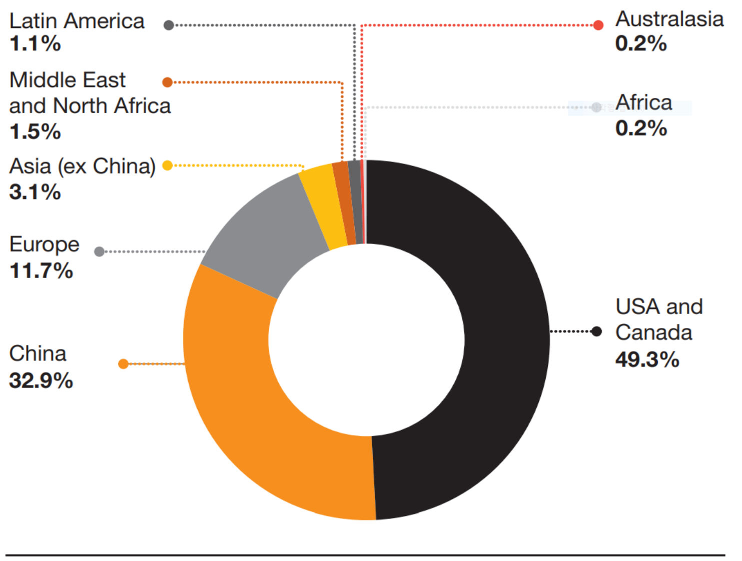
*Data for 2023 is current through the third quarter of the year.
Source: Pitchbook, PwC analysis

It is amply recognised by sources that investments in climate technologies are heavily concentrated in some developed markets and China (PwC, 2020).[28] In reality, such disproportionate capital distribution is not a surprising discovery. The ASEAN bloc has definitely remained the neediest in relation to capital inflows and investments in climate responsive technologies, and the gap is being further exacerbated by the unexpectedly prolonged post-pandemic emergence.

Such unfavourable situation is further compounded by the global economic slowdown coupled with the geopolitical unrest, inflation and interest rate hikes. According to the World Bank, in its recent Global Economic Prospects report (2022), a prolonged period of soaring inflation and frail growth is expected, potentially leading up to severe repercussion on low- and middle-income economies.[29] Investors and enterprises are hesitant to enter into markets that have yet to recover from the utterly catastrophic pandemics, halting the planning of investments into developing and emerging markets. All in all, while developing and emerging markets are disproportionately severely impacted by the economic hardship, the recent downward trajectory of venture funding deals is evident across the globe.

[27] PwC. 2023. The State of Climate Tech 2023: How can the world reverse the fall in climate tech investment?
[28] PwC. 2020. The State of Climate Tech 2020: The next frontier for venture capital.
[29] World Bank. 2022. *Global Economic Prospects, June 2022*. Washington, DC: World Bank. doi: 10.1596/978-1-4648-1843-1.

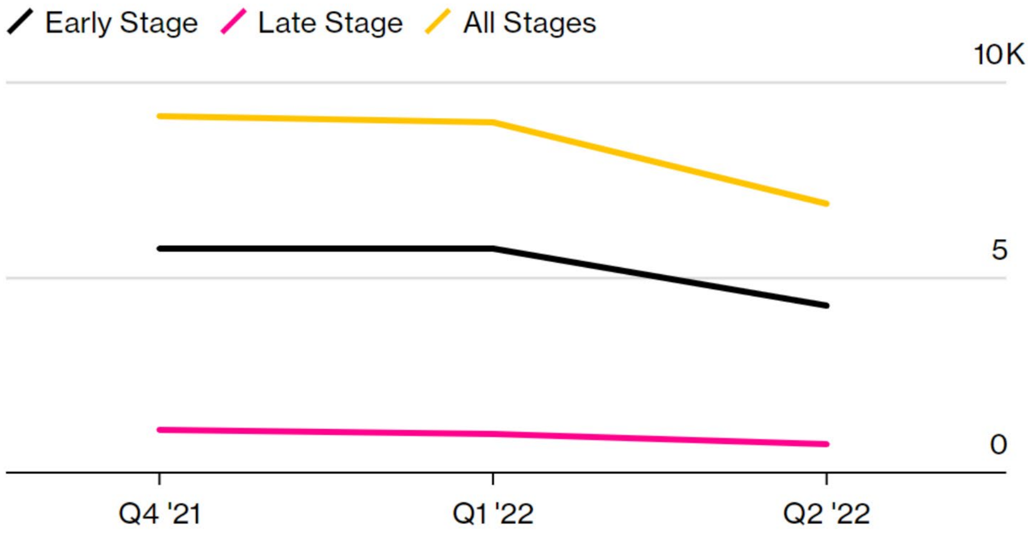
A Geographical Split of Climate Technology Start-up Investments



* Source: PwC (2020, p.33)

The programme, in particular Climate Technopreneurship Fund (CTF), also faces challenges in attracting co-financiers in times of such turbulence in addition to delays in the GCF board approval which would not have hindered potential co-investors to allocate investment proceeds as per their business strategy and outlook otherwise. In light of such imperative investment needs, the GCF engagement as an anchor investor with a first loss risk capital is well-aligned with its mission of assisting developing countries in leapfrogging climate challenges. Such support from the GCF could not have come at a more opportune period, where climate technology acceleration in the emerging economies would call for catalytic, patient capital that would work as a de-risking mechanism to accelerate private investment in decarbonisation and climate resiliency. KDB will join forces with the GCF to catalyse capital for the five ASEAN countries in dire need.

Slowdown of Venture Investment at All Stages (Q4 '21- Q2 '22)



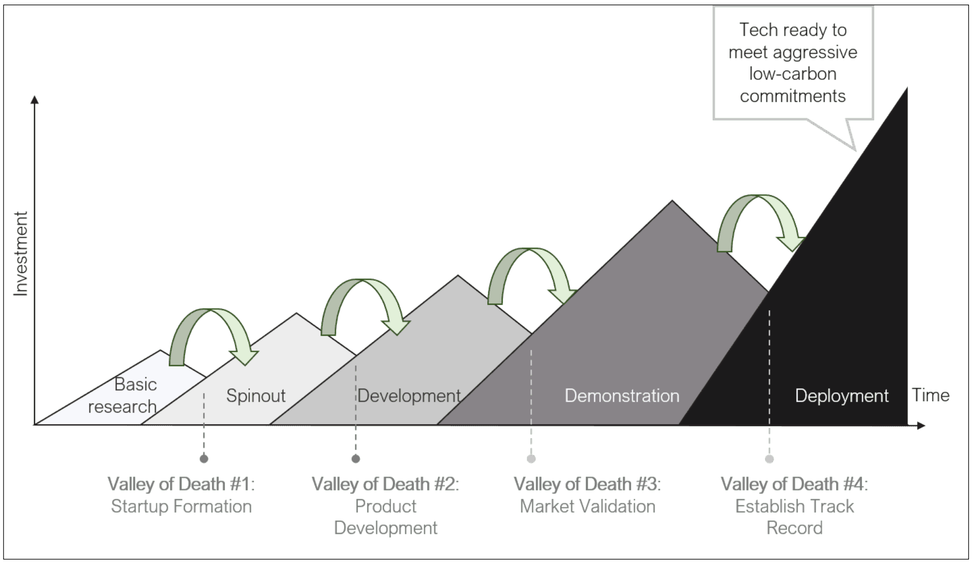
Source: CB Insights
*Q2 '22 data is as of 6/23/22. Early stage includes Series A and earlier. Late stage includes Series D and later.

Financing Barriers Holding Back Climate Technology-enabled Solutions from Flourishing across the ASEAN Bloc. Finance poses a strong barrier in keeping technology-enabled climate solutions from reaching commercialisation. Low-carbon and climate responsive technologies is in need of not only longer investment horizons but also a more holistic cost-benefit approach justifying enhanced resilience, health-related benefits, prospective energy savings, and evaded ecological harm (Pigato et al., 2020).[30] Obviously, dedicated long-term and sufficient patient funding is necessary for entrepreneurs over the course of the early stages of development. Unfortunately, such funding needs become more acute, chronic in fact, when it comes to entrepreneurs with little to no track record in the developing world with a perceived higher risk. A more severe funding gap is identified with entrepreneurs with innovative – in other words, not commercially proven – technology-driven climate solutions, and the level of acuity has increasingly risen as the climate change phenomenon intensifies in frequency and magnitude.

The programme extends over the three challengeable facets that drive entrepreneurs forced into a corner: i.e. newly set up JVs or partnerships with little to no track record (facet 1) in emerging markets (facet 2) for technology-intensive actions (facet 3). At this juncture, GCF has an inevitable role as the world's largest climate financier to perform for the sake of great strides in developing and emerging markets where technology-driven climate solutions are highly insufficient, but so desperately needed. The programme, by enabling the development of low-carbon, environmentally-friendly businesses, could in turn contribute to the economic stabilisation of the ASEAN region and beyond in the long run.

The Programme Shall Invite Long-Standing, Patient Funding Needed to Bridge Four Valleys of Death in Accelerating Climate Responsive Technology Business Deployment. The programme will bridge four valleys of death that under-resourced entrepreneurs are predestined to cross as their climate technology business blossoms: (1) start-up formation; (2) product development; (3) market validation; and (4) track record establishment (Wang and Yee, 2020).[31] Components 1 and 2 plan to address the first and second valleys of death with grant proceeds via which entrepreneurs benefit from start-up JV formation and market-fit product invention under a collaborative R&DB growth platform.

Climate Technology's Four Valleys of Death



(Source: Wang and Yee, 2020)

[30] Pigato, Miria A., Simon J. Black, Damien Dussaux, Zhimin Mao, Miles McKenna, Ryan Rafaty, and Simon Touboul. 2020. *Technology Transfer and Innovation for Low-Carbon Development*. International Development in Focus. Washington, DC: World Bank. doi:10.1596/978-1-4648-1500-3.
[31] Wang and Yee. Climate Tech's Four Valleys of Death and Why We Must Build a Bridge. *Third Derivative*, accessed on June 17, 2020.

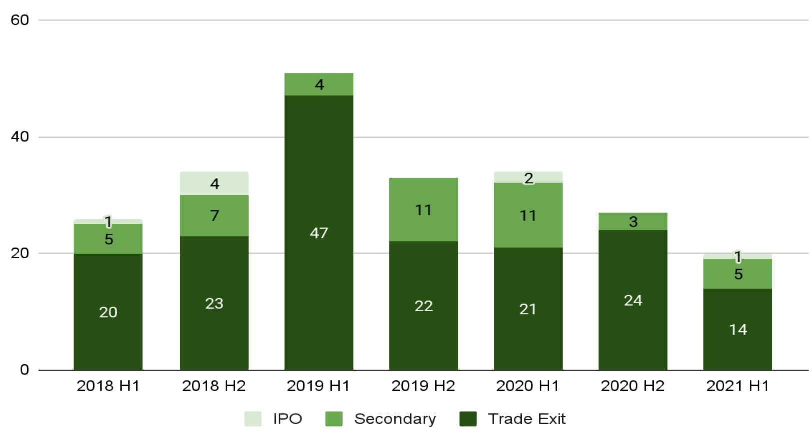
Rendering and Invigorating ASEAN Venture Capital Financing with Catalytic First-Loss Capital. A catalytic first-loss instrument is a must to render venture capital investments appealing and crowd in private capital in a nascent but growing climate technology-themed investment market. In actual, real-world experiences reveal a commonly practiced venture capital market trajectory that private parties, such as venture capitalists, would absorb the first loss up to a pre-negotiated threshold – as a matter of fact, pre-ordered by institutional investors of a stronger bargaining power – prior to the commencement of risk sharing, which has led to skewed capital flows towards stable and low-risk safe markets. In this regard, GCF's catalytic role is vital as the world's largest climate financier; under the global climate finance architecture, no alternative distinguishes itself with catalytic finance at scale. The programme anticipates that GCF's catalytic role be distinctive beyond the above-mentioned default approach so that it can eventually render the ASEAN-centric investments in the first-ever climate technology transfer initiative.

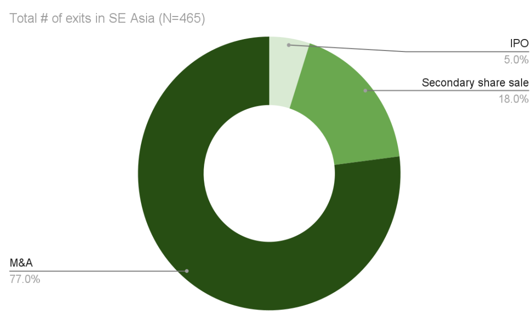
As an investment vehicle of Component 3, the Climate Technopreneurship Fund (CTF) of USD 200 million will be structured with a catalytic first-loss instrument that is funded by the GCF equity proceeds. The GCF's first-loss equity funding as a junior tranche will be a strong signal for the ASEAN-wide VC market by enhancing the fund's credit worthiness, thereby resulting in successful private capital mobilisation under a challenging theme of the programme: i.e., 'climate technology', 'start-up JVs', and 'emerging markets.' The unprecedented first-loss protection mechanism will leverage much larger volumes of co-financing capital than the fund could mobilise on its own in the absence of the GCF's catalytic first-loss capital, laying the foundation for investment flows into underserved or untouched climate technology-driven markets.

B.6. Exit strategy (max. 500 words, approximately 1 page)

Evidence-based Exit Strategy. The GP shall discern the most viable strategy for exiting each business. While exit strategies are necessarily specific to each investment, the GP expects around 10% of its fund portfolio to achieve one of three forms of exit, generating varying amounts of proceeds relative to the investment: i.e., trade sale, secondary sale, and IPO, according to the GP's analyses on start-up exit data collected on its own, as shown below.

<u>Historical Exit Trends in Southeast Asia (2013-2020 Cumulative) and Liquidity Events</u>





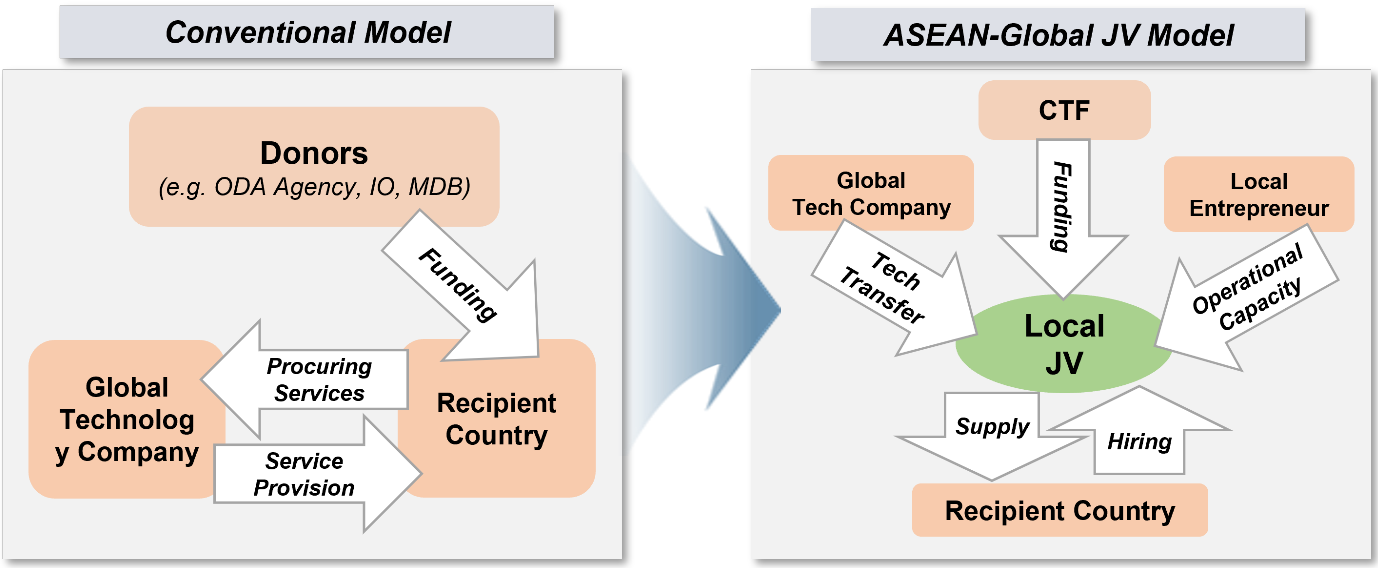
(Source: CB Insight)

The most likely scenario for exit is via trade sale to a strategic acquirer, consistent with historical trends for the region. It is deemed that a "local-to-local in-industry sale," where a local company in the same industry acquires the portfolio company, will be the most likely outcome of the fund's investments. Alternatively, but less commonly, the fund may realise its investments through secondary share sales to investors coming into investment rounds subsequent to the establishment of the fund. Despite the dearth of IPO exits to date for Southeast Asian companies, wherever feasible, the fund will work with its investment portfolio to guide these companies toward becoming publicly listed on a capital market exchange. The GP will consider the following key factors when deciding on an exit strategy: the extent of the fund's investment control, additional capital required by the portfolio company, suitability of acquisition by a strategic third party, public market appetite, and urgency for liquidity.

Post-Exit Sustainability in Pursuit of a Genuine Paradigm Shift. An answer to post-exit sustainability lies in the programme's core concept itself – JV (or partnership) formation. Most of the conventional international development cooperation projects used to subcontract global technology companies and/or suppliers so that a substantial amount of aid proceeds from donors have been given to foreign entities in the form of service fees. In particular, one-time basis grant-funded initiatives are quite often demanding to sustain when initial funding is over with personnel turnover and an exit of foreign entities, like global consulting firms and technology advisory providers. The programme therefore devises a proposal of replacing the conventional trajectory with locally owned JVs as the enablers that ensure sustainability at the local level. As a core instrument of sustainable businesses, ASEAN-Global JVs (or partnerships) with strong local ownership will run the marathon, not sprints, beyond the life of funding so as to be instrumental in bringing about the impacts and benefits of innovation even after the GCF and KDB exit.

The programme's sole private investment component, i.e., Climate Technopreneurship Fund (CTF) under component 3, will bolster the confidence of ASEAN-Global JVs (or partnerships) in ensuring sustainability from contextual adjustments of technology solutions to replication of climate businesses at scale. Sustainability of the programme participants and their businesses will be architected through the long-term structure of the CTF, under which the GP will orchestrate components to optimise support measures for building the skillset and resilience of JVs (or partnerships) and bring them in a formal reporting and auditing mechanism. The JVs will undergo additional due diligence by the GP at a transaction level to ensure that the partnership has been established on a sound legal, reputational, and financial foundation and, adheres to the necessary anti-money laundering, anti-terrorism, anti-bribery, and anti-corruption, and FCPA laws, among other compliance requirements. In addition to actively engaging with each of the developed partnership, the GP will make continuing efforts to attract new investors and create networking opportunities for the programme's ASEAN participants and potential investors and stakeholders. With this, the fund-backed climate responsive ventures will prove the ingenuity of the JV model centred on a fundamental proposition that bankability is key to post-exit sustainability.

Beyond Conventional Model: JV as a Local Sustainable Enabler



C. FINANCING INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount	Currency
	104.471	million USD ($)

GCF financial instrument	Amount	Tenor	Grace period	Pricing
(i) Senior loans	Enter amount	Enter years	Enter years	Enter %
(ii) Subordinated loans	Enter amount	Enter years	Enter years	Enter %
(iii) Equity	83.75	11 years	3 years of extension possible	13.9% IRR (indicative)
(iv) Guarantees	Enter amount	Enter years		
(v) Reimbursable grants	Enter amount			
(vi) Grants	20.721			
(vii) Results-based payments	Enter amount			

(b) Co-financing information	Total amount	Currency
	116,743,900	million USD ($)

Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
GP	Equity	2	million USD ($)	Enter years Enter years	Enter%	junior (Subordinate to GCF)
LPs	Equity	114.25	million USD ($)	Enter years Enter years	Enter%	senior
KDB	In kind	0.1	million USD ($)	Enter years Enter years	Enter%	Options
GGGI	In kind	0.3939	million USD ($)	Enter years Enter years	Enter%	Options

(c) Total financing (c) = (a)+(b)	Amount	Currency
	221.2149	million USD ($)

(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	
	Please explain if any of the financing parties including the AE would benefit from any type of guarantee (e.g. sovereign guarantee, MIGA guarantee). N/A. The programme does not benefit from any type of guarantee mechanism. *Please also explain other contributions such as in-kind contributions including tax exemptions and contributions of assets.* In addition to the indicative financing plan above, the programme additionally anticipates cash and/or in-kind contributions from participating global and local enterprises in buying the JV (or partnership) ownership interests: e.g., capital, physical assets, and IP rights. *Please also include parallel financing associated with this project or programme (refer to the co-financing policy).* The programme anticipates potential parallel financing from various sources, including but not limited to Korea International Cooperation Agency (KOICA), the Korean Ministry of SMEs and Startups (MSS), Food and Agriculture Organisation

(FAO) of the United Nations, and Korea Meteorological Institute (KMI). Each entity that expressed interest with a supportive willingness has been operating under different timelines and procedures for budgeting and disbursement approval with differential funding windows and implementation tracks (e.g., fast-track aligned with the GCF processing for the Funded Activities). In pursuit of investment at scale, a series of practical discussions on parallel finances between KDB and these potential partners have been in progress.

C.2. Financing by component

Component	Output	Indicative cost million USD ($)	GCF financing		Co-financing**		
			Amount million USD ($)	Financial Instrument	Amount million USD ($)	Financial Instrument	Name of Institutions
Component 1	Output 1	6.171	6.071	Grants	0.1	Grants	KDB (in-kind)
Component 2	Output 2	9.5	9.5	Grants			
Component 3	Output 3	200	83.75	Equity	114.25	Equity	LPs
					2	Equity	GP
Component 4	Output 4	4.5531	4.5531	Grants			
PMC		0.9908	0.5969	Grants	0.3939		GGGI (in-kind)
Indicative total cost (USD)		221.2149 million	104.471 million		116.7439 million		

** Above figures are indicative with being rounded.*

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?	Yes ☒ No ☐
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☒ No ☐

The main purpose of the programme is to support the five ASEAN countries in building capacity in order to be able to uptake climate responsive technologies from the global supply chain, self-deploy, and sustain the technologies at scale in pursuit of a genuine technology transfer. In other words, the kernel of this proposal is technology development (customisation) and transfer. In particular, the programme takes a market-driven approach in the belief that success – i.e., self-sustainability – can be achieved only through impact investing-oriented acceleration and consulting services.

In fact, the local markets have witnessed an oversaturated supply of early-stage start-up support services with the generosity of international donors, but graduates from such one-off grant provisions have not led to incoming investments at scale; a problematic status quo. With that in mind, the programme requests grant and equity proceeds of the GCF to deploy the funds appropriately throughout the whole systemic process of technology innovation amongst a range of interacting actors; above all things, devised business models to materialise in the nascent markets. Below is an indicative list of prospective activities, in that R&DB acceleration consulting services are personalised for individual JV teams. Additional detail is provided in both the Demand-driven and Supply-driven Feasibility Studies of Annex 2.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's Initial Investment Framework.

D.1. Impact potential (max. 500 words, approximately 1 page)

Outcome Estimation via Virtual Demand-Supply Technology Matchmaking. As specific technologies, project proponents, and their product specifications are to be identified after JV (or partnership) matching – i.e. little information available on technical specifications, performance, costs, and baseline, the proposal estimates climate impacts with an investment portfolio scenario of the CTF with the total capitalisation of USD 200 million plus the additional programme costs that support the delivery of the mitigation and adaptation outcomes. The forward-looking scenario results from ex-ante technology match between demand and supply assessments and adjusted allocation of investment volumes to virtually matched technologies. See Annex 17, 22, 23, and 24 for detailed description on chosen methodologies for calculation, additionality of assessed technologies against individually set baselines, assessment of common practices, and indicative estimation on potential outcomes by country.

The programme will support new ventures and their technology transfer activities over 39 individual technologies across seven technology groups in five countries. The CTF aims for equal funding distribution across the five countries, but there is still considerable uncertainty about how the funding will be distributed across the technology groups and individual technologies. At the same time, insufficient information – e.g., venture growth and sales data – is available at this programme designing and development stage to evaluate a specific portfolio of investments and technology interventions within the technology groups.

The dimensions of the sectors, measures, and ventures will not be known until R&DB acceleration services and investments are fully implemented; the value of these mitigation and adaptation impacts is extremely difficult to predict at this juncture. As a result, the calculated ex-ante mitigation and adaptation impacts are considered conservative at this stage. However, the mitigation impacts and adaptation beneficiaries will be closely monitored and managed on an annual basis in accordance with the MRV protocol and implementation plan (to be developed as Component 4.3.1 deliverable).

Ex-ante Impact Estimates – Mitigation. Benchmarks were identified from publicly available information. Where available, benchmarks specific to the five countries represented in the funding proposal or specific to Southeast Asia were selected. In some cases, only global average or project-specific benchmarks were available. An effort was made to find the most recent benchmark data available.

Sources of benchmark data include, in order of preference:

1. Benchmarks based on aggregated cost and emission impact data from multiple projects. Examples include data from IRENA, IEA, and other trade and technology associations and NGOs.
2. Marginal emissions abatement cost curves from NGOs, trade and technology associations, and similar organizations
3. Peer-reviewed academic publications
4. Documents from the implementation of individual projects or programs, including GCF funding proposals, case studies, CDM project design documents, and other similar sources

For each technology group, two individual technologies were selected. Where possible, technologies with the broadest applicability across the five countries were selected. For each individual technology, two benchmarks were identified using the criteria described above.

Selected Individual Technologies for Benchmarking

Technology Group	Selected Individual Technology	Applicability				
		C	I	L	P	V
Renewable Energy Generation	Solar energy	X	X	X		X
	Wind energy			X		X
Transmission, Distribution, and Electricity Storage Systems	Off-grid Energy Systems	X		X	X	
	Battery Storage	X			X	
Low-emission transport	Electric Vehicle infrastructure	X		X	X	
	Electric Vehicles	X	X	X	X	X
Residential and Industrial Energy Efficiency Technologies	Lighting	X		X	X	X
	HVAC	X		X	X	X
Waste Management and Treatment	Composting	X	X			X
	Mechanical-Biological Treatment	X	X	X	X	X

In some cases, additional calculations and assumptions were required to transform the available benchmark data into the appropriate units. In other cases, additional data, such as the grid emission factors for the five countries, were needed to complete the calculations. In all cases, the assumptions are documented in the model and the source of data is identified and linked (if available online). Wherever benchmark development required the selection from multiple values or a range of values, values resulting in conservative impact estimates (*i.e.,* a higher USD per tCO2e of mitigated emission) were selected.

For each technology group, an average of the benchmarks representing the selected individual technologies was calculated. The total mitigation impact was then estimated by applying a series of assumptions from the Funding Proposal about the distribution of the total funding amount in the proposal.

- It was assumed that 50% of the total funding would be distributed to mitigation technologies, with 50% being distributed to adaptation technologies; especially, at least 50% (or more) of the CTF funding would be allocated to the GCF adaptation result areas. This resulted in an estimate of approximately $100M in funding distributed to mitigation.
- It was assumed that the funding would be distributed equally among the five countries (20% each).
- An impact multiplier of 2.2x was applied to the mitigation funding allocation to account for the fact that the fund will take only a partial stake in the ventures and that their capitalization will grow during the fund life.

Ex-ante Mitigation Impact Estimate Model Results (Mitigation)

Model Results	Total	Per Country
Funding allocated for mitigation (USD)	$100,000,000	$20,000,000
Fund-Level Mitigation Efficiency Benchmark (USD/tCO2e)	$134.17	
Expected Mitigation Impact (tCO2e)	1,639,681	327,936

The ex-ante impact estimates generated by this model are likely to be highly conservative due to the nature of the programme relative to the benchmarks used. Rather than investing in individual mitigation projects, the programme, through its venture acceleration and capital investments, is seeking to build enterprises that will persist in the market following the fund's exit, and that will help create a market ecosystem for the technologies and a multiplier effect for their deployment and implementation. It presents a conservative measure of 1,639,681.

RESULT AREAS	lifespan GHG emission reduction	Implementation period GHG emission reduction
m1	521,717 tCO2eq	191,296 tCO2eq
m2	1,043,433 tCO2eq	382,592 tCO2eq
m3	74,531 tCO2eq	27,328 tCO2eq
total	1,639,681 tCO2eq	601,216 tCO2eq

Ex-ante Impact Estimates – Adaptation. The methodology developed for the ex-ante estimates is a bottom-up approach based on a benchmark study of approved project examples (full proposals approved by the GCF, as well as from the Asian Development Bank and the World Bank). The methodology includes some examples of mitigation technologies that explicitly generate adaptation benefits or co-benefits, as clarified in full proposals or other benchmark sources used for the mitigation ex-ante impact estimate. From reviewing actual financed projects, it was possible to identify the efficiency figure (US dollars invested per beneficiary) from each example and then find the average efficiency figure across multiple projects to settle on a programme-level benchmark figure. The figure was used as the denominator for dividing the total programme capital for adaptation (technical/venture accelerator assistance and plus the venture fund capital, multiplied by 2.2. – the venture capital model multiplier, as explained in annex 22).

This resulted in the estimated impact – the number of direct and indirect program beneficiaries, disaggregated by gender. The estimated number of beneficiaries should be seen as a reference point rather than a minimum target or projected ceiling, given project specific intervention and contextualisation are essential to assess adaptation impacts.

The steps used to implement this method are as follows:

1. Identify a representative sample of project examples for project technologies (refer to Section B1 of the Funding Proposal and Annex 23 for details on technology groups and individual technologies).
2. Calculate each project example's efficiency figures (for direct and indirect beneficiaries).
3. Calculate the average efficiency figures (direct and indirect beneficiaries) for each adaptation technology group (or mitigation technology group with adaptation co-benefits) included in the KDB funding proposal.
4. Calculate the average efficiency figures (direct and indirect beneficiaries) for all technology groups considered in the steps above (aggregate figure from steps 1-3 above).
5. Divide the total amount of programmatic and investment adaptation capital of the R&DB Programme (based on a targeted programme and capital allocation of 50% to adaptation and 50% to mitigation) by the efficiency figures found in step four to obtain the final ex-ante impact estimate – the number of direct and indirect beneficiaries - at the programme level.
6. Apply figures from country statistics on the percentage of female population to identify the number of female direct and indirect beneficiaries.

The detailed modelling is presented in the Annex 22 spreadsheet. The model's characteristics and boundaries, assumptions, and limitations are shown below.

Model Boundaries

- The adaptation outputs are delivered through financed climate technology-led interventions (ventures applying and bringing to market equipment, tools, services) with related capacity building to deploy and maintain the technologies. Therefore, TA/capacity building and investment capital figures are both utilised for total adaptation spend. Project management, supervision, monitoring, and reporting expenses are also included, as these are intrinsic to the venture capital fund's implementation and compliance with GCF requirements.
- Projects from all four adaptation and cross-cutting technology groups included in the Funding Proposal (Agricultural Technologies & Practices, Water Management & Treatment, and Waste Management & Treatment) are benchmarked. In addition, as mentioned above, benchmarks and project examples from mitigation technology groups that provide adaptation co-benefits were also included in the model.

Model Assumptions

- Total fund investment - USD 220,000,000 - the total investment size (programmatic and investment capital) multiplied by the venture capital financial multiplier of 2.2. The amount from this step is then multiplied by the correspondent percentage allocation to adaptation (50%) and by the percentage allocation to mitigation (50%).
- Country-by-country allocation – equal distribution between the five programme countries.
- Female population: 49.9% (average from the population of the five countries, according to the World Bank Data)
- Indirect beneficiaries: for project examples that did not disclose the number of indirect beneficiaries, we applied the average difference (2.1) between direct and indirect beneficiaries found across the project examples that disclose such information. Such average calculation discounts the outliers. Further, this average was not applied to the mitigation technologies examples with adaptation co-benefits, as the examples from this group did not provide data on indirect beneficiaries.
- Seven adaptation project technologies have been captured –as representative technologies, i.e., those with significant demand according to the country Demand Analysis (detailed in Annex 2). As for the mitigation technology groups, data was found for six individual technologies with adaptation co-benefits.
- In some cases, a single project example offered interventions targeting more than one adaptation or mitigation technology group. In those cases, the total beneficiaries and project amount were equally divided between the individual technologies to find the efficiency figure for their respective technology groups.
- Project examples for benchmarking were drawn from Southeast Asia whenever possible.

List of Representative Adaptation Technologies for Benchmarking

Technology Group	Individual adaptation technologies
Agricultural Technologies & practices	Climate Resilient Agriculture (precision farming, integrated farming, conservation agriculture, sustainable rice cultivation, agro-forestry, crop-diversification, adapted plants and animal genetics/breeding, pest management, water harvesting, retention, and efficient irrigation RE-powered equipment (i.e., pre- and post-harvesting processing)

	Aqua/mariculture development (Mapping changes in the range of fish species, monitoring of fish stocks, Improved allocation of areas for fishing, Changes to fishing gear and fleet, Adjusting fishing fleet composition, RE-powered equipment, i.e., for storage, and processing)
Water Management & Treatment	Water treatment (small scale)
	Water treatment (large scale)
Waste management & treatment (adaptation)	Large-scale waste collection and treatment (i.e., mechanical-biological treatment, aerobic/semi-aerobic digestion, composting), including or not the conversion of gases to energy
	Large-scale waste collection and treatment (i.e., mechanical-biological treatment, aerobic/semi-aerobic digestion, composting), including or not the conversion of gases to energy

List of Mitigation Technologies with Adaptation Co-Benefits

Mitigation tech group with adaptation benefits	Individual Technology
Renewable Energy Generation	Solar
Transmission, distribution and Electricity Storage	Off-grid energy systems
	Battery storage
Residential and Industrial EE Technologies	Lighting
Waste management & treatment (cross-cutting)	Composting
	Mechanical Biological Treatment

Ex-ante Adaptation Impact Estimate Results

Direct beneficiaries (total)	Direct beneficiaries (female)	Indirect beneficiaries (total)	Indirect beneficiaries (female)
1,180,881	589,260	1,132,408	565,071

"Innovation and Additionality Tool (IAT) Scoring Methodology" with a Robust Project Selection Investment Tool (Annex 21). As per the adjusted ex-ante simulation, the best impact scenarios are 1,138,106 tCO_2eq and 1,180,881 direct beneficiaries (ca. 50% of female ratio). By programme nature, estimated impacts expect backward-looking discrepancies, depending on unexpected demand-supply technology match, the fund's real-time risk tolerance, and investing time horizon. While paying heed to such potential discrepancies, the programme shall take three-layered actions to achieve intended, or higher, mitigation and adaptation targets: (i) readiness and acceleration support for JV candidates to extract measurable impacts; (ii) CTF Investment Committee's robust Project Selection Tool; and (iii) third-party independent verification on submitted impacts. Once a technology-enabled climate project is determined, the CTF will monitor invested JV's self-reporting on impacts throughout the fund lifespan, and will annually report outcomes to KDB, which will be verified by periodic site visits of KDB, both in communication with the GCF and in collaboration with executing entities and other engaging partners.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

Paradigm Shifting Fundamental Transition Sought. The programme shall initiate, support, and drive the five countries to grapple with innovation-triggered transformation towards low-carbon and climate-resilient development pathways, completely away from climate-damaging and highly inefficient technologies. In order to achieve this, it does intend to go beyond one-off action, ensuring a fundamental change in perception, structural adjustment for innovation, and new market segments, through customisation, scaling, and replication of transferred technologies.

It is believed that the programme will contribute to a genuine paradigm shift in three fashions. The first and foremost contribution is to introduce a term, *climate technology*, to local entrepreneurs and line ministries – what it refers to, why it is important and how both public and private sectors can shape the relevant ecosystems together. In fact, consultations during the PPF implementation have already initiated such deliberations amongst local stakeholders as they seized the concept within individual national contexts. Second, such heightened awareness will allow the emergence and growth of essential ecosystems such as regulators, policies, agencies, committees, innovation clusters, and VCs with contextual expertise, in the medium to long term phase. Further, the process may give rise to the ASEAN-unique ecosystem culture on the ground of shared values amongst ASEAN innovators, akin to that observed in Silicon Valley. Lastly, the programme will prove the workable innovation investment model which will serve as a standard financing model to encourage the community, including those not participating in the initiative, to move together.

(1) **Cambodia.** There are barriers to education and access to information, particularly for women, due to prevailing social norms related to gender, and there is limited education in science, technology, engineering, and mathematics (STEM). This results in a limited understanding of climate technology and its economic, social, and environmental benefits. To foster access to markets and an enabling environment for the innovation of climate mitigation and adaptation technologies, greater support is needed in the form of technology transfer, education, and finance – exactly what the proposed project aims to offer.

(2) As for **Indonesia,** the programme will kick start the advancement of traditionally under-tapped equity investment mechanisms in the country's start-up incubation and acceleration landscape. This truly remains a niche area, and thus has not garnered much attention from major investors. Thus, the programme's investment vehicle, CTF, will revamp already attractive start-up ecosystems of the country with great potential for a higher leap, shifting away from the current business-as-usual carbon lock-in pathway while replacing climate-damaging industries – e.g., mining, deforestation-linked commodity production – with alternative solutions enhanced by transferred technologies.

(3) **Laos**-centred activities are to be implemented under the national-level strategic direction towards innovative, sustainable, and climate-resilient economy, with a strong emphasis on enabling entrepreneurial ecosystem formulation. Deploying risk tolerance capital, the programme shall create successful climate business cases of JVs (or partnerships) which should function as unprecedented best practices, trigger the green and climate market creation, and inspire policy makers to brainstorm contextually workable solutions for a true paradigm shift.

(4) As for the **Philippines**, 'much-needed interventions' will be carried out, considering that the country has relevant institutions and passionate groups of players in the market but limited awareness of climate responsive technology hinders a more-sophisticated climate innovation leap. The programme's packaged interventions will gather young and fast learning entrepreneurs to uptake and sponge transferred technologies, delivering on a holistic approach on what parts of the climate technology customisation value chain should be further filled and strengthened.

(5) **Vietnam.** Even for the emerging economy with vibrant moves amongst pioneering peers, the climate technology ecosystem still remains quite small and scattered with only a handful of key stakeholders. The programme shall support the nascent yet vibrant ecosystem to awaken innovation potentials at a paradigm-shifting scale, through business acceleration services with a climate lens and substantial capital. This will provide a solid basis for enhanced regulatory frameworks with attractive policy incentives to attract global innovators beyond the lifetime of the programme.

Along with overcoming similar and/or different barriers by country, the programme ensures the following potentials – e.g., scaling up and replication, knowledge sharing and learning, formulation of enabling ecosystem including the regulatory framework and policies, and a comprehensive contribution to a low-carbon and climate-resilient growth pathway at a national and regional levels – that are instrumental to achieving a paradigm shifting transition across the NOL countries.

- **Potential for scaling up and replication:** The programme's basic concept is expected to have significant scaling and replication potential. The proposed intervention with GCF grant and first loss equity funding will pilot acceleration plus customised JV/technology transfer model. Through a successful rollout and lessons learnt from this intervention, the programme can be up-scaled in other developing and emerging countries while leveraging funding from various sources including the governments, MDBs, climate funds, bilateral donors, also private sector including impact investors and foundations. The market-driven platform for venture incubation/acceleration and VC investment as well as multi-country aggregation and programmatic approach demonstrated though this programme will be key in devising and implementing similar programmes in Africa and Latin America.

- **Potential for knowledge sharing and learning**: Introducing brand new terms and concepts, climate technology and acceleration, to the key local ecosystem builders is important. The market-driven intervention also embraces TA activities that will contribute to national-level and ASEAN-wide knowledge sharing and learning under Activity 4.1. and 4.3, for example. This grant-based supportive intervention aims to enhance the awareness of the relevant space, and promote and strengthen national/regional networks for climate technology innovation in the middle of developed communication channels.

- **Contribution to the creation of an enabling environment**: Component 4, a grant-supported TA element, will contribute to strengthening entrepreneurial ecosystems for climate technology and innovation in the five NOL countries by formulating the relevant network framework that is key to acceleration, building capacities of national accelerators and startup/business support entities, establishing national strategies, and offering policy and regulatory recommendations. These activities will allow the emergence and growth of essential enabling ecosystems such as regulators, policies, agencies, committees, innovation clusters, and VCs with contextual expertise, in the medium to the long-term phase. Eventually, all the activities and their outputs are anticipated to give rise to the ASEAN-unique ecosystem culture on the ground of shared values amongst ASEAN rooted innovators.

- **Contribution to the regulatory framework and policies**: The TA component will provide policy and regulatory recommendations which are fitting into each country under Sub-Activity 4.2.3. This will contribute to creating an enabling environment and business ecosystem which will ensure a genuine sustainability of the programme beyond the programme lifespan. Government officials and policy makers will actively involve throughout the programme implementation, contributing to coordinating a high-level inter-ministerial discussion and drafting policy recommendations.

- **Overall contribution to climate-resilient development pathways consistent with relevant national climate change adaptation strategies and plans**: A successful roll out of the programme will accelerate a genuine transition towards climate-resilient development pathways by strengthening national climate entrepreneurial ecosystems and by transferring of technology-enabled climate solutions into five NOL countries. Climate technologies for transfer, localisation, and investment will be selected in line with national climate change adaptation strategies and plans of each country. NDAs and their line ministries of all the NOL countries will actively engage with the programme activities as a member of the Project Steering Committee at each SMU.

D.3. Sustainable development (max. 500 words, approximately 1 page)

Gender Co-benefits through the Emergence of Gender Mainstreamed Innovation Ecosystem with Talented Female Entrepreneurs. Besides the primary climate responsive impacts, the programme anticipates wide-ranging co-benefits that will contribute to the sustainable development of the five partner countries. As becoming sustainable does not happen overnight, this ASEAN region-wide climate technopreneurship initiative shall trigger the tuning of the entire ecosystem and market. Therefore, ensuring positive externalities are so comprehensive that it is actually demanding to single out certain co-benefits. Co-benefits may vary and differ by transferred technology and applied industry, and a technology generally brings about different kinds of benefits and impacts across the ecosystem value chain. As an example, climate-resilient agricultural technologies are prioritised by all the five countries as shown in Table 2. Such advanced technology-enabled agricultural projects can improve efficiency in agricultural land use by applying sustainable agricultural techniques into degraded areas (environmental), guarantee food security in climate-affected vulnerable communities (social), increase income generation of climate vulnerable farmers with better productivity (economic), and build women's capacity at higher level of supply chain such as processing and distributing benefits to women equally (gender co-benefits). Considering such uncertainty, the programme only concentrates on gender co-benefits that are crystal clear at present.

Alignment with the UN Sustainable Development Goals (SDGs). In that the overall ecosystem is to be covered under the programme, the technology transfer trial expects alignment with all of the 17 SDGs, in both direct and indirect ways. Real dynamics reveal that associations between or within SDGs and co-benefits are collectively exhaustive and not mutually exclusive. Nevertheless, the proposal shall anticipate and sort the directly linked alignment of 10 SDGs by gender co-benefit potential as below; e.g., SDG 10 (Reduced Inequalities) is excluded below, but the programme's endeavours to promote economic inclusion of all regardless of sex and ethnicity will contribute to eradicating inequality.

Alignment with SDGs and Potential Co-benefits

Alignment with 11 SDGs	Gender Co-benefits *(Environmental Co-benefits – TBD per Tech)*
SDG 2. Zero hunger with food security & nutrition	• Emergence of women-led/owned SMEs under gendered ownership and management structure • Girls and women's enhanced access to healthcare services, clean water and sanitation facilities, etc. • Women's active participation in climate technology relevant industries • Increased incomes amongst girls and women in agricultural areas vulnerable to climate change impacts • Women's advancement in skilled technology and engineering jobs
SDG 3. Good health and well-being	
SDG 5. Gender equality	
SDG 6. Clean water and sanitation	
SDG 7. Affordable and clean energy	
SDG 9. Industry, innovation and infrastructure	
SDG 11. Sustainable cities and communities	
SDG 13. Climate action	
SDG 15. Protect and restore forests, biodiversity	
SDG 17. Partnerships for the SDGs	

In Pursuit of Market-oriented Sustainability – JV as a Key Agent. The programme has a desire for JVs with full financial and intellectual supports to grow them as pioneers with a mission of greening the ASEAN emerging market. Contextually adjusted long-standing business operation of technology-transferred JVs (or partnerships) will function as the key market-driving agent for sustainable development, activating a number of co-benefits such as local job creation, expanded role for women in enterprise ownership and leadership roles, decent work with fair and safe working conditions, and local resource development – e.g., in terms of human resources, raising potential for under-resourced grassroots innovators to acknowledge what must be done and how the problem can be resolved, a vibrant gender-balanced society with talented women and girls' aspirations, and social inclusion of vulnerable communities with enhanced access to energy. Proactive interplay of the motivated and technology-transferred entrepreneurs will be central to the market's development and advancement, driving the national uplift in various fields of development. Furthermore, additional sustainability co-benefits include the incapacitation of the local JVs on climate technology innovation in the five countries to equip themselves with and be able to operationalise a proper environmental and social management system (ESMS) in accordance with the international safeguard standards for future climate technology innovation even after the CTF exit.

D.4. Needs of recipient (max. 500 words, approximately 1 page)

Outcomes of studies and analyses (see Annex 2) conclude that needs of the five participating countries are classified into three overall categories, though the needs vary by country, in both depth and feature: (i) weak regulatory framework; (ii) limited access to finance; and (iii) nascent or immature climate technology and innovation ecosystem. The programme shall embark on needs-specific technology transfer activities for five different countries, concentrating on (i) the establishing of a regulatory and policy framework, tailored to climate technopreneurs; (ii) addressing the financing gap through private sector capital crowd-in and equity investments at large; and (iii) nurturing a robust innovation ecosystem by awareness enhancement and capacity building assistance.

- **Cambodia'**s corporate structure and business ecosystem is orientated around SMEs, accounting for circa 70% of employment, 99.8% of companies, and 58% of GDP. Surrounded by weak institutional environments for technology innovation, SMEs with a significant footprint have faced challenges, some of which include lack of skilled human resources and poor access to finance as most of them lack the necessary assets to secure credit. The missing middle backbone group, which are too high-risk to be attractive to global investors, will be served by the programme in various manners such as training local talents to be equipped with technology.

- **Indonesia.** Aligned with unprecedented challenges posed by COVID-19 and reallocation of climate priorities and budgets, in 2020, only 13% of the financing needs outlined in the NDC Mitigation Roadmap were addressed. With that, a vibrant private sector engagement with investment crowd-in into climate technology deployment is essential for Indonesia to meet its NDC commitments. In recognition of a harsh reality that the most marginalised group is more likely to experience loss and damage by shifting climate, the fastest adoption of climate technology solutions is urgent. There is no time to wait for the right technologies to mature through traditional cycles of trials and errors.

- **Laos** is seeking to improve overall sustainable management of national resources. Challenges in doing so have undermined the country's long-term development, exacerbated by double disasters of climate change and the pandemic crisis. The Lao PDR government is committed to enhancing access to finance for local entrepreneurs as well as strengthening regulatory and institutional arrangements. The programme will assist the government's novel mission by unlocking the role of under-resourced private sectors via technology trade at a corporate level, devising an innovation-intervened management framework for the country's previous resources at an institutional level, and accompanying both public and private to run a race towards 2050 Net Zero emissions at last.

- **The Philippines'** needs arise due to (i) absence of a centralized repository of information and indicators and data on SMEs regarding climate change adaptation and mitigation; (ii) investment gap; (iii) limited availability of capacity building initiatives for climate technology firms; (iv) impact of the pandemic, community quarantine, and the resulting disruption in the supply chain and distribution channels; (v) low accessibility of public services designed to support entrepreneurs; and (vi) limited support on climate innovation activities. Filipino entrepreneurs will take advantage of the investment opportunities and incentives throughout the programme, witnessing improving market conditions and other enabling circumstances for the climate technology innovation leap.

- **Vietnam** is in need of responding to climate-relevant natural hazards like heat waves, drought, and floods that accelerate over-exploitation of non-substitutable natural resources, like water, at an alarming pace. Meanwhile, bottom of the pyramid groups is disproportionately exposed to risks: primarily residing in the Mekong River Delta, the Red River, and the Central Coast. Although investment in climate actions is urgent to minimise adverse climate impacts, limited availability of climate responsive technologies along with the funding gap for technology entrepreneurs has constructed institutional bottlenecks for climate business deployment. The initiative shall therefore stimulate innovation to trickle benefits down to all, including the most vulnerable, in the economy.

D.5. Country ownership (max. 500 words, approximately 1 page)

Country Ownership at the Heart of the Climate Technopreneurship Initiative. Ensuring ownership of the five countries is at the centre of the innovation proposal which cannot materialise unless it is driven and sustained by the countries themselves. Since the ideation stage, two courses of action have been adopted to strengthen the principle of country ownership and accountability as follows; (i) national strategy and policy review; and (ii) stakeholder consultation and local ownership-centred SMU governance structure.

(1) **Country Planning and Strategies Review**: An extensive literature review was conducted over the course of extracting a prospective list of priority technologies by country (Table 2). In addition to the five countries' Nationally Determined Contributions (NDC) and Technology Needs Assessment (TNA), comprehensive studies on primary policy documents were explored to be aligned with each country's directions and priorities, including but not limited to the following list. For more, see the Demand-driven FS of Annex 2.

Cambodia	*Rectangular Strategy IV 2018-2023, National Strategic Development Plan 2019-2023, National Green Growth Roadmap (2010), National Policy on Green Growth (2013), National Green Growth Strategic Plan 2013-2030, Cambodian Climate Change Strategic Plan 2014-2023, and National Science, Technology and Innovation Policy 2020-2030*
Indonesia	*2020-2024 National Medium-Term Development Plan (RPJMN), the GCF Country Programme Document (CPD, 2020), a National Adaptation Planning – Executive Summary (2019), NDC Adaptation Roadmap (2021), NDC Mitigation Roadmap (2020), 2020-2024 Climate Resilience Development Policy (2021), 3rd Biennial Update Report (2022), and a review on climate change references throughout sectoral strategic plans such as forestry, energy, industry, waste, agriculture*
Laos	*9th National Socio-Economic Development Plan 2021-2025 (NSEDP), National Strategy on Climate Change, National Green Growth Strategy (2018), and National Adaption Programme of Action to Climate Change (NAPA)*

Philippines	*Republic Act 10771: Philippine Green Jobs Act of 2016, Republic Act 9729: Climate Change Act of 2009, Republic Act 9501: Magna Carta for Micro, Small and Medium Enterprises of 2007, Republic Act No. 10644: Go Negosyo Act of 2014, Republic Act No. 11337: Innovative Start-up Act, and Republic Act No.11293: Philippine Innovation Act (PIA), Republic Act 10174: Sustainable Finance Framework*
Vietnam	*National Strategy on Climate Change (2011), National Green Growth Strategy for the period 2021-2030, with a vision to 2050, National Action Plan on Climate Change (NAPCC) for the period 2012 to 2020, Plan for the Implementation of the Paris Agreement on Climate Change (PIPA), National Action Plan for the Implementation of 2030 Agenda for Sustainable Development, Viet Nam's National Climate Change Adaptation Plan (NAP)*

(2) **Local Ownership-centred Governance Structure as a Result of Wide-Ranging Stakeholder Consultations**: KDB has closely engaged with multiple stakeholders from governments, ministries, relevant public entities, state-owned enterprises, VCs, investors, start-up assisting services firms, and international donors present on the ground to CSOs, in partnership with the GGGI equipped with a firm local presence, since the concept designing stage (see Demand-driven FS of Annex 2 for consultation methods and a full list of tapped stakeholders).

Such wide-ranging stakeholder consultations resulted in the SMU governance structure that ensures strong country ownership. The NDAs of the five countries will function as members of the SMU Steering Committee to offer high-level policy guidance and strategic advice over the course of executing country-specific activities, while supporting partnership coordination with entities attached to ministries, enterprises, VCs, CSOs, academia, etc. Undoubtedly, this governance structure will ensure throughout the implementation phase of the programme (see Demand-driven FS of Annex 2 for SMU governance details with information on engaging entities by country).

D.6. Efficiency and effectiveness (max`. 500 words, approximately 1 page)

GCF First-loss Funding as the Catalyst for the Scale-up of Capital in the ASEAN Market. First-loss equity capacity of the GCF is critical in enabling the CTF to meet expected results directed at catalysing innovation-driving venture capital at scale across ASEAN, home to the five attractive markets. In the absence of the GCF's first-loss equity capital, the CTF will face difficulty in mobilising private sector-led investment capital in the middle of the regional post-pandemic recession when investment flows have been exceptionally low. With the GCF, the world's largest climate financier, on board as the anchor investor, funding sources from the private sector can be secured, complementing the GCF mandate on financing climate action.

A successful mobilisation with unprecedented volumes of resources will incite keen attention to the initial closing trial throughout the region with few precedents. Leveraging the initial achievement, the CTF shall arrange the second round of co-investment post-initial closing towards USD 200 million, where KDB shall be a catalytic financier who will re-direct investors' interest in this move to spur additional private sector capital as one of Asia's leading financiers. The multi-closing method is the most viable avenue for catalysing the largest volumes of private-oriented VC finance, thereby achieving the scale needed to fill the innovation funding gap present in ASEAN countries. Besides KDB's investment leveraging power, the GCF will remain significant throughout the fund's lifespan. Such significant GCF commitment will strengthen the fund's credit worthiness in the region-wide venture investment market especially in volatile time of economic downturn and stagnation.

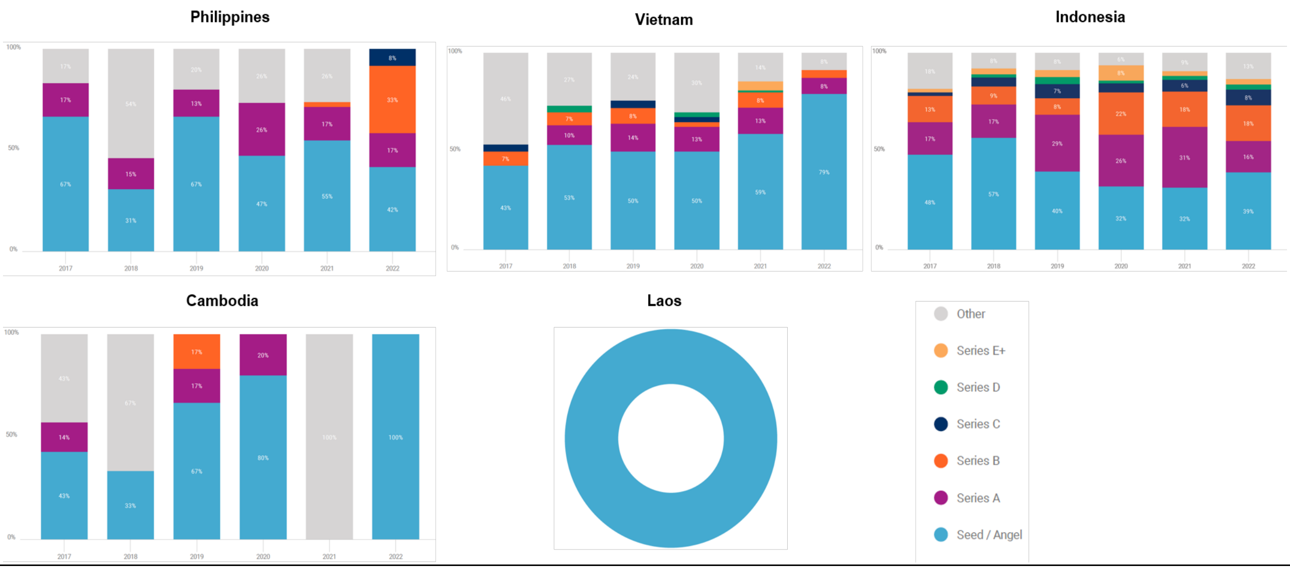
Co-financing Efforts for the Optimal Efficiency and Effectiveness. Institutional, corporate, and private investors are being sought to participate in the programme as limited partners. The consortium, and the GP in particular, is utilising its network and credibility as a successful general manager to initiate conversations and gauge interest and achieve the objectives and goals of the Climate Technopreneurship Fund (CTF). Potential co-investors for the CTF can be comprehensively mapped into following categories once attractive commercial terms and conditions are confirmed and fixed: (i) global investors with previous experience in committing to funds of similar nature, scale, or magnitude (e.g., Temasek and its subsidiaries);(ii) local financial institutions across the five NOL countries; (iii) global and local enterprises that are currently operating (or planning to do so) in at least one of the five NOL countries; (vi) public or private institutions and investors committed to addressing ESG-related issues (e.g., investors and proponents developing technology on Carbon Capture Utilization and Storage); and (v) financial investors, impact investors, or fund of funds that could complement the programme, aligning with the GCF's objectives. It is anticipated that the programme will attract the majority of limited party investors after the technology providers have been identified so that the potential industries of the CTF can be better marketed.

As part of the ongoing investment attraction efforts, KDB plans to organise a series of GCF-themed climate technology showcases and networking events (KDB NextRound), targeting potential investors globally and corporate venture capital firms equipped with a presentation of exemplary technology solutions for climate resilience and other potential investors.

Counterfactual Baseline Comparison. At the outset, it should be emphasised that, without the GCF's participation, the baseline is zero because private financiers, not to mention global technology providers, would have low levels of incentives to strengthen budding innovation ecosystems on climate: i.e. far from easy. The preliminary climate technology market assessment (see Demand-driven FS of Annex 2 for details) has observed that, in spite of the region's growth of digital innovation and technology start-ups, bottlenecks and gaps in the climate innovation ecosystem exist – e.g., lack of climate-focused start-up support intermediaries, absence of legal incentives for start-ups and SMEs on climate technology. In light of this, the GP of the CTF, with the GCF's funding support, targets to execute investments tailored to individual country-specific contexts and structured into specific stages, spanning seed, early (Series A), and growth (Series B and above) stage opportunities, in order to achieve an economic rate of return. See below.

<u>Investment Stage Analytics by Country</u>

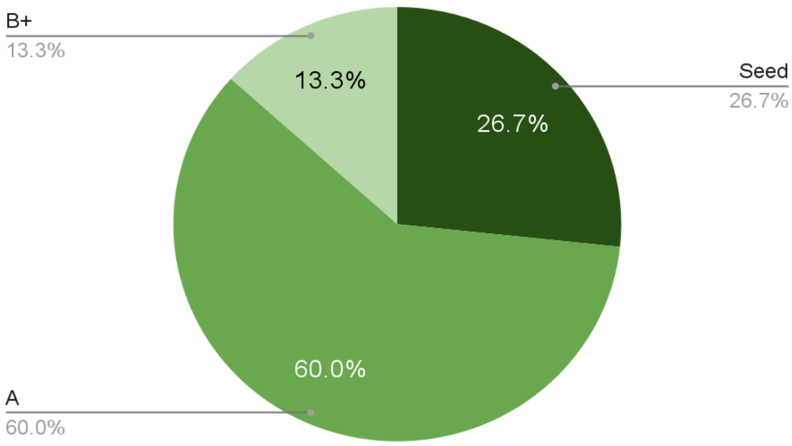
(Source: CB Insight, customised analytics by GP)



Anticipated Investment Stage Focus by Country

	Indonesia	Vietnam	Philippines	Cambodia	Laos
Seed			○	●	●
Series A	○	●	●	○	
Series B+	●	○			

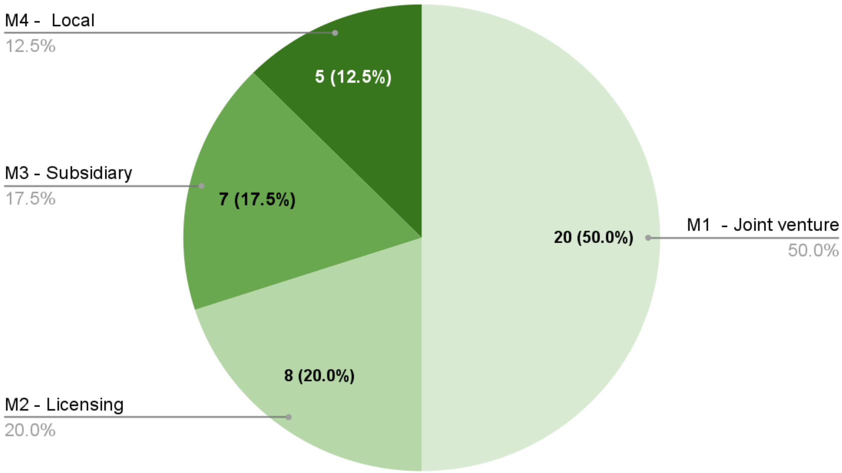
CTF Anticipated Deal Breakdown by Investment Stage



CTF Anticipated Investment Stage Focus by Technology Transfer Model

	Model 1	Model 2	Model 3	Model 4
Seed	●			●
Series A	●		●	
Series B+		●		

CTF Anticipated Deal Count by Technology Transfer Model



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While these are to be adjusted for investment complexity, shifting market trends, and exit considerations, in the long run, the CTF is expected to attain a 7-13.9% rate of return (subject to success rate and returns accordingly as per Annex 3), backed by that ambitious GCF-KDB partnership which will be at the core of cultivating prime, innovative climate technology solutions for the ASEAN region and beyond. See Supply-driven FS of Annex 2 for more.

E. LOGICAL FRAMEWORK

This section refers to the project/programme's logical framework in accordance with the ***GCF's Integrated Results Management Framework*** *to which the project/programme contributes as a whole, including in respect of any co-financing.*

E.1. Project/Programme Focus

Please indicate whether this proposal is for a mitigation or adaptation project/programme. For cross-cutting proposals, select both.

☒ Reduced emissions (mitigation)

☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

This section of the logical framework is meant to help a project/programme monitor and assess how it contributes to the paradigm shift described in section D.2 above by applying three assessment dimensions - scale, replicability, and sustainability.

Accordingly, for each assessment dimension (see the definition per assessment in the accompanying guidance note), describe the current state (baseline) and the potential scenario (target) and rate the current state (baseline) by using the three-point-scale rating (low, medium, and high) provided in the guidance note. Also describe how the project/programme will contribute to that shift/ transformation under respective assessment dimensions (scale, replicability and sustainability). In doing so, please refer to section B.2(a) (theory of change).

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Workable innovative technology-enabled climate solutions rarely exist. A few precedents are mostly small-scale trials, not qualified to be a scalable business model for go-to-market.	Low	A genuine paradigm shift will be achieved by benefiting from economies of scale, in that catalysed VC funds at scale will spur emergence and implementation of likely scalable climate businesses which would otherwise have been the left-behind areas or beneath the surface in the five emerging markets where small-sized proceeds cannot afford expensive tech-driven projects at scale.	The programme will scale the VC fund (CTF) to achieve economies of scale by catalysing private investment crowd-in, anchored by GCF de-risk financing. This intervention is projected to enable the selected forward-looking businesses to advance their scalable capabilities and reach a full-blown commercialisation point in the five countries.

Replicability	No climate technology-transferred business model has been proven to deserve replicability, so that there is a remote yet distinct possibility of replication.	Low	Once a role model climate technology transferred business is identified to work contextually, the model with contextual and technological adjustments will be replicated on a nation-wide and further ASEAN region-wide basis.	The intervention – climate technology transfer model and VC fund risk-sharing structure – is a first of its kind initiative with an ambition to create replicable best practices, so that lessons learnt will guide followers in driving replication across the ASEAN market beyond the programme's target geography.
Sustainability	Climate-responsive innovation ecosystem, key to sustainability, is immature, even though the vulnerability and the type of gaps differ across target countries.	Low	Strengthened innovation ecosystems will foster fast and fit deployment of climate technology solutions, which will lead up to technological leapfrogging of the five countries with inspiration of a genuine ASEAN-wide paradigm shift.	The programme will propel market-driven ecosystem formulation with bankability-centred incentives, in order to encourage key local players with ownership interests to be accountable for continued impacts and benefits after the programme exit.

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

Select appropriate IRMF core and supplementary indicators to monitor project/programme progress. More than one IRMF (core and or supplementary) indicators may be selected as applicable for each GCF results area and project/programme outcome (as defined in the table in section B.2(b)). If IRMF indicators are unable to measure any given project/programme outcomes, project/programme-specific indicators should be developed under section E.5 (project/programme specific indicators).

Below figures are all indicative based on lots of assumption layers and virtual scenarios: for detailed assumptions and scenario, please see Annex 22 and 23.

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final[32]	
All result Areas – MRA 1,2,3 (1,639,681 tCO2eq)	Core 1: GHG emissions reduced, avoided or removed/sequestered	CTF reporting to KDB with third-party independent verification on the annual basis on the annual basis	0	-	Implementation period GHG emission reduction (11 years): 601,216 *tCO2eq*	Zero baseline is assumed because there exist no JVs (outcome producers) in the absence of the acceleration programme. GHG mitigation estimates are presented in detail in

[32] The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

					Implementation period GHG emission reduction by country: 120,243*tCO2eq* Lifespan GHG emission reduction (30yrs): 1, 639,681 *tCO2eq* Lifespan GHG emission reduction by country: 327,936 *tCO2eq*	Annex 22 – (4) Mitigation Spreadsheet.
MRA1 Energy generation and access	Core 1: GHG emissions reduced, avoided or removed/sequestered	CTF reporting to KDB with third-party independent verification on the annual basis on the annual basis	0	-	Implementation period (11yrs): 191,296 *tCO2eq* Total lifespan (30yrs): 521,717 *tCO2eq*	Zero baseline is assumed because there exist no JVs (outcome producers) in the absence of the acceleration programme. GHG mitigation estimates are presented in detail in Annex 22. Mid-term target is assumed zero in regards to 5-year investment period.
	Supplementary 1.2: Installed energy stoarage capacity	CTF reporting to KDB with third-party independent verification on the annual basis	0	TBD	TBD	

	Supplementary 1.3: Installed renewable energy capacity	CTF reporting to KDB with third-party independent verification on the annual basis	0	TBD	TBD	- Implementation Period GHG emission reduction: 417,306 *tCO2eq* - Annual GHG emission reduction: 37,936 *tCO2eq* - Lifespan GHG emission reduction (30 years): 1,138,106 *tCO2eq* TBD – Once sub-project selection is to be done, specific mid-term/final targets will be provided.
	Supplementary 1.4: Renewable energy generated	CTF reporting to KDB with third-party independent verification on the annual basis	0	TBD	TBD	
MRA2 Low-emission transport	Core 1: GHG emissions reduced, avoided or removed/sequestered	CTF reporting to KDB with third-party independent verification on the annual basis	0	-	Implementation period (11yrs): 382,592 *tCO2eq* Total lifespan (30yrs): 1,043,433 *tCO2eq*	
	Supplementary 1.5 Improved low-emission vehicle fuel economy	CTF reporting to KDB with third-party independent verification on the annual basis	0	TBD	TBD	
MRA3 Buildings, cities, industries and appliances	Core 1: GHG emissions reduced, avoided or removed/sequestered	CTF reporting to KDB with third-party independent verification on the annual basis	0	-	Implementation period (11yrs): 27,328 *tCO2eq* Total lifespan (30yrs): 74,531 *tCO2eq*	
	Supplementary 1.1: Annual energy savings	CTF reporting to KDB with third-party independent verification on the annual basis	0	TBD	TBD	

All result Areas – ARA 1, 2 (1,180,881, female 50%)	Core 2: Direct and indirect beneficiaries reached	CTF reporting to KDB with third-party independent verification on the annual basis	0	-	Expected adaptation outcome in total: [Direct beneficiary] 1,180,881 (female, 50%) Country breakdown: 236,176 for each country [Indirect beneficiary] 1,132,408 (female, 50%) Country breakdown: 226,481 for each country	Zero baseline is assumed because there exist no JVs (outcome producers) in the absence of the acceleration programme. Adaptation estimates are presented in detail in Annex 22 - (5) Adaptation Spreadsheet.
ARA1 Most vulnerable people and communities	Core 2: Direct and indirect beneficiaries reached	CTF reporting to KDB with third-party independent verification on the annual basis	0	-	[Direct beneficiary] 224,367 (women 50%) [Indirect beneficiary] 215,157 (women 50%)	Zero baseline is assumed because there exist no JVs (outcome producers) in the absence of the acceleration programme. Detailed assumptions (linkages amongst hazards, risks, and envisioned activities) prescribed in Annex 23. Mid-term target is assumed zero in regards to 5-year investment period.
	Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new	CTF reporting to KDB with third-party independent verification on the annual basis	0	TBD	TBD	

	climate-resilient livelihood options					TBD – Once sub-project selection is to be done, specific mid-term/final targets will be provided.
	Supplementary 2.5: Beneficiaries (female/male) adopting innovations that stregthen climate change resilience	CTF reporting to KDB with third-party independent verification on the annual basis	0	-	[Direct beneficiary] 224,367 (women 50%) [Indirect beneficiary] 215,157 (women 50%)	
ARA2 Health, well-being, food and water security	Core 2: Direct and indirect beneficiaries reached	CTF reporting to KDB with third-party independent verification on the annual basis	0	-	[Direct beneficiary] 956,514 (women 50%) [Indirect beneficiary] 917,250 (women 50%)	
	Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options	CTF reporting to KDB with third-party independent verification on the annual basis	0	TBD	TBD	
	Supplementary 2.2: Beneficiaries (female/male) with improved food security	CTF reporting to KDB with third-party independent verification on the annual basis	0	TBD	TBD	

	Supplementary 2.3: Beneficiaries (female/male) with more climate-resilient water security	CTF reporting to KDB with third-party independent verification on the annual basis	0	TBD	TBD	
	Supplementary 2.5: Beneficiaries (female/male) adopting innovations that stregthen climate change resilience	CTF reporting to KDB with third-party independent verification on the annual basis	0	-	[Direct beneficiary] 956,514 (women 50%) [Indirect beneficiary] 917,250 (women 50%)	

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Select at least two relevant IRMF core (enabling environment) indicators to monitor and elaborate the baseline context and project/programme's targeted outcome against the respective indicators. Rate the current state (baseline) vis-à-vis the target scenario and select the geographical scope of the outcome to be assessed. Describe how the project/programme will contribute towards the target scenario. Refer to a case example in the accompanying guidance to complete this section.

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner	Institutional and regulatory frameworks as well as government initiatives are weak, and highly insufficient for promoting innovation and entrepreneurship with climate solutions.	low	*Relevant institutional and regulatory frameworks will be established and strengthened in NOL countries by adoption of proposed policy and regulatory recommendations.*	The programme will assess gaps, barriers, and opportunities of the existing policy and regulations to foster innovative climate actions, and develop recommendable policy and regulatory for NOL countries to consider and adopt.	Multi-countries
Core Indicator 6: Degree to which GCF	Apt technologies to combat climate change	low	*Access to transferred disruptive technologies*	The programme will facilitate global-ASEAN	Multi-countries

investments contribute to technology deployment, dissemination, development or transfer and innovation	are scarce or insufficient for the demand in local markets which cannot afford high costs of technology imports from wealthy countries.		*will empower and uplift local capacity to self-devise and deploy context-specific climate solutions.*	exchange for local entrepreneurs to purchase and digest technologies not yet on the ASEAN market.	
Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level	Climate tech-specific acceleration and VC markets are nascent or in need of innovation shifts, to be ready for taking up and absorbing climate tech-enabled solutions.	low	*Market-driven entry point, i.e. acceleration, will contribute to market formation and/or sophistication, instrumental to paradigm shifting transformation across ASEAN beyond the nation-wide basis.*	Its market-focused intervention in both financing and ecosystem building will open or mature climate tech-specific acceleration and VC markets on an economically viable scale.	Multi-countries
Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards	Infant/emerging local VC markets lack awareness on climate tech theme, in-depth understanding on structured equity financing, and best practices.	low	*Growing awareness and obtained track record of the five markets will be instrumental in bringing about a true technology innovation nation-wide and ASEAN-wide.*	The programme will spur proactive exchange of expertise, ideas, and experiences amongst Global-ASEAN partners, via knowledge sharing platforms.	Multi-countries

E.5. Project/programme specific indicators (project outcomes and outputs)

*This section should list out project/programme-specific performance indicators (outcomes and outputs) that are not covered in sections above (E.1-E.4). List down tailored indicators to monitor /track progress against relevant project/programme results (outcomes/outputs). AEs have the freedom to decide against which outcomes they would like to set project/programme specific indicators. If any co-benefits are identified in sections B.2(a)(b), and D.3, AEs are encouraged to add and monitor co-benefit indicators under the "**Project/programme co-benefit indicators**" section in table below. Add rows as needed.*

Please number each outcome and output as shown below to indicate association of outputs to the contributing outcome. The numbering for outputs under this section should correspond to the output numbering in annex 4 (detailed budget plan).

Below country breakdown of the mid- and final-targets are indicative. During the implementation period, it will be revised according to country context.

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Output 1. Country-driven shortlist of local JV candidates ready for climate technology uptake (i.e., NOL countries will shortlist their own champion entrepreneurs with climate technology capacity)	# of local entrepreneurs (JV candidates) supported by N-CEAP of Component 1	Quarterly SMU reporting to KDB on N-CEAP	0	50 in the Philippines; 40 in Cambodia, Indonesia, and Viet Nam; 15 in Laos	50 in the Philippines; 40 in Cambodia, Indonesia, and Viet Nam; 15 in Laos	Mid-term and final targets are the same, as activities of Component 1 will be done before the mid-term schedule.
	# of local entrepreneurs (JV candidates) shortlisted for Global Acceleration of Component 2	N-CEAP Innovation challenge demo day event, in addition to Quarterly SMU reporting to KDB on shortlist information (with a possible KDB/GCF visit)	0	20 in Indonesia, the Philippines, and Viet Nam; 12 in Cambodia; 6 in Laos	20 in Indonesia, the Philippines, and Viet Nam; 12 in Cambodia; 6 in Laos	Mid-term and final targets are the same, as activities of Component 1 will be done before the mid-term schedule.
Output 2. Climate tech transfer between Global -ASEAN JVs with investment readiness (i.e. JVs with climate tech-driven solutions are ready for investment)	# of global technology entrepreneurs equipped with tech-enabled climate solutions	Global Acceleration that entails JV match-making with N-CEAP graduates from Component 1, in addition to the Acceleration Secretariat's reporting to KDB	0	5-10 in Indonesia, the Philippines, and Vietnam; 4-5 in Cambodia and Lao PDR	5-10 in Indonesia, the Philippines, and Vietnam; 4-5 in Cambodia and Lao PDR	Conservative estimation, as several technologies may be transferred for a single JV or partnership entity.
	# of gender mainstreamed or committed Global-ASEAN JVs (or	Global acceleration advisory Secretariat's reporting to KDB on JV (or partnership) selection based on Global	0	5-10 in Indonesia, the Philippines, and Vietnam; 4-5 in	5-10 in Indonesia, the Philippines, and Vietnam; 4-5 in	Approx.6-8 expected per country but certain cases may not go through Component 2.

	partnerships) with investment readiness	Acceleration / CTF Investment criteria		Cambodia and Lao PDR	Cambodia and Lao PDR	
Output 3. Climate Technopreneurship Fund with private capital crowd-in	Volume of catalysed capital from private (by result area of MRA1-3, ARA1-2)	GP's quarterly reporting to KDB on initial, subsequent, and final commitments for the CTF	0	(USD million) MRA1: 18.6, MRA2: 37.2, MRA3: 2.325, ARA1: 11.625, ARA2: 46.5	(USD million) MRA1: 18.6, MRA2: 37.2, MRA3: 2.325, ARA1: 11.625, ARA2: 46.5	USD 116.25 million in total, based on first loss equity of USD 83.75 million (Result area breakdown is indicative, based on ex-ante assumptions for an indicative climate outcome estimation) Mid-term and final targets are the same, due to a final close before the mid-term schedule
	Volume of catalysed capital from private (by NOL country)	GP's quarterly reporting to KDB on initial, subsequent, and final commitments for the CTF	0	USD 23.25 million per NOL country	USD 23.25 million per NOL country	USD 116.25 million in total (Country breakdown is indicative, based on an ex-ante assumption of equal distribution by country) Mid-term and final targets are the same, due to a final close before the mid-term schedule
	# of investments	GP's quarterly reporting to KDB on CTF Investment Committee (IC) approval	0	40 (ca. 8 per NOL country)	40 (ca. 8 per NOL country)	Mid-term and final targets are the same, due to all investments to be done before the mid-term: 5-year investment period
	# of gender-sensitive investments	GP's quarterly reporting to KDB on CTF Investment Committee (IC) approval based on	0	20 (ca. 4 per NOL country)	20 (ca. 4 per NOL country)	50% gender-sensitive investment rule applies as per GAP (Annex 8).

		CTF investment criteria, in addition to gender M&E				Mid-term and final targets are the same, due to all investments to be done before the mid-term: 5-year investment period
	CTF's expected rate of return	GP's quarterly reporting to KDB & LPs on CTF financial and investment information	0	n/a	7-13.9%	An indicative figure at the fund termination timeline as per Annex 3 modelled based on two scenarios (best/BAU)
Output 4. Climate technopreneurship to be powered as scale under strengthened relevant ecosystem	Level of relevant tech innovation ecosystem improvement	AE scorecard (0= status quo, 1= little improvement, 2=mid-level improvement, 3=high-level advancement)	0	2	3	AE scorecard sets the status quo at zero, regardless of varying status quo and degree of improvement
	Detailed M&E implementation plan report	M&E report submission to KDB after Steering Committee confirmation	0	1	1	Detailed M&E must entail country-specific details with endorsement from the National SMU Steering Committee under quality control of the Regional SMU Mid-term and final targets are the same, as it will be submitted before the mid-term schedule.
	Annual M&E Report	Annual reporting to KDB to ensure a success of N-CEAPs, with ex-post impact assessment	0	5 annual M&E reports	5 annual M&E reports	Impact includes progress on N-CEAPs and co-benefits.
	# of local experts and trainers trained (female ratio, too)	Periodic SMU reporting to KDB on TA activities		200 in total (ca. 40 per country – 40(C), 100(I), 16(L), 100(P), 60(V))	200 in total (ca. 40 per country – 40(C), 100(I), 16(L), 100(P), 60(V))	Mid-term and final targets are the same, as TA activities will be carried out before the mid-term schedule.

				Women 30%	Women 30%	
	Recommendations on national strategy or policy endorsed by the government of NOL countries	Submission of endorsed national strategies and policy proposal to KDB	0	1	1	Every submission to be done by individual NOL countries, with country fit recommendations. Mid-term and final targets are the same, as it will be submitted before the mid-term schedule.
Project/programme co-benefit indicators *(cannot be identified unless a target technology is chosen)*						
Outcomes/outputs	**Indicator**	**Means of Verification (MoV)**	**Baseline**	**Target**		**Assumptions / Note**
				Mid-term	**Final**	
Co-benefit 1 (Gender): Gender inclusive and gender empowering innovation deployment across NOL countries under the programme that promotes women and girls become central agents and major beneficiaries – i.e., selected women-led entrepreneurs under the programme, in accordance with Gender Action Plan (GAP).	# of locally hired gender expert in each National SMU, which promotes gender expertise on the ground	Annual SMU reporting to KDB on co-benefits with submission of duties randomly requested by the GCF	0	At least 5 locally hired gender experts in NOL countries	At least 5 locally hired gender experts in NOL countries	Compliance to Gender Action Plan (GAP)
	# of local women-led SMEs to be selected as the programme beneficiary	Annual SMU reporting to KDB on co-benefits with submission of duties randomly requested by the GCF	0	At least 39 in total (30% representation per country)	At least 39 in total (30% representation per country)	Compliance to Gender Action Plan (GAP)
	Responsible GAP implementation	Annual SMU reporting to KDB on co-benefits with submission of duties in addition to M&E on GAP implementation	0	5 (Annual reporting to KDB)	5 (Annual reporting to KDB)	Compliance to Gender Action Plan (GAP) with gender responsive TA activities rolled out nationally
Co-benefit 2 (Environment): Varying environmental co-	TDB	Annual SMU reporting to KDB on co-benefits with submission of duties	0	TBD	TBD	Co-benefits to be specified as per project – e.g., water, disaster risk

benefits to NOL countries, as per JV business		randomly requested by the GCF				management, clean energy, and so on.
Co-benefit 3 (Economic): Emergency of climate technology driven entrepreneurs as pioneer industry players	# of locally owned/registered climate technology driven entrepreneurs	Annual SMU reporting to KDB on co-benefits with submission of duties randomly requested by the GCF	0	40	40	Mid-term and final targets are the same, as activities will be carried out before the mid-term schedule.

E.6. Project/programme activities and deliverables

All project activities should be listed here with a description and sub-activities. Significant deliverables should be reflected in annex 5 implementation timetable. Add rows as needed.
Please number the activities as shown below to indicate association of activities to the related outputs provided above in section E.5. Similarly, please number sub-activities as shown below to associate to the related activity.

Activities	Description	Sub-activities	Deliverables
Activity 1.1. Set up a Regional and National Sustainability Management Unit (SMU)	With a Regional SMU at the GGGI Seoul HQ, each country will establish a customised National SMU to lead implementation and enable effective delivery of outputs in consultation with NDAs and key local stakeholders.	Sub-activity 1.1.1. Set up a Regional SMU in GGGI Seoul HQ Sub-activity 1.1.2. Set up a National SMU in each country	Regional SMU governance structure National SMU(s) governance structure (Gender expertise secured through at least 1 recruitment or partnership) Partnership agreement with local implementing partner(s)
Activity 1.2. Launch sourcing strategies to identify qualified local JV candidates	As a country's top-class sourcing unit, SMUs will undertake promotional communication and outreach activities to build a pool of qualified talents in priority industries and technologies with a gender investing focus.	Sub-activity 1.2.1. Develop a robust communication and sourcing strategy Sub-activity 1.2.2. Prepare necessary platforms and materials for sourcing JV candidates Sub-activity 1.2.3. Launch outreach activities	Documented communication and sourcing strategy submitted to the Global Acceleration Advisory Secretariat Offline/online promotional channels/ materials/events/platforms per country to be shared with the Global

		Sub-activity 1.2.4. Narrow down the pool of local JV candidates for acceleration readiness	Acceleration Advisory Secretariat and NDAs List of selected candidates of local entrepreneurs, to be selected for N-CEAPs (30% female-founded, female-headed, or more than 50% women-employed)
Activity 1.3. Offer tailor-made acceleration readiness services to nurture local JV candidates	SMUs will design the fittest N-CEAPs in partnership with local / regional mentors. Under the N-CEAPs, cohorts of shortlisted entrepreneurs will be trained and top talents will be finally selected as JV candidates through a competitive mechanism.	Sub-activity 1.3.1. Design the National Climate Entrepreneur Programme (N-CEAPs) Sub-activity 1.3.2. Train selected candidates for acceleration readiness Sub-activity 1.3.3. Organise a competition to shortlist JV candidates for global acceleration	N-CEAP design document including list of mentors/trainers/experts, activity outlines and gender mainstreaming manual N-CEAP progress reporting to the Global Acceleration Advisory Secretariat Selected finalists ready to digest global level JV matching and business acceleration approved by the Global Acceleration Advisory Secretariat
Activity 2.1. Set up the Global Acceleration Advisory Secretariat	The Global Acceleration Advisory Secretariat will be set up to manage the implementation of the global component and coordinate communication between consortium partners and relevant stakeholders.	Sub-activity 2.1.1. Implement the global component with necessary logistics Sub-activity 2.1.2. Manage multi-dimensional stakeholder communication Sub-activity 2.1.3. Conduct regular M&E and reporting to KDB	Operational manual including E&S/gender components (gender expertise secured through the partnership with Gaia Consult Inc.) submitted to KDB Annual programme execution plans including budget allocation, timelines, and logistical details established Annual performance reporting on the progress of acceleration activities to KDB

Activity 2.2. Launch global sourcing strategies to attract global technology innovators	By leveraging the global innovation network, the Secretariat will source and shortlist global technology businesses with climate-related solutions viable for Global-ASEAN technology transfer.	Sub-activity 2.2.1. Establish a Global-ASEAN pool of global technology innovators Sub-activity 2.2.2. Assess the suitability and feasibility of global technology innovators Sub-activity 2.2.3. Select the finalists for Global-ASEAN technology transfer	Diagnostic assessment on the technological suitability and feasibility results shared with the Investment Committee (IC) List of qualified global technology providers shortlisted upon rounds of reviews and interviews
Activity 2.3. Offer collaborative R&DB acceleration services for JVs	With selected teams on board, a series of tailor-made consulting sessions will take place, to enhance their technical capacity and sharpen go-to-market strategies, leading up to JV/partnership creation between global technology providers and local entrepreneurs.	Sub-activity 2.3.1. Perform diagnostic analyses on participating JV candidates Sub-activity 2.3.2. Offer tailor-made acceleration services for JV candidates Sub-activity 2.3.3. Achieve Global-ASEAN technology transfer with best fit mechanisms Sub-activity 2.3.4. Support investment attraction	Diagnostic assessment results and acceleration consulting plan for JVs List of JVs (or partnerships) enabling and accelerating climate technology transfer Regional event composed of demo day/networking/investment attraction event
Activity 3.1. Execute the CTF formation with raising investment capital	With necessary documentation including term sheets, the investment fund vehicle and related entities will be established, accompanied by closing the GCF into the CTF and securing private investment capital	Sub-activity 3.1.1. Establish the fund vehicle and related entities Sub-activity 3.1.2. Close the GCF into the CTF Sub-activity 3.1.3. Fundraise private investment capital	In-depth legal DD results LPAs with Side Letters Potential investors identified and LP commitments endorsed
Activity 3.2. Execute investments with portfolio management as per investment criteria	Through the process of building up the investment pipeline and performing due diligence respectively, the GP will execute investment, offer structured support and guidance to portfolio companies in areas ranging from capital planning to business modelling for their performance enhancement, and conduct reporting to KDB (semi-annual reporting on GCF relevant	Sub-activity 3.2.1. Perform due diligence on proposed businesses Sub-activity 3.2.2. Build, revise, and manage investment portfolios Sub-activity 3.2.3. Monitor performance of portfolio companies with structured support Sub-activity 3.2.4. Conduct regular reporting to KDB	Legal, financial, technology, and E&S due diligence on target companies Investment memo presented to the IC Fund-level performance report and analytic documentation Quarterly reporting to KDB and LPs

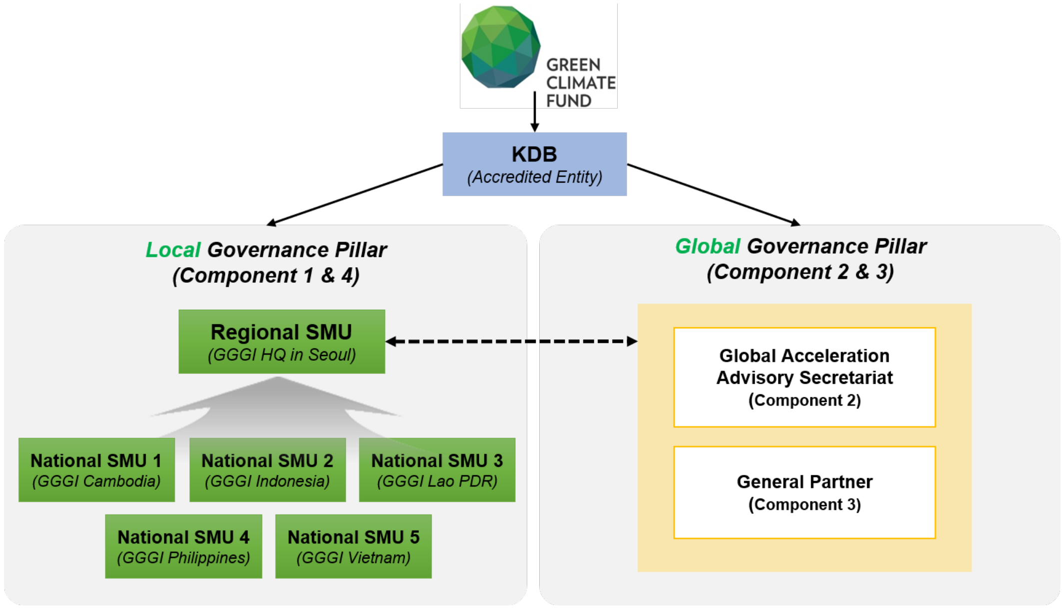
	obligations) and investors (quarterly fund-level performance).		
Activity 3.3. Achieve successful exits with a reasonable rate of return	With mid- and long-term exit strategies calculated, the GP will perform exits (e.g., trade sale, M&A, IPOs, etc.), distribute proceeds and returns, and subsequently execute the fund's post-exit obligations backed by legal counsel, terminating the CTF.	Sub-activity 3.3.1. Realise exits with a reasonable rate of return Sub-activity 3.3.2. Distribute proceeds and returns Sub-activity 3.3.3. Terminate the CTF Sub-activity 3.3.4. Perform the fund's port-exit obligations	Quarterly reporting on exit performance to KDB and LPs Transfer / purchaser details Documentation on necessary disclosure to regulatory parties CTF termination reporting to KDB on completion of post-exit obligations
Activity 4.1. Formulate the best workable network framework	SMUs will set up national networks and a communication hub to connect local greenpreneurs and start-up assistance organisations nationwide and disseminate knowledge products.	Sub-activity 4.1.1. Promote national networks for climate technology innovation Sub-activity 4.1.2. Develop a communication hub with knowledge sharing activities	Online/offline communication channel for networking/knowledge sharing (with at least 30% of women and other vulnerable group representation)
Activity 4.2. Build capacities to enable and strengthen the relevant ecosystem	SMUs will build capacities of national accelerators by training experts and trainers for the N-CEAPs while making recommendations on national strategies and regulatory policies to strengthen climate technopreneurship ecosystem.	Sub-activity 4.2.1. Build capacities of national accelerators to sustain N-CEAPs (1.3.1) Sub-activity 4.2.2. Establish the national strategy on climate technopreneurship promotion Sub-activity 4.2.3. Offer regulatory and policy recommendations	Documentation with a list of national trainers/experts trained for N-CEAP (including gender responsive TA activities) Documentation on national strategies and policy recommendations submitted to NDAs and KDB

Activity 4.3. Promote sustainability via the National SMUs	SMUs will undertake M&E, support ex-post impact assessments, and regularly report to KDB to ensure the successful management of the N-CEAPs and regional initiatives.	Sub-activity 4.3.1. Develop a detailed M&E implementation plan Sub-activity 4.3.2. Conduct regular reporting to KDB Sub-activity 4.3.3. Support ex-post impact assessments Sub-activity 4.3.4. Coordinate with the Regional SMU and the Global Acceleration Advisory Secretariat	Detailed M&E plans submitted to the Global Acceleration Advisory Secretariat and KDB Regular country-by-country M&E reporting including co-benefits to KDB Reporting on need-based assistance for impact assessment to the Regional SMU and the Global Acceleration Advisory Secretariat

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

Reporting to the GCF. As an accredited entity, KDB shall carry out monitoring, reporting, and evaluation duties to the GCF in the forms of the initial annual performance report (APR), interim independent evaluation report, programme completion report (final APR), final independent evaluation report, financial information report on a semi-annual basis and audited financial statements in accordance with the terms and conditions of the AMA and FAA. Information reported to the GCF will be prompt, reliable, precise, and transparent, since all the funded activities will be monitored and KPI associated data and evidence will be collected through a strategic two-track governance system of Executing Entities (GGGI for Components 1 and 4, NHIS for Component 2, and NH ARP for Component 3) under KDB oversight and supervision, along with independent third party audits to ensure quality management over the course of a committed period as shown in the "Two-track Monitoring, Reporting, and Evaluation Governance System" under which GGGI HQ and its

Two-track Monitoring, Reporting, and Evaluation Governance System



local offices across NOL countries and NHIS are to function as important partner entities accountable for reporting necessary information to KDB. The programme shall utilise its Executing Entities' strong local and global presence and capacity to undertake monitoring, reporting, and evaluation arrangements in compliance with GCF standards. KDB's site visits, if possible with the GCF Secretariat (i.e., Portfolio Management Division), will be carried out on both an annual and a random basis, to verify reported information and confirm alleged alignment with key indicators and three-point scale rating.

Financial Reporting to Investors. Reporting to KDB, the CTF is to monitor performance of invested portfolio companies and offer financial information to limited partners on a quarterly basis during the fund's lifespan. At the JV corporate level, portfolio companies are to submit reporting on business performance, climate impacts, and E&S/gender safeguards compliance as committed. Based on expertise of investment trends and insights, the GP and partnering investment advisors shall verify the submitted reporting contents through both routine and random manners to achieve proper monitoring, evaluating, and reporting milestones as planned under the programme.

Transparent Information Disclosure of Invested Sub-Projects for GCF Stakeholders via a Systemic Reporting. For more specific details on the CTF-related reporting, the GP will aim to send to each LP (a) Annual reports within ninety (90) business days of 31 December, the annual audited accounts prepared based on applicable financial standards (including certain information on Related Party Transactions); and (b) Quarterly reports within sixty (60) business days of each quarterly date a report comprising (i) details of investments purchased and of investments disposed of; and (ii) a statement of the investments and other property and assets of the CTF together with a brief commentary on the progress of investments. With the importance of proper reporting acknowledged as an essential responsibility on sub-projects to be financed and managed in a sense of accountable manner to all the GCF stakeholders including LPs and the wider beneficiaries that do include NDAs and line ministries, the programme is committed to provide multi-functional reports if deemed necessary in compliance with the GCF standards.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

The risks outlined below are the primary risks and associated mitigations identified at a programme level for the programme's implementation. At an investment level, each of the JV (or other partnership) candidates for the CTF will undergo a stringent risk assessment and management review, as well as technology, legal, financial, and other due diligence procedures prior to investment as parts of components 2 and 3 of the programme. The JV (or partnership) candidates will review the results of assessed risk and enact appropriate mitigation measures prior to being submitted to the Investment Committee for review.

Selected Risk Factor 1

Category	Probability	Impact
Technical and operational	High	Medium

Description

The programme may face planning, implementation, and execution delays due to the on-going COVID-19 pandemic, associated waves of high numbers of infections, and related travel restrictions and delays in communication and coordination.

Mitigation Measure(s)

The three mitigation measures below will bring the impact from medium to medium-low.

1. To reduce the delay and lost time in its communication with local stakeholders, KDB has early on established a strategic partnership with the GGGI, to not only mobilise each of its local offices and native staff in each of the five countries with the respective language skills and cultural background, but also utilise its expansive network within each of the countries' regional and national governments and ministries.
2. KDB has also converted many of its meetings with the programme's various stakeholders (e.g., local country stakeholders including NDAs, consortium members, GP, consultants, etc.) to virtual platforms (e.g., Zoom, WebEx, Google Meet, etc.) to leverage the resources available to encourage timely communication and efficiency within and across country borders and time zones. Preparations to accommodate virtual acceleration services and consulting services as parts of component 1 and 2 are also being made to not delay any of the JV candidates in receiving the needed services in becoming investment-ready businesses.
3. As the pandemic continues through 2022, the programme plans to incorporate more local consultants who are already on the ground in the respective countries to conduct additional stakeholder consultations and the JV (or other partnership) candidate due diligence services for the local candidates.

Selected Risk Factor 2

Category	Probability	Impact
Technical and operational	High	Medium

Description

The programme may face difficulties in identifying appropriate technology companies to establish investments (in forms of JVs or other partnerships), given the nascent nature of low-carbon and climate-resilient solutions and the difficulty of technology commercialisation.
Mitigation Measure(s)
The three mitigation measures below will bring the impact from medium to medium-low. 1. Executing entities (esp. NHIS and NH ARP) will team up with global players who will support all the pooling, sourcing, and matchmaking activities whenever intervention is necessary, leveraging a strong global network and relevant partnership foundations. 2. Coupled with relevant investment and company operating experience applicable to the fund, the GP will leverage its strong networks of investors, technology partners, and industry experts for identifying climate sensitive companies and securing actionable pipelines. At the same time, the GGGI will strengthen the existing network by bringing in local ecosystem stakeholders across the ASEAN region. 3. A set of criteria encompassing technical feasibility and a minimum threshold of mitigation and adaptation impacts will be set up to validate appropriate candidates.

Selected Risk Factor 3		
Category	**Probability**	**Impact**
Technical and operational	Low	Low
Description		
As the programme plans to provide tailor-made acceleration and consulting services to JV (or other partnership) candidates or JVs themselves based on individual diagnostic analysis, the executing of the acceleration services across components 1 and 2 may not meet the urgency and timeline of limited partner investor needs.		
Mitigation Measure(s)		
The two mitigation measures below ensure the impact remains low. 1. During initial discussions, the GP will share with potential investors the overarching goals of the programme and CTF, as well as the timeframe to ensure that there is a clear understanding that for this programme, and ultimately the sought-after paradigm shift for the ASEAN region, to be successfully achieved, patient capital is required. 2. KDB and the GP of the programme aim to bring on a diverse portfolio of companies across different growth stages to participate in the programme and CTF. The targeted allocation of investments across seed, series A, B, B+ investment types, described in further detail in Annex 2 (Supply-driven FS), is designed to establish a more diversified portfolio for investors across industries and timeframe of returns. With such endeavours, by creating the best possible climate solutions going to market over the first few years of the fund execution, the programme shall narrow the gap with timeline of limited partner investor needs as much as possible.		
Selected Risk Factor 4		

Category	Probability	Impact
Technical and operational	Medium	Low

Description

As a 11-year life investment instrument with diverse cross-sectoral investors and stakeholders, the CTF may face challenges with the fund's long-term governance in light of volatile economic conditions and complexity over the duration, diverse interests, and geographies of its investments.

Mitigation Measure(s)

The three mitigation measures below ensure the impact remains low.

1. The GP will have governance mechanisms and strategies aligned with national long-term economic development plans of respective target countries, leveraging its extensive experience managing private investment vehicles for public and private institutional investors.
2. The CTF will be structured according to industry best practices for long-term fund management including legal provisions, monitoring and reporting requirements of the environmental and social management system (ESMS), and governance requirements for private investment vehicles.
3. The CTF shall leverage the GGGI's solid local presence and network across the ASEAN region. The GP will keep up with the updated local dynamics, and will be informed of the top events and stories of the season in close communication with the Regional Sustainability Management Unit (SMU) which governs five National SMUs in partner countries.

Selected Risk Factor 5

Category	Probability	Impact
Other	Medium	Low

Description

The programme presents challenges associated with the investment portfolio over limited access to information and potential misconduct by portfolio companies, as well as difficulty of attracting follow-on investments.

Mitigation Measure(s)

The three mitigation measures below ensure the impact remains low.

1. The GP will establish communication channels with the investment portfolio to implement regular reporting practices in mitigating risks related to insufficient and/or misleading information provisions.
2. The GP will assist the investment portfolio with diversified strategies to guide them towards adequate capital trajectory with milestones for their sustained growth.
3. In close communication with the GP, the SMUs (GGGI) and the Team NH (centred on NHIS) will offer agile assistance to tackle such risks, leveraging their own expertise for the sake of the programme success – not just for their own components: a local presence and stable network with key local partners for the GGGI; and diplomatic actions as a government agency.

Selected Risk Factor 6

Category	Probability	Impact
Forex	Low	Low

Description

As the programme will invest into five different countries, there may be impacts of dealing with various local currencies and fluctuations in exchange rates with the investments.

Mitigation Measure(s)

Given the likely structure of the CTF in Singapore and investment flow into and out of the respective programme countries with different local currencies, the impact of foreign exchange will primarily be upon repatriation in the form of dividends or upon exit. After the GCF Board approval, the programme in the meantime will perform analysis for the financial risks for investments in project-specific and CTF-level financial models, which can include local currency to USD likely long-term swap rates, for instance. While using these rates could provide adequate comfort that project financials are able to cope with any potential future currency volatility, other measures to minimise the forex risk could be proposed after a customised due diligence at GP's own discretion. This ensures the impact remains low.

Selected Risk Factor 6

Category	Probability	Impact
ML/FT	Low	Low

Description

Risks associated with money laundering, terrorist financing, or other prohibited practices, throughout the process of mobilising and transferring funds via multiple layers of partner entities with weaker anti-money laundering controls and a lack of traceability

Mitigation Measure(s)

The three mitigation measures below ensure the impact remains low.

1. Given KDB, equipped with a strong internal compliance system under a strict/regular guidance of the Korean government as the member of the Financial Action Task Force (FATF), will ensure that the programme adheres to the GCF's Anti-Money Laundering (AML) and Countering the Financing of Terrorism (CFT) Policy in place, as well as other compliance mechanisms to mitigate the risk of money laundering, terrorist financing, or prohibited practices, all executing and partnering local entities are subject to the same policies as well.
2. The CTF is to be operational under the best practice governance mechanism by undertaking investments only after thorough screening and due diligence that include AML/CFT and Know Your Customer (KYC) procedures to prohibit the consideration of an investment with potential and present clients along with their previous transaction records against international and/or local watch lists. Above all, careful attention will be paid when it comes to investment countries in the Grey List.
3. Lastly, legal agreements will require investors to affirmatively confirm that the sources of capital of their investments into the CTF does not come from any prohibited sources.

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

The Programme has developed its own Environmental and Social Management System (ESMS), encompassing both CTF and non-CTF components. CTF component pursues maximisation of sustainability outcomes beyond "do not harm" principles. Non-CTF components (Component 1, 2, and 4) are extension of the fund's environmental and social (E&S) risk management and due diligence. As such, non-CTF components of the ESMS (in Components 1 and 2) aim to support readiness of the applicant JVs as they will reduce rejection rates in the CTF approval stage in Component 3. Non-CTF components also aims to generating co-benefits of capacitating the candidate JVs in E&S management performance, regardless of the fund application results. Importantly, ESMS of the Programme is in line with the Revised Environmental and Social Policy of the GCF (Adopted in B.BM-2021/18).

Mainstreaming Environmental and Social Safeguards (ESS) into the Programme Implementation, CTF is committed to Supporting Projects with E&S Risk Category B/I-2 and below. While micro scale sub-project investments of the CTF mostly anticipate minimal or no adverse social & environmental (E&S) risks – Category C/I-3, the programme set up an environmental and social management system (ESMS) to ensure systematically promoting sustainable co-benefits of the funded activities and safeguarding the ecosystem services, environment, and the communities of all the five NOL countries from environmental and social risks associated with the funded activities beyond the medium (moderate) levels.

The programme proactively follows the mitigation hierarchy to avoid, minimise, and mitigate any risks and impacts. Where avoidance, minimisation, or mitigation measures are not available or sufficient, and where there is sufficient evidence to justify and support viability, remediation and restoration will be required before adequate and equitable compensation of any residual risks and impacts. The ESMS (particularly, for the CTF component) ensures compliance with the respective countries' safeguard system (EIA/E&S laws, regulations, and procedural requirements), GCF ESS/Sexual Exploitation, Abuse, Harassment (SEAH), IFC PS (KDB adherence to it as per the Equator Principles) and Exclusion List, the EHS Guidelines of the World Bank Group, and core labour standards of the International Labour Organisation (ILO), while integrating them into the business cycle to manage any associated E&S risk. The programme will not finance any project classified as high risks (Category A/I-1) and/or likely to pose adverse impacts to the indigenous peoples (IP) in line with the IFC PS 7 (see ESMS Appendix E of the IP screening checklist and Appendix M of the Indigenous Peoples Plan Framework).

CTF's E&S Performance Management is Integrated to the Fund Risk Management. The CTF E&S and Gender Compliance Team will carry out routine ESS compliance duties throughout the entire fund operation process: from the fund application (E&S screening and categorisation), fund appraisal and approval to contracting (E&S review summary and E&S covenant preparation, as required), to *ex-poste* due diligence (compliance monitoring and evaluation of JV-specific regular E&S progress report for Category B activities, recommendation on remedial/corrective actions to grievances/unexpected occurrence of regular/justified risks/outstanding cases). For atypical proposal cases where E&S risks are uncertain and the nature of the risks are highly complex or unknown (i.e., in case the proposal involves a relatively new/emerging climate technologies, and there are not sufficient data base to assess the risks and impacts in advance), the CTF E&S and Gender Compliance Team (ESGCT) will refer the case to the Expert Advisory Committee (EAC/a sub-committee on ESS matters) for consultative procedures to determine the risk category and required E&S management measures (ESIA, ESMP, or any other topical management plans).

In case a certain E&S issue poses potentially high risks to investors and host countries, necessarily including vulnerable groups such as the indigenous communities, the deprived, and women and girls, the E&S Risk Management Committee (ESRMC) will be convened *ad hoc*, inviting the representatives of the stakeholder groups. For the highest level of risk – unjustified, i.e., hard to remedy within the ESMS and the Covenant, so that the CTF is required to consider the cessation or cancellation of investment agreement, the CTF senior management shall be activated to address the concerning red flags. In the latter case, an external audit will be strictly carried out with KDB's direct engagement.

The programme is organised as a fit combination of: (i) preparatory stages for the CTF application (Components 1 and 2); (ii) CTF investment as Component 3 (from application, approval, implementation of the approved climate projects, and monitoring and evaluation); (iii) a simultaneous and/or parallel activities on the ground for local ecosystem capacity building (Component 4).

Accelerating ESS Capacity Enhancement of Shortlisted JVs for Investment Readiness. Given that most of the local applicant companies and some of global innovators might not be properly equipped with a strong level of ESMS, the R&DB readiness and acceleration (components 1 and 2) shall assist local and/or JV level entities in strengthening their preliminary ESMS and climate impact status by providing diagnostic-based advisory support. In addition, upon request, the applicant JVs have access to readiness-stage ESS and climate impact consulting services: i.e., scoping, preliminary E&S screening, and categorisation. E&S advisory services as part of the non-CTF components are extension of the fund's E&S risk management and due diligence, as they will reduce rejection rates in the CTF approval stage in component 3, while generating co-benefits of capacitating the candidate JVs in E&S management performance, regardless of the fund application results.

Information Disclosure & Grievance Redress Mechanism (GRM). For Category B projects, JVs may be required to establish a separate stakeholder engagement and commitment plan (SECP) including the information disclosure plan in accordance with the GCF IDP-aligned ESMF: the frequency and level of disclosure may vary in proportion to the level and nature of the E&S risks and impacts. This will be fine-tuned by the CTF E&S and Gender Compliance Team's (ESGCT) review, consultation, and capacity building support. Also, all of the JVs are required to develop and operate the GRM in accordance with the GCF standards, as KDB, GGGI, SMU, and the GP can also be alternative receptors of grievances in the intermediate executing platform. Grievances lodged at the JV or country SMU level shall be reported through regular procedures to the CTF, which will be primarily reviewed by the CTF E&S Compliance Team. A high-risk grievance will be reported to the E&S Risk Management Committee for their attention, and will be subject to the fund-level adjudication as a critical part of the CTF risk management mechanism. Depending on the nature of grievances and findings of the resultant audit (or internal investigation), the fund's decision may be disclosed to the public through a website and other accessible means to country stakeholders for the sake of transparency and accountability.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

Gender Assessment (GA) and Gender Action Plan (GAP) Established to Overcome the Challenges Facing Women and Girls in STEM and Entrepreneurial Space. The programme's GA and GAP have been devised based on a dynamic process of consultations with key local stakeholders with literature review and feedback communication. GA extracted findings asset that women's entrepreneurial ecosystem seems limited to small and micro-sized enterprises, characterized by low profitability and informality, across the target countries; few women entrepreneurs in the formal sector is attributable to (i) social and cultural constraints, (ii) access to business and markets, and (iii) lack of financing. While limited progress has been recently observed in girls' higher education in all the five countries, women's participation in science, technology, engineering, and math (STEM) has been identified to be significantly less than that of men's. S&T enterprises in the region with cultural stigmatisation seems culturally less likely to welcome women as a counterpart. Women also seem underrepresented in the ICT sector and relevant areas requiring a capacity for innovation. In light of this, women in the five recipient countries need to be given prioritised opportunities to become equal partners for climate technology innovation and entrepreneurship, to empower themselves, and to benefit from the programme as the active key agency: be it a direct recipient and a primary beneficiary of the programme through evolvement of the funded activities.

Gender Benefits. Gender benefits of the proposed programme has two prongs: On one hand, the very act of financing and supporting the climate tech start-ups and JVs will benefit women by reducing their climate vulnerability and enhancing their resilience. Another prong is the programme design (through gender mainstreaming) itself. Most importantly, gender plays a key role as one of the CTF investment criteria (see Table 3); only JVs (or partnerships) screened by the CTF Gender Compliance Team shall grasp the investment opportunities at scale.

As per the CTF investment criteria, the programme deliberately targets female technology leaders and gender-friendly (women-founded, headed or more than 50% employees being women) firms as prioritised beneficiaries of the CTF funding. Selection of the local candidate firms for the CTF will target at least 30% of the selected firms being gender-friendly (outcome statement of component 1 in GAP) and CTF shall target 50% of the fund portfolio companies being either gender-friendly or committing to establish and fulfil their own gender mainstreaming action plans as an investment agreement condition. (outcome statement of component 3 in GAP). Component 4 (TA) of the proposed programme shall focus on nurturing and supporting the creation of enriched gender-inclusive tech-driven entrepreneurship ecosystem.

Integrating Gender Considerations into the Entire Programme since the Programme Inception, Maximising the Synergy Effects with E&S Safeguards Specialists. Since a scratch outline of the concept, the programme has been designed, advanced, and sophisticated with a strategic gender focus, driven by a team of KDB, GGGI, and external gender experts. It has resulted in programme components equipped with measures and tools for nurturing talented women to be skilled human resources and encouraging their proactive engagement surrounded by the region's innovation leap; e.g., Under component 1, partnership with local academic institutions was first inspired with an aim to employ female students majoring in STEM subjects, recognising realistic constraints of forming women-led SMEs (JVs) in component 2 as a result of feasibility studies.

Above all things, the programme plans the establishment and operation of an independent E&S and Gender Compliance Team (ESGCT) in pursuit of the best gender risk management mechanism in close communication with ESS experts, on which the climate technology acceleration initiative will be excellently operational with female talents accounting for 50% of the programme beneficiaries and creation of lots of women-led/owned tech companies in the five countries' nascent innovation markets.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

Adequate Financial Management at the Programme's Core. As a legally authorised accredited entity under the Accreditation Master Agreement (AMA) effective since re-accreditation, KDB will supervise and oversee the whole capital flow of the programme, pursuant to the Funded Activity Agreement (FAA) to be made upon the Board approval on the funding proposal. The programme's financial resources will be channelled, allocated, transferred, managed, monitored, audited, and sustained by a set of legal and contractual arrangements, pursuant to the FAA provisions; KDB pre-sets up the programme-specific structure of multiple legal and contractual arrangements as depicted in the Capital Flow with Legal and Contractual Arrangements of Section B, to be able to direct the alignment of the FAA umbrella setting and control all the contracting parties. Meanwhile, KDB shall receive, channel, hold, administer, monitor, return, and record the GCF proceeds in an independent GCF account in accordance with the AMA and FAA, so as to avoid a mix of investments sourced from KDB or third parties.

In addition to KDB's overarching financial oversight and supervision at a programme level, the programme builds partnership arrangements with global/local investors through intermediation of the GP team, and with local ecosystem stakeholders through intermediation of the GGGI. In that some of them or their sources of capital may come from countries in the Grey List and prohibited sources under AML/CFT guidance, KDB will strengthen stricter due diligence and legal arrangements, surely including AML/CFT covenants, with the executing entities. In particular, the programme has endeavoured a tie with local partner entities under formally regulated systems of finance, in order to ensure adequate management of the proceeds at a project level. Designated local partner entities will function as safety nets to address conventional financiers' heightened concerns over traditional lack of transparent capital flows due to immature financial management systems and informal economic environments in the investment receiving countries. Legal or contractual agreements to be entered into between the GGGI and local acceleration partner entities will stipulate roles, obligations, and responsibilities as robust covenants to ensure transparent and accountable management of financial resources in compliance with the FAA, in terms of, including but not limited to, information disclosure, reporting, anti-money laundering, and counter terrorist financing.

Executing entities will follow KDB's guidance for financial management. In fact, KDB once had operated and managed the GCF Project Preparation Facility (PPF) proceeds in partnership with GGGI and its engaging consortium partners over 11 months, which was fully audited by an independent external auditor and transparently confirmed with a clearance from the GCF Secretariat. In short, as for this team-up, such operation and management capacity of the GCF proceeds have been already proven. KDB, GGGI, PwC (and its consortium members), and the GP team all are equipped with an internal oversight mechanism that includes financial control and internal/external audit, evaluation, investigation, and other management support system to maintain various financial transactions in different forms in line with the global standard, which is transparently disclosed under several internal and external supervisory layers.

Procurement Undertaken in Compliance with KDB Procurement Policy, Proven to be Aligned with the GCF Standard. Upon launching the PPF in July 2021, KDB issued a request of proposals (RfP) to receive competitive bid offers. The two-month long open tender process was carried out in compliance with KDB procurement policy aligned with the *Act on Contracts to Which the State is a Party* as a state-owned bank under scrutiny by oversight authorities as well as civil parties, which abides by the GCF procurement guidance: re-confirmed by the GCF re-accreditation assessment in 2021-2022. The contract was awarded to a PwC-led consortium, composed of PwC's global legal services network, investment advisory firms, and sustainability experts, whose bid met the requirements in the RfP. In addition, NH ARP was assessed and selected as per KDB internal regulation on the fund manager selection.

The programme is accordingly to procure services of the pre-selected consortium members along with the GGGI and its global/local partner entities. As to country-specific selection of local partner entities, during the PPF phase, the GGGI advertised terms of reference (ToR) as per its procurement policy, approved by the GCF standard as Delivery Partner, while the NDAs' recommendations and advice were reserved via consultation. In rolling out activities on advancing climate technology business and the respective ecosystems, relevant, multi-disciplinary stakeholders at both local and global levels are to be adequately procured upon the discretion of the pre-selected consortium members as per KDB guidance, to facilitate the programme execution. For instance, in case technical expertise on a specific climate technology (e.g., weather forecasting, carbon capture, utilisation and storage) is required during the due diligence process, advisory services from specialised technical agencies may be procured on an ad hoc basis.

Procurement Supervision and Oversight. If additional procurement is required in addition to partnering entities procured during the PPF implementation, all executing entities will follow KDB's guidance for procurement – Article 6 (Procurement Method) of the Guidance for GCF-Funded Projects approved by the GCF, while pursuing their own procurement policy and rules only in case theirs are stricter in terms of threshold and method as for GGGI and using the existing service providers of NH ARP. All the procurement activities will be reported to KDB – so accordingly to GCF – while being screened as per regular supervision and oversight duties as Accredited Entity of the programme.

G.4. Disclosure of funding proposal

Note: The Information Disclosure Policy (IDP) provides that the GCF will apply a presumption in favour of disclosure for all information and documents relating to the GCF and its funding activities. Under the IDP, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Information provided in confidence is one of the exceptions, but this exception should not be applied broadly to an entire document if the document contains specific, segregable portions that can be disclosed without prejudice or harm.

Indicate below whether or not the funding proposal includes confidential information.

☐ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☒ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

☒	Annex 1	NDA no-objection letter(s) **(template provided)**
☒	Annex 2	Feasibility study - and a market study, if applicable
☒	Annex 3	Economic and/or financial analyses in spreadsheet format
☒	Annex 4	Detailed budget plan **(template provided)**
☒	Annex 5	Implementation timetable including key project/programme milestones **(template provided)**
☒	Annex 6	E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3): **(ESS disclosure form provided)** ☐ Environmental and Social Impact Assessment (ESIA) or ☐ Environmental and Social Management Plan (ESMP) or ☒ Environmental and Social Management System (ESMS) ☒ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
☒	Annex 7	Summary of consultations and stakeholder engagement plan
☒	Annex 8	Gender assessment and project/programme-level action plan **(template provided)**
☒	Annex 9	Legal due diligence (regulation, taxation and insurance)
☒	Annex 10	Procurement plan **(template provided)**
☒	Annex 11	Monitoring and evaluation plan **(template provided)**
☒	Annex 12	AE fee request **(template provided)**
☐	Annex 13	Co-financing letter of intent, if applicable **(template provided)**
☒	Annex 14	Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

☐	Annex 15	Evidence of internal approval **(template provided)**
☐	Annex 16	Map(s) indicating the location of proposed interventions
☒	Annex 17	Multi-country project/programme information **(template provided)**
☐	Annex 18	Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
☐	Annex 19	Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
☐	Annex 20	First level AML/CFT (KYC) assessment
☒	Annex 21	Operations manual + Project Selection Tool
☒	Annex 22	Assessment of GHG emission reductions and adaptation benefits (ex-ante impact modelling)
☒	Annex 23	Climate basis and additionality analysis
☒	Annex 24	Country-level climate rationale
☒	Annex 25	EE Capacity Assessment

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

No-objection letter(s) issued by the national designated authority(ies) or focal point(s)

KINGDOM OF CAMBODIA
Nation Religion King

Ministry of Environment
Nº : 595 MoE

Phnom Penh, 13 April 2022

Mr. Yannick Glemarec
Executive Director
Green Climate Fund Secretariat
Songdo International Business District
G-Tower, 175 Art Center-daero
Yeonsu-gu, Incheon 22004
Republic of Korea

Re: Funding proposal for the GCF by the Korea Development Bank (KDB) regarding Collaborative R&DB Programme for Promoting the Innovation of Climate Technopreneurship

Dear Mr. Glemarec,

We refer to the programme titled *Collaborative R&DB Programme for Promoting the Innovation of Climate Technopreneurship* as included in the funding proposal submitted by the Korea Development Bank (KDB) to us on 8 March 2022.

The undersigned is the duly authorized representative of the Ministry of Environment, the National Designated Authority of Cambodia.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The Royal Government of Cambodia has no-objection to the programme as included in the funding proposal;
(b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Cambodia;
(c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Please, **Mr. Glemarec**, accept the assurances of my high consideration and regards.

Sincerely yours,

SAY Samal
Minister of Environment
Chair of the National Council for Sustainable Development

Morodok Techo Building (Lot 503) Tonle Bassac, Chamkarmorn, Phnom Penh, CAMBODIA, Phone: 023 213 908 / 023 220 369. Fax: 023 212 540

MINISTRY OF FINANCE OF THE REPUBLIC OF INDONESIA
FISCAL POLICY AGENCY

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Ref. : S-51/KF/2022 08 April 2022

Mr. Yannick Glemarec
Executive Director
Secretariat of the Green Climate Fund
175, Art center-daero
Yeonsu-gu, Incheon 406-840
Republic of Korea

Subject : Funding Proposal for the GCF by Korean Development Bank regarding Collaborative R&DB Programme for Promoting the Innovation of Climate Technopreneurship

Dear Mr. Glemarec,

We refer to the program of Collaborative R&DB Programme for Promoting the Innovation of Climate Technopreneurship submitted by Korean Development Bank to us on 10 March 2022.

The undersigned is the Chairman of Fiscal Policy Agency, Ministry of Finance as the National Designated Authority of Indonesia.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Indonesia has no-objection to the programme as included in the funding proposal;

(b) The programme as included in the funding proposal is in conformity with Indonesia's national priorities, strategies and plans;

(c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in comformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the program.

We acknowledge that this letter will be made publicly available on the GCF website.

Yours faithfully,

Ditandatangani secara elektronik
Febrio Nathan Kacaribu
Chairman

Lao People's Democratic Republic
Peace Independence Democracy Unity Prosperity

2309

Ministry of Natural Resources and Environment

No:................./MONRE
Vientiane dated: 3 May 2022

To: The Green Climate Fund (GCF)

Subject: No-Objection Letter - Funding proposal for the GCF by "the Korean Development Bank (KDB)" regarding "Collaborative R&DB Programme for Promoting the Innovation of Climate Technopreneurship"

Dear Madam/Sir,

We refer to the programme "Collaborative R&DB Programme for Promoting the Innovation of Climate Technopreneurship" in Lao People's Democratic Republic (Lao PDR) as included in the funding proposal submitted by "the Korean Development Bank (KDB)" to us on 24 February 2022.

The undersigned is the duly authorized representative of the Ministry of Natural Resources and Environment (MONRE), the National Designated Authority of Lao PDR.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our non-objection to the programme as included in the funding proposal.

By communicating our non-objection, it is implied that:

a) The government of Lao PDR has no objection to the programme as included in the funding proposal.
b) The programme as included in the funding proposal is in conformity with Lao PDR's national priorities, strategies and plans.
c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the program as included in the funding proposal has been duly followed and confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Sincerely,

Phouvong Luangxaysana
Vice Minister
National Designated Authority of Lao PDR

Republic of the Philippines
DEPARTMENT OF FINANCE
Roxas Boulevard Corner Pablo Ocampo, Sr. Street
Manila 1004

MR. YANNICK GLEMAREC
Executive Director
Green Climate Fund (GCF)
175 Art Center-daero
Yeonsu-gu, Incheon
Republic of Korea

SUBJECT : **No Objection Letter (NOL) for the Funding Proposal "Collaborative Research, Development and Business (R&DB) Programme for Promoting the Innovation of Climate Technopreneurship" for the GCF**

Dear **Mr. GLEMAREC:**

We refer to the programme titled "Collaborative Research, Development and Business (R&DB) Programme for Promoting the Innovation of Climate Technopreneurship" as included in the funding proposal submitted by the Korea Development Bank (KDB).

The undersigned is the duly authorized representative of the Department of Finance, the National Designated Authority of the Republic of the Philippines.

Pursuant to GCF decisions B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The Government of the Republic of the Philippines has no-objection to the programme as included in the funding proposal;

(b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of the Government of the Republic of the Philippines; and

(c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Very truly yours,

BENJAMIN E. DIOKNO
Secretary of Finance

SEP 13 2022

DEPARTMENT OF FINANCE
REPUBLIC OF THE PHILIPPINES
019634

Ministry of Planning and Investment

Socialist Republic of Viet Nam

To: The Green Climate Fund ("GCF")

Hanoi, 22, June, 2022

Re: Funding proposal to the GCF by the KDB regarding Collaborative R&DB Programme for Promoting the Innovation of Climate Technopreneurship

Dear Sir/Madam,

We refer to the programme titled *Collaborative R&DB Programme for Promoting the Innovation of Climate Technopreneurship* in Viet Nam as included in the funding proposal submitted by Korea Development Bank to us on May, 2022.

The undersigned is the duly authorized representative of the Ministry of Planning and Investment, the National Designated Authority of Viet Nam.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Viet Nam has no-objection to the programme as included in the funding proposal;

(b) The programme as included in the funding proposal is in conformity with Viet Nam's national priorities, strategies and plans;

(c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Le Viet Anh
GCF NDA of Viet Nam
Ministry of Planning and Investment

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Collaborative R&DB Programme for Promoting the Innovation of Climate Technopreneurship
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Private
Accredited entity	Korea Development Bank (KDB)
Environmental and social safeguards (ESS) category	Category I-2
Location – specific location(s) of project or target country or location(s) of programme	Cambodia, Indonesia, Lao PDR, Philippines, and Vietnam
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	Friday, June 14, 2024
Language(s) of disclosure	English, Bahasa Indonesia, Khmer, Lao, and Vietnamese
Explanation on language	These are the official languages of the target countries. The Philippines uses English as its official language.
Link to disclosure	English: https://www.kdb.co.kr/wcmscontents/pdf/2024_ESMS_English.pdf Bahasa Indonesia*: https://www.kdb.co.kr/wcmscontents/pdf/2024_ESMS_Bahasa.pdf

	Khmer*: https://www.kdb.co.kr/wcmscontents/pdf/2024_ESMS_Khmer.pdf Lao*: https://www.kdb.co.kr/wcmscontents/pdf/2024_ESMS_Lao.pdf Vietnamese*: https://www.kdb.co.kr/wcmscontents/pdf/2024_ESMS_Vietnamese.pdf
Other link(s)	https://www.kdb.co.kr/index.jsp → Investor Relation → Sustainability → Green Climate Fund → Disclosure of Environmental and Social Information (here) → No.3
Remarks	An ESMS consistent with the requirements for a Category I-2 programme is contained in the "Environmental and Social Management System"
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	
Description of report/disclosure on accredited entity's website	Indigenous Peoples Plan Framework (IPPF) / Friday, June 14, 2024
Language(s) of disclosure	English, Bahasa Indonesia, Khmer, Lao, and Vietnamese
Explanation on language	These are the official languages of the target countries.
Link to disclosure	English: https://www.kdb.co.kr/wcmscontents/pdf/2024_ESMS_English.pdf Bahasa Indonesia*: https://www.kdb.co.kr/wcmscontents/pdf/2024_ESMS_Bahasa.pdf Khmer*: https://www.kdb.co.kr/wcmscontents/pdf/2024_ESMS_Khmer.pdf Lao*: https://www.kdb.co.kr/wcmscontents/pdf/2024_ESMS_Lao.pdf Vietnamese*: https://www.kdb.co.kr/wcmscontents/pdf/2024_ESMS_Vietnamese.pdf
Other link(s)	https://www.kdb.co.kr/index.jsp → Investor Relation → Sustainability → Green Climate Fund → Disclosure of Environmental and Social Information (here) → No.3
Remarks	Indigenous Peoples Plan Framework (IPPF) is contained in the ESMS, and translated for affected local stakeholders.

Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Friday, June 14, 2024
Place	ESMS (both hard and soft copies for the vulnerable peoples' convenience) will be made available at Global Green Growth Institute (GGGI) Country Offices in the five NOL countries, given that GGGI is the Executing Entity of the programme. <u>GGGI Cambodia</u> 4th Floor, Ministry of Environment, Building 503, Sangkat Tonle Bassak, Khan Chamkar Mon, Phnom Penh, Cambodia <u>GGGI Indonesia</u> Jl. Taman Patra Raya No. 10, RT 5/RW 4, Kuningan Timur, Kecamatan Setiabudi, Jakarta Selatan 12950, Indonesia <u>GGGI Lao PDR</u> Ministry of Planning and Investment (MPI), 3F, Corner of Souphanouvong Rd. and Sithane Rd., P.O Box 5745, Vientiane Capital, Lao PDR <u>GGGI Philippines</u> 25th Floor, Menarco Tower, 32nd Street, Bonifacio Global City Taguig, Metro Manila, 1634 Philippines <u>GGGI Viet Nam</u> Room 311, Ministry of Planning and Investment bldg. 65 Van Mieu Street, Dong Da District, Ha Noi, Viet Nam
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, July 15, 2024

* Subsequent to the disclosure of this form to the Board and active observers on June 14, 2024 (KST), the following updates have been made: The translation of the Environmental and Social Management System (ESMS) into the local languages of the target countries (Bahasa Indonesia, Khmer, Lao, and Vietnamese) have been made available on June 21, 2024.

Note: This form was prepared by the accredited entity stated above.

Independent Technical Advisory Panel's assessment of FP240

Proposal name:	Collaborative R&DB Programme for Promoting the Innovation of Climate Technopreneurship
Accredited entity:	Korea Development Bank (KDB)
Country/(ies):	Cambodia, Indonesia, Lao People's Democratic Republic (the), Philippines (the), Viet Nam
Project/programme size:	Medium

I. Summary

Brief on the programme	This proposal is a resubmission from the thirty-seventh meeting of the Board (B.37). The Collaborative R&DB Programme for Promoting the Innovation of Climate Technopreneurship targets the transfer of innovative climate technology solutions and empowers local entrepreneurship acceleration ecosystems in five member states of the Association of Southeast Asian Nations: Cambodia, Indonesia, Lao People's Democratic Republic, Philippines and Viet Nam. This research and development bank (R&DB) programme will create an incubation and acceleration programme for global technology innovators and local entrepreneurs. It will also establish a c. USD 201.2 million Climate Technopreneurship Fund (CTF) to crowd in private capital. The programme will be accompanied by various technical assistance activities targeted at building capacity on the ground, including policy support to enable technology transfer, and an incubator and accelerator which will find and incubate local companies to enable them to receive and scale up technology from global climate technology companies. The programme is visionary in its attempt to showcase a comprehensive intervention to facilitate climate technology transfer in South-East Asia and has heeded the call of the Conference of the Parties to the United Nations Framework Convention on Climate Change secretariat (COP to the UNFCCC) in attempting to make clean technologies available to entrepreneurs in South-East Asia.
Targeted programme size	USD 226.7 million
Proposed GCF involvement	USD 104.47 million (USD 83.75 million in equity; USD 20.72 million in grants)

Recipient of GCF assistance	For the equity component, which will create a technopreneurship fund, the finance will flow directly to the Climate Technopreneurship Fund through the Korea Development Bank, the accredited entity (AE). KDB will be the recipient of the grants and will distribute the funds to the executing entities (EEs) and local partners responsible for delivering the technical-assistance activities.

The programme has four components which will be managed by different EEs.

Component		Executing entity	Funding (USD)	Type of source funding (USD)
1	Country-driven acceleration Readiness	KDB, Global Green Growth Institute (GGGI)	8.6 million	Grant
2	Global acceleration for collaborative R&DBs	NH Investment & Securities (NHIS)	9.5 million	Grant
3	Climate Tech Fund (CTF)	NH Absolute Return Partners (NH ARP)	201.2 million	Equity
4	Technical assistance & project management costs	KDB, GGGI	7.5 million	Grant
Total			**226.8 million**	**201.2 million (equity)** **25.5 million (grant)**

II. Assessment of the independent Technical Advisory Panel

2.1 Impact potential *Scale: Medium*

1. The independent Technical Advisory Panel (iTAP) recognizes and welcomes KDB taking its first step in responding to the call of the COP to the UNFCCC to promote technology transfers in developing countries. This is especially impressive as KDB is a recognized entity in this space globally and has played a significant role in technology transfer to and from the Republic of Korea.

2. Given the call of the COP to the UNFCCC and the role of the Republic of Korea in the successful transfer of technology for both the Republic of Korea and countries worldwide, this is a welcome proposal.

3. The expected impact target of the programme is also well received: (i) mitigation (avoidance of greenhouse gas emissions) amounting to 1,050,976 tonnes of carbon dioxide equivalent; and (ii) adaptation targeting 7,403,487 direct and indirect beneficiaries over the lifespan of the programme, with 78 per cent of the funding expected to go towards adaptation

measures and 22 per cent towards mitigation. The funding proposal states that 50 per cent of the beneficiaries of the proposed programme are assumed to be women.

4. The iTAP notes that this programme aims to be a market-making intervention, which, by its very nature, means causality is difficult to prove in terms of climate-impact outcomes. Therefore, the iTAP considers the paradigm shift potential of this programme to be its most significant factor.

5. **Tailored solutions to the region:** In comparison to the previous two submissions**,** the programme has become more focused owing to its prioritization of sectors for each country**:** The funding proposal includes a table listing 7 eligible prioritized technology needs and 35 technology transfers allocated differently for each of the target countries. The iTAP welcomes the additional work performed by NH ARP, the General Partner (GP), in tailoring target technologies and sectors to specific countries, which improves the focus in each country and increases the chances of success.

6. A more tailored financing approach to the different markets, and the different types of companies within them, which prioritizes only two or three technologies in each market, will have a higher chance of success.

7. The programme (i) **is still complex in both its design and execution**; but, (ii) **has significantly improved the long-term alignment** of the executing entity, NH ARP, and its parent company, NHIS (the limited partner (LP)), through respective contribution targets to the CTF of USD 2 million by the GP and USD 10 million by the LP.

(a) More specifically, in terms of complexity, multiple interfaces would exist in the proposed programme between numerous executing entities which have never worked together before, and which have very different mission statements, capacities, expertise and experience – this naturally raises execution concerns. Such concerns are not insurmountable, but they do require a realistic plan from the AE on how such risk will be managed. The plan of the AE is to have multiple coordinating management units between each executing entity and itself, between country and regional teams, and between the investment teams and governments. Some level of coordination is important in any programme of this size, but the high level of coordination envisaged in this programme between the EEs seems to arise from a failure of design. Component 1, as the key building block of creating local investable companies, should be created and housed within the GP, which can then directly hire/partner with local and international incubators in different countries and in different sectors, or build its own internal capacities to cater to larger markets.

(b) An important indicator of alignment of interest is to have at least one of the EEs having at-risk capital in the proposed programme alongside GCF capital. This will ensure that, notwithstanding the complexity of the programme and the challenges ahead, the EE will have enough capital at risk that it will put the right resources and time into fixing whichever features in the initial design of the programme may not be fit for purpose. The iTAP welcomes the intention of the GP to contribute at least USD 2 million to the CTF and the intention of its parent company, NHIS, to invest up to USD 10 million as an LP.

8. Execution risk in terms of timelines

(a) The timelines do not seem realistic for achieving 40 investments and exiting those within 11 + 3 years. Considering that day 0 is GCF approval, it would be followed by fundraising and building out the team (1–2 years); identification of both global technology companies (1 year) and local companies (1 year); necessary changes to local government regulation (1–5 years); identification, incubation and acceleration of successful local companies (1–5 years); set up of the joint ventures (presumably 1 year); scale up of investment (4 years); and then exit (2-3 years) – a total of 12-22 years. Some of these steps may be performed in parallel, but no discussion of the timeline risks and the possible scenarios (upside, downside) was presented to the iTAP to evaluate. Whilst we applaud the drive to accelerate these transformations and their impact, the programme has not provided adequate analysis or assessment of timeline and execution risk for this programme.

2.2 Paradigm shift potential *Scale: Medium*

9. Technology transfer towards developing countries is a visionary goal and the iTAP congratulates the AE on its high aspirations. However, one of the main barriers identified for successful technology transfer is the adoption and enforcement by the receiving countries of a licensing framework. The solution provided is to make this framework compulsory. But it is not clear who will make it compulsory and who will enforce it. It remains unclear to the iTAP how such a compulsory licensing framework would work in practice, and how the AE would be able to make it compulsory for the respective governments. It is also unclear how such a framework would be enforced and as such the iTAP does not see how the international technology providers would be assured that their intellectual property would be protected.

10. In addition, it is unclear how the necessary local licensing and regulatory changes would happen and what support, if any, the different governments would provide for such changes. The iTAP acknowledges that the funding proposal package development has relied heavily on national designated authorities and their recommended local players, but that this structure seems very generic: everyone is **in favour of** the transfer of technology in principle, but there is no information on a pre-agreed framework under which specific technologies and their licensing requirements would be adopted by the governments of the participating countries.

11. Ultimately, the programme may face difficulties in identifying appropriate technology companies to establish investments (in the form of joint ventures or other partnerships), given the fact that low-carbon and climate-resilient solutions are in the early stages of development and the difficulty of technology commercialization.

12. Notwithstanding the lack of specifics, the iTAP commends the addition of three different modalities for accelerating technology transfer in South-East Asian countries, which could be more appropriate for different stages of the target companies and countries.

2.3 Sustainable development potential *Scale: Medium*

13. Creating an ecosystem to support the acceleration and incubation programme will spur the emergence of climate technology entrepreneurs as pioneer industry players and thereby the emergence of local climate technology companies. The programme aims to provide support for 40 locally owned and registered climate technology entrepreneurs, and it has potential to create a wide range of green job opportunities in the local markets.

2.3.1. Gender co-benefits

14. The programme will focus on nurturing and supporting the creation of an enriched gender-inclusive technology-driven entrepreneurship ecosystem. At least five local gender experts will be hired in each national Stakeholder Management Unit (SMU), who will promote gender expertise on the ground. The programme will target female technology leaders and gender friendly firms as prioritized beneficiaries of the CTF and as the local joint venture candidates. This has the potential to increase women’s participation in the climate technology industry.

2.3.2. Social and environmental co-benefits

15. Creating joint ventures between local companies and global technology providers will ensure that indigenous climate technology and knowledge are fully considered when selecting the climate technology innovation in the target countries. The climate technology products and services will aim to deliver improvements in air, water and soil quality, and to support sustainable livelihoods through conservation and biodiversity. These positive impacts will lead to improvements in health and safety, access to education and cultural preservation.

2.4 Needs of the recipient *Scale: Medium*

16. The needs of the recipients have been very thoroughly researched but not necessarily assessed, analysed and prioritized. Outcomes of studies and analyses (see annex 2 to the funding proposal) are broad and merely conclude that the needs of the five participating countries are classified into the following three overall categories (although the needs vary by country): (i) weak regulatory framework; (ii) limited access to finance; and (iii) nascent or immature climate technology and innovation ecosystem.

17. In addition, from the description of the studies performed, it remains unclear whether technology transfer is the right solution for many of the small and medium-sized enterprises (SMEs) in most of the countries targeted. Many of the SMEs may benefit more from climate financing rather than agreeing to and entering into sophisticated and complex technology-transfer agreements in order to access finance.

18. The programme will promote needs-specific technology-transfer activities for the five different countries, concentrating on (i) establishing a regulatory and policy framework, tailored to climate technopreneurs; (ii) addressing the financing gap through crowding in private-sector capital and equity investments; and (iii) nurturing a robust innovation ecosystem by enhancing awareness and providing capacity-building assistance.

(a) **Cambodia:** The corporate structure and business ecosystem of Cambodia is oriented around SMEs, which account for around 70 per cent of the country's employment, 99.8 per cent of its companies and 58 per cent of its gross domestic product. There is a weak institutional environment for technology innovation, so SMEs that are large enough, have faced challenges, including a lack of skilled human resources and poor access to finance, because most of them lack the necessary assets to secure credit. The missing middle "backbone group" of SMEs, which are too high risk to be attractive to global investors, will be served by the proposed programme in various ways, such as training local talent to make full use of technology.

(b) **Indonesia:** Following the unprecedented challenges posed by the coronavirus disease 2019 (COVID-19) pandemic and reallocation of climate priorities and budgets, in 2020 only 13 per cent of the financing needs outlined in the nationally determined contribution (NDC) mitigation road map were addressed. It is therefore clear that vibrant private-sector engagement with investment to crowd in and assist with climate technology deployment is essential for Indonesia to meet the commitments made in its NDC;

(c) **Lao People's Democratic Republic**: The country is seeking to improve the overall sustainable management of its national resources. The challenges in doing so have undermined the country's long-term development, exacerbated by climate change and the COVID-19 crisis. The Government of the Lao People's Democratic Republic is committed to enhancing access to finance for local entrepreneurs as well as strengthening regulatory and institutional arrangements. The programme will assist in the government's mission by unlocking the role of the private sector via technology trade at the corporate level, and devising an innovation-driven management framework for the country's existing resources at the institutional level;

(d) **Philippines:** The Philippines' needs arise as a result of (i) the absence of a centralized repository of information and indicators and data on SMEs regarding climate-change adaptation and mitigation; (ii) investment gaps; (iii) the limited availability of capacity-building initiatives for climate technology firms; (iv) the impact of the COVID-19 pandemic, community quarantine, and the resulting disruption in the supply chain and distribution channels; (v) low accessibility of public services designed to support entrepreneurs; and (vi) limited support on climate innovation activities. The country's entrepreneurs will take advantage of the investment opportunities and incentives throughout the programme, with the aim of improving market conditions and other enabling circumstances for the climate technology innovation leap; and

(e) **Viet Nam:** The country needs to respond to climate-driven natural hazards such as heatwaves, drought and floods, which accelerate the overexploitation of non-substitutable natural resources, such as water, at an alarming pace. Meanwhile, people who are at the bottom of the economic pyramid in Viet Nam are disproportionately exposed to risks: primarily those residing in the Mekong River Delta, the Red River and the Central Coast regions. Although investment in climate actions is urgently required to minimize adverse climate impacts, the limited availability of climate-responsive technologies along with the funding gap for technology entrepreneurs has resulted in institutional bottlenecks for climate business deployment. The programme will therefore stimulate innovation so benefits will permeate down to all, including the most vulnerable in the economy.

2.5 Country ownership *Scale: High*

19. The iTAP would like to highlight the expansive country ownership plans of the AE.

20. Country-specific analyses have been conducted through an extensive literature review, covering each country's NDC and technology needs assessment, as well as other national policies and plans, to ensure that the programme is aligned with the country's priorities. A number of strategic early programme reviews have been conducted, including two feasibility studies (one demand-driven and one supply-driven), and a stakeholder engagement plan and a gender assessment have been completed.

21. The AE has held extensive stakeholder consultations with the governments, ministries, relevant public entities, state-owned enterprises, venture capital firms (VCs), investors, start-up assisting service firms, and international donors as well as civil society organizations since the concept design stage. The wide-ranging stakeholder consultations resulted in the creation of an SMU governance structure to ensure strong country ownership.

22. The national designated authorities of the target countries will function as members of the SMU Steering Committee to offer high-level policy guidance and strategic advice over the course of executing country-specific activities and will support partnership coordination among the key stakeholders.

2.6 Efficiency and effectiveness *Scale: Medium*

23. The estimated **cost per tonne of carbon dioxide equivalent** for the total requested GCF amount dedicated to mitigation (22 per cent of the total GCF request) of USD 22.99 million (including guarantee and grant) is USD 22.06.

24. GCF funding **could** catalyse private investments of USD 120.75 million from KDB, general partners, limited partners, Korea International Cooperation Agency, Asia–Europe Meeting SMEs Eco-Innovation Center, and local intermediary entities. Overall, the co-investment ratio on the total GCF investment is 1:1.16.

25. The mobilization of investors is difficult to assess given the fundraising risk associated with a GP which manages a similar fund of USD 15 million and whose capital came mainly from its parent company (i.e. the GP does not have significant experience fundraising from outside sources). However, the iTAP acknowledges and applauds the tranching of GCF capital disbursement in line with fundraising milestones, which mitigates fundraising risk for component 2 (The fundraising of the CTF).

26. However, there is no similar tranching of the grant capital awarded to Component 1 or any performance-based milestone that the executing entity (GGGI) needs to meet to access further GCF grant capital. This adds to the execution risk for the programme.

27. The upstream work of component 1 is essential for the creation of the appropriate pipeline for the CTF. According to the information provided by GGGI during the written questions-and-answers, "GGGI will team up with various third-party partners – for example, sourcing local companies together with top-tier local VC/Private equity firms with abundant pipelines, and training local entrepreneurs with international/local business experts". If the executing entities – GGGI and KDB – do not incubate enough investable companies for CTF, the whole programme may fail. When the iTAP shared our concerns the AE and the EE responded that they are mitigating this risk by creating coordinating committees to facilitate the multiple interfaces KDB itself will be the main coordinator and GGGI the executor of the pipeline: building its content and incubating local companies. However, such a loose arrangement increases the likelihood of execution risk down the line. Clearer accountability upfront is needed on how the value chain of companies and projects will funnel through to the CTF and its GP, with more power being given to the GP to define and agree terms for the roles of the other components' executing entities (components 1), key performance indicators, including the ability of the GP to replace such entities if they do not perform.

28. The iTAP welcomes the involvement of GGGI in this proposal given its extensive work with target country governments in policy and international cooperation. GGGI has a strong track record of driving legislative, administrative, and regulatory change in diverse developing country contexts employing a green growth and climate responsive lens, with over 300 policies adopted across Asia, Africa, Europe, Latin America and the Caribbean, Middle East and North Africa, and the Pacific. GGGI has accumulated a variety of working experience and built-up, close ties with key ministries (including national designated authorities in the programme's target countries) as the proven delivery partner for the GCF Readiness and Preparatory Support Programme, which supports the target governments' capacity, including the regulatory set-up and policy design.

29. However, GGGI does not *currently* have a strong track record of incubating companies, which scale up and raise substantial capital to be able to meet the enormous climate challenges of the target countries. GGGI has an average annual total budget of USD 100 million (only a fraction of which is for incubation), but it has nonetheless been able to mobilize USD 3 million of additional capital into the companies it has trained and incubated. Although the iTAP acknowledges that companies take time to scale up and attract further capital, it is difficult to evaluate GGGI's performance.

30. NHIS and NH ARP (as the GP of CTF) on the other hand, have such experience and are the right party to directly determine what their pipeline should look like. Rather than go through an intermediary, NHIS and NH ARP can hire directly flexible, country-specific incubators and technical assistance specialists and/or combine with building out their own incubation team. Moreover, it would be more cost and time efficient, outcome targeted, and specialized to have the GP in charge of component 1 than going through a large international company such as GGGI with large overhead costs, and without strong track record in choosing and scaling local ventures companies which become commercially successful.

31. Lastly, having the EE of the incubation of the pipeline (component 1) as a different entity to the main Fund (CTF) (component 3) as a pre-ordained set up by KDB may be too inflexible: These two EEs have never worked together, they have different commercial orientations, requiring the two very different organizations to understand each other's cultures, constantly coordinate investment strategies which will change and adapt with time, requiring unnecessary complex coordination via the SMUs and KDB. This level of coordination and alignment has undoubtedly increased the level of GCF capital/fees required to run this program. This decrease in efficiency and effectiveness would have been understandable if the trade-off was that Component 1 was run by a nimble, commercial, and deeply experienced entity with a track record in finding and incubating the right companies for technology transfer and scale.

III. Overall remarks from the independent Technical Advisory Panel

32. The proposed KDB programme has been one of the most ambitious programmes the iTAP has evaluated and the iTAP congratulates KDB for the programme's most challenging and inspirational aims. The iTAP welcomes the significant changes in design and execution of the proposed programme for that proposed at B.36, which make this a solid proposal.

33. The iTAP commends the AE for improving the programme since its initial review during the lead-up to B.36. It welcomes the improved focus associated with reducing the number of sectors and most importantly the engagement of a fund manager (NH ARP) with relevant experience for component 3 (although that experience includes being a fund manager of USD 15 million as opposed to the current proposed fund (CTF) of USD 200 million).

34. Despite the above-mentioned welcome improvements, there are significant items from the B.36 assessment which were not addressed – and getting the design of the programme, its execution and, especially, the accountability of its executing entities right is particularly important because of the importance of the role of GCF in supporting and ensuring the success of technology-transfer initiatives.

35. The programme is still very complex and high risk with multiple executing entities which have never worked together – all held together by an AE whose main role will be coordinating a complex and expensive institutional set-up which could have been simplified by designing component 1 around the GP. The GP could have used the USD 8.5 million in grants to hire the support they need directly & flexibly, rather than having GGGI be an intermediary. In this regard, the proposed programme risks being too expensive, inefficient, and ineffective. *To mitigate this concern, we have outlined a recommended condition below, which attempts to address potential non-performance by GGGI. This condition is very similar to the already existing performance-based condition put on NH ARP by GCF to disburse capital based on fundraising milestones. The iTAP's additional condition would similarly ensure that the incubation capital is awarded to GGGI is disbursed in three tranches, conditional on them incubating a minimum viable percentage (30%) of investable companies for CTF, evaluated every two years. It aims to ensure that the EE of Component 1 performs in terms of KPIs agreed before they receive further capital.*

36. In summary, such a visionary concept needs to marry complexity and innovation in goals on the one hand with simplicity, focus and deep expertise in implementation to mitigate execution risk. The diversity of South Asian markets in terms of capacity to adopt technology and the varied profile of the companies within each market, the broad spectrum of technologies and sectors, the regulatory frameworks and licensing regimes, the many stakeholders involved, and the multitude of executing and advisory agencies to the programme highlight a high executing risk to this proposed programme in terms of fundraising, investment execution and achieving the stated climate outcomes.

37. As such, the iTAP recommends that the Board approve this funding proposal conditional on:

(a) Condition to be met prior to the first Disbursement of GCF Non-Reimbursable Funds in respect of Component 1:

(i) Delivery to GCF by the Accredited Entity, in form and substance satisfactory to the GCF Secretariat, of evidence that the amount of disbursement requested is not more than one third (1/3) of the total GCF Non-Reimbursable Funds budgeted for Component 1 of the Funded Activity;

(b) Condition to be met prior to the second Disbursement of GCF Non-Reimbursable Funds in respect of Component 1:

(i) Delivery to GCF by the Accredited Entity, in form and substance satisfactory to the GCF Secretariat, of evidence that the amount of disbursement requested is not more than one third (1/3) of the total GCF Non-Reimbursable Funds budgeted for Component 1 of the Funded Activity;

(ii) Delivery to GCF by the Accredited Entity, in form and substance satisfactory to the GCF Secretariat, of evidence that 30% of the beneficiaries under Component 1 meet the investment criteria of the CTF Fund and assessed through Components 1 and 2 as having high potential to make the CTF Fund investment pipeline;

(c) Condition to be met prior to the third Disbursement of GCF Non-Reimbursable Funds in respect of Component 1:

(i) Delivery to GCF by the Accredited Entity, in form and substance satisfactory to the GCF Secretariat, of evidence that 30% of the beneficiaries under Component 1 (excluding those beneficiaries counted to meet the condition for the second Disbursement) meet the investment criteria of the CTF Fund and assessed through Components 1 and 2 as having high potential to make the CTF Fund investment pipeline;

38. **Include the following covenant in the Funded Activity Agreement:** If the Accredited Entity has not satisfied the conditions for second or third Disbursement of GCF Non-Reimbursable Funds in respect of Component 1, respectively, within two (2) years of the date of the previous disbursement of GCF Non-Reimbursable Funds in respect of Component 1, it shall consult with the GCF Secretariat and submit a restructuring proposal in respect of Component 1.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP240)

Proposal name:	Collaborative R&DB for Promoting the Innovation of Climate Technopreneurship
Accredited entity:	Korea Development Bank (KDB)
Country/(ies):	Cambodia, Indonesia, Lao PDR, Philippines, Vietnam
Project/Programme size:	Medium

Impact potential
AE appreciates iTAP assessment that grasped the programme's core theme and unique features and considered the paradigm shift potential as the most significant factor.
Paradigm shift potential
AE appreciates iTAP comments that will be carefully considered for the proper programme's implementation.
Sustainable development potential
AE appreciates iTAP assessment that recognised various kinds of co-benefits, which will be maximised for the NOL countries' leap.
Needs of the recipient
AE appreciates iTAP assessment that paid attention to the needs of the five individual countries. With consideration on your comments, AE shall optimise the Funded Activities in line with the needs as much as possible.
Country ownership
AE appreciates iTAP assessment that recognised a strong country ownership based designing and implementation plan of the programme.
Efficiency and effectiveness
AE appreciates iTAP assessment that considered the execution of Component 1, willing to take recommendations and comments into account.
Overall remarks from the independent Technical Advisory Panel:

AE appreciates iTAP' consideration and efforts that all aim at the programme success. With regards to proposed conditions, we shall consult with the Secretariat for specific and practical arrangements.

Gender documentation for FP240

GENDER ASSESSMENT & GENDER ACTION PLAN

COLLABORATIVE R&DB PROGRAMME FOR PROMOTING THE INNOVATION OF CLIMATE TECHNOPRENEURSHIP

GAIA CONSULT INC.
(주) 가이아컨설트

March 2024

Table of Contents

1. Introduction

Programme Overview

The Collaborative R&DB Programme for Promoting the Innovation of Climate Technopreneurship is a multicounty project aiming to support entrepreneurs with climate technology-enabled solutions that will accelerate the low carbon and climate resilient transition across Southeast Asia, underpinned by the success of pilots in Cambodia, Indonesia, Laos, the Philippines and Vietnam. Comprised of four key components. Components 1 (Country Driven Climate Acceleration Preparedness) and Component 4 (Technical Assistance) focus on acceleration readiness and the ecosystem for local climate technology firms, respectively. Components 2 (Global Acceleration for Climate Technopreneurs) and Component 3 (Climate Technopreneurship Fund, CTF hereafter) are oriented more towards investment-ready JVs. (For more details of the proposed Programme, see the Funding Proposal.)

Gender Assessment: Objectives

This gender assessment aims to provide a rapid assessment of the gender situation in Cambodia, Indonesia, Laos, the Philippines and Vietnam; identifying gender equality and women's empowerment constraints, opportunities and recommendations to ensure gender inclusiveness and gender empowerment through the proposed Programme. To do so, country-specific gender analyses were carried out in the backdrop of the related national legal and institutional framework: Through the desk reviews and a series of consultative meetings with the relevant key stakeholders (with gender specialists of KDB (AE) and the GGGI (EE of the Component 1 and Component 4), representatives of NHIS and NH ARP (EE of the Component 2 and Component 3) and other delivery partners (Acceleration Advisory Consortium group for Component 2 of the Programme)[1], the general situation of gender equity, gender perspective on climate change and the country's responses, climate technopreneurship (or generally business entrepreneurship), particularly of the women were examined. This assessment also presents the gender landscape of identified priority sectors/technology that are likely to be funded through the Programme with their respective impacts on gender equality and empowerment, as well as promising intervention points through the Programme design.

This assessment also informed the development of the Gender Action Plan (GAP) for the Programme. GAP is an important implementation tool to mainstream the overall execution of the Programme in accordance with the GCF Gender Policy (2019).

Key findings

In a nutshell, the report finds that women and girls (esp. in remote rural areas) in all the five target countries, i.e. Cambodia, Indonesia, Lao PDR, the Philippines and Vietnam, are more vulnerable to climate change, disproportionally affected and, to a varying degree among the countries, largely isolated from the (climate and general) technology-based entrepreneurship ecosystem. Having said that, innovative scale-up of enterprises related to climate adaptation and mitigation technologies will benefit women as much as men, if not higher, as they are more vulnerable to climate change-related challenges and risks. In general, sectors/technology areas that girls and women are either more employed (such as in agriculture, where female farmers are a major (although mostly unrecognized and informal) workforce in the four countries except in the Philippines) or girls' and women's livelihood could be substantially improved through the Programme (such as water supply and wastewater management, climate-smart agriculture, etc.) are areas that are strongly recommended for investment.

In all five countries, the women's entrepreneurial ecosystem is limited to small and microenterprises. Compared with their male counterparts, women-led businesses and enterprises are often characterized by lower profitability and constitute in majority informal sectors. Access to formal credit systems is

[1] For overall governance structure of the Programme, pls see Figure: Governance Structure between Local and Global Components in Section B.3 of the Funding Proposal of this proposed Programme.

often limited due to the respective countries' social and cultural norms and (in some locations) religious precepts. In some countries, STEM education often favours male students and technology-based entrepreneurship support are predominantly enjoyed by men.

Nevertheless, there are signs of improvements: In Vietnam and the Philippines, more women are taking ICT and engineer careers, which had been dominated by men. In Lao PDR, women are leading agents for MSMEs in street vending and retailing businesses. Across the five countries' women are active participants in the agricultural sector, whose rich know-hows and experiences need to be tapped for climate technology entrepreneurship in the agricultural sector.

Recommended Action Plan

The Programme has an explicit goal of proactively inviting women climate technology and business entrepreneurs in the five countries and ensuring the funded JV activities shall clearly mainstream gender into their implementation process in line with the Gender Action Plan (GAP) of this Programme (Chapter 5 of this report).

The following actions were integrated into the designs of the Programme:

1) **Gender Component in CTF's Investment Criteria**

The Programme prioritizes the "women-led enterprises/businesses" as CTF recipients and the Fund's Investment Criteria include "Gender Mainstreaming" Criteria, as follows:

Table 1. CTF Investment Criteria: Gender Component[2]

9	Gender mainstreaming	**[JV formation dimension]** ✓ At least 50% of the CTF-invested JVs are "Women-led JVs" that are defined in Condition 1 below, or alternatively; ✓ JVs are requested to present a gender mainstreaming commitment plan (GMCP) to the CTF. **[Condition 1 "Women-led JV" Definition]** It constitutes the following conditions. The conditions below will be screened from the selection process of local entrepreneurs. The women-led SME candidates shall be prioritized for selection.: ▪ SMEs with at least 26% of the total capital is owned by women; ▪ At least one woman participates on the Board of Directors and/or different types of boards or internal committees, such as ESG committees, among others (CEO, COO, CTO), and/or; ▪ SMEs with more than 30% of the workforce are females[3].

[2] For the full content of the CTF Investment Criteria, see Table of the same title in Section B3 of the Funding Proposal of this Programme.
[3] CTF's definition of the "women-led JV" is based on the IFC's definition of "women owned enterprise," according to which: "An enterprise qualifies as a woman-owned enterprise if it meets the following criteria:
(A) ≥ 51% owned by woman/women; OR
(B) ≥ 20% owned by woman/women; AND (i) has ≥ 1 woman as CEO/COO/President/Vice President; AND (ii) has ≥ 30% of the board of directors composed of women, where a board exists. " (Source: International Finance Corporation (IFC), IFC'S Definitions of Targeted Sectors, available at: https://www.ifc.org/wps/wcm/connect/industry_ext_content/ifc_external_corporate_site/financial+institutions/priorities/ifcs+definitions+of+targeted+sectors).
Vietnam has its own definition of "Women-owned small and medium enterprises (SME)" as "an SME having one or more women owns at least 51% of its charter capital and at least a woman is the executive director of this enterprise. (Source: Law 04/2017/QH14 on assistance for small and medium sized enterprises, available at:https://vanbanphapluat.co/law-04-2017-qh14-on-assistance-for-small-and-medium-sized-enterprises) In Lao PDR such definition has not been found. In the Philippines, Indonesia and Cambodia, the above-referred IFC definition is being used. (Source: HAWASME, Mekong Business Initiative, Women-owned Small and Medium-sized Enterprises in Viet Nam: Situation Analysis and Policy Recommendations, 2016, available at: https://www.economica.vn/docs/publications-and-reports/17)

- ✓ The above conditions shall be screened from the local SME selection process at Component 1. The women-owned SME candidates shall be prioritized for selection.

[Condition 2] The funded SMEs (esp. which do not meet the Condition 1 above), need to be ready to commit to gender mainstreaming. A proactive **Gender mainstreaming commitment plan (GMCP)** [4] needs to be developed and presented as part of the application package to CTF.

- ✓ Proactive measures for gender mainstreaming include, among others:
  - Set specific **employment and (esp. high-value skill) training targets for girls and women** with concrete timelines.
  - Reflect gender inclusiveness in the **design/business plans of the JV enterprises** (along the climate technology RD&B processes) by, e.g. intentionally targeting the female population as major beneficiaries or active agents of the enterprises.
- ✓ At Component 2 (Acceleration Program), the candidate JV/SMEs will be provided, upon request, a customised guidance by the CTF gender expert(s) (as part of the Acceleration Advisory Consortium Group) to establish GMCP.

[CTF-funded Enterprise dimension]

- ✓ All the CTF investment JVs/activities are requested to develop concrete gender performance indicators and generate tangible gender-related results proven.
- ✓ The minimum requirement for CTF-supported JV enterprises are: ① Gender-disaggregated impacts to be presented, monitored and evaluated; ② A (simplified) Gender Assessment and Gender Action Plan(*) to be prepared, submitted as part of the CTF application packages. *(*At minimum a standard format (Appendix 3 of Annex 8)) need to be filled in and submitted.)*
- ✓ Overall gender performance shall be monitored and evaluated throughout the JV implementation (Component 3) against the submitted GMCP and GAP.
  - Prioritisation scoring system for a winning gender-balanced portfolio
  - Higher gender scores will be weighted higher in selection and investment in cases where scores are identical in other criteria

Given the varying circumstances across the five countries, through the consultation with the KDB and GGGI gender specialists, it was agreed that the CTF's definition shall be more relaxed than the IFC's definition and also include the female employment components, to promote the women's employment to the applicant JVs.

[4] The funded SMEs meeting the Condition 1 are also encouraged to develop and implement the GMCP. GMCP is to be presented as part of the Simplified Gender Analysis (GA) & Gender Action Plan (GAP) (General Template) esp. Section I-2 (Entity-level Gender Mainstreaming Commitment Plan).

2) Gender-related Capacity Building & Technical Support

Component 1 (Country Driven Acceleration Readiness) will profile the gender aspects against the above-presented 3 conditions for the CTF's "women-led JV" definition (in Table 1 above.) of all the participating local firms in the five countries. All the applicant firms will be requested to prepare gender profiling by the above-referred criteria. Those who are not meeting the "women-owned" firm status will be required to prepare gender mainstreaming plan for the firm level. All the applicants for the CTF will also be required to establish JV/enterprise levels gender mainstreaming strategies and action plan (Gender Action Plan) through Component 2 (Global Acceleration for Collaborative RD&B) activities. The gender commitments by each of the applicant JVs shall be submitted as part of the CTF application package as an important component for the Fund's investment decision-making in Component 3 (CTF). Component 4 (TA) shall focus on nurturing and supporting the creation of enriched tech entrepreneurship ecosystem development from a gender inclusiveness perspective. Women in Cambodia, Indonesia, Lao PDR, the Philippines, and Vietnam need to be given opportunities to become equal partners for climate technology innovation and entrepreneurship, to empower themselves and benefit from the Programme as active participants be them airect recipients of the Programme support, primary employees and technicians, or the primary beneficiaries of the Programme through evolvement of the funded activities.

The National Climate Entrepreneur Accelerator Programme (N-CEAPs) (Component 1 of the Programme) will ensure the local women-led entrepreneurs to strengthen their readiness to participate in a JV partnership with a global entrepreneur(s) (Component 2) and execute the climate technology JV enterprises through the CTF support (Component 3). All the applicant JV enterprises will be required to carry out a gender profiling and prepare and execute a gender mainstreaming commitment plan (GMCP).

Country Sustainable Management Unit (SMUs) in the respective five countries and a gender specialist of the CTF (its Environmental & Social and Gender Compliance Team, ESGCT) shall provide capacity building and technical support for each JV participant to prepare their gender action plan and implement them in line with the CTF's Gender Action Plan (GAP) (which is in accordance with the GCF Gender Policy (2019)[5]) woman entrepreneurs and their climate technology enterprises/businesses in the five countries would be empowered to enjoy rare opportunities to spearhead in the global level climate technology innovation, upscale their businesses beyond their location and countries to the ASEAN level in the long term.

[5] https://www.greenclimate.fund/document/gender-policy

2. Methodology

This Gender Assessment is largely based on desktop research and close consultation with the key stakeholders including the Accredited Entity (KDB) and the Executing Entities (GGGI for Component 1 and Component 4, and Representatives of NHIS and NH ARP for Component 2 and Component 3), as well as potential delivery partners (Acceleration Advisory Consortium group for Component 2 of the Programme).[6] Following are some of the key consultation activities carried out in relation to the GA and GAP preparation:

Assessment: Analysis and Structure

The following sections include each of the five countries' legal and institutional framework and development analysis related to gender equality and women's empowerment. Areas with gaps and high potential for women's participation are also highlighted with a view to identify effective intervention points to maximize benefits to the women. This paper also views the issue of climate change vulnerability, climate technopreneurship (or business entrepreneurship in general, in case the information on the former is lacking) from a gender perspective and identifies root causes for the persistent gender inequality as well as signs of progress in the five countries.

Country assessment has been made through a dynamic process: A series of consultation meetings with KDB (AE), Global Green Growth Institute (GGGI, EE for Components 1 and 4) and NHIS and NH ARP (EE for Components 2 and 3), and potential delivery partners (Acceleration Advisory Consortium group for Component 2 of theProgramme. All the EEs of the Programme Components were regularly consulted through call conferences and face-to-face meetings. Data have been gathered and opinions were sought from the inception state of the assessment preparation. Extensive discussions have been made with the GGGI Gender specialist and gender team throughout the preparation of this report, who have also coordinated country stakeholders in the review and feed backing processes (See Appendix 4 below for detailed activities carried out). As EE of Components 1 and Component 4, GGGI will oversee executing GAP for the respective components targeting girls and women in the five countries (and beyond to ASEAN level in the future through the rollout.)

The last section of each country assessment includes the gender analysis of some of the key sectors/technology that have been identified by each country for funding support through GGGI's pre-feasibility study (country demand analysis): Potential effects of the investment on gender perspective and feasible measures for gender inclusiveness and empowerment are recommended. In the end, the Study presents key recommendations for integration into the Gender Action Plan (GAP) of the Programme.

[6] For overall governance structure of the Programme, pls see Figure: Governance Structure between Local and Global Components in Section B.3 of the Funding Proposal of this proposed Programme.

3. Country Specific Assessment

CAMBODIA

- **Legal and Institutional Setup and International Commitment of Cambodia**

International Agreements on Women's Rights. Cambodia has enacted several international agreements and signed several international declarations related to violence against men, women, and children. Starting from the Universal Declaration of Human Rights (1948), Cambodia ratified the United Nations Convention on the Elimination of all Forms of Discrimination against Women (CEDAW) in 1992 and ratified its Optional Protocol in 2010, the UN Declaration on the Elimination of Violence Against Women (1993), UN Declaration on the Rights of Indigenous People (UNDRIP) (2017), and the UN Security Council Resolutions (SCR) 1325, 1820 and 1888 related to women, peace and security. At the regional level, the country signed the Declaration of the Advancement of Women in the ASEAN Region in 1988, which focuses on the promotion and implementation of women's equitable and effective participation in all fields and at all levels, and the Declaration on the Elimination of Violence Against Women in the ASEAN Region (2012).

Institutional Arrangements. For the promotion of gender equity, the government of Cambodia established the Ministry of Women's Affairs (MOWA), the Cambodia National Council for Women, the Technical Working Group on Gender (since 2004 and chaired by MOWA, with the UNDP and the Japan International Cooperation Agency as co-facilitators), several Gender Mainstreaming Action Groups in the government line ministries, and Women and Children's Consultative Committees (WCCCs) at all levels of subnational government.[7]

At the national level, Cambodia has made progress in addressing domestic gender inequality through legislative initiatives, national action plans, education programs, and establishing resources for prevention and response. Nowadays, the main components of the country's framework related to women's rights include:

- The Constitution of the Kingdom of Cambodia states that "all forms of discrimination against women shall be abolished. The exploitation of women in employment shall be prohibited. Men and women are equal in all fields, especially with respect to marriage and family matters…" (Article 45)

- The 2005 Law on Domestic Violence and the Protection of Victims

- The Law on the Suppression of Human Trafficking and Sexual Exploitation in 2008

- The National Action Plan on Prevention Violence Against Women II 2014-2018 (NAPVAWII). It was developed and approved by the Royal Government of Cambodia to provide multi-institutional operational plans addressing domestic violence, rape and sexual violence, and violence against women with increased risk

- The National Strategic Development Plan (NSDP) 2019-2023[8], which includes a specific section on "Enhancing Implementation of Population Policy and Gender Equity", in the national category outcome of Human Resource Development

- The Rectangular Strategy for Growth, Employment, Equity and Efficiency Phase IV 2018-2023. It is the "Socio-Economic Policy Agenda" of the Political Platform of the Royal Government of the Sixth Legislature of the National Assembly and specifically mentions

[7] WCCCs were established by a decree of the Ministry of Interior in December 2009.
[8] National Strategic Development Plan (NSDP) 2019-2023 - OD Mekong Datahub. (2019). Retrieved from Opendevelopmentcambodia.net website: https://data.opendevelopmentcambodia.net/laws_record/national-strategic-development-plan-nsdp-2019-2023.

gender equality, in the first rectangle on Human Resource Development, enhancing the need for mainstream women's social-economic situation and strengthening their role in the society.[9]

- The Neary Rattanak strategic plan IV 2019-2023 is a five-year strategic plan for gender equality and women's empowerment in Cambodia. This plan outlined MOWA's central role in assessing, reforming, and implementing policy to: mainstreaming gender in development policies and plans in all sectors and at all levels; mainstreaming women's entrepreneurship through expanded education, technical, and vocational training for women; and uplifting social morality, the values of women, and the Cambodian family through investments in gender equity and strengthened partnerships.

Women and Climate Change. The Cambodia Climate Change Strategic Plan 2014-2023 includes the objective to reduce sectoral, regional, gender vulnerability and health risks related to climate change impacts. Cambodia's MOWA has formed a Gender and Climate Change Committee in 2016 with the Asian Development Bank's assistance[10], which has the key role to gather information on gender and climate change, conduct studies on the impact of climate change on women and children, and build climate change capacity in the ministry's departments. In line with the updated Cambodia's Nationally Determined Contribution (NDC), all Ministries recognized the priority of targeting women's participation in climate action, by pursuing ranges from 15% to 70%.

In collaboration with relevant ministries, such as the Ministry of Agriculture, Forestry and Fisheries (MAFF), the Ministry of Environment (MOE), MOWA facilitates a mechanism for gender mainstreaming with climate change nexus at policy levels[11]. Political commitments in this climate change-gender nexus are reflected in the National Strategic Plan on Gender and Climate Change 2014-2023, Master Plan for Gender and Climate Change (2018-2030)[12] and Action Plan on Gender and Climate Change 2019-2023. In particular, the National Strategic Plan on Gender and Climate Change 2014-2023 proposes policy measures covering both adaptation and mitigation, cross-cutting agendas and institutional, organizational and human capacity development, research and development, financing, monitoring and evaluation.[13]

Although Cambodia has been working hard to reduce gender gap in diverse sectors, there remains a way to go. According to the United Nations, with an effort, gender gaps in education in Cambodia has been reduced. However, with the advent of the COVID-19 pandemic, progress has been slowed down or in some cases reversed as in the case of gender wage gaps and increased gender-based violence. Traditional bias of positioning women in informal, under-paid or unpaid work such as domestic and care work has been reported to be deepened[14].

- **General Situation Analysis of Cambodia**

Cambodia is a predominantly rural society, with a population of approximately 16 million people (2019), whose 51.19% are women,[15] with only 39.4% living in urban areas. Over the past decades, Cambodia's gross domestic product has increased at a remarkable rate reaching 7% in the period of 2018 and 2019. However, Covid-19 crisis hit the country's economy hard and its GDP decreased by -3.1% in 2020. The economy of the country is heavily reliant on tourism, agriculture, the garment industry, construction,

[9] Rectangular Strategy Phase 4 of the Royal Government of Cambodia - OD Mekong Datahub. (2020). Retrieved from Opendevelopmentcambodia.net website: https://data.opendevelopmentcambodia.net/laws_record/rectangular-strategy-phrase-4-of-the-royal-government-of-cambodia.

[10] Ministry of Women Affair (2020) Retrieved from the official website: https://www.mowa.gov.kh/en/detail/5542

[11] https://ncsd.moe.gov.kh/sites/default/files/2019-06/MASTER%20PLAN%20ON%20GENDER%20AND%20CLIMATE%20CHANGE.pdf & https://www.uncclearn.org/wp-content/uploads/library/final-digital_cambodia_report.pdf

[12] Cambodia National Council for Sustainable Development (NCSD) (2018). Retrieved from the official website: https://ncsd.moe.gov.kh/sites/default/files/2019-06/MASTER%20PLAN%20ON%20GENDER%20AND%20CLIMATE%20CHANGE.pdf

[13] FAO (2021). Retrieved from Fao.org website: https://www.fao.org/faolex/results/details/en/c/LEX-FAOC182425/.

[14] https://cambodia.un.org/sites/default/files/2022-03/Gender%20Deep%20Dive%20-%20CCA%20Cambodia_V6_010322_LQ.pdf

[15] Cambodia National Institute of Statistics. (2019). Cambodia Socio-Economic Survey 2019-20. Retrieved from Nis.gov.kh website: https://www.nis.gov.kh/index.php/en/14-cses/86-cambodia-socia-ecomonic-survey-2019-20.

and the real estate sector. Trade restrictions and international travel bans heavily have affected the overall economy.

With the higher portion of the population having been vaccinated, COVID-19 confirmed cases have significantly dropped from the end of 2021[16]. However, there are up-and-downs during 2022. Still, the confirmed patients are few. As the global COVID-19 situation has been eased, Cambodia's major industries such as garment, travel goods, and footwear exports have been resilient. In addition, tourism also has been doing well since late 2021. In that sense, the World Bank predicts 4.8% of economic growth in 2022[17]. In turn, besides the influence of COVID-19, due to the global economic downturn, declining in exports is projected, and economic growth in 2023 is not projected to be as good as in 2022.[18]

Cambodia scored 0.593 on the Human Development Index (HDI) in 2021, falling into the medium human development group with a rank of 148 (2020)[19]. In the same year of 2021, the Gender Development Index (GDI), which measures gender gaps in human development achievements on health, knowledge and living standards, classifies Cambodia as a medium equality (Group 3 countries)[20] with a score of 0.926. The OECD's Social Institutions and Gender Index 2019 (SIGI)[21] which focuses on the discrimination against women in the formal and informal laws, social norms, and practices, Cambodia ranked "low" in terms of gender discrimination in social institutions (score 0.3).In particular, the country ranked high on discrimination in the "restricted civil liberties" (41%).[22]

Gender inequalities in Cambodia is also reflected in governance, education, and employment. This is largely due to (among others), gender roles, culture/religious/social norms.

Gender equality in the labour market in Cambodia has been fluctuating over time and worsened since 2020: the female labour force participation rate (against male labour force) has fallen from 88% in 2019 to 86% in 2020 and 2021 respectively[23]. Women in Cambodia continue to face a gender pay gap with an income per capita 3,697$ for women and 4,822$ for men.[24] Considering a large portion of women's labour is not captured in official data as most of the women's work remains in informal sector and temporary types of employment, the reality may be even starker. Women are mostly employed in the primary sector, with a progressive increase in the secondary and tertiary. Women represent 74% of the agricultural workforce while they own only 15.4% of the recorded agricultural land area (2017 Cambodia Socio-Economic Survey).[25] Women also spend significant time on unpaid domestic and care work, which affects the time available to engage in paid work. Marital status is the most important

[16] WHO (2022). COVID-19 Joint WHO-MOH Situation Report 90. Retrieved from https://www.who.int/cambodia/internal-publications-detail/covid-19-joint-who-moh-situation-report-90
[17] World Bank (2022). Retrieved from website: https://www.worldbank.org/en/news/press-release/2022/12/07/cambodia-s-economy-is-recovering-but-could-face-headwinds-world-bank-report-says#:~:text=The%20World%20Bank%20forecasts%20economic,Update%20for%20December%202022%20said.
[18] Europa (2021). Retrieved from website: https://knowledge4policy.ec.europa.eu/publication/cambodia-economic-update-living-covid-special-focus-impact-covid-19-pandemic-learning_en
[19] The reported country ranking is only available in the year 2020 latest.
[20] What is the advantage of grouping countries into five GDI groups instead of ranking them according to the absolute deviation from parity? | Human Development Reports. (2020). Retrieved January 3, 2022, from Undp.org website: http://hdr.undp.org/en/content/what-advantage-grouping-countries-five-gdi-groups-instead-ranking-them-according-absolute.
[21] The SIGI covers four dimensions of discriminatory social institutions, spanning major socio-economic areas that affect women's lives: Discrimination in the family; Restricted physical integrity; Restricted access to productive and financial resources; and Restricted civil liberties. https://www.genderindex.org/sigi/ .
[22] OECD. Social Institutions and Gender Index 2019. Available at: https://www.genderindex.org/ranking/.
[23] https://data.worldbank.org/indicator/SL.TLF.CACT.FM.ZS?locations=KH. The graph indicates that the female to male labour participation ratio had been more equal prior to 2000, which ranged between 94% and 96%. It is not clear the main reason for significant fall since 2000s. This may be due to the data accuracy and sufficiency (i.e., technical issue in data collection) or any other substantial factors.
[24] UNDP. (2021). Gender Development Index. Retrieved from https://www3.weforum.org/docs/WEF_GGGR_2021.pdf.
[25] ADB. Cambodia Agriculture, Natural Resources, and Rural Development Sector Assessment, Strategy, and Road Map. (2021). Retrieved from https://www.adb.org/sites/default/files/publication/718806/cambodia-agriculture-rural-development-road-map.pdf.

determinant of women's wage employment, given the fact that married women are 38% less likely to be in paid employment.[26]

The literacy rate for the population aged 15 years and older in Cambodia is slowly increasing reaching 84% in 2021[27], while inequality persists in these numbers with 88% of males reaching the adult literacy rate[28] compared to 80% of females (2021)[29].

Physical and sexual violence against women is reported to be persisting. As indicated in the Cambodia Demographic and Health Survey 2014, 20% of women aged 15-49 were exposed to physical violence at least once in their lifetime, and 6% of them had experienced sexual violence at least once since they were born. Further concerns come from sex trafficking, in the 2018 Global slavery index reported that Cambodia has over 260,000 victims of human and sex trafficking, including minors.

Because of persistent gender inequalities, COVID-19 had a disproportionate impact on women and girls in Cambodia. Especially lives of Cambodian women who generally earn less, save less, and hold insecure and informal jobs, or live near and in poverty have been worsened by the pandemic. The ability of being engaged in paid work has also been constrained by the closure of schools during the pandemic spread in 2020 and 2021 along with an increased demand for women's unpaid care responsibilities. During the same period, education quality for girl children and women was equal especially in rural areas due to the gender digital divide, with women's and girls' more limited access to the Internet. Restricted movements such as quarantine, isolation, and closed schools during the peak of the pandemic were reported to have increased the risk of child abuse, gender-based violence and sexual exploitation globally. Women and girls in Cambodia also faced escalating risks of domestic violence due to tensions in the household from food and economic insecurity.[30]

- **Climate Change and Gender in Cambodia**

Cambodia is particularly vulnerable to climate change, with variable and extreme weather patterns such as severe rainfalls and droughts, ranking 12th most climate-vulnerable country in the Climate Risk Index from 1999-2018.[31] The estimated loss and damage from climate change was 1.5 billion dollars in 2015 alone, forcing many households into a cycle of debt caused by borrowing money as a coping strategy during natural calamities and emergencies.

Women in Cambodia are increasingly seen as more vulnerable than men to the impacts of climate change because of their connection to poverty and higher dependence on natural resources. Women, especially in times of crisis and post-crisis, also end up being overburdened by the responsibility of caring for their family members, including children and the elderly.[32]

Following are some of the general snapshots of the Cambodian women's vulnerability to climate change from the perspective of the four major sectors, i.e., agriculture, renewable energy, water and waste management. Thus, proactive gender mainstreaming in these sectors is expected to enhance women's resilience to climate change.

Women and Agriculture. Climate disasters have been found to be particularly damaging to the agricultural sector, rural communities, and therefore rural women, which represents 74% of the

[26] The Gender Wage Gap in Cambodia | UNDP in Cambodia. (2021). Retrieved from UNDP website: https://www.kh.undp.org/content/cambodia/en/home/library/the-gender-wage-gap-in-cambodia.html.
[27] https://data.worldbank.org/indicator/SE.ADT.LITR.ZS?locations=KH
[28] https://data.worldbank.org/indicator/SE.ADT.LITR.MA.ZS?locations=KH
[29] https://data.worldbank.org/indicator/SE.ADT.LITR.FE.ZS?locations=KH
[30] Gender Equality and COVID-19. (2020). Retrieved October 24, 2021, from Sweden Abroad website: https://www.swedenabroad.se/fr/ambassade/cambodia-phnom-penh/current/news/gender-equality-and-covid-19/.
[31] Eckstein, D., Künzel, V., Schäfer, L., & Winges, M. (2018). GLOBAL CLIMATE RISK INDEX 2020. Retrieved from https://germanwatch.org/sites/default/files/20-2-01e%20Global%20Climate%20Risk%20Index%202020_14.pdf.
[32] Cambodia Women's Resilience Index. (2019, December 23). Retrieved from ActionAid Cambodia website: https://cambodia.actionaid.org/publications/2019/cambodia-womens-resilience-index-2019#downloads.

agricultural workforce and produce 80% of Cambodia's food.[33] Agriculture and natural resources are the primary sources of income for most women in the country. Between 1996 and 2013, floods damaged 67% of all the paddy fields in Cambodia, while drought caused a 31% loss of the total crop during the same period. Patriarchal gender norms in Cambodia tend to create better opportunities for men in commercial or higher-income agricultural activities. 1 out of 5 rural households is accounted for as female-headed households. They are reported to be facing significant challenges in obtaining irrigation water and agricultural land[34].

Vulnerability of Cambodian rural women, especially rural female-headed households, has been worsened by their limited capacity and opportunities to diversify agricultural practices and to lessen dependency on climate-sensitive natural resources. One of the reasons for Cambodian rural women's limited coping capacity is their limited access to knowledge over new agricultural production and post-production techniques and technologies compared with their male counterparts. Furthermore, disaster prediction and preparedness information are often not shared with the female members of the rural communities in Cambodia, reducing their chances of adaptation and, in some cases, survival. There is a clear need to capacitate and empower Cambodian rural women to be better equipped with the knowledge and skills for climate-resilient agriculture as well as disaster preparedness.

Women and Water. Sustainable and reliable water supply and sanitation services remain scarce in Cambodia's rural areas. Only 11% of rural homes had access to piped water in 2017, even though 73% of urban households had improved water supply. Only 56% of rural households had access to improved sanitation, and about 41% of rural residents still practice open defecation, which can cause diarrhea and other public health problems.[35] Urban-rural divide in access to basic drinking water services and safely managed drinking water services are also stark in Cambodia. As of 2018, while about 90% of urban population can access drinking water services, and 56% of them to managed drinking water services, only about 65% of the rural population has access to drinking water services, and only 17.6% to managed drinking water services respectively[36].

Data on mortality rate attributed to unsafe water, unsafe sanitation and lack of hygiene is not available in Cambodia. Nevertheless, shortage and unclean water provision affects girls and women more severely. Lack of sanitation facilities for girls and women, which makes them sharing facilities with men and boys (or having to travel to remote outer locations for defecation) increases the risk of abuse and assaults, endangering their personal safety. Women and girls also have specific hygiene needs, without which women and girls would risk health hazards.

The impacts of drought, flood, sea-level rise, and extreme weather conditions have a significant effect on ecosystems and water supply, degrading natural resources while imposing severe consequences for mental health and human wellbeing. In general, the lack of accessible water nearby also burdens girl children and women as they are culturally responsible for fetching water. This will demand more time and physical labour from girls and women[37].

The increased weather variability has the highest impact on vulnerable people and women, causing the water supply infrastructure to be destroyed, resulting in a loss of water resources and changes in water

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[33] ADB. Cambodia Agriculture, Natural Resources, and Rural Development Sector Assessment, Strategy, and Road Map. (2021). Retrieved from https://www.adb.org/sites/default/files/publication/718806/cambodia-agriculture-rural-development-road-map.pdf. . According to representatives of the Ministry of Agriculture, Forestry, and Fisheries (MAFF), 64 percent of women in Cambodia are involved in agriculture. (https://asiafoundation.org/2019/12/04/water-gender-and-poverty-in-cambodias-stung-chinit-watershed/)
[34] https://asiafoundation.org/2019/12/04/water-gender-and-poverty-in-cambodias-stung-chinit-watershed/
[35] ADB. (2019, September 30). Help Cambodia Expand Access to Water Supply, Sanitation Services. Retrieved from Asian Development Bank website: https://www.adb.org/news/adb-help-cambodia-expand-access-water-supply-sanitation-services.
[36] https://databank.worldbank.org/source/world-development-indicators
[37] As women are specifically responsible for water in the home, the Ministry of Rural Development of Cambodia is training women to repair wells and promote local sanitation. While these measures are important, they largely support women in serving the home and do not increase women's economic independence or capacity. (Water, Gender, and Poverty in Cambodia's Stung Chinit Watershed, 2019)

quality, while droughts affect freshwater security.[38] Improving access to modern industrially processed water service to remote rural areas will significantly benefit female rural farmers extending proper water and sanitation infrastructure in urban slum areas would significantly enhance the quality of life and even survival of the low-income households and especially girls and women in slum areas.

Women and Waste Management. The waste sector is responsible for approximately 2.1% of GHG emissions in Cambodia, as of 2016. Solid waste and wastewater management is an ongoing challenge in the country with increasing levels of waste generation. Waste composition is also changing with an increasing share of non-degradable waste such as plastics, metals, and electronic waste. There is a need for additional waste segregation, treatment and reuse infrastructure and management systems. Rapid economic and urban development, combined with a growing middle class, has had a substantial impact on waste generation in Cambodia's major cities and waste collection and management is increasingly recognized as a major issue in the country. Currently, households, local companies and factories frequently dispose of their waste by burning it or dumping it in public places, streams, and deserted locations, posing public health threats and soil, water and other environmental pollution risks.

While there is a general lack of data on the actual amount of recycled waste in Cambodia, recycling activities of limited scale are observed in Phnom Penh. Several private companies are exploring the business potential targeting both organic and inorganic waste. Both men and women have a low level of awareness about the waste sector and recycling.[39]

Informal female workers as garbage collectors and dismantlers in urban areas, such as Phnom Penh, are the key actors in waste recycling in a sector largely lacking central organization.[40] While this provides previous income generation opportunities to low-income Cambodia women[41], lack of segregation of toxic and hazardous materials, and unregulated unhygienic recycling practices pose serious threats to occupational safety and health. Only a limited amount of recyclable waste is collected nationwide, and domestic recycling activities in Cambodia are limited due to the lack of recycling industry, recycling infrastructures and a market for recyclable materials and recycled products. Therefore, some of the recyclable wastes and materials are exported to neighbouring countries for recycling purposes.

- **Climate Technopreneurship and Gender in Cambodia**

Cambodia's entrepreneurship ecosystem is behind the lower middle-income average as well as Cambodia's ASEAN peers, globally ranking 113th out of 137 countries in the 2018 Global Entrepreneurship Index (GEI).[42]

Given the relatively recent development of climate technopreneurship, specific country data related to the field are not available. The empower initiative of UNEP and UN Women, aims to support women's entrepreneurship in the renewable energy sector, and it is partnering with the NCDD's and organizations such as SHE Investments, Nexus for Development and SNV to implement pilot interventions,

[38] KINGDOM OF CAMBODIA. CAMBODIA CLIMATE CHANGE STRATEGIC PLAN 2014 -2023 2013 National Climate Change Committee. Retrieved from https://www.cambodiaip.gov.kh/DocResources/ab9455cf-9eea-4adc-ae93-95d149c6d78c_007729c5-60a9-47f0-83ac-7f70420b9a34-en.pdf.
[39] GIZ. (2019). Partnership Ready Cambodia: Waste management. Retrieved from https://www.giz.de/en/downloads/GBN_Sector%20Brief_Kambodscha_Waste_E_WEB.pdf.
[40] Heinrich Böll Stiftung. Veins of Phnom Penh: Urban waste pickers as the best infrastructure for recycling waste. (2018). https://kh.boell.org/en/2018/12/17/veins-phnom-penh-urban-waste-pickers-best-infrastructure-recycling-waste.
[41] They are locally called "ad chais", mainly comprised of less physically fit and aged women who often cannot fit into more regularized work settings in shop assistants, factory workers or cleaners at construction sites. They comprise the most societally marginalized groups./ [41] Heinrich Böll Stiftung. Veins of Phnom Penh: Urban waste pickers as the best infrastructure for recycling waste. (2018). https://kh.boell.org/en/2018/12/17/veins-phnom-penh-urban-waste-pickers-best-infrastructure-recycling-waste.
[42] ENTREPRENEURIAL CAMBODIA. Cambodia Policy Note Public Disclosure Authorized Public Disclosure Authorized Public Disclosure Authorized Public Disclosure Authorized. Retrieved from https://openknowledge.worldbank.org/bitstream/handle/10986/30924/128265-REVISED-Nota-Entrepreneruship-web.pdf?sequence=5&isAllowed=y.

influencing policy and providing training, mentorship, and access to finance for female entrepreneurs in Cambodia in renewable energy to improve their lives and livelihoods.

Historically, there has been a time when Cambodian women were pushed into entrepreneurship: During Pol Pot regime many women were induced to widowhood due to the loss of their husbands by the Pol Pot genocide from 1975 to 1979. At present, women entrepreneurs own 62% of microenterprises. It is notable that women-owned businesses in Cambodia remain at a largely micro level in the informal sector (26% of women's SMEs in Cambodia are informal) on subsistence basis.

A greater level of Cambodian women's economic engagement, however, it does not always imply that they are able to fully embrace equal opportunities with their male counterparts. Barriers exist to Cambodian women in doing business, especially in non-agricultural sectors, such as manufacturing, service and knowledge sector. The key constraints for women entrepreneurs even in the formal sector are:

- **Social and cultural constraints**: Women's economic empowerment and progress in Cambodia are hampered by societal norms and traditional gender relations. Even though women nominally hold equal rights to men, they are still viewed as having a lower social status than their male counterparts. Buddhism, the country's major religion which accounts for 97% of the population, also has a strong influence on gender roles as well propagating the belief that it is a sin to be born as a woman. According to this idea, women need to cultivate more merits in the present life to be born as a man in the next life.[43] Despite the fact that Cambodian society has become more open and women have assumed increasing leadership roles in the economy, with a rate of female top managers at 57.3% (2016), women's economic freedom and prospects are still limited due to their subjugation to males under the *Chbab Srey,* a Khmer code of conduct for women.

- **Access to Business Markets**: Less access to higher education and abovementioned social norms favouring men reduce Cambodian women's opportunities to network and acquire the knowledge needed to formalize their business, which is one of the greatest obstacles to their access to market, scaling up their businesses and enterprises, especially for international trade. In general, women in Cambodia usually have less access to and time for networking and building business partnerships due to their family responsibilities, which is an additional barrier for their business development. Women are less likely to formally register their enterprises because the processes are time-consuming, and the procedures complicated. Alternatively, they often opt to keep their businesses informal and micro- or small-scale due to the lack of incentives for formalization and the high costs involved (e.g., taxes).

- **Lack of Financing:** Lack of financing was reported to be a top challenge for all SMEs in Cambodia, and many women-owned SMEs have been funded from scratch using their own savings without access to external finances. Venture capitals are not a familiar concept for Cambodian women, although organizations like *SHE investments* are attempting to take on that role.[44] Slightly more women than men borrowed from a bank or an MFI. Using informal sources and personal savings can help enterprises get started and survive for a while, larger access to capital is required for the firm to develop. Only 26% of women entrepreneurs run large businesses, and their profits are on average lower than men's.[45] Lack of financial literacy also

[43] Opinion: The Need to Promote Cambodian Women's Participation in Political Decision Making. (2021). Retrieved October from Cambodianess.com website: https://cambodianess.com/article/opinion-the-need-to-promote-cambodian-womens-participation-in-political-decision-making,
[44] SHE investments is a well-known Programme in providing female entrepreneurs with micro- to medium-sized enterprises with incubators and accelerators Programmes, giving mentoring, networking, and financing services. https://www.sheinvestments.com/.
[45] National Institute of Statistics. (2014). Cambodia intercensal economic survey. Retrieved from https://www.stat.go.jp/info/meetings/cambodia/pdf/c14ana02.pdf.

aggravates women's vulnerability in terms of collateral availability and cost of funding, which often require specific support and mentoring to fulfill the banks' requirements.[46]

Innovation is an important aspect of the success of many entrepreneurial enterprises, and so it is for climate technology. The relative weakness of Cambodia's national innovation system (NIS)[47] is reflected in its Global Innovation Index: The country ranks 109th out of 131 countries (2021), being at the bottom within the ASEAN countries.[48]

The severe underrepresentation of women in technopreneurship is felt across many developing economies where common structural factors affect women's participation in the tech industries. For Cambodia, additional pressures, institutional and socio-cultural constraints prevent young women from pursuing career-advancing opportunities in the tech ecosystem.[49] Young women who complete post-secondary education in related STEM or IT fields stay in lower-paid and low-skilled work sectors.[50] Female researchers in R&D (per million people) account for only 26.82 % (2015) compared to their male counterparts.[51] Researchers suggest that the ecosystem is dominated by men, particularly mentors and start-up founders who act as role models. In one survey, over 80% of tech start-up founders who participated in the study were male. [52]

The CTF's investment criteria explicitly include the prioritization of the "women-owned" SMEs and encourage the other applicant enterprises to develop gender mainstreaming plans. Through this, the proposed Programme aims to shift the traditional cultural norms imposed on Cambodian women and facilitating Cambodian women entrepreneurs' access to finance and markets.

The Programme will address each of the above-listed constraints in the following manner: First, the CTF's upfront adoption of the gender mainstreaming component as part of its Investment Criteria (see Table 1 above) will gradually thaw away the social and cultural constraints Cambodian women suffer by inducing change in the public perception on the women's roles in climate technology and entrepreneurship. Strong promotion of women's roles within the JV enterprises as well as the leadership roles of the invested companies (through capacitation support in Component 1 and Component 2 as well as prioritization for CTF approval in Component 3) will also strengthen women entrepreneurs and women-led enterprises to build business networking and partnerships, enabling women to have better access to business markets and financing.

- **Priority Sectors and Gender Implications in Cambodia**

Areas for priority intervention through the proposedProgramme in Cambodia are, among others: Climate resilient agriculture/ Water resource management/ Infrastructure retrofit to address the shifting climate.

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[46] Exploring the Opportunities for Women-owned SMEs in Cambodia. Retrieved from https://www.ifc.org/wps/wcm/connect/9e469291-d3f5-43a5-bea2-2558313995ab/Market+Research+Report+on+Women_owned+SMEs+in+Cambodia.pdf?MOD=AJPERES&CVID=mOU6fpx.

[47] The national innovation systems (NIS) approach stresses that the flows of technology and information among people, enterprises and institutions are key to the innovative process. Innovation and technology development are the result of a complex set of relationships among actors in the system, which includes enterprises, universities and government research institutes. (OECD). https://www.oecd.org/science/inno/2101733.pdf.

[48] Global Innovation Index 2021: Which are the most innovative countries? (2021). Retrieved January 3, 2022, from Wipo.int website: https://www.wipo.int/global_innovation_index/en/2021/.

[49] Assessment of Digital Literacy for Employability and Entrepreneurship among Cambodian Youth | UNDP in Cambodia. (2020). Retrieved October 17, 2021, from UNDP website: https://www.kh.undp.org/content/cambodia/en/home/library/assessment-of-digital-literacy-for-employability-and-entrepreneu.html.

[50] Tsang, T., & Poum, C. (2018). FACTORS AFFECTING WOMEN ENGAGING IN TECH CAREERS IN CAMBODIA Rapid Design Research. Retrieved from http://www.development-innovations.org/wp-content/uploads/2018/07/Factors-Affecting-Women-Engaging-in-Tech-Careers-in-Cambodia-Report.pdf .

[51] Cambodia | UNESCO UIS. (2021). Retrieved September 10, 2021, from Unesco.org website: http://uis.unesco.org/en/country/kh?theme=science-technology-and-innovation.

[52] Responsible Finance Forum (2018), "Cambodia's Vibrant Tech Start-up Ecosystem" https://responsiblefinanceforum.org/wp-content/uploads/2019/11/Cambodias-Vibrant-Tech-Start-up-Ecosystem-2019.pdf.

The agricultural sector is a key driver for Cambodia's economy, making up an estimated 21.1% of the country's GDP (2019). It is a female-labour intensive sector (74% of the workforce). Engaging women to learn agricultural production and post-production processing skills and knowledge would be particularly beneficial for Cambodian female farmers' economic empowerment.

Since the COVID-19 pandemic, however, technology for the agriculture sector has emerged, along with government efforts in promoting digital business and mitigating the impact of the pandemic. There is growing attention to innovative technology solutions in the agriculture sector. Agritech start-ups are reported to be springing up (marking 10 in number by 2021).

Several start-ups in Cambodia are working to innovate agriculture, by linking farmers to the agro-chains through online platforms[53]. By providing online platforms to competent woman-led agricultural enterprises will significantly increase the female farmers' income generation potential.

Clean and sufficient water provision would directly ease girls and women in rural areas from house chores and ensure their basic health, sanitation, and safety. A number of international organisations and donors, such as the AFD, JICA, KfW Development Bank, Oxfam and the World Health Organization are supporting for improving the country's water supply and waste management. Often Most water supply and wastewater treatment projects in rural areas are implemented by international NGOs, which often work in concert with local NGOs. There would be ample opportunities to empower women in this endeavour[54].

Introducing solid waste and wastewater management technologies has also a substantial potential for benefiting Cambodian women. The county has no large-scale waste treatment facilities, and that waste collection practice is limited. Recycling sector is under-developed and, if any, largely reliant on informal garbage collectors. Small-scale enterprises have emerged to tackle waste management in the country, but it is insufficient to cover the rapid growth in waste generation. In most cities in Cambodia, municipal solid and organic wastes are being collected and deposited without effective environmental and hygiene regulations. Most often garbage collectors are informally employed and constitute the lowest income level group mostly by women who migrated from rural areas. Industrial, toxic, and hazardous materials are often not segregated exposing the workers to serious public health threats and environmental pollution.

In Oct 2018 Waste Management Strategy and Action Plan of Phnom Penh 2018-2035 was established. Focusing on 3 Rs (Reduce, Reuse and Recycle), the Plan aims to promote environmentally sound management of waste disposal and mitigate impact on environment and human livelihood (Action Area 3) while involving both formal and informal private recycling sector players. Given the nature of the informal practices on the ground where vulnerable women being involved, it is desirable to consciously mainstream gender in strengthening the waste management system in Cambodia by empowering the women in this sector.

Promoting women-led start-ups in the sectors of managing organic waste, and recycling plastic would offer the opportunity for GHG emission reduction and pollution mitigation, while economically empowering Cambodian women in the circular economy of waste and resources. The Program may consider the example of "Eco-Plastic", a start-up which is recycling plastic to make asphalt roads. The company was founded by a young Cambodian woman[55].

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[53] See Cambodia's Ecosystem for Technology Startups, Sopheara Ek & Paul Vandenberg, June 2022 (Country Report No.1/ Ecosystems for Technology Startups in Asia and the Pacific) / Cambodia's Ecosystem for Technology Startups (adb.org)
[54] Partnership Ready Cambodia: Water supply and wastewater treatment, Global Business Network Programm/https://www.giz.de/de/downloads/GBN_Sector%20Brief_Kambodscha_Wasser_E_WEB.pdf
[55] https://kr-asia.com/combating-climate-change-through-cleantech-in-cambodia

INDONESIA

- **Legal and Institutional Setup and International Commitment of Indonesia**

International Agreements on Women's Rights. Indonesia has ratified and signed several international conventions, treaties, and plans of action on gender equality. Essential among them are:

- United Nations Convention on the Political Rights of Women, ratified by Law 68/1958
- United Nations Convention on the Elimination of all Forms of Discrimination against Women (CEDAW), ratified by Law number 7/1984
- Optional Protocol to the CEDAW, signed by the Government in 2000
- International Labour Organization's (ILO) Convention number 100 on Equal Remuneration for Men and Women Workers for Work of Equal Value, ratified by Law number 80/1957
- International Covenant on Economic, Social and Cultural Rights, signed by the Government in 2006
- International Covenant on Civil and Political Rights signed by the Government in 2000

The country also committed to following the recommendations of the 1994 Copenhagen Declaration on Social Development, the 1995 Cairo International Conference on Population and Development, and the 1995 Beijing Platform for Action.[56] At the regional level, the country signed on the Declaration of the Advancement of Women in the ASEAN Region in 1988, which focuses on the promotion and implementation of women's equitable and effective participation in all fields and at all levels, and the Declaration on the Elimination of Violence Against Women in the ASEAN Region (2012).

Institutional Arrangements. In 1978, the Government of Indonesia established the Ministry for Women's Empowerment, which was later changed into the Ministry of Women Empowerment and Child Protection (MoWECP), as the national entity to ensure the implementation of gender equality and women's empowerment. In 1998, the Indonesian Government founded the National Commission of Violence Against Women, while also developing different focal points on women and children under relevant Ministries/Agencies including the Coordinating Ministry of Human Development and Culture, MoWECP, Ministry of Social Affairs, Ministry of Health, Ministry of Home Affairs, Ministry of Manpower, and the National Police.[57]

At the national level, the 1945 Constitution of the Republic of Indonesia is the basic legal foundation for women's rights in Indonesia. The Presidential Instruction No. 9/2000 on Gender Mainstreaming in National Development instructed all government bodies to implement gender mainstreaming for planning, formulation, implementation, monitoring, and evaluation of national development policies and programs. The Law No. 8/2012 on General Elections introduced the requirement for political parties to have a minimum of 30% women candidates to be elected to the parliament.[58]

Gender considerations are also integrated into the National Long-Term Development Plan of 2005–2025. The quality of life and role of women and children in various development fields are low: Violence, exploitation, and discrimination against women and children are reported to be high. The

[56] ADB. (2006). Country Gender Assessment Indonesia. Retrieved from website: https://www.adb.org/sites/default/files/institutional-document/32231/cga-indonesia.pdf.
[57] Women Organizing for Change in Agriculture and Natural Resources Management (2012). A Guidance Note to Integrate Gender in Implementing REDD+ Social Safeguards in Indonesia. UN-REDD Programme. Available at: http://www.wocan.org/resources/guidance-note-integrate-gender-implementing-redd-social-safeguards-indonesia.
[58] United Nations Human Rights Council (2017). Indonesia: National report submitted in accordance with paragraph 5 of the annex to Human Rights Council resolution 16/21.

Plan aims to strengthen the institutions and network for gender and children mainstreaming at the national and regional levels, and increase the availability of gender data and statistics.

In 2016, the Ministry of Women Empowerment and Child Protection (MoWECP) launched 3Ends Programme, a flagship Programme targeting addressing violence against women and children, human trafficking, and removing barriers to economic justice for women. The MoWECP supports women's economic empowerment through the promotion of a home-based industry plan. The Ministry's programme is based on the Ministerial Regulation No. 2/2016 on General Guidance of House-Based Industry for Improving Family Welfare via Women's Empowerment.[59]

In Indonesia, legal restrictions on women's ability to act and right to asset ownership still exist. The Law No. 1 of 1974 on Marriage impedes women from serving as the head of a household, creating further discrimination in terms of divorce rights. Indonesia does not grant widows equal inheritance rights and sons and daughters have different inheritance rights. Indonesia is reported to lack legislation, civil remedies or criminal penalties to protect against sexual harassment. There is no protection under domestic-violence legislation for unmarried partners.

Religion and customs are also important in Indonesia's legal system, generating legal pluralism. While the civil law system is the foundation, coexists with that of customary laws (Hukum Adat or Adat Recht) and Shari'a law[60] coexists in practices in some locations. In principle, the adoption of either Adat law or Shari'a law should be consistent with the prevailing Indonesian law and regulations. However, inconsistencies exist in terms of marriage and inheritance.[61]

Women and Climate Change. In the updated submission of the National Nationally Determined Contribution (NDC) to UNFCCC 2021, Indonesia committed itself to map gender mainstreaming in climate change in all development sectors. Enhancing the role of women in development and strengthening women's capacity and leadership in climate change has been recognized as a priority for social and livelihood resilience.

The Directorate General of Climate Change Control (DJPPI) of the Ministry of Environment and Forestry, a female official, is the key policy-making body on the topic of climate change in Indonesia. Also, a number of female officials holds the positions of Directors of Climate Change Adaptation and of Climate Change Mitigation.[62]

- **General Situational Analysis of Indonesia**

As of 2021 the population of Indonesia is 273.5 million people, of whom 49.7% are women.[63] As part of the G20, Indonesia is considered one of the world's emerging economies and a newly industrialized country.

According to the Human Development Index 2021, the country ranks 114th out of 191 studied countries and territories (value 0.705). This is 34% improvement from the 1990 value, mainly mirroring the country's improvements in life expectancy at birth, mean and expected years of schooling, and gross

[59] Indonesia. (2017, February 20). National report submitted in accordance with paragraph 5 of the annex to Human Rights Council resolution 16/21: Retrieved from United Nations Digital Library System website: https://digitallibrary.un.org/record/1637627?ln=es.
[60] Adat law is a set of unwritten rules that apply to certain groups of people based on their traditions and customs, which might cover informal dispute resolution procedures. Shari'a law applies to Muslim's localities, where it governs areas such as marriage, inheritance, and finance.
[61] Indonesia: Closing the credit gap for women entrepreneurs | White & Case LLP. (2017, November 20). Retrieved October 26, 2021, from Whitecase.com website: https://www.whitecase.com/publications/article/indonesia-closing-credit-gap-women-entrepreneurs.
[62] http://ditjenppi.menlhk.go.id/tentang-kami-ppi/profil-pejabat.html
[63] OECD. (2021). OECD iLibrary | OECD Economic Surveys: Indonesia 2021. Retrieved from Oecd-ilibrary.org website: https://www.oecd-ilibrary.org/economics/oecd-economic-surveys-indonesia-2021_fd7e6249-en . While Indonesia is relevantly free from the practices of gender-selective abortion and excess female mortality compared with other Southeast Asian countries, it is reported in some peripheral region of the country, with strong patrilocal traditions, indicates higher gender ratio discrepancy among minors under the age of 15, which may indicate the persistent practice of selective abortion and excess female mortality. See: https://www.niussp.org/health-and-mortality/gender-bias-births-child-mortality-indonesiapeut-parler-de-discrimination-sexuelle-en-indonesie/

national income per capita. Margin of improvement was greater for women than men. Accordingly, the country's Gender Development Index in the same year (value 0.941) marked slightly below the global average.[64] Notwithstanding, in the OECD's Social Institutions and Gender Index 2019 (SIGI)[65] of Indonesia, the level of discrimination against women was assessed as "high" (with a score of 41.6%): The worst outcomes in terms of Indonesia's gender inequalities were in the area of "discrimination in the family" and "restricted civil liberties".[66]

In 2020, the female representation in the national parliaments was about 20.3% of the available seats, a substantial increase from the previous year (2019 rate: 17.4%). Indonesia is behind its minimum target of 30% women elected to the parliament. In general, female representation in public decision-making remains relatively low.[67] Indonesian women are less likely to participate in public affairs and primary decision-makers in private affairs are often males. Only 15.46% of the households nationwide are headed by women.[68]

Since the 1970s, women's education started to improve. By 2020 the literacy rate for the above- 15-year-olds marked 97% for males and 95% for females.[69]

A significant gender gap exists in employment and evidence of gender discrimination persists in the labour market. In 2019, the employment rate of women in Indonesia was 52.2%, while men had a rate of 79.3%.[70] Wage and income gaps remain large with males at 69.7% and females at 51.7%, respectively. Women are also more likely than men to work in the informal sector: 81.8% of the women's employment is in the informal sector compared to 79.4% of men, where a significant number of them are un(der-)paid workers.[71]

The economic recession after the Covid-19 pandemic outbreak affected the Indonesian labour market and the national unemployment rate increased by 1.8%, up to 7.1% between August 2019-2020.[72] Women are disproportionately more affected by the current economic recession, mainly because of the extra burden of unpaid work, massive layoffs in female-dominated service sectors, such as retail and tourism, and light manufacturing, such as clothing. Low-income women working in the informal sector are mostly self-employed, or are employed on informal and insecure contractual arrangements. They are at the highest risk of losing their income source during the pandemic, suffering the most.

As the global pandemic situation has been eased, the Indonesian economy is also recovering. Indonesia, representatively Bali, is one of the attractive tourism destinations, the tourism sector has particular concerns, and related businesses started to boost again.[73] Although the pandemic situation has not finished, President Joko Widodo announced the removal of large-scale restrictions on crowds and people's movements after observing the past 10 months' improvements[74]. Due to the COVID-19 shock, internet or digital platforms enlarged and many firms utilizing those technologies have increased.

[64] Gender Development Index (2022) Retrieved from: https://hdr.undp.org/gender-development-index#/indicies/GDI
[65] The SIGI covers four dimensions of discriminatory social institutions, spanning major socio-economic areas that affect women's lives: Discrimination in the family; Restricted physical integrity; Restricted access to productive and financial resources; and Restricted civil liberties. https://www.genderindex.org/sigi/ .
[66] OECD. Social Institutions and Gender Index 2019. Available at: https://www.genderindex.org/ranking/.
[67] United Nations Human Rights Council (2017). Indonesia: National report submitted in accordance with paragraph 5 of the annex to Human Rights Council resolution 16/21.
[68] Badan Pusat Statistik. (2021). Survei Sosial Ekonomi Nasional (Susenas), 2019 Kor. *Harvard Dataverse*. https://doi.org/doi:10.7910/DVN/9T0QN1.
[69] Literacy rate, adult total (% of people ages 15 and above) - Indonesia | Data. (2021). Retrieved November 2, 2021, from Worldbank.org website: https://data.worldbank.org/indicator/SE.ADT.LITR.ZS?locations=ID.
[70] OECD. (2021). OECD iLibrary | OECD Economic Surveys: Indonesia 2021. Retrieved from Oecd-ilibrary.org website: https://www.oecd-ilibrary.org/economics/oecd-economic-surveys-indonesia-2021_fd7e6249-en.
[71] World Bank. 2015. Country Partnership Framework for the Republic of Indonesia for the Period 2016–2020. Jakarta. website: http://documents.worldbank.org/curated/en/195141467986374707/ pdf/99172-REVISED-assem-Bank-Indonesia-Country-Partnership-Framework-2016-2020.pdf.
[72] World Bank. (2020). Indonesia. Retrieved from World Bank website: https://www.worldbank.org/en/country/indonesia/overview
[73] https://www.worldbank.org/en/news/press-release/2021/12/16/indonesia-economy-grew-in-2021-despite-covid-19-will-accelerate-in-2022-world-bank-report-says
[74] https://english.alarabiya.net/coronavirus/2022/12/30/Indonesia-lifts-all-coronavirus-related-restrictions

Almost every industry shows sales drops during the pandemic period. When it comes to the gender-related indexes, female managers dominant firms have a greater fall in sales compared to male managers dominant ones[75].

- **Climate Change and Gender in Indonesia**

Indonesia ranked the 77th most vulnerable country to climate change globally in the Climate Risk Index from 1999-2018.[76] The temperature increase, intense rainfall, sea-level rise, and the threat to food security are some of the key climate change-related challenges. Low-income women in rural regions, forest areas, or coastal areas are found to be most vulnerable. In particular, single mothers and widowed women are especially vulnerable due to weak social safety nets.[77]

Women and Agriculture. Agriculture, forestry, and fisheries are the three largest economic sectors in Indonesia, making 13.7% of the country's GDP (2020).[78] Data from the Indonesian Peasant Union show that **around 70–80% of workers in agriculture are women**.[79] Women participate in all aspects of agriculture, including farming, animal husbandry, and fishing. The ecosystem degradation from climate change has a significant impact on the survival of rural women which carry a greater share of the burden due to the damage of agricultural land and fisheries.

Rising sea levels, combined with weather variability, rising sea surface temperature, and other negative effects, expose fishing households to the possibility of losing their foundation for livelihood, prompting some to migrate. Women from poor fishing households are often burdened more.

Women and Water. The JMP global database (2020) reported that 92% of the population has access to safely managed and basic service drinking water, while 86% have access to basic sanitation.[80] However, water and sanitation deficits exist across Indonesia's thousands of islands, posing challenges for the country. While there is plenty of fresh water, population growth, urbanization, and expanding agriculture are putting a strain on available water resources, quality, and systems. As a result, 63% of the low-income Indonesian households lack access to safe drinking water, and 34% of them lack adequate sanitation.

Women are also the principal water collectors and providers in the household: two out of every five Indonesian families delegate water responsibility to women. Water-fetching is becoming an increasing burden borne by girls and women with worsening water shortage. Climate Techpreneurship and Gender in Indonesia

Indonesia is the world's fourth most populous country. It is the fifteenth largest economy[81] and currently undergoing a period of significant economic transition with a GDP growth rate averaging 4.87% from 2000 to 2021. Indonesia has become a fertile ground for business start-ups and the digital economy. Multiple support programs, initiatives, and organizations, such as hubs, accelerators, and incubators, are hosted by government agencies, donors, and start-up networks to nurture start-ups and

[75] https://openknowledge.worldbank.org/bitstream/handle/10986/37584/IDU087850cba0b204043f608dea019acef5f2be1.pdf?sequence=5&isAllowed=y
[76] Eckstein, D., Künzel, V., Schäfer, L., & Winges, M. (2018). GLOBAL CLIMATE RISK INDEX 2020. Retrieved from https://germanwatch.org/sites/default/files/20-2-01e%20Global%20Climate%20Risk%20Index%202020_14.pdf.
[77] FAO. (2021). Food and Agriculture Organization of the United Nations. Retrieved from Fao.org website: http://www.fao.org/publications/card/en/c/CA6110EN/.
[78] World Bank. (2020), "Agriculture, forestry, and fishing, value added (% of GDP) - Indonesia".
[79] FAO. (2021). Food and Agriculture Organization of the United Nations. Retrieved from Fao.org website: http://www.fao.org/publications/card/en/c/CA6110EN/.
[80] JMP. (2019). Retrieved from Washdata.org website: https://washdata.org/data.
[81] Per Nominal GDP (2020).

entrepreneurial spirit, provide managerial and technical support, and essentially build an active community of entrepreneurs. In 2019, it ranked the second-biggest digital economy in Southeast Asia.[82]

Nevertheless, the number of entrepreneurs in the country is still very low (around 2%) compared to its neighbouring countries (6% in Malaysia, 5% in Thailand, and 7% in Singapore) in 2017[83] . Indonesia's entrepreneurship ecosystem84 ranks 75th out of the 137 countries in the 2019 Global Entrepreneurship Index (GEI), where 99% of them were micro-, small- and medium enterprises until recently.[85]

As of 2013, 43% of formal SMEs in Indonesia have been estimated to be female-owned.[86] The World Bank estimated that **most women-owned enterprises in Indonesia are informal and unregistered (97%) and characterized by a small size.**

The key constraints for women entrepreneurs in Indonesia are:

- **Societal and Cultural Norms:** Compared to businesses owned by men, women-owned SMEs face constraints that are imposed by women`s dual responsibility for the business on the one hand, and family and household on the other. In the technology industry, women are still perceived to be unable to handle multiple functions and tasks that have been dominated by men.[87] The number of female top managers was 22.1% in 2015 and female-owned enterprises tend to be smaller in size.

  Women's interactions with non-family members are often perceived to be undesirable. The 1974 Marriage Law preserves men's traditional status as household heads, setting males to the role of principal decision-makers, particularly in business and other public affairs. It is also reported that Islamic norms also influence on women doing business, particularly in Muslim rural regions. As a result, rural women are less keen to do business compared to their male counterparts.[88]

- **Access to Business Markets:** Indonesian women in general have lower education level than their male counterparts. Vocational training is often found only in big cities, limiting opportunities to those residing in remote, rural areas. Business skill training is often taken by men due to the cultural perception that business training for women is inappropriate. Girls and women are often burdened with their familial duties and the limited freedom of movement.

[82] WIT. (2021). Programme Pertumbuhan Ekonomi Hijau (Green Growth Programme) mendukung Indonesia dalam mewujudkan pertumbuhan ekonomi hijau yang dapat mengurangi kemiskinan serta memastikan inklusi sosial, kelestarian lingkungan dan efisiensi sumber daya. Retrieved October 21, 2021, from Green Growth website: http://greengrowth.bappenas.go.id/en/towards-the-future-of-climate-technology-and-entrepreneurship-in-indonesia/.

[83] Genoveva, Genoveva. (2019). THE INFLUENCE OF ENTREPRENEURIAL CULTURE ON ENTREPRENEURIAL INTENTION AMONG BUSINESS STUDENTS. Firm Journal of Management Studies. Retrieved from ResearchGate website: https://www.researchgate.net/publication/334281264_the_influence_of_entrepreneurial_culture_on_entrepreneurial_intention_among_business_students

[84] Entrepreneurial ecosystem is defined such as a dynamic community within a geographic region, composed of varied and inter-dependent actors (e.g. entrepreneurs, institutions and organizations), factors (e.g. markets, regulatory framework, support setting, entrepreneurial culture), and process which evolves over time and whose actors and factors coexist and interact to promote new enterprise creation through the spirit of entrepreneurship. https://www.researchgate.net/publication/336037404_The_entrepreneurial_ecosystem_to_foster_competitiveness_among_enterprises_a_national_level_analysis.

[85] Bambang Hermanto, & Suryanto, S. E. (2017, January 26). Entrepreneurship Ecosystem Policy in Indonesia. Retrieved from ResearchGate website: https://www.researchgate.net/publication/345769314_Entrepreneurship_Ecosystem_Policy_in_Indonesia.

[86] Women-owned SMEs in Indonesia: A Golden Opportunity for Local Financial Institutions Market Research Study. (2016). Retrieved from https://www.ifc.org/wps/wcm/connect/260f2097-e440-4599-91ec-e42d45cf3913/SME+Indonesia+Final_Eng.pdf?MOD=AJPERES&CVID=lj8qhPY.

[87] Ramadhan Triwijanarko. (2018, October 7). Women's and Gender Stereotypes in Indonesia Tech Industry. Retrieved from Medium website: https://ramadhantriwijanarko.medium.com/womens-and-gender-stereotypes-in-indonesia-tech-industry-de71bf216ab0.

[88] Tahi, T., & Tambunan, H. (2017). Women Entrepreneurs in MSEs in Indonesia: Their Motivations and Main Constraints. International Journal of Gender, 5(1), 2333–2603. https://doi.org/10.15640/ijgws.v5n1p9.

Poor security for women and girls' safety is another barrier. Women entrepreneurs often lack relevant information and suffer from complicated procedures, long processes, and high costs.[89]

- **Lack of Financing:** Indonesian males use more banking services (45%) than women (42%). A higher proportion of females (49%) use informal financial services, compared to males (29%). Women use informal financial services – especially in terms of saving and borrowing – mostly to meet their short-term needs by asking support from local providers as *arisans*.[90]

- **Limited Ownership to Assets as Collateral:** In Indonesia, men and women do not have equal ownership rights over moveable and immoveable properties. Most female entrepreneurs do not own fixed assets as usually homes and vehicles are registered with male family members. This prevents woman entrepreneurs from accessing (formal) credit for their businesses. Indonesian women report a lack of collateral as their top challenge in accessing finance. Only 21% of women have these assets registered in their names, even though 88% of the households had buildings or land that may be used as collateral.[91]

Indonesia ranks 87th Global Innovation Index (2021) for the 3rd year in a row, falling under the lower middle-income group. The country falls short compared to its lower middle income-group neighbours in ASEAN, such as Vietnam (44th), and the Philippines (51st). [92] In the current digitalization and SME ecosystem of Indonesia, the role and contribution that women play are still unclear.

The Global Entrepreneurship Monitor (2018/2019) pictured Indonesia with a significantly low relative levels of innovation for women entrepreneurs, with a gender gap of 50% or more compared to men.[93] Over the past decade, women's participation in the labour market has stagnated. The gender gap has remained largely unchanged. In rural areas, younger women are opting out of the labour market altogether. In 2019, only 14% of women were working in high-tech industries such as information technology and communications compared to 31% of men. [94] Increasing women's participation in STEM is particularly urgent in Indonesia.

In 2018, female top managers were 22.1% and female researchers in R&D (per million people) were recorded at 44.87%.[95] In these occupations, women are paid less than men. The Global Gender Gap Report 2021 reports a wide gap in economic participation and opportunity between men and women in the country: A sharp drop was observed in the share of women in senior roles, from 54.9% in 2020 to 29.8% in 2021[96].

Many women still face inequalities like affordability, access to sound digital structure, and availability that stop them from effectively adopting the latest digital tools. However, women SMEs are at the forefront of adopting digital tools to stay afloat during the pandemic. Even more encouragingly, 82%

[89] Tahi, T., & Tambunan, H. (2017). Women Entrepreneurs in MSEs in Indonesia: Their Motivations and Main Constraints. International Journal of Gender, 5(1), 2333–2603. https://doi.org/10.15640/ijgws.v5n1p9.

[90] Arisan is a social and money-saving gathering, which represents an alternative to bank loans and other forms of credit. For unbanked Indonesians, Arisan can be the only way to get any financial service that would give them a lot of advantages as it does not charge interest. https://journals.sagepub.com/doi/pdf/10.1177/0973005219898922. / Survey on Financial Inclusion and Access (SOFIA) Focus Note on Gender. (n.d.). Retrieved from https://www.bappenas.go.id/files/5615/1668/2564/SOFIA_Focus_Note_Gender_Nov_2017.pdf.

[91] World Bank. (2016). Women entrepreneurs in Indonesia: a pathway to increasing shared prosperity. Retrieved October 25, 2021, from undefined website: https://www.semanticscholar.org/paper/Women-entrepreneurs-in-Indonesia-%3A-a-pathway-to-Arsana-Alibhai/169022dd12b7b6987aebaf31fdf757afa6a74e5c.

[92] Global Innovation Index 2021: Which are the most innovative countries? (2021). Retrieved January 3, 2022, from Wipo.int website: https://www.wipo.int/global_innovation_index/en/2021/.

[93] GEM. (2020). GEM 2018 / 2019 WOMEN'S ENTREPRENEURSHIP REPORT. Retrieved from GEM Global Entrepreneurship Monitor website: https://www.gemconsortium.org/report/gem-20182019-womens-entrepreneurship-report.

[94] Marshan, J. (2020, November 15). Will Indonesia's 4.0 revolution leave women behind? - Indonesia at Melbourne. Retrieved from Indonesia at Melbourne website: https://indonesiaatmelbourne.unimelb.edu.au/will-indonesias-4-0-revolution-leave-women-behind/#_ftn1.

[95] Indonesia | UNESCO UIS. (2021). Retrieved from Unesco.org website: http://uis.unesco.org/en/country/id?theme=science-technology-and-innovation.

[96] Global Gender Gap Report 2021. (2021). Retrieved from World Economic Forum website: https://www.weforum.org/reports/ab6795a1-960c-42b2-b3d5-587eccda6023/in-full/gggr2-benchmarking-gender-gaps-findings-from-the-global-gender-gap-index-2021

of women SME owners as surveyed by UN Women and Pulse Lab Jakarta use digital solutions and find them helpful in juggling work and domestic responsibilities.

The Programme will address each of the above-listed constraints in the following manner: First, as in the case of Cambodia, the CTF's upfront adoption of the gender mainstreaming component as part of its Investment Criteria (see Table 1 above) will gradually change in the public perception on the women's roles in businesses and technology sector in general. All the CTF-supported JVs will strengthen their gender mainstreaming by either increasing women's employment or improving welfare programme for female employment and/or providing other types of gender mainstreaming efforts (See Appendix 3. Simplified Gender Analysis (GA) & Gender Action Plan (GAP) (General Template): Section 1-2. Entity's Gender Mainstreaming Commitment Plan (GCMP) below.) Gender-friendly business management and improved women's welfare as a labour condition will improve the general perception of women's labour within the formal business sector in the country over time.

There are some structural issues barring Indonesian women's equal participation in business and technology entrepreneurship in Indonesia, such as the 2017 Marriage Laws, favouring male household heads as beneficiaries to formal credit services, and some religious dogma negating active roles of women in formal business sector. The country SMU may bring these legal/policy-related issues to the attention of the country NDA and other relevant government entities of Indonesia as part of Component 4 (Country Capacity Building) activities.

Strong promotion of women's roles within the JV enterprises as well as the leadership roles of the invested companies (through capacitation support in Component 1 and Component 2 as well as prioritization for CTF approval in Component 3) will also strengthen women entrepreneurs and women-led enterprises to build business networking and partnerships, enabling women to have better access to business markets and financing.

- **Priority Sectors and Gender Implications in Indonesia**

Areas for priority intervention through the proposed Programme in Indonesia are, among others: Health Biomonitoring & Trend monitoring of infectious diseases exacerbated by rising temperature and extreme weather events/ Water harvesting/ Domestic wastewater recycling/ Climate-resilient crop development.

Since 2010, Indonesia's government has been spending more to support the government's new health-related strategic goals, including digitalization of the sector. The COVID-19 pandemic since early 2020 accelerated the trends for a digital transformation of the country's health sector. Access to basic health services in Indonesia is gender-biased and is compounded by other factors such as geographical location, education level and income level. Thus, little-educated, low-incomed girls and women in remote areas are benefiting the least from the health services. In addition, due to societal and cultural norms, some female health issues, such as menstruation and reproductive health, are often tabooed and not discussed in public. This makes access of the Indonesian girls' and women's access to the health services more difficult. Through the digitalization of public health services and better monitoring of disease trends, access for girls and women is expected to improve in the long run.

With increasing water shortage related to climate change-induced extreme weather conditions, rainwater harvesting is increasingly being employed both in urban and rural settings. Being a primary water collector of a household, girls and women will be lifted from household chores with better water resource management. Treatment of water for safer and cleaner domestic water provision shall also benefit girls and women's health and hygiene and even improve their safety.

Agriculture, forestry, and fisheries are key drivers of the Indonesian economy and agriculture is a female-dominated sector (70-80%). Thus, the adoption of crop (rice) resilient to drought and flood and technology for mariculture development could have a major impact on Indonesian women. Persistent rural poverty and malnutrition, as well as rising urbanisation, make Indonesia struggle in improving agriculture, fisheries, and forestry profitability and resilience to climate resilient, women should be considered as primary target recipients in this sector. Given the feminisation of agriculture in the country, the Programme should use the competitive advantage of women's rural knowledge and should involve them in advancing innovation and entrepreneurship in Indonesia. This would alleviate poverty and food insecurity, especially for rural and coastal households in Indonesia.

LAO PDR

- **Legal and Institutional Setup and International Commitment of Lao PDR**

International Agreements on Women's Rights. Lao PDR is a signatory since 1981 to the Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW). A recent Capacity Assessment on CEDAW implementation in Lao PDR was conducted by the Association for Development of Women and Legal Education (ADWLE) in 2016. The Lao Government has also established interagency mechanisms, to promote and protect human rights as follows:

- The National Steering Committee on the Preparations for the Ratification and Implementation of the International Covenants on Human Rights;
- the National Steering Committee on the Preparations for the UPR (Universal Periodic Review);
- the National Steering Committee on reporting and implementation of the International Convention on the Elimination of all Forms of Racial Discrimination;
- the National Commission for the Advancement of Women (NCAW);
- the National Commission for Mothers and Children (NCMC), the National Committee for Disabled People;
- the National Committee for Rural Development and Poverty Alleviation and the National Committee Against Human Trafficking.[97]

At the regional level, the country signed on the Declaration of the Advancement of Women in the ASEAN Region in 1988, which focuses on the promotion and implementation of women's equitable and effective participation in all fields and at all levels, and the Declaration on the Elimination of Violence Against Women in the ASEAN Region (2012).

At the national level, Lao PDR has a strong legal framework for promoting gender equality. The revised Constitution of 2003 explicitly states that women and men have equal rights in all spheres - political, social, cultural and in the family. Women's equal rights are also stipulated in the Family, Land and Property Laws; the Labour Law; the Electoral Law; and the Penal Law.

- The Law on the Development and Protection of Women (2004) is the most specific Lao legislation articulating equal rights and access for women. The law provided a framework for several subsequent laws defining women ́s rights and served also as a basis to form the National Commission for the Advancement of Women and Mother-Child.
- The Small and Medium Enterprises Development Plan 2016-2020: Endorsed by Prime Minister Decree No. 253/PM in 2017, its main objectives are: to promote productivity,

[97] https://uprdoc.ohchr.org/uprweb/downloadfile.aspx?filename=1517&file=EnglishTranslation

technology, and innovation; to enhance access to finance for Small and medium-sized enterprises (SMEs);, to enhance access to business development services; to enhance SMEs access and expansion to domestic and international markets; to develop entrepreneurs, and; to create an enabling environment for establishment and operating business of SMEs and tax incentives.

Women and Climate Change. Lao PDR established the National Green Growth Strategy (NGGS) till 2030 in 2018. Through the NGGS, the country is committed to pursuing sustainable green growth development that makes the best use of its natural and human resources in line with its Vision 2030,10-year Socio-Economic Development Strategy to 2025, 8th five-year National Socio-Economic Development Plan, the Sustainable Development Goals, and the Paris Climate Change Agreement. The Strategy identified the promotion of the advancement of women's participation to be an important focus and promotes vocational training, labour skill development and creation of jobs for women, encouraging and promoting the participation of women in economic, political and sociocultural activities.

- **General Situation Analysis of Lao PDR**

Lao PDR is a country with a population of about 7.4 million people (2021). 83% of the population lives in rural areas and two-thirds rely on subsistence agriculture. Lao PDR has enjoyed high economic growth, with a GDP growth rate averaging 7.8% over the last decade[98] and falling poverty rates from 33.5% to 23.2%.[99] However, much of this growth has come from the unsustainable use of the country's natural resources, which has not translated into a commensurate decline in poverty. Women's access to productive and financial resources, especially in land ownership and use rights are restricted and so is women's participation in leadership and access to education and employment.

In 2019, Lao PDR scored 0.613 on the Human Development Index (HDI), ranking 137th out of 189 countries, grouping the country into the medium human development level category.[100] Despite consistent economic growth, many challenges remain for better distribution of the economic benefits. Lao PDR has a Gender Development Index (GDI) of 0.927, classified as a country with "medium to low" gender equality (Group 5).[101] Lao PDR's Social Institutions and Gender Index (SIGI) scored to be low (with a score of 0.262) in gender discrimination in social institution.[102]

The political representation of women in the National Assembly has grown by nearly 20% since 1990. It's among the highest in the region. In Lao PDR women make up less than 5% of high-level government officials. 89% of village chiefs are men. Women are still under-represented in senior government positions with 10% occupancy in 2018.

In Lao PDR, the working population is made up of an equal share of women and men, but generally, women occupy the lower rungs of the labour market. The share of women in wage employment is lower in all sectors at 35%[103]. ILO estimates that in Lao PDR, 64.3% of female employment is in the agricultural sector, 26.9% is in the services sector, and 8.8% is in the industries sector. Of the total employed females, over 87.5% are self-employed[104].

[98] GDP growth (annual%) – Lao PDR: https://data.worldbank.org/indicator/NY.GDP.MKTP.KD.ZG?locations=LA.
[99] World Bank Group (2015). Drivers of Poverty Reduction in Loa PDR https://www.worldbank.org/en/country/lao/publication/drivers-of-poverty-in-lao-pdr
[100] UNDP (2020). Gender Inequality Index-Human Development Reports: http://hdr.undp.org/sites/default/files/hdr2020.pdf
[101] Human Development Report (2013). The Rise of the South: Human Progress in a Diverse World. (2013). Retrieved from http://hdr.undp.org/sites/default/files/Country-Profiles/KHM.pdf.
[102] OECD. Social Institutions and Gender Index 2019. Available at: https://www.genderindex.org/ranking/.
[103] https://www.undp.org/sites/g/files/zskgke326/files/migration/la/Country-Analysis-Report-Lao-PDR-2015.pdf
[104] https://www.unicef.org/laos/media/4831/file/IMPACT%20OF%20COVID-19%20ON%20REIMAGINING%20GENDER.pdf

Decision-making of female business owners in managing the businesses is often shared within their family members such as husbands. Women's decision-making is not so high in public contracts and signing legal documents (only 58% and 57% respectively of the men contracting).

The gender gap in education has narrowed[105]. The literacy rate for the population aged 15 years and older in Lao PDR has also improved, reaching 84% in 2020 on average. However, gender disparity persists: 90% for males vs. 79% for females (2020).[106]

Having said that, Laotian women are active economic agents, marking 70% of the country's economic migrants. An increasing number of women are migrating to Thailand as workers mainly in the service sector and as domestic workers. A UNODC study estimates that 35% of women migrants to Thailand are victims of sexual exploitation. Physical and sexual violence persist, and domestic violence is a silent but widespread problem: 1 in 5 women is reported to be a victim in Lao PDR.

Because of pre-existing gender inequalities, COVID-19 had a disproportionate impact on women and girls in Lao PDR. The COVID-19 pandemic and its corresponding mitigation measures have significantly increased women and girls' exposure to gender-based violence (GBV). Female workers were more likely to lose their source of income due to COVID-19. The economic implications of this crisis are skewed towards jobs and sectors where women are overwhelmingly represented such as agricultural and service sectors. The impact of the crisis on these sectors, combined with the after-effects of severe weather events on the agricultural sector in 2019, must have left a substantial impact on women[107].

Women and Agriculture. In Lao PDR a third of the country's GDP (29.9%) is generated in the agricultural sector and over 75% of the population is engaged in agricultural activities. **Women in Lao PDR are key players in agriculture,** with more than 50% of the agricultural population being women since 1980.[108] As high as 70% of the total female employment is concentrated in the agricultural sector.[109] Women traditionally manage marketing of agricultural products and livestock production. They are typically given key responsibility for food security in the family.

However, Women's high level of participation in agricultural work in Lao PDR is not always a source of women's empowerment. As stated by the World Bank and ADB's 2012[110] gender assessment report for Lao PDR, FAO's Country Gender Assessment of Agricultural and the Rural Sector (2018)[111], and ASEAN State of Gender and Climate Change Report (2022)[112], women's role in agriculture is significant, but often undervalued. Increased agricultural productivity and opportunities for off-farm jobs, often taken by female agricultural workers, are helping to pull some households, in selected areas, out of poverty.

Lao PDR is transitioning from subsistence farming to commercial agriculture. This transition has brought benefits for some rural communities, but it has also exacerbated gender disparities. Scaling up for commercialization in the agriculture sector often resulted in dispossession and deprivation of small farming landowners of their productive land, including female land holders. This worsened the vulnerability of small-scale farmers, esp. those who cannot find employment off-farm.

[105] https://gggi.org/wp-content/uploads/2018/10/Gender-and-Green-Growth-Policy-Brief-Lao-PDR.pdf
[106] World Bank (2021). Literacy rate https://data.worldbank.org/indicator/SE.ADT.1524.LT.FE.ZS?locations=LA
[107] https://lao.unfpa.org/sites/default/files/pub-pdf/covid-19_impact_assessment_lao_pdr-_brief_women_1_0.pdf
[108] UNDP (2016). Human development report 2016: Lao People's Democratic Republic. http://hdr.undp.org/sites/all/themes/hdr_theme/country-notes/LAO.pdf
[109] OECD (2021). Strengthening Women's Entrepreneurship in Agriculture in ASEAN Countries https://www.oecd.org/southeast-asia/regional-Programme/networks/OECD-strengthening-women-entrepreneurship-in-agriculture-in-asean-countries.pdf
[110] World Bank and ADB (2012). Country Gender Assessment for LAO PDR Reducing Vulnerability and Increasing Opportunity https://www.adb.org/sites/default/files/institutional-document/33755/files/cag-lao-pdr.pdf
[111] https://www.fao.org/3/CA0154EN/ca0154en.pdf
[112] https://asean.org/wp-content/uploads/2022/08/State-of-Gender-Equality-and-Climate-Change-in-ASEAN_FINAL-1.pdf

As noted by the World Bank's 2016 working paper on the feminization of agriculture,[113]commercial agriculture is, however, opening new opportunities for women to undertake paid employment outside the family farm, particularly through participation in 'non-traditional export crop production' (as contract farmers or direct wage employees). Formal contractual arrangements to secure fair work conditions for female agricultural workers need to be strengthened.

Women and Water. About 80% of Lao PDR's water flow materializes during the rainy season and the rest during the dry season. Most of the domestic water supply in Lao PDR is adequately supplied by streams and rivers. Most people in rural areas obtain their water supplies from streams, rainwater, and water from public standpipes or wells for drinking. Bathing and laundry washing usually take place in streams or at public standpipes. [114] Nevertheless, some parts of the country experience water shortages during the dry season, particularly in the headwaters of some rivers and areas far from perennial rivers. General pressure on water resources of the country is compounded by Lao PDR's incompatible legal and institutional capacity to manage its rapid economic development and population growth in recent years. New issues are flooding and drought, which have started to significantly impact the country in recent years. One of the challenges to Lao PDR is to expand its irrigation capacity (including water storage and distribution infrastructure) to overcome localized temporary water shortages in a country so richly endowed with water resources.

The Lao Social Indicator Survey 2017 reported that 48.7% of those responsible for gathering water for the family are women along with girls 9.4% (Table 2). The disparity is even steeper in (esp. remote, without road) rural areas: women and girls in rural areas are bearing a much heavier burden of water-fetching for the family. This takes significant time away from their education and other activities that could lead to better jobs, as well as income generation.

Table 2. Person Collecting Water in Lao PDR in 2017[115]

	Percentage of household members without drinking water on premises	Number of household members	Person usually collecting drinking water						
			Woman (15+)	Man (15+)	Female child under age 15	Male child under age 15	DK/Missing/ Members do not collect	Total
Total	**17.7**	**104,851**	**48.7**	**19.8**	**9.4**	**2.9**	**19.2**	**100.0**
Area								
Urban	4.5	32,178	29.1	28.2	3.8	1.8	37.0	100.0
Rural	23.5	72,674	50.3	19.1	9.9	2.9	17.7	100.0
Rural with road	22.2	61,970	47.2	19.9	9.9	3.1	19.8	100.0
Rural without road	31.1	10,704	63.3	15.8	9.6	2.2	9.1	100.0

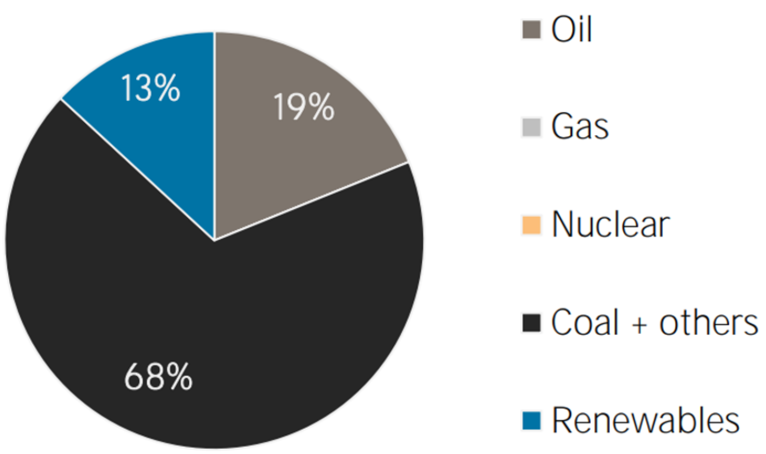
Water quality is often substandard due to the lack of proper water treatment facilities in rural areas and poor sanitation conditions often result in illness of the individuals.[116] The country's open defecation rate is the second-highest in the region, often leading to ground water pollution. Lao PDR and the World Bank have been working to supply rural areas with facilities to reduce open defecation.

[113] World Bank (2016). Feminization of Agriculture in the Context of Rural Transformations: What is the evidence? https://openknowledge.worldbank.org/bitstream/handle/10986/25099/ACS20815.pdf?sequence=6&isAllowed=y
[114] Ministry of Natural Resources and Environment (2012). Lao Environment Outlook 2012. https://wedocs.unep.org/bitstream/handle/20.500.11822/8485/-LAO%20Environment%20Outlook%202012-2012Lao_EO_2012.PDF?sequence=3&isAllowed=y
[115] 2017 the Lao Social Indicator Survey. https://laopdr.un.org/
[116] World Bank (2016). Improving Rural Sanitation in Lao PDR https://www.worldbank.org/en/country/lao/publication/improving-rural-sanitation-in-lao-pdr

- **Climate Change and Gender in Lao PDR**

Lao ranks 45th in the Global Climate Risk Index (CRI) 2019 and 52nd in CRI from 2000 to 2019[117]. Climate change results in hotter seasons, heavy floods, droughts, abnormal cold and other irregular weather patterns to which crops are vulnerable. Crop crises because of extreme weather can result in food shortages, familial stress, and even occurrence of domestic violence.[118]

Women are mainly responsible for household water supply and energy for cooking, yet their voices are soften excluded from local and national decision-making processes in energy policies and coping plans to climate change.



Source: IRENA https://www.irena.org/

Figure 1. Lao PDR Total Primary Energy Suppy in 2018

Lao PDR's entrepreneurship ecosystem is behind the lower middle-income average as well as Lao PDR's ASEAN peers, ranking 102nd out of the 137 countries in the 2019 Global Entrepreneurship Index (GEI).[119] World Bank reports indicate that **SMEs in Lao PDR are a major employer, but their productivity is relatively low. While accounting for 81% of the total labour force in 2010, SMEs in Lao PDR contribute only about 16% to GDP.** Men account for most workers in the formal sector, and as a result have greater access to social protections, exempting female workers whose majority remain in the informal sector.

Data is not available on the number of firms and micro-entrepreneurs operating in the informal economy in the country, such as street vendors, tuk-tuk drivers, handymen, construction workers, etc. and the contribution of the informal SMEs as an economic sector is largely unknown. **Informal street vendors take predominance as Lao women's business patterns. Women represent more than 90% of vendors in fresh food markets throughout the country.**

Meanwhile, women-owned SMEs registered in Lao PDR are also predominantly made up of small firms. By sectors, women-owned enterprises mainly function in the retail and wholesale trade, and only to a

117 The Global Climate Risk Index identifies the extent to which countries have been affected by extreme weather events. /https://germanwatch.org/sites/default/files/Global%20Climate%20Risk%20Index%202021_1.pdf

118 FAO (2018). Country gender assessment of agriculture and the rural sector in Lao people's democratic Republic. https://www.fao.org/3/CA0154EN/ca0154en.pdf.

119 The Global Entrepreneurship and Development Institute (2019). Global Entrepreneurship Index 2019 https://thegedi.org/wp-content/uploads/2020/01/GEI_2019_Final-1.pdf.

small extent in the industrial sector. In small firms, women are more engaged in the retail and wholesale trade sector than in the industrial sector. While direct data on women's entrepreneurship in the technology sector is not available, it is judged to be low.

Given the quite recent development of climate technopreneurship, specific data related to the field are not available for Lao PDR.

Women's economic participation is heavily skewed to the unpaid jobs or informal sector. The number of male government employees is 2.5 times higher than female employees.[120] Wage payment rates in 2018 was just 13% for women, compared to 25% for men.[121] Limited education opportunities impact Lao women's capacities and skills necessary for starting a business and managing it efficiently and profitably. The barriers for women are even steeper for technology-related businesses.

- Female workers on average earn only two-thirds the monthly wage of their male counterparts. Increasingly women are taking up conventional male occupations such as plumbing and electrical engineering etc. for better income and enterprise development, whose market demand is steadily on the rise.

- **Access to Business and Markets:** Women suffer lower level of education and cultural norms on women's roles in the households are limiting women's opportunities to network and acquire the knowledge needed to formalize their business and to access to market, especially for international trade. Women spend four times the amount of time on housework per day (2.6 hours) compared with men (0.6 hours). Moreover, there are legal restrictions for female entrepreneurs, as reported in the World Bank's "Women, Business and the Law" (2015) evaluated there are differences between women and men of the same marital status in 21 areas in the country's legal system.

- **Lack of Financing**: Access to reliable and long-term finance and credit poses structural challenges to business enterprises of all sizes in Lao PDR, even more difficult to the SMEs in informal sectors, where women business owners are more skewed to. It is reported that in 2014, less than 20 % of SMEs had access to long-term credit. This poses difficulty for Lao SMEs to grow and compete against other SMEs in Asia.[122] Female entrepreneurs in Lao PDR are also less likely to have a bank account or credit line. Women market vendors in Vientiane regularly report the lack of access to finance and credit as a key barrier to their economic success.[123]

Lao PDR ranked 117th out of 132 countries in the Global Innovation Index 2021.

Women-owned enterprises have been rising in recent years. According to the World Bank's enterprise surveys, enterprises owned by women had increased from 39% in 2009, to 42% in 2012, and to 43% in 2016.[124] The country has a comparatively high proportion of female top managers: 43.1% (2018).[125] While quantity of the women's business enterprises has increased over time, further study is

[120] Lao Statistics Bureau, Laos Population and Housing Census 2015

[121] World Bank Group. 2017. World Bank Group Country Partnership Framework for the Lao People's Democratic Republic, 2017-2021 (Vol. 2). Washington, DC: World Bank Group. http://documents.worldbank.org/curated/en/381061497445645649/pdf/116234-WB-v2-116p-CPF-Long-ReportEng-V6-final-LR.pdf

[122] World Bank, 2014. 'Lao PDR's Small and Medium Enterprises to Benefit from New World Bank Financing'. Press Release, June 9, 2014. Available from: http://www.worldbank.org/en/news/pressrelease/2014/06/09/lao-pdr-small-medium-enterprises-benefit-new-world-bank-financing

[123] UN Women. 2017. The Situation of Women Market Vendors in Vientiane: A Baseline Report. Vientiane: UN Women. http://www2.unwomen.org/-/media/field%20office%20eseasia/docs/publications/2017/01/women-marketvientiane.pdf.

[124] Parliamentary Institute of Cambodia (2018). Overview of Women's Entrepreneurship in Micro, Small and Medium Enterprises in Lao PDR: https://pic.org.kh/images/2019Research/20190128%20Overview%20of%20Women%E2%80%99s%20Entrepreneurship%20in%20Micro,%20Small%20and%20Medium%20Enterprises%20in%20Lao%20PDR_Phetchinda%20Keovilay.pdf

[125] Firms with female top manager (% of firms) - Vietnam, Cambodia, Indonesia, Philippines, Lao PDR | Data. (2021). Retrieved from Worldbank.org website: https://data.worldbank.org/indicator/IC.FRM.FEMM.ZS?locations=VN-KH-ID-PH-LA.

required to understand whether these have led to benefit women with high value-added jobs in formalized sector with safety and security.

Few Start-up incubators and accelerators exist in Lao PDR. Women's Entrepreneurial Centre (WEC) is one of the rare business incubators targeting women.

The Programme will address each of the above-listed constraints in the following manner: First, CTF requires all the Fund-supported JV enterprises to proactively increase female employment (esp. for decent jobs), improve welfare conditions for female employment (See Appendix 3. Simplified Gender Analysis (GA) & Gender Action Plan (GAP) (General Template): Section 1-2. Entity's Gender Mainstreaming Commitment Plan (GCMP) below.) Gender-friendly business management and improved women's work conditions (equitable with male workers in terms of contractual terms and payment etc.) will encourage more women's participation in the formal employment sector in the country over time.

Strong promotion of women's roles within the JV enterprises as well as the leadership roles of the invested companies (through capacitation support in Component 1 and Component 2 as well as prioritization for CTF approval in Component 3) will also strengthen women entrepreneurs and women-led enterprises to build business networking and partnerships, enabling women to have better access to business markets and financing.

- **Priority Sectors and Gender Implications in Lao PDR**

Areas for priority intervention through the proposed programme in Lao PDR are, among others: Climate-resilient crop development with insurance scheme/ Livestock disease control & pest management/ Sustainable Forest management and protection/ Wastewater treatment/ Water resource management.

Given the high level of women's involvement in the agricultural sector, investment in climate-resilient technology has a potential to positively impact female farmers and agricultural workers' livelihood. To overcome the existent gender disparity in access to agri-technology education and related entrepreneurship, the Programme needs to consciously promote female farmers' participation in the funded enterprises related to the climate-resilient agriculture.

Enhanced livestock management systems would also benefit female livestock farmers in remote areas when they are provided required technical skill training for disease control and pest management.

As Laotian girls and women hold higher responsibilities as water collectors for the family, investments in climate-resilient water supply and quality management, along with flood and drought management technologies would contribute to improving gender equality by easing them from their physical chores and time wasted. Wastewater treatment will improve overall sanitation conditions, and this will in particular, positively affect girls and women's health and safety.

PHILIPPINES

- **Legal and Institutional Setup and International Commitment of the Philippines**

International Agreements on Women's Rights. The Philippines is one of the earliest signatories to the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW). The Philippines also ratified the Optional Protocol to the CEDAW on November 12, 2003. Over the last two decades, the Philippines submitted six State reports and subsequently three NGO shadow reports which were accomplished by 47 NGOs to the CEDAW in compliance with the obligations of State Parties.

The country also ratified in 1967 the International Convention on the Elimination of All Forms of Racial Discrimination, and the International Covenant on Economic, Social and Cultural Rights in 1974. The Philippines also ratified in 1990 the Convention on the Rights of the Child, the Optional Protocol to the Convention on the Rights of the Child on the involvement of children in armed conflict in 2003 and the Optional Protocol to the Convention on the Rights of the Child on the Sale of Children, Child Prostitution and Child Pornography in 2002 respectively.

At the regional level, the country signed on the Declaration of the Advancement of Women in the ASEAN Region in 1988, which focuses on the promotion and implementation of women's equitable and effective participation in all fields and at all levels, and the Declaration on the Elimination of Violence Against Women in the ASEAN Region in 2012.

At the national level, the Philippine Constitution (1987) stipulates equal protection of women in its Article II Section 12 indicates "It [the State] shall equally protect the life of the mother and the life of the unborn from conception." In addition to that Section 14 from the same Article II states "[The State] recognizes the role of women in nation building and shall ensure the fundamental equality before the law of women and men."

Magna Carta of Women (Republic Act No. 9710): As a comprehensive women's human rights law, the Act was enacted in 2009. The Act stipulates that "the State shall take all appropriate measures to ensure the opportunity of women, on equal terms with men and without any discrimination to represent their government at the international level and to participate in the work of international organizations." The Act seeks to eliminate discrimination through the recognition, protection, fulfillment, and promotion of the rights of Filipino women, especially those in the marginalized sectors of the society. The Act requires institutionalization for the **Gender and Development (GAD),** a gender mainstreaming strategy in all the major government institutions. Thus, in accordance with existing guidelines, each of the relevant agency within the government need to prepare specific agenda and allocate and execute gender budget.[126]

The Philippine Plan for Gender-Responsive Development (PPGD) 1995-2025 was adopted in 1995, through Executive Order 273. It contains a long-term vision of women's empowerment and gender equality and translates the Beijing Platform for Action (BPFA) into policies, strategies and programs and projects for Filipino women. This 30-year perspective plan ensures that women-friendly policies take root and flourish.[12]

The National Commission on the Role of Filipino Women (NCRFW) was established in 1975 by the Magna Carta of Women with an aim to ensure and advance women's rights on policies and programs. NCRFW was renamed in 2009 as the Philippines Commission on Women (PCW), with expanded mandates of: monitoring and overseeing the implementation of the Magna Carta and capacitating the government agencies to execute their respective gender mainstreaming agenda.

RA No. 8760, ("General Appropriations Act (GAA) On Programs/Projects Related to Gender and Development (GAD)") was enacted in 2000 to mandatorise all the concerned government entities to submit their GAD plan to the PCW. Each plan submitted by the agencies was evaluated based on its

[126] Section 27 of RA No. 8760. To quote: "All concerned government entities shall submit their GAD plan to the National Commission on Women for review. They shall likewise submit annual reports to Congress, the Department of Budget and Management (DBM), National Commission on Women (NCW), indicating the accomplishments and amounts utilized to implement programs/projects/activities addressing gender issues and women empowerment. The evaluation of agencies utilization of the GAD budget shall be performance based."
See also Section 36 and Section 36-B of the RA No. 9710. To quote: "Section 36: Gender Mainstreaming as a Strategy for Implementing the Magna Carta of Women: All government departments, including their attached agencies, offices, bureaus, state universities and colleges, government owned and controlled corporations, local government units and all other government instrumentalities shall adopt gender mainstreaming as a strategy to promote women's human rights and eliminate gender discrimination in their systems, structures, policies, programs, processes and procedures."
& "Section 36-B: Creation and/or Strengthening of the GAD Focal Points (GFP): All concerned government agencies and instrumentalities mentioned above shall establish or strengthen their GAD Focal Point System (GFPS) or a similar GAD mechanism to catalyze and accelerate gender mainstreaming within the agency."

performance. Since 1995, all government agencies are mandated to allocate at least 5% of their regular budget to their respective GADs.

The Gender Equality Plan and Women's Empowerment (GEWE) Plan 2019-2025: This plan is the third time-slice framework plan in support of the PPGD 1995-2025. The plan covers the Philippines Development Plan (PDP) 2017-2022 and the remaining years of the PPGD 1995-2025. This Plan set out the following Strategic Goal Areas (SGAs), as directly relevant to our proposed program as follows:

SGA 1: Expand economic opportunities for women (in trade, industry, services/agriculture, fishery and forestry/ labour and employment).

SGA 2: Accelerated Human Capital Development through investment in gender equality and women's empowerment (in health and nutrition/ education/ shelter)

SGA 5: Expand opportunities for women's participation, leadership and benefits in science, technology, ICT, infrastructure, and energy.

While the Filipinas' STEM education is comparatively faring well compared with the global average and other neighbouring countries in the region (marking 43% of the enrolment in STEM education being women in 2020), the Plan notes the need to further strengthen women's participation and benefiting in the science, technology and innovation (STI) and ICT sector, especially after the COVID-19 situation where "STEM jobs and professional work have fared relatively well during the COVID-19 crisis due to their suitability to remote work."[127] Thus the Plan, among others, aims to support technology and innovation for SMEs and entrepreneurs affected by the COVID-19 crisis and continue to promote more women into STEM fields of study and STEM-related careers. The Philippines Science and Education Institute (SEI) will take the lead in GAD in STEM Education program and increase the number of women scholarship programs for S&T education[128].

Other laws related to gender mainstreaming and women's empowerment include, among others:

- Prohibition on Discrimination Against Women (Republic Act No. 6725): This prohibits discrimination with respect to terms and conditions of employment solely based on sex. Under this law, any employer favouring a male employee over a female in terms of promotion, training opportunities, and other benefits solely on account of sex is considered discrimination.

- Anti-Violence Against Women and Their Children Act of 2004.

- Act Providing Assistance to Women Engaging in Micro and Cottage Business Enterprises, and for Other Purposes (Republic Act No. 7882), 1995.

- Act to Promote Entrepreneurship by Strengthening Development and Assistance programs to Micro, Small and Medium Scale Enterprises Amending for the Purpose (Republic Act No. 6977). This act is otherwise known as the Magna Carta for Small and Medium Enterprises and for other purposes as amended by RA No. 8269 and 9501 (Known for short, "Magna Carta for Micro, Small and Medium Enterprises (MSMEs)". It provides all possible assistance to Filipino women in their pursuit of owning, operating and managing small business enterprises.

- The Go Negosyo Act or Republic Act 10644 (titled "An Act Promoting Job Generation and Inclusive Growth through the Development of Micro, Small and Medium Enterprises") strengthens MSMEs to create more job opportunities in the country, by establishing "negosyo

[127] ADB Brief: COVID-19 is No Excuse to Regress on Gender Equality. November 2020. Retrieved from https://w ww.adb.org/sites/default/files/publication/651541/covid-19-no-excuse-regress-gender-equality.pdf
[128] Updated Gender Equality and Women's Empowerment Plan 2019-2025 (April 2022, Philippine Commission on Women) / https://library.pcw.gov.ph/wp-content/uploads/2022/07/PCW-Updated-Gender-Equality-and-Womens-Empowerment-Plan-2019-2025-2022.pdf

centers" nationwide to support MEMEs through various business advisory services, business registration assistance, business information and advocacy services etc.

- Magna Carta for Countryside and Barangay Business Enterprises or Republic Act 6810 promotes ease of doing business in the countryside and municipalities for small and micro-enterprises.

- 105-Day Expanded Maternity Leave Law (RA 11210) (fully titled, "An Act Increasing the Maternity Leave Period to one hundred five (105) days for Female Workers with an Option to Extend for an Additional Thirty (30) Days without Pay and Granting an Additional Fifteen (15) Days for Solo Mothers, and for Other Purposes.")

- Solar Parents Welfare Act (RA 8972) (fully titled, "An Act Providing for Benefits and Privileges to Solar Parents and Their Children, Appropriating Funds therefor and for Other Purposes."), sets out a comprehensive Programme for social development and welfare services for sole parent and their children, including, among others, flexible work arrangements, and parental leave, livelihood development services, educational and housing benefits[129].

- Anti-Sexual Harassment Act of 1995.

The Philippines Commission on Women (PCW), as a pivotal agency of the Philippines Government that is mandated to ensure the implementation of the country's Magna Carta of Women (Republic Act No. 9710), which has a focus area on "Micro, Small and Medium Enterprises Development Sector," is currently assessing the immediate effects of COVID-19 on the women micro-entrepreneurs (WMEs) across the priority regions of the project, to establish need-based measures to back up their gaps and losses and economically empower the women MSME owners and entrepreneurs[130].

Another important project funded by the Government of Canada is **"GREAT Women Project 2**" (also named, the Supporting Women's Economic Empowerment in the Philippines Project), which seeks to improve the economic empowerment of women micro-entrepreneurs (WMEs) and their workers through (1) improving the competitiveness and sustainability of women's micro-enterprises and (2) improving the enabling environment for women's economic empowerment, which PCW participates as the lead executing agency131. Great Women Project 2 also completed profiling and enrolment of 831 WMEs with its 15,770 employees, of which 75 percent or 11,760 were female workers across the regions in the country[132].

Women and Climate Change. The Climate Change Act of 2009 of the Philippines recognizes the need to identify the "differential impacts of climate change on men, women, and children." The Act aims to incorporate a gender-sensitive, and pro-poor perspective in all climate change and renewable energy efforts, plans and programs, mandating local governments to carry out capacity-building Programme with a focus on women and children, especially in rural areas. The National Climate Change Plan of the Philippines highlights the need to mainstream gender considerations in (ii) research and development (R&D); (ii) planning and policy-making; (iii) knowledge and capacity development; and (iv) enhancing women's participation in climate change adaptation. The government has been committed since 1995 to allocate 5% of its annual budget for GAD.

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[129] ESCAP/APCICT Case Study: Women ICT Fronteir Initative (WIFI) in the Philippines/ https://www.unapcict.org/sites/default/files/2019-12/WIFI%20Case%20Study%20-%20The%20Philippines%20-%20Final.pdf

[130] https://pcw.gov.ph/micro-small-and-medium-enterprises-development/

[131] The project also involves and involves different national government agencies such as the Department of Trade and Industry (DTI), Department of Agriculture (DA), Department of Science and Technology (DOST) and (since 2017) the Food and Drug Administration (FDA), the Department of Information and Communications Technology (DICT), and the University of the Philippines Diliman Extension Program in Pampanga (UPDEPP) for technical capacity building activity implementation, as well as private sector partners such as Bayan Academy, ECHOsi Foundation, National Confederation of Cooperatives (NATCCO), Socio-Economic Development Program Multi-Purpose Cooperative (SEDP-MPC), and other individual SMEs/entrepreneurs. / See more details: https://pcw.gov.ph/great-women-project/

[132] This could potentially provide a valuable DB for identifying the potential female-led SMEs to take part in the proposed Programme.

A gender-sensitive approach has been adopted also in the National Framework Strategy on Climate Change in April 2010.

The Philippine Green Jobs Act of 2016 or Republic Act 10771 aims to foster low-carbon, resilient sustainable growth, and decent job creation by providing incentives to businesses that generate green jobs. However, there is no gender component in the Act.

The country established the Climate Change Commission, as a focal point to the UNFCCC. The Commission has issued Resolution 2019-01 on the implementation of the National Climate Risk Management Framework. It carries out several capacity-building programs for national government agencies, higher educational institutions, and local government units ensuring gender balance of participants, trainers, experts and coordinators involved.

The first Nationally Determined Contribution (NDC) document submitted by the Philippines government proposed to the UNFCCC stipulates the importance of women's roles in tackling climate change challenges. The Philippines is developing an NDC. The country is still yet to complete and organize its national climate change gender-responsive statistics for targeted, fit for the purpose for adaptation and mitigation measures.

- **General Situation Analysis of the Philippines**

The Philippines has made significant progress in empowering women and advancing gender equality. Since the Government introduced the Constitution in 1987 affirming women and men equality, there have been serious efforts to mainstream gender concerns in government policies and programs.[133]

Based on the 2021 Census of Population by the Philippine Statistics Authority, 50.4% of the total population were males while 49.6% were females which resulted in a sex ratio of 101 males for every 100 females. In the age groups 0-45, males outnumbered females. Females, on the other hand, outnumbered their male counterparts in the older age groups of 55 years old and over.

The Human Development Index (HDI) of the Philippines for 2020 values at 0.718 which placed the country in the high human development category[134]. With this, the Philippines ranks 107th out of 189 countries and territories. In terms of the Gender Development Index,[135] the country falls under Group 1 countries, with high equality in HDI achievements on health, knowledge and living standards (with a score of 1.007). **High level status of human development of the Philippine citizens is contrasted to the "very high" status of gender-related discrimination** reported in SIGI index in 2019. The highest level has been identified in the category of **discrimination in the family and restricted access to productive and financial resources** (both at 67%).

Poverty continues to be concentrated in specific segments of the population. The poverty incidence among the population increased to 23.7% during the first half of 2021 from 21.1% in the same period of 2018. This translates to 3.9 million more Filipinos living in poverty. i.e., farmers (31.6%), fishers (26.2%), children belonging to families with income below the official poverty threshold (23.9%), the self-employed and unpaid family workers (18%) and women from poor families (16.6%) for the year 2018.[136]

The Philippines ranked 104th out of 160 countries based on the Gender Inequality Index (GII) value of 0.430. This is based on the following statistics of the Philippines: 28% of parliamentary seats are held

[133] ADB (2004). Country Gender Assessment PHILIPPINES. https://think-asia.org/bitstream/handle/11540/6347/pass-Country%20gender%20assessment%20-%20Philippines%20Jan04.pdf?sequence=1
[134] UNDP (2020). Human Development Report 2020 The next frontier Human development and the Anthropocene: http://hdr.undp.org/sites/default/files/hdr2020.pdf
[135] UNDP (2019). Gender Development Index. Retrieved from http://hdr.undp.org/sites/default/files/2020_statistical_annex_table_4.pdf.
[136] The Philippine Statistics Authority (2020). Poverty Statistics for the Basic Sectors in 2018 https://psa.gov.ph/content/farmers-fisherfolks-individuals-residing-rural-areas-and-children-posted-highest-poverty

by women; for every 100,000 live births, 121 women die from pregnancy-related causes; adolescent birth rate is 54.2 births per 1,000 women of ages 15-19; and female participation in the labour market is 46.1% compared to 73.3% for men.

It is remarkable that in the Global Gender Gap Report 2018 of the World Economic Forum (WEF), the Philippines is top 8th in the list of the most gender-equal countries out of the 149 countries assessed. **Filipinas are found to excel in educational attainment in all primary, secondary, and tertiary levels of education**. **This also explains the higher level of participation of women than men in the Philippines as professional and technical workers. Also, in the Philippines, women and men are already equally likely to attain managerial positions.** This marks the country with stark difference from the rest of the target countries. [137]

However, **the rapid deterioration in women's health and survival is alarming.** Compared with the score in 2006, the country plummeted in its gender parity in this area in 2018, from 1st to 43rd in the country rankings. Further study is required to understand the underlying causes for this. Also noteworthy is **the persistence of wide gap in wage level between men and women despite women's parity in education attainment as men**: the labour market in the country is still dominant with male workers (with only 51.4% of women workers vs. 76.9% male), ranking 106th out of 149 countries assessed and the income disparity between genders remains.

Women and Agriculture. Filipinas have developed innovative solutions to challenges posed by climate change and natural disasters. Conservation farming – in which coffee and root crops are interspersed with rice to have the most efficient water conservation and land use possible – generates income from local sales and increases crop diversification, protecting against the risk of crop failure due to altered weather patterns. Hazard mitigation is more difficult to integrate into indigenous communities who are in rural areas, with heavy dependence on small-scale agriculture and natural resources.[138]

Agrarian reform laws and guidelines have increasingly recognized and protected the rights of rural women to land and other productive resources.[139] However, weak implementation of existing land laws and guidelines, customary and discriminatory practices, lack of information and ineffective gender mainstreaming strategies have blocked gender equality in agriculture and the rural sectors, particularly regarding land rights.[140] Rural women's limited registration in the farmers' databases, which traditionally consist of male farmers who are heads of the family, is a barrier to gender equality in agriculture.

Agriculture is regarded as a male-dominated sector, and officially there are more male farmers registered than female farmers. The agricultural labour force is mainly composed of men at 77% while only 23% of the employed workers are women.[141] Unpaid care work is heavily dominated by women. **Women tend to be less integrated in value chains than men. Their lack of mobility and access to markets, reinforced by social norms, prevent them from linking with buyers, suppliers, and other actors in the chain. In agriculture value chains, women are predominantly active in subsistence farming and food for family consumption, whereas men primarily focus on cash crops.**[142] **Women farmers face greater disadvantages in accessing agricultural and rural advisory services.** They are less likely to be targeted for extension services, as many extension agents still do not recognize women as farmers. Research shows that despite their primary role in the family's food security, **only 36% of**

[137] https://www3.weforum.org/docs/WEF_GGGR_2018.pdf
[138] https://www.preventionweb.net/files/globalplatform/5cd6e23ca5361International_Disaster_Risk_Reduction_Strategies.pdf
[139] FAO (2018). Country gender assessment of agriculture and the rural sector in lao people's democratic Republic https://www.fao.org/3/ca1345en/ca1345en.pdf
[140] National Confederation of Small Farmers and Fishers Organizations (PAKISAMA). (2016). Analysis of Land Tools in the Philippines using Gender Evaluation Criteria. http://asianfarmers.org/wp-content/uploads/2015/06/Corral_289-mar2015.pdf
[141] Philippine Statics Authority (2018). https://psa.gov.ph/content/male-farm-workers-are-paid-higher-female-farm-workers
[142] Quisumbing, A., Heckert, J., Faas, S., Ramani, G., Raghunathan, K., & Malapit, H. (2021). Women's empowerment and gender equality in agricultural value chains: evidence from four countries in Asia and Africa. *Food Security*, *13*(5), 1101-1124.

women farmers have access to irrigation, 29% to seeds, 26% to training, 23% to extension services, 21% to fertilizers and seeds subsidy, 20% to pest control management, 20% to calamity assistance and 14% to financial assistance. According to a study, if women farmers were given the same level of support as their male counterparts, food production by women could increase by 25% and total national food production by 1.5% to 3%.[143]

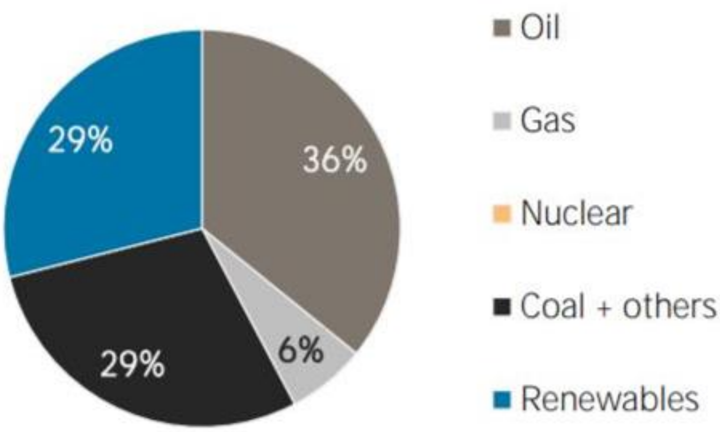
A 2021 publication on Gender and Value Chains from GIZ notes that in Southeast Asia, women are also more involved in small trade, particularly in agricultural goods.[144] The Philippine Centre for Postharvest Development and Mechanization (PhilMech), an attached agency to the Department of Agriculture, has been active in the design, fabrication, and commercialization of gender-friendly technologies to provide both male and female farmers with access to machines that will help them increase their productivity.[145]

Women and Energy. Access to electricity is still a dream for about 11 million Filipinos. In Southeast Asia, the Philippines ranks as having the third highest number of people who are energy deficient. Looking at the data from the World Bank (2019), the electricity penetration rate is 95.6%. However, even for poor families who do have access to electricity, paying for electricity is a constant struggle—not surprising for a country whose electricity rates are among the highest in the world, second only to Japan in Asia.

The primary energy supply is mainly based on fossil fuels like oil (36 %), gas (6%) and coal and others (29%), following IRENA data of 2018. Renewable energy, particularly bioenergy and geothermal have a limited share of 29%, and consequently the energy sector results as the largest contributor to greenhouse gas emissions in the country.[146]

Women in the Philippines are the primary consumers of energy and are most affected by the lack of clean, affordable, and reliable energy. This underscores the value of women’s perspective in policies which, ultimately, contributes to improved robustness of decision-making. As a major influence on family and household decisions, women play an active role in implementing energy projects. Having power at home allows the mother of the house to use time-saving equipment such as rice cookers and washing machines, among other things. Furthermore, the availability of energy in isolated rural areas means they don't have to travel half a day to the main island to recharge their phones and flashlights. Women's financial opportunities have also increased because of having electricity, as they may work longer hours.[147]

[143] National Confederation of Small Farmers and Fishers Organizations (PAKISAMA). (2016). Analysis of Land Tools in the Philippines using Gender Evaluation Criteria. http://asianfarmers.org/wp-content/uploads/2015/06/Corral_289-mar2015.pdf
[144] GIZ (2021). Promotion of Sustainable Agricultural Value Chains in ASEAN. https://www.giz.de/en/downloads/giz2021-en-agri-asean-factsheet.pdf
[145] The Philippine Centre for Postharvest Development and Mechanization: https://www.philmech.gov.ph/?page=gender-corner&subPage=gad_role
[146] IRENA. (2018). Indonesia ENERGY PROFILE. Retrieved from https://www.irena.org/IRENADocuments/Statistical_Profiles/Asia/Indonesia_Asia_RE_SP.pdf.
[147] Gender perspectives on energy poverty and energy democracy in The Philippines. (2020). Retrieved from Library.fes.de website: https://library.fes.de/TouchPoint/singleHit.do?methodToCall=showHit&curPos=4&identifier=2_SOLR_SERVER_136434115.



Source: IRENA https://www.irena.org/

Figure 2. Philippines Total Primary Energy Supply in 2018

The Philippines has a female chairperson of the Philippine National Renewable Energy Board (NREB) since her appointment by the Philippine President on 28 March 2019.[148] She steers the performance of the Board in its mandate of monitoring the implementation of a program established under the Philippine Renewable Energy Act of 2008 (Republic Act No. 9513) and in formulating recommendations for new policies to the Department of Energy (DOE) toward achieving the renewable energy targets of the country. The Department of Energy launched the gender toolkit for the energy sector to respond effectively to gender issues and to bolster equal rights and shared responsibilities between men and women.[149]

Women and Water. Results of the 2020 Annual Poverty Indicators Survey show that most families had an improved source of drinking water (97.5%). **Residents in rural areas are more likely to have an unimproved source of drinking water than those in urban areas (9.1% vs. 2.6%).** However, four in every five families (80.2%) did not practice any method or treatment in ensuring that drinking water is safe to drink. About nine in every ten families (93.9%) had basic service level with drinking water from an improved source with roundtrip collection time of not more than 30 minutes. Families with basic drinking water service level in urban areas (96.9%) was higher than in rural areas (90.6%). In 2020, about four out of five families (80.4%) had a basic sanitation service level or used an improved sanitation facility not shared with another household.[150] Women and their households also need water for bathing and personal hygiene, laundry, and cooking – which require water that has not been polluted by pesticides or fertilizers. In view of the gender division of labour in Filipino households, **women have a greater stake in safe water supply. They use the water in their housework, and they care for household members who may fall ill due to contaminated water.**

[148] https://usea.org/article/women-energy-monalisa-c-dimalanta
[149] Department of Energy (2016). DOE GENDER TOOLKIT for the Energy Sectorhttps://www.apec.org/docs/default-source/Publications/2017/5/Guidelines-to-Develop-Energy-Resiliency-in-APEC-Off-Grid-Areas/TOC/Annex-8-Philippines-DOE-Gender-Toolkit.pdf
[150] Philippine Statistics Authority (2020). https://psa.gov.ph/sites/default/files/%5BONSrev-cleared%5D%202020%20APIS%20Final%20Report_rev1%20wo%20comments_ONSF3_signed.pdf

- **Climate Change and Gender in the Philippines**

The Philippines is one of the world's most vulnerable populations to the effects of climate change, ranking 2nd in the Climate Risk Index from 1999-2018. [151] When the United Nations Office for Disaster Risk Reduction (UNISDR) evaluated disaster-prone countries in 2015, the Philippines ranked third. This is due, in part, to its geographic location as an archipelago with an exposed coastline, and its position in the Pacific Ring of Fire. Extreme weather events like flooding, typhoons, and earthquakes are common; on average, eight or nine tropical storms make landfall each year. The frequent occurrence of natural disasters poses challenges for a developing nation such as the Philippines, which lacks the financial resources and organizational capacity for comprehensive disaster risk reduction.

Women play an integral role in this localized disaster risk reduction and management, stemming from their traditional position as community leaders, as well as the mobility of the male labour force in the region. Within the remaining population, women are the key drivers in all aspects of community organizing and developing resilience. Inequality is often worsened by various risks and hazards that can affect the situation of the poor, the vulnerable, and the severely marginalized. Super typhoon Yolanda (Haiyan) in 2013 affected approximately 16 million individuals, half of whom are women and girls.[152]

Women are increasingly seen as more vulnerable than men to the impacts of climate change, because of their connection to poverty and dependence on natural resources.

- **Business Enterprises and Gender in the Philippines**

The Philippine Statistics Authority indicates that in 2015 there were around 900,000 registered enterprises in the country. An estimated 10% of these businesses were classified as small, and less than 1% were classified as medium. Less than 1% of businesses in the country were classified as large, while the vast majority (90%) were composed of micro-enterprises. It is estimated that there are around 28,000 SMEs in the Philippines.

The Philippine entrepreneurship ecosystem is middle-income average as well as its ASEAN peers, ranking 86 out of 137 countries in the 2019 Global Entrepreneurship Index (GEI), while the Global Innovation Index ranks the country 51st out of 132 countries (2021). **The Philippine start-up ecosystem has strengthened over the years with the recent rise in the number of incubators and accelerators and other support providers** such as Investing in Women, Philippines Women Entrepreneur Network and WOCAN which are specifically targeting women start-ups.[153] Currently, disaggregated data on the nature of business (i.e., whether they are start-ups on climate change adaptation and mitigation) are not available. There are no available gender statistics on SMEs ownership in the Philippines.

While 77.3% of males are economically active participating in labour force, women are only 50.1%. The only type of workforce activity where:

- the participation of women was higher compared to men in unpaid family work, where the women's participation rate was at 14%, while the men's participation was at 7% in 2015. Women's participation tends to be generally low across all forms of economic activity. Only

[151] Eckstein, D., Künzel, V., Schäfer, L., & Winges, M. (2018). GLOBAL CLIMATE RISK INDEX 2020. Retrieved from https://germanwatch.org/sites/default/files/20-2-01e%20Global%20Climate%20Risk%20Index%202020_14.pdf.
[152] Women and the Environment, Beijing Platform for Action (BPfA+20) Philippine Progress Report
[153] Francisco, Jamil Paolo and Canare, Tristan. 2017. The Challenges To SME Market Access in the Philippines and the Role of Business Promoting SME Competitiveness in the Philippines (2020). International Trade Centre. Retrieved from https://apfcanada-msme.ca/sites/default/files/2019-01/The Challenges to SME Market Access the Philippines and the Role ofBusinessAssociations_edited.pdf.

1.7% of women in the workforce are employers while male counterparts were 3.5%. Female top managers are only 29.9% (2019).[154]

- **Social and cultural constraints:** Although the 2009 Magna Carta for Women (MCW, R.A. 9710) ensures women's equal rights in all matters relating to marriage and family relations – inclusive of owning, acquiring, and administering property, married women are applied exceptions: The Family Code states that in case of disagreement over conjugal property, "a husband's decision shall prevail." The Philippine Commission on Women is currently working to repeal this discriminatory article of the Family Code.[155] Policy on land distribution by the Philippine government is virtually disadvantageous to women. Prioritization of the beneficiary groups to the government's land distribution is reported to favour men in effect.

- **Access to Business Markets**: About 92% of all enterprises in the Philippines were at micro-level, where women were most owners and operators. About 60% of women's enterprises were under performing due to lack of management/business support and poor access to finance. Many female micro-entrepreneurs are reported to be reluctant to formalize their businesses in response to government efforts in this regard, due to concerns about official interference and the greater costs of meeting the requirements of formal registration.[156]

- **Lack of Financing:** The Women in Development and Nation Building Act (Republic Act No. 7192) stipulates equal treatment between women and men regarding opening a bank account and obtaining credit, loans and ensures equal access to public and private sector grants for agricultural credits, equal rights to enter various financial contractual relations. Again, married women in the Philippines are discriminated against. According to the Philippine Tax Code (1997), in case both spouses earn wages, the husband is deemed the household head by default, eligible for tax exemptions. Although legally not required, some financial institutions require and expect a male partner to co-sign any financial contracts.[157] Women entrepreneurs often seek micro-finance schemes, but it is reported to be restrictive and complicated in procedures. This makes women seek other more expensive means for credits. Due to their lack of collateral, a bank loan is often inaccessible to women.[158]

- The government has taken a range of measures to increase women's financial literacy and access to credit – especially for rural women and female entrepreneurs, such as: contract standardization, lowered transaction costs, incentive schemes, flexible payment schedules, etc. In addition, the Philippines Commission on Women implements the "GREAT Women Project" which stands for Gender Responsive Economic Actions for the Transformation of Women.[159]

These barriers discourage women from pursuing their business. QBO and DTI's Start-up Pinay Programme aims to mitigate this issue. Start-up Pnay is an initiative that aims to create a diverse, inclusive Start-up landscape in the Philippines by empowering women founders, leaders, enablers, and explorers. Through the Start-up Pinay Programme, QBO and DT aim to take meaningful steps towards shifting gender norms, especially on the role of women in the workplace. Impact Pioneers Network is

[154] Firms with female top manager (% of firms) - Vietnam, Cambodia, Indonesia, Philippines, Lao PDR | Data. (2021). Retrieved from Worldbank.org website: https://data.worldbank.org/indicator/IC.FRM.FEMM.ZS?locations=VN-KH-ID-PH-LA.

[155] In addition, according to the Civil Code, young wives under the age of 21 is not posited to properly equipped for all acts of civil life: its article 39 states that "a married woman, twenty-one years of age or over, is qualified for all acts of civil life, except in cases specified by law." Married men are not applied the age limit. See: https://www.genderindex.org/wp-content/uploads/files/datasheets/2019/PH.pdf

[156] ESCAP (2018). Women's Entrepreneurship: Lessons and Good Practice Women's Entrepreneurship: Lessons and Good Practice (unescap.org)

[157] OECD. SIGI Philippines. (2019). Retrieved from https://www.genderindex.org/country/philippines/.

[158] ADB (2008), Paradox and Promise in the Philippines: A Joint Country Assessment, Asian Development Bank, Mandaluyong City, Philippines, https://www.adb.org/documents/paradox-and-promise-philippines-joint-country-genderassessment.

[159]OECD. SIGI Philippines. (2019). Retrieved from https://www.genderindex.org/country/philippines/.

also a recently launched activity by USAID Philippines which also supports women enterprises as an angel investing network.[160]

The Programme will address the above-listed constraints in the following manner: First, relatively disadvantaged female farmers' status in the Philippines could be improved when agricultural "women-led enterprises" are prioritized and supported by the CTF. In case a CTF-supported enterprises/JVs involve involuntary resettlement associated with the land acquisition, the CTF shall apply the GCF Safeguard policy on involuntary resettlement, and its interim safeguard standards (such as IFC Performance Standard 5 on Land Acquisition and Involuntary Resettlement (2012) and its Guidance Note 5[161]) and ensure equal treatment between male and women as compensation and other benefit beneficiaries[162].

Given the equal/higher education level of women in the Philippines, promoting (by prioritizing for CTF approval) the "women-led enterprises" will substantially capacitate the female entrepreneurs in the country to remove their structure barriers to the formal finances and credits as well as by strengthening their business network and partnership to the global level and address the limitations of many female-led enterprises stagnant in the small and micro-scale levels.

- **STEM and Higher Education, and Gender in the Philippines**

The Philippines has, according to UNESCO, an adult literacy rate of 98.18%. Only in a few countries **female literacy rates are above male literacy rates**. In the Philippines, 98.24% of all women aged 15 years and older are literate, compared to 98.12% of men. 75.6% of adult women have reached at least a secondary level of education compared to 72.4% of their male counterparts. In the Philippines, there are more female graduates than males. The average Tertiary school enrolment was 35.48% in 2017. Among those with baccalaureate degrees, there were more females (56.0%) than males (44.0%). Similarly, among those with post-baccalaureate courses, females (59.9%) outnumbered males (40.1%). Regarding STEM and higher enrolment rate, Filipinas reach around 18%.[163]

Meanwhile, the World Economic Forum recorded an improvement in economic participation and opportunity because of increased wages for women's estimated income and equal pay for similar work. **The Philippines ranks higher than any other Asia-Pacific country except New Zealand. However, qualified women have less opportunities to be promoted to the top managerial positions and female workers with the same educational qualification and responsibilities are being paid less than men.**

The Philippines is facing an oversupply in information technology (IT) and needs more students to go into the science, technology, engineering, and mathematics (STEM) fields. The undersupply of STEM-trained workers is most apparent in the life sciences, physical sciences, mathematics, statistics and engineering by 2025. The Philippine Institute for Development Studies (PIDS) estimates the supply-demand gaps in 2025 at 13,964 workers in the life sciences, 569,903 in engineering, 9,689 in the physical sciences, and 13,285 in math and statistics.[164] **In the Philippines, women in STEM-related industries are often in low-skilled positions and are less likely to progress professionally.**

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[160] US Embassy in the Philippines. (2021 Nov 22). USAID launches two activities to support micro, small, and medium enterprises i n the Philippines. US Embassy in the Philippines. Retrieved from https://ph.usembassy.gov/usaid-launches-two-activities-to-support-micro-small-and-medium-enterprises-in-the-philippines/.

[161] https://www.ifc.org/wps/wcm/connect/topics_ext_content/ifc_external_corporate_site/sustainability-at-ifc/policies-standards/performance-standards/ps5

[162] In addition, if it is clear that the female farmers and residents are socio-economically and climate disaster-wise more vulnerable than men, the CTF invested enterprises shall also integrate some special consideration of the female farmers and resident groups as the vulnerable groups, in accordance with the GCF Gender Policy (2019).

[163] Marshan, J. (2020, November 15). Will Indonesia's 4.0 revolution leave women behind? - Indonesia at Melbourne. Retrieved from: https://indonesiaatmelbourne.unimelb.edu.au/will-indonesias-4-0-revolution-leave-women-behind/#_ftn1 .

[164] BusinessWorld Online. Philippines facing oversupply in IT graduates, STEM shortage. Retrieved from: https://www.bworldonline.com/philippines-facing-oversupply-in-it-graduates-stem-shortage/.

The ILO has also estimated that **women in the Philippines are 140 percent more likely to lose their job due to automation than men.**

- **Priority Sectors and Gender Implications in the Philippines**

Areas for priority intervention through the proposed Programme in the Philippines are, among others: Off-grid renewable energy (RE) and battery storage for health facilities in climate disaster vulnerable areas / RE-powered mobile water treatment and desalination/ Technologies for crop tolerance to extreme weather events.

Women play an active role in impacting energy consumption. Improved availability of affordable clean energy, as solar PV and other green energy technologies would affect mainly the time spent by women and girls in doing house chores, electricity bills and the indoor pollution. The proposed Programme also had better **target women and young entrepreneurs in gaining the technical skills to enter the renewable energy sector. Given the high level of education of women, there is a high potential for women's participation in RE enterprises in the Philippines.**

Agriculture is mostly a male-dominant sector, where Filipinas' farms are only 23% whose majority are of small-scale substantial agriculture type. In the Philippines, women often have less access to agricultural services (such as irrigation, seeds, training, etc.) and land ownership than men. The Programme needs to target women as beneficiaries of the commercialization and diffusion of agricultural technologies and have a concrete impact on women's livelihood and food security.

VIETNAM

- **Legal and Institutional Setup and International Commitment of Vietnam**

International Agreements on Women's Rights. Vietnam is a signatory to the Universal Declaration of Human Rights (1948) and most of the key international instruments on gender equality. Vietnam participated in the Beijing World Conference on Women (1995) and follow-up meetings. Vietnam was one of the first countries to ratify the Convention on the Elimination of Discrimination Against Women (CEDAW) in 1982 but it has not signed the Optional Protocol to CEDAW; Operational Program to the International Covenant on Economic, Social and Cultural Rights; and Operational Protocol to the International Covenant on Civil and Political Rights.

At the regional level, Vietnam signed on the Declaration of the Advancement of Women in the ASEAN Region in 1988, which focuses on the promotion and implementation of women's equitable and effective participation in all fields and at all levels. The country also signed the Declaration on the Elimination of Violence Against Women in the ASEAN Region in 2012.

Institutional Arrangements. The central government agency responsible for overseeing gender equality initiatives is **the National Committee for the Advancement of Women (NCFAW)**, which coordinates gender mainstreaming as an inter-ministerial committee. Under NCFAW have been also created the **Committees for the Advancement of Women (CFAWs)** in all ministries/agencies and in all 64 provinces of the country.

The **Vietnam Women Union (VWU**) is a mass organization with strong links to the government. VWU also works closely with CFAWs at the grassroots level and implements projects to improve women's lives and contribute to their economic development. In addition, at the central level, there are a number of central legislative (i.e. the National Assembly) and administrative and management agencies, as well as the mass party organization for women. At sub-national (provincial) levels, a Department of Labour-Invalids and Social Affairs (MOLISA), Department of Culture, Sports and Tourism (MOCST), Provincial Women's Union, Provincial Committee for the Advancement of Women exist. The Sub-provincial (district and commune) levels, there also exist similar structures.

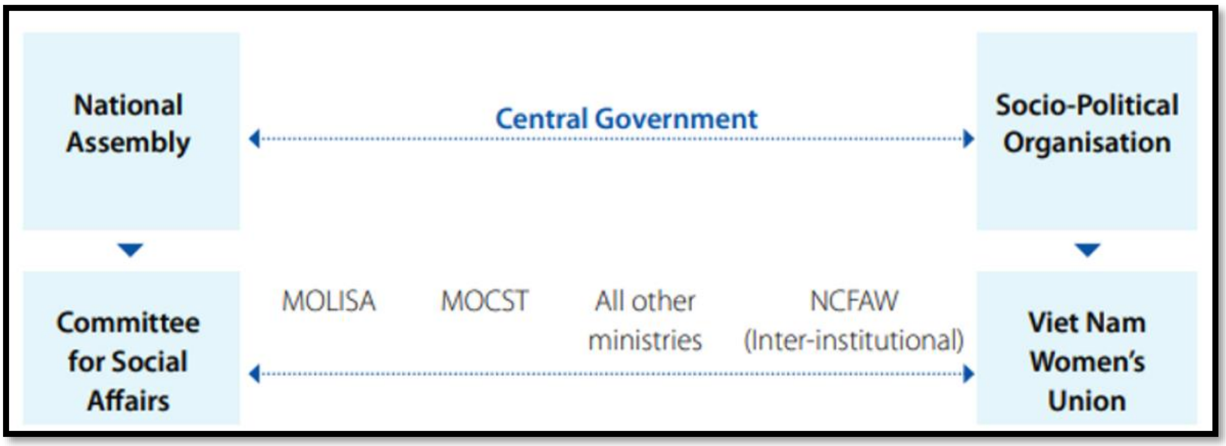

Figure 3 -Vietnam's National Institutional Structure for Gender Equality at the Central Level
(Source: UN Women, Country Gender Equality Profile Vietnam 2021[165])

At the national level, Vietnam has a solid legal framework in place to promote gender equality. The Constitution (1946) proclaims equality between men and women in terms of economic opportunities, political engagement, land tenure, property ownership, marriage, and family. After the ratification of the CEDAW in 1982, the Vietnamese government has taken important steps to ensure that legislation supports gender equality as follows:

- Law on Marriage and Family (2000) upholds the principle of equality in ownership and inheritance in case of divorce and death.

- Law on Gender Equality (2006)

- **The Land Law of 2013** makes it compulsory to have both husband's and wife's names on land use certificates and house ownership documents.

- **Decree No.56/2009/ND-CD (on assistance to the development of small- and medium-sized enterprises)** prioritizes assistance program for SMEs owned by women and SMEs which uses more female employees.[166] However, it is reported that there is no official regulation that defines a women-owned MSEs[167].

- **The Law on Domestic Violence Prevention and Control came into effect in 2008** as one of the most proactive measures taken towards ending many forms of GBV at home. A number of projects on GBV have also been implemented nationwide, including the Programme for Prevention and Response to Gender-Based Violence (2016-2020) and Vision to 2030; the

[165] https://asiapacific.unwomen.org/sites/default/files/Field%20Office%20ESEAsia/Docs/Publications/2021/10/vn-CGEP_Full.pdf
[166] See Article 5 of the Decree.
[167] "There is no official definition of women-owned SMEs in Vietnam. The General Statistics Office of Vietnam uses the gender of the top manager as a proxy (i.e., enterprises managed by a woman are considered to be women-owned). There is a strong correlation between a female top manager and majority female ownership. According to the World Bank Group Enterprise Surveys 2015, there is 79% correlation between having a female top manager and majority female ownership; this study found 92% correlation." (Women-owned enterprises in Vietnam: Perceptions and Potential. (2017). Retrieved from https://www.ifc.org/wps/wcm/connect/86bc0493-78fa-4c7d-86ec-5858aa41fa1a/Market-study-on-Women-owned-enterprises-in-Vietnam_Eng_v1.pdf?MOD=AJPERES&CVID=l-Yi6gF.)

National Action Plan on Prevention of Domestic Violence (2014-2020); the Project on the Reduction of Domestic Violence in Rural Areas of Vietnam (2015- 2020); the Annual Plan to Minimize Teen Marriage and Consanguineous Marriage in Ethnic Minorities in 2018; and a Programme to support gender equality among ethnic minorities in 2018-2025.[168]

- **The Law on Support for Small and Medium-sized Enterprises (2017)** stipulates putting priorities to small-and medium-sized enterprises that are owned by women or employing more female workers. This is the very first time that the country's laws and regulations explicitly mentioned the importance of women-owned SMEs[169].

- Currently, **the National Strategy on Gender Equality (2021- 2030)** sets specific targets for gender equality and meeting the Sustainable Development Goals (SDGs) by 2030. Some of those targets include: by 2025, 60% and by 2030, over 75% of state management agencies and local administration at all levels shall have female leaders; increase female employment engaged in paid work to 50% by 2025 and to approximately 60% by 2030; and reducing women's average hours of housework and domestic care without wage to 1.7 times in 2025 and 1.4 times in 2030 compared to men.[170]

Women and Climate Change. Vietnam's updated Nationally Determined Contribution (NDC) was submitted in November 2022 to the UNFCCC with specific references to the gendered impacts of climate change. However, a comprehensive integration of gender in the policy frameworks underpinning the implementation of the updated NDC is lacking and there are no explicit gender equality targets or measures identified to address women's vulnerabilities in climate change. The updated NDC of Vietnam also highlights the need for new technologies and livelihood diversification for rural women and stresses that gender inequalities should not be exacerbated by the introduction of such technologies.[171]

- **General Situation Analysis of Vietnam**

Vietnam is one of the fastest-growing economies in the world. The 2019 Population and Housing Census reported a demography of 96.2 million people, with 65.6% of Vietnam's population being located in rural areas.[172] Vietnam has been praised for its achievement of gender targets and the government's strong commitment to promote gender equality and women's empowerment.

Such efforts have paid off with improved rankings in several global indices, including the Human Development Index and the Gender Development Index. In 2020, Vietnam's HDI was 0.704 falling under the high human development category. It is a substantial improvement from 0.475 in 1990. The country now ranks 117th out of 189 countries and territories. The 2020 female HDI value was 0.703 (0.705 for males), resulting in a Gender Development Index (GDI) of 0.997, which positively places Vietnam into Group 1 countries.[173] Vietnam has also taken tremendous steps toward becoming a lower-middle-income country and reducing poverty and achieving a low unemployment rate. Considering the

[168] VIETNAM'S VOLUNTARY NATIONAL REVIEW ON THE IMPLEMENTATION OF THE SUSTAINABLE DEVELOPMENT GOALS. (2018). Retrieved from https://sustainabledevelopment.un.org/content/documents/19967VNR_of_Viet_Nam.pdf.
[169] Article 5 of the Law, under the heading of "Principles of Support for small- and medium-sized enterprises" include the following provision: "5. […] In case many small- and medium-sized enterprises are eligible for support prescribed in this Law, priority shall be given to those owned by women or employing more female workers."
[170] Government of Vietnam. (2021). Resolution No. 28/NQ-CP the National Strategy on Gender Equality in 2021 - 2030. Retrieved from https://english.luatVietnam.vn/resolution-no-28-nq-cp-dated-march-03-2021-of-the-government-on-promulgation-of-the-national-strategy-on-gender-equality-in-the-period-of-2021-203-199301-Doc1.html.
[171] The State of Gender Equality and Climate Change in Vietnam (EmPower – Women for Climate Resilient Societies). Retrieved from https://asiapacific.unwomen.org/.
[172] Macrotrends. Vietnam Rural Population 1960-2021. (2021). https://www.macrotrends.net/countries/VNM/Vietnam/rural-population#:~:text=Vietnam%20rural%20population%20for%202020,a%200.06%25%20decline%20from%202016.
[173] As identified by UNDP, Group 1 comprises countries with high equality in HDI achievements between women and men (absolute deviation of less than 2.5 %).

OECD's Social Institutions and Gender Index 2019 (SIGI),[174] the level of discrimination against women is regarded as "low" (25%). Following the parameters of the index, the categories with higher discrimination were related to "restricted access to productive and financial resources" and "restricted physical integrity", scored at 32% and 31% respectively.[175]

The country still faces a gender imbalance regarding political decision-making. In 2020, the female representation in the national assembly was at 26.7%, with little improvement from 2009 (25.8%). Women are underrepresented across government bodies.[176]

Vietnam has a relatively high labour force participation rate (76.2%) with an average women's participation of 73.2% (2018). The country emphasizes the importance of labour force participation by women. While being highly economically active, women also bear a disproportionate amount of family responsibilities. In Vietnam, women's challenge in the labour market is the quality of that employment rather than the share of employment.:[177]

In 2019, the agriculture sector accounted for 36.1% of female employment; Combined with the services sector, these represented the largest sector of employment for women (36.8%); while roughly one-fourth (25.4%) of women in Vietnam work in the industrial sector, mostly in manufacturing.[178] Since the COVID-19 pandemic outbreak, labour force participation has fallen sharply for both women and men, but women bore the bigger damage. According to an April 2020 analysis by the International Labour Organization (ILO), 64% of workers in the sectors that were hardest hit by the economic impacts of the pandemic in the early months in Vietnam, namely accommodations and food; manufacturing; wholesale, retail trade, and repair; transport and communication; and arts and entertainment were women. It is also reported that the COVID-19 pandemic has worsened gender inequalities and GBV in Vietnam.

Vietnam has made remarkable progress in eradicating illiteracy and providing universal education, with national statistics indicating almost gender parity in the attendance of girls and boys in general education. The literacy rate of the population aged 15 years and above is 95% on average (female 93.6% and male 96.5%), which is largely due to the implementation of literacy programs and reduced gender inequality in terms of access to education, as stipulated in the Action Plan on Gender Equality in Education 2016-2020.[179]

- **Climate change and Gender in Vietnam**

Vietnam ranked 38th in the Global Climate Risk Index for 2019 and 13th in the Climate Risk Index for 2000-2019[180]. Over the past 30 years, hundreds of victims have been generated by climate change-related disasters while incurring an annual national economic loss of approximately 1.5% of GDP. Vietnam's most vulnerable areas are located in the Mekong Delta, the Red River Delta and the central coast, where ecosystems, biodiversity and water resources are at risk and related economic sectors such as agriculture, public health and infrastructure are increasingly more vulnerable. The poor, ethnic

[174] The SIGI covers four dimensions of discriminatory social institutions, spanning major socio-economic areas that affect women's lives: Discrimination in the family; Restricted physical integrity; Restricted access to productive and financial resources; and Restricted civil liberties. https://www.genderindex.org/sigi/ .

[175] OECD. Social Institutions and Gender Index 2019. Available at: https://www.genderindex.org/ranking/.

[176] Statista. Vietnam: proportion of seats held by women in national parliament. (2020). https://www.statista.com/statistics/730331/Vietnam-proportion-of-seats-held-by-women-in-national-parliament/.

[177] Theme, G. (n.d.). AUSTRALIA-WORLD BANK GROUP STRATEGIC PARTNERSHIP IN VIETNAM. Retrieved from https://openknowledge.worldbank.org/bitstream/handle/10986/34895/Perceptions-of-Gender-Disparities-in-Vietnam-s-Labor-Market.pdf?sequence=1&isAllowed=y.

[178] Gender and the labour market in Vietnam* An analysis based on the Labour Force Survey Main messages ILO Brief. (2021). Retrieved from https://www.ilo.org/wcmsp5/groups/public/---asia/---ro-bangkok/---ilo-hanoi/documents/publication/wcms_774434.pdf.

[179] The State of Gender Equality and Climate Change in Vietnam. (2021). Retrieved from https://asiapacific.unwomen.org/-/media/field%20office%20eseasia/docs/publications/2021/04/publication_Vietnam%20report_digital2%20(1).pdf?la=en&vs=4800.

[180] Germanwatch. Global Climate Risk Index 2021. Available at: https://www.germanwatch.org/sites/default/files/Global%20Climate%20Risk%20Index%202021_2.pdf

minorities, the elderly, women, children, and people with disabilities are among those who bear the burden of the negative impacts of climate change.[181]

Women and Agriculture. According to a report conducted by FAO (2019), agriculture contributes 21% of Vietnam's total GDP and employs more than 7% of the country's workforce. **Women are an important labour force in agricultural production, especially in rural areas, where about 63% of the women work in agriculture in the agricultural sector.[182] Agriculture is the sector that will be hardest hit by climate change in Vietnam** and women are disproportionately impacted because they make up a large slice of the workforce. Their per capita income is relatively lower than males'.

Regarding rural women, the Vietnamese government is working on providing a preferential credit instrument to support women in agriculture. However, it is reported that men are the foremost beneficiaries of the agricultural process mechanization as men are more involved in machinery-related tasks like tillage, harvesting, processing and transport. It is reported that in Vietnam, climate change also intensifies women's double burden in having to fulfill both productive and reproductive roles.[183]

Women and Energy. Over the past few decades, Vietnam has emerged as an important oil and natural gas producer in the area. According to the IRENA (2018), coal-fired power and oil led primary energy supply for the country.[184] At the same time, however, Vietnam's renewable energy sector is one of the most vibrant in the Southeast Asia region. Renewable energy sector accounts for 24% (mostly hydropower and bioenergy) of the total primary energy supply in the country.[185] The electricity penetration rate is more than 99% (WB, 2019).[186]

While the country is providing almost full electrification to their households,[187] clean energy solutions are not available to everyone, and women are considered more vulnerable in this regard. As they take on traditional household roles, including cooking and fuel collection, their health is more affected by indoor pollution and climate change might extend the travel distance to get fuel because of the increased scarcity of natural resources. IRENA (2018) reported that only 64 % of the population has access to clean cooking methods.[188]

Gender roles make women the focal point of energy consumption in households, while they are still marginalized in terms of decision-making at the community level about energy and they are not well integrated into the energy job market because most jobs in the manufacturing, construction, and engineering are traditionally seen as suitable only for men, thus women are significantly underrepresented.

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[181] Eckstein, D., Künzel, V., Schäfer, L., & Winges, M. (2018). GLOBAL CLIMATE RISK INDEX 2020. Retrieved from https://germanwatch.org/sites/default/files/20-2-01e%20Global%20Climate%20Risk%20Index%202020_14.pdf.

[182] FAO, Country Assessment Report. Gender in Agriculture and Rural Development of Vietnam. 2019.

[183] The State of Gender Equality and Climate Change in Vietnam (EmPower – Women for Climate Resilient Societies). Retrieved from https://asiapacific.unwomen.org/-/media/field%20office%20eseasia/docs/publications/2021/05/empowerpolicy%20brief%20for%20stakeholdersstate%20of%20gender%20equality%20%20climate%20change%20viet%20nam.pdf?la=en&vs=119 .

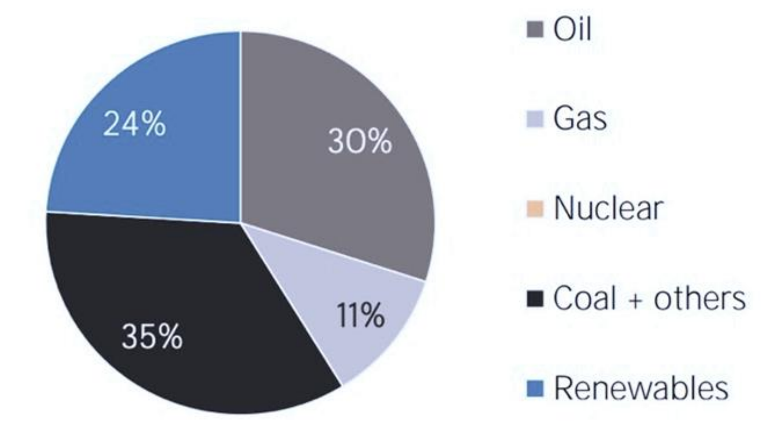
[184] Total primary energy supply (TPES) is the total amount of primary energy that a country has at their disposal.

[185] IRENA. (2018). Vietnam Energy Profile. Retrieved from https://www.irena.org/IRENADocuments/Statistical_Profiles/Asia/Viet%20Nam_Asia_RE_SP.pdf.

[186] Access to electricity (% of population) - Cambodia, Indonesia, Lao PDR, Vietnam, Philippines. (2019). https://data.worldbank.org/indicator/EG.ELC.ACCS.ZS?locations=KH-ID-LA-VN-PH.

[187] Vietnam Briefing. (2020, November 12). Renewables in Vietnam: Current Opportunities and Future Outlook. Retrieved from Vietnam Briefing News website: https://www.vietnam-briefing.com/news/vietnams-push-for-renewable-energy.html/.

[188] IRENA. (2018). Vietnam Energy Profile. Retrieved from https://www.irena.org/IRENADocuments/Statistical_Profiles/Asia/Viet%20Nam_Asia_RE_SP.pdf.



Source: IRENA, https://www.irena.org/

Figure 4 - Vietnam Total Primary Energy Supply in 2018

Women and Water. Vietnam has a dense river network of 2,360 rivers with a length of more than 10 km. Households' access to improved water supplies increased from 65% in the year 2000 to 95% in 2017, while access to basic sanitation jumped from 52% to 84% during the same period.[189] Nevertheless, water quality remains an issue all over the country because of environmental pollution in the river basins and natural disasters, which affect the nation's public health. The Ministry of Natural Resources and Environment stated that almost 80% of the diseases in Vietnam are caused by polluted water.[190]

Whereas the rivers traversing Vietnam should provide an abundant supply of water, the country experiences uneven distribution of water resources, imbalance in water demand and water storage capacity, and unsustainable exploitation and use of water resources, especially from the agricultural sector.[191]

Water insecurity due to climate change increases the workload of women, especially those working in small-scale agriculture as they have to spend more time on tillage water watering and protecting crops against diseases. At the same time, water quality and scarcity due to climate change impact women as being the traditional primary water collectors, and due to women's health-specific need for clean water during menstruation, pregnancy and caregiving.

Women and Waste management. Urban solid waste collection and treatment rates account 86.5% (2019), while the rural rate is down to 63.5% (2019). However, currently, 71% of municipal solid waste is still being buried, mostly in an unsanitary manner.

Women represent 35-40% of formal waste collectors while they dominate the informal sector with a 65% participation rate, often living and working under dangerous and unhealthy conditions. In this sector, men have higher paid professions in the informal waste economy, being dealers and owners of recycling activities, but also in the formal waste economy, such as truck drivers and managers.[192]

[189] Water, sanitation and hygiene in Vietnam. (2020, February). Retrieved 2021, from Unicef.org website: https://www.unicef.org/vietnam/reports/water-sanitation-and-hygiene-viet-nam.
[190] Water, sanitation and hygiene in Vietnam. (2020, February). Retrieved from Unicef.org website: https://www.unicef.org/vietnam/reports/water-sanitation-and-hygiene-viet-nam.
[191] The State of Gender Equality and Climate Change in Vietnam (EmPower – Women for Climate Resilient Societies). Retrieved from https://asiapacific.unwomen.org/-/media/field%20office%20eseasia/docs/publications/2021/05/empowerpolicy%20brief%20for%20stakeholdersstate%20of%20gender%20equality%20%20climate%20change%20viet%20nam.pdf?la=en&vs=119 .
[192] The role of gender in waste management- GA Circular. (2021). Retrieved from GA Circular website: https://www.gacircular.com/publications/.

Women and Education. Regarding tertiary education, Vietnam has a total gross enrolment ratio of 28.5% (2016), with more female than male students attending school in upper secondary and tertiary levels: 31.7% vis-à-vis 25.5% (2020).[193]

However, women's participation in science, technology, engineering, and mathematics (STEM) education is less than half of that of men's (15.38% vs. 31.19%), underrepresenting women in the sector of Information Communication Technology (ICT), and areas requiring a capacity for technological innovation.[194] This reflects the traditional expectations impressed upon girls and boys when it comes to choosing a course of study or occupation.

According to UNESCO, among students enrolled in tertiary education, men are considerably more likely to specialize in engineering, manufacturing, construction, and services, while women are more likely to specialize in social sciences, education, humanities, and the arts. Additionally, women often choose vocational training programs that lead to traditionally women-dominated occupations, limiting their career prospects in non-traditional and emerging roles and other economic opportunities. [195] Moreover, women achieving higher education tend still to be clustered in lower-paid manufacturing jobs, garment and footwear industries, the service sector, education field, and paid domestic work, earning 10-12% less than their male counterparts.

- **Business Entrepreneurship and Gender Dynamics in Vietnam**

The entrepreneurial ecosystem of Vietnam is by far one of the most advanced among the Mekong countries and ranked 87th out of 137 countries in the 2018 Global Entrepreneurship Index (GEI).[196] **Over the past three decades, Vietnam has shown steady signs of progress in increasing its number of women entrepreneurs**. Since 1990, with the introduction of the Private Enterprises Law, the Vietnamese legislation has become more gender-inclusive, with women-owned SMEs being mentioned for the first time in the 2017 Law on Support for Small and Medium-sized Enterprises.

The statistics, however, are not very encouraging if we take a deeper look at women's entrepreneurial participation. Based on the 2014 GSO Enterprise Census, **Vietnam's women own about 21% of formal enterprises, the majority (57%) are microenterprises, 42% are SMEs and only 1% or 854 are large enterprises.** Since 2020, half of these businesses were partially suspended or dissolved given the start of COVID-19 and, in the first quarter of 2020, the number of women-owned enterprises reported a revenue decrease of 75%, or more, which is double of losses compared to men-owned enterprises.[197]

An IFC's study in 2017 on the needs and opportunities for women-owned SMEs in Vietnam, stated that **Vietnam's investment climate for women is generally supportive**. According to it, Vietnam has a long history of women in business, and compared to many similar economies, there are less inequalities between males and women's legal positions in doing business. Whereas it is assumed that women and men share the same problems and experiences in running small companies, there are a multitude of challenges faced by female entrepreneurs in Vietnam that affect women's ability to realize their

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[193] Vietnam | UNESCO UIS. (2019). Retrieved from Unesco.org website: http://uis.unesco.org/en/country/vn?theme=science-technology-and-innovation.
[194] The State of Gender Equality and Climate Change in Vietnam. (n.d.). Retrieved from https://asiapacific.unwomen.org/-/media/field%20office%20eseasia/docs/publications/2021/04/publication_Vietnam%20report_digital2%20(1).pdf?la=en&vs=4800.
[195] UNESCO. (2021). A Complex Formula: Girls and Women in Science, Technology, Engineering and Mathematics in Asia. Retrieved from Unesco.org website: https://unesdoc.unesco.org/in/documentViewer.xhtml?v=2.1.196&id=p::usmarcdef_0000231519&file=/in/rest/annotationSVC/DownloadWatermarkedAttachment/attach_import_90fa1973-d643-4e74-b84d-e54e7cd3db86%3F_%3D231519eng.pdf&updateUrl=updateUrl2920&ark=/ark:/48223/pf0000231519/PDF/231519eng.pdf.multi&fullScreen=true&locale=en#STEM%20Study%20Final%20Draftv3.indd%3A.154648%3A1417.
[196] Entrepreneurship & Business Statistics | GEDI. (2018). Retrieved from Thegedi.org website: https://thegedi.org/global-entrepreneurship-and-development-index/.
[197] Thi, T. (2021, July 23). Training and support for female entrepreneurs in Vietnam: What do women want and need? Retrieved from Brookings website: https://www.brookings.edu/blog/education-plus-development/2021/07/23/training-and-support-for-female-entrepreneurs-in-Vietnam-what-do-women-want-and-need/.

entrepreneurial ambitions. [198] Thus, even though men are relatively more likely to be employed in the informal sector (the informal employment rate in 2019 was 67.2% among women, and 78.9% among men), women are overrepresented among disadvantaged categories of informal workers.

Vietnam's labour market still has areas of **gender-segregation, with women heavily represented in marketing, office and support work, human resources, and accounting, rather than in technology and innovation-oriented sectors.** Further obstacles include lack of market information and opportunities for business development. The challenges of balancing business with family-care constrain women's time to participate in networking and learning activities and further grow their businesses.[199] Research conducted by ActionAid Vietnam reported that on average, Vietnamese women spend 105 minutes more than Vietnamese men on unpaid care work. The intra-household unequal division of labour between men and women constitutes a considerable brake on women's labour participation and returns to livelihoods, limiting their opportunities for training, networking, and promotion.[200]

Lack of Financing: In general, women entrepreneurs find it more difficult to access credit. Even though female entrepreneurs bring in average annual revenues like men's. 42 % of women-owned SMEs had difficulty accessing financial resources, which is a key barrier to business development. At the same time, the small and micro-size nature of women's enterprises along with the fact that these kinds of businesses mostly lack adequate business planning and accounting systems make it difficult for them to qualify for loans.[201] Women's capacity to access finance is also influenced by cultural factors. Usually, collaterals are needed to fulfil the borrowing requirements and many women entrepreneurs are unable to fulfil these criteria since their names are mostly not included on the state-issued land use right certificates (LURC).[202] The LURC legislation of Vietnam stipulates that both men and women shall have equal access to land usage and rights, but in practice, women have less land rights and smaller farm holdings. Consequently, women business owners keep relying on informal sources of finance, while accessing legal financial products remains a privilege for large businesses. [203]

Provincial Women's Union, provincial People's Party, and other government departments have developed several action plans on women's entrepreneurship and renewable energy in supporting the NDCs, Green Growth action plan, and other SDG targets. The National Women's Start-up Program, led by the Vietnam Women's Union, also created a special category for renewable energy in 2020, with UNEP's support. Likewise, the EmPower project, led by UNEP, supports another women entrepreneur's initiative in three provinces in Vietnam to use renewable energy as a means for developing climate-resilient livelihoods.[204]

Innovation is an important aspect of the success of many entrepreneurial enterprises, and so it is for climate technology. Vietnam continues to be an example for other developing countries in treating innovation as a national priority and is heavily interested in building and promoting its own NIS. In 2020, Vietnam had a ranking of 44 in the Global Innovation Index (2021), heading the lower middle-

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[198] Women-owned enterprises in Vietnam: Perceptions and Potential. (2017). Retrieved from https://www.ifc.org/wps/wcm/connect/86bc0493-78fa-4c7d-86ec-5858aa41fa1a/Market-study-on-Women-owned-enterprises-in-Vietnam_Eng_v1.pdf?MOD=AJPERES&CVID=l-Yi6gF.

[199] #ClosingTheGap Mekong Country Report Vietnam Entrepreneurial Ecosystem Assessment Commissioned on behalf of: Dutch Good Growth Fund (DGGF): Investment funds local SMEs. (2019). Retrieved from http://mekongbiz.org/wp-content/uploads/2019/05/CTG-CountryReport_Vietnam_Final.pdf.

[200] How much does it cost women for men to be the family's backbone? (2017). Retrieved from ActionAid Việt Nam website: https://Vietnam.actionaid.org/en/publications/2017/how-much-does-it-cost-women-men-be-familys-backbone.

[201] TAF, VWEC, & MBI. (2018). Needs Assessment of Women-Owned Small and Medium-Sized Enterprises in Vietnam.

[202] #ClosingTheGap Mekong Country Report Vietnam Entrepreneurial Ecosystem Assessment Commissioned on behalf of: Dutch Good Growth Fund (DGGF): Investment funds local SMEs. (2019). Retrieved from http://mekongbiz.org/wp-content/uploads/2019/05/CTG-CountryReport_Vietnam_Final.pdf.

[203] THE STUDY A REVIEW OF THE IMPLEMENTATION OF SMALL AND MEDIUM ENTERPRISES (SMES) SUPPORT LEGISLATION AND THE CAPACITY BUILDING NEEDS AND TRAINING SERVICES FOR WOMEN-OWNED SMES AND WOMEN ENTREPRENEURS IN VIETNAM Hanoi 12-2020. (n.d.). Retrieved from https://Vietnam.un.org/sites/default/files/2021-02/UNW_Review_Eng%20Full_18.12.2020_6.pdf.

[204] The State of Gender Equality and Climate Change in Vietnam. (2021). Retrieved from https://asiapacific.unwomen.org/-/media/field%20office%20eseasia/docs/publications/2021/04/publication_Vietnam%20report_digital2%20(1).pdf?la=en&vs=4800.

income group and has made the most significant progress over time.[205] This is also reflected in the number of female researchers in R&D (per million people), where Vietnam has seen a positive outcome of 44.8% (2015).[206]

In Vietnam the Programme will address the above-listed constraints in the following manner:

Given the equal/higher education level of women in Vietnam, promoting (by prioritizing for CTF approval) the "women-led enterprises" will substantially capacitate the female entrepreneurs in the country to remove their structure barriers to the formal finances and credits as well as by strengthening their business network and partnership to the global level and address the limitations of many female-led enterprises stagnant in the small and micro-scale levels. Gender stereotyping in the labour market and bias of women's roles in the climate technology sector and aggressive business entrepreneurship require a long-term perception change of the Vietnams public. Generating a successful case(s) of the climate technology innovative RD&B case out of locally based "women-led enterprises" through the Programme will trigger this perception change in the long term.

- **Priority Sectors and Gender Implications in Vietnam**

Areas for priority intervention through the proposed Programme in Vietnam are, among others: Climate scenario modelling with data set analysis, risk detection, and assessment/ climate-smart agriculture/ Inter-reservoir operation skills for water-efficient irrigation operation/ Environmental Technology for Water.

Climate-smart agricultural activities include, among others, smart water and irrigation management; adoption of improved crop varieties; agroforestry; intercropping trees with crops; sustainable land management; agricultural waste treatment such as integration of biogas technologies in livestock production; and improved agro-climate information services[207]. Investments in climate adaptation and mitigation technologies in the agricultural sector would also be particularly beneficial for Vietnamese women. Agriculture contributes 21% of Vietnam's total GDP, with a female employment rate of about 63%. The Programme should consider women's climate vulnerability and climate risk exposure, while Vietnamese women also lack decision-making power regarding the choice of crops, technologies or commercial development.

Waste management could also be considered a priority sector for Vietnamese women. In the country, waste collection and treatment systems are already developed and women account for 35-40% of formal waste collectors, dominating the informal sector with a 65% participation rate. Investment in recycling technologies and infrastructure in Vietnam could contribute to the female labour migration from the informal to formal sector, improving their income generation and work conditions. Promoting women-led start-ups in the sector of organic waste management, plastic recycling and water would offer the opportunity for GHG reduction and pollution mitigation while empowering women in Vietnam.

In Vietnam, gender mainstreaming into the policies and plans in agriculture, waste management, water management and renewable energy sectors need to be strengthened: Currently. there are no gender equality targets or measures identified to address women's vulnerabilities or lack of agency in climate change. The proposed Programme could consider strengthening the capacity of the relevant government entities and ministries to strengthen their gender mainstreaming capacities at policy and plan levels, while developing sector-specific guidelines for gender mainstreaming[208].

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[205] VIETNAM. (2020) Retrieved from https://www.wipo.int/edocs/pubdocs/en/wipo_pub_gii_2020/vn.pdf.
[206] Vietnam | UNESCO UIS. (2019). Retrieved from Unesco.org website: http://uis.unesco.org/en/country/vn?theme=science-technology-and-innovation.
[207] Climate-Smart Agriculture (CSA) in Vietnam/ https://cgspace.cgiar.org/bitstream/handle/10568/96227/CSA_Profile_Vietnam2.2.pdf?sequence=1&isAllowed=y
[208] The State of Gender Equality and Climate Change in Viet Nam (2021)/ https://asiapacific.unwomen.org/en/digital-library/publications/2021/04/the-state-of--gender-equality-and-climate-change-in-viet-nam

4. Final Considerations and Recommendations

The following are the summaries of the key findings and recommendations from this gender assessment.

Agricultural Sector

In Cambodia and Indonesia, women contribute to more than 70% of the agricultural workforce, followed by Vietnam (63%) and Lao PDR (over 50%). Only in the Philippines, agriculture is male dominated sector (with women's participation only being 23%). Across the countries, climate hazards threaten the livelihoods and food security of millions of smallholder farmers and agribusiness who depend on agriculture in our selected countries, especially women".[209]

The proposed Programme targets climate technology RD&B support for all five countries. Prioritization of the sector will significantly affect the girls and women in the countries. Also, the potential is high to engage women entrepreneurs given the high level of women's participation in the sector.

Water Sector

Gender sensitivity extends to the water sector, water-related natural catastrophes, and water-borne illness outbreaks. Indonesian women, for example, are the principal water collectors and providers in their home: two out of every five Indonesian families delegate water responsibility to women. In Lao PDR, 48.7% of women and 9.4% of girls oversee gathering water for the household. Gender responsibilities related to water have also been found to be strongly defined in rural areas, especially for Laotian and Cambodian rural women. Nevertheless, positive improvements can be recognized in the access to water supplies and sanitation, where in Vietnam, households' access to improved water supplies increased from 65% in the year 2000 to 95% in 2017.

Women's role in water management activities needs to be strengthened with proper training and skill development to cope with climate change-related water disasters, such as floods and droughts. Women can become central agents for Sustainable development and management of water supplies and water treatment. Failure to do so would worsen the water contamination putting the lives of millions of men, women, boys, and girls at risk.

Energy Sector

In all five countries, clean energy solutions are not available to everyone and women are considered less privileged. Given the traditional household roles of cooking and fuel collection, women's health is more affected by indoor pollution, and women's time is overspent in collecting fuel because of the increased scarcity of natural resources. At the job market level, renewable energy jobs fall under the manufacturing, construction, and engineering sectors, where women are less represented in all of the five countries analysed.

While most directly affected by the types and quality of the energy sources, girls and women are not active agents in energy transitions in all the five countries, due to gender bias in education (especially in STEM, ICT and energy subjects) as well as cultural tendencies to educate male children to higher levels. The proposed Programme needs to proactively invite women's participation in renewable energy-related enterprises.

[209] The term can be defined as such when women have to take on more and more agricultural work due to the migration of men to the city in search of urban employment opportunities.

Waste Management Sector

Women's limited participation is also observed across the countries in environmentally friendly solutions in waste management. In Indonesia, fewer women participate in the waste collection chain and in the formal waste sector. Waste collection is seen as a men's job because the collectors travel fair distances away from home on trucks and carry heavy loads. In Vietnam, women represent 35-40% of formal waste collectors, but 65% of them are in the unregulated informal sector, which are underpaid, unhygienic and often dangerous. In Cambodia and Lao PDR, solid waste management has recently raising attention. All in all, the Programme had better invest in lifting these specific barriers faced by women to make waste disposal and recycling initiatives a good opportunity for women's employment and more proactive participation in the countries' transition to a circular economy.

Women's Business Entrepreneurship In all five countries, the women's entrepreneurial ecosystem seemed to be limited to small and microenterprises, characterized by low profitability and informality. Common barriers are: (i) social and cultural constraints, (ii) access to business markets, and (III) lack of financing. Remarkable are the strong gender norms, particularly in Cambodia and Indonesia.

Regarding the (ii) access to the business market, in all the countries' labour markets, there is gender segregation, with women heavily represented in the agricultural and sales sectors, rather than technology and innovation-oriented ones. Women entrepreneurs mostly lack market information and opportunities for business development. The challenges of balancing business with family care constrain women's time to participate in networking and learning activities and further grow their businesses.

In Indonesia, women's access to financial programs is limited compared to men's access. Indonesian males use more banking services (45%) than women (42%) and a higher proportion of females (49%) use informal financial services, compared to males (29%). Men and women do not have equal ownership rights over moveable and immovable property, making it difficult for women to have collateral for lending from financial institutions. In Lao PDR female entrepreneurs are also less likely to have a bank account or credit line. In the Philippines, although legally not required, some financial institutions expect a male partner to co-sign any women's financial contract. Filipino women entrepreneurs seeking micro-finance schemes, mostly encounter restrictive and complicated procedures, which makes them seek other more expensive means for credit. Due to their lack of collateral, a bank loan is often inaccessible to Filipinas.

STEM and Higher education

In Indonesia, Philippines and Vietnam the level of literacy rates is comparatively high, with no significant signs of gender inequality. On the other hand, Cambodia and Lao PDR had low literacy rate with marked gender differences

In Indonesia, Philippines and Vietnam, women's tertiary enrolment rate is from 6 to 9 percentage points higher than the male one. Nevertheless, women's participation in science, technology, engineering, and mathematics (STEM) education has been found to be significantly less than men's. More broadly, women seem underrepresented in the sector of information communication technology (ICT) and other areas related to innovation in all five countries.

This reflects the traditional expectations impressed upon girls and boys when it comes to choosing their study path or occupation. According to UNESCO, among students enrolled in tertiary education, men are considerably more likely to specialize in engineering, manufacturing, construction, and services, while women are more likely to specialize in social sciences, education, humanities, and the arts. Additionally, women often choose vocational training programs that lead to traditionally women-dominated occupations, limiting their career prospects in non-traditional and emerging roles and other economic opportunities. Moreover, women achieving higher education tend still to be clustered in lower-paid jobs, earning less than their male counterparts. [210]

Regarding education, the key barriers for women's enrolment and STEM education have been identified as mostly related to gender roles and lack of financial resources. Particularly vulnerable to those gender stigmas are rural women and girls, which are more likely to experience gender-based violence and early marriage, and being prone to premature withdrawals from schooling.

Decision-making & Empowerment

It is evident that numerous women in Southeast Asia are in underprivileged situation in terms of access and benefits from natural resources and environmental decision-making processes in comparison to men. Women in the five countries are not exceptions.

Recommendations: Acknowledging the key findings related to women and climate entrepreneurship in the countries of analysis, the following recommendations should be taken into account while implementing the Collaborative R&DB Programme for Promoting the Innovation of Climate Technopreneurship:

Component 1 **Country-Driven Climate Acceleration Readiness**	- Local SME selection shall prioritize candidate SMEs in each of the five target countries which meet the "women-owned SME." (Defined as: SMEs with one of the three conditions: ① at least 26% of the total capital owned by women; ② at least one woman participates on the Board of Directors and/or different types of boards or internal committees, such as ESG committees, among others (CEO, COO, CTO), or; ③ with more than 30% of the workforce are females) - Each country SMU is recommended to hire one gender specialist, under the supervision of GGGI Gender Specialist. - Country-specific acceleration support programs (N-CEAPs) should include gender awareness component and support the participating firms to assess their own gender mainstreaming readiness status.
Component 2 **Global Acceleration for Collaborative R&DB**	- Every selected JVs will submit its own gender mainstreaming strategy (at company level), detailed gender assessment and action plan for their CTF Proposal, fit-for-purpose. To support this: - the Programme should set up a fund-level gender expertise in the Global Acceleration Advisory Secretariat. - Integrate gender component to collaborative R&DB services: Fund-level Gender expertise assists participating JV candidates to prepare gender analysis (GA) and gender action plan (GAP) in accordance with the GCF's Gender Policy (B.24/12 adopted in Nov 2019

[210] UNESCO. A Complex Formula: Girls and Women in Science, Technology, Engineering and Mathematics in Asia. (2021). https://unesdoc.unesco.org/in/documentViewer.xhtml?v=2.1.196&id=p::usmarcdef_0000231519&file=/in/rest/annotationSVC/DownloadWatermarkedAttachment/attach_import_90fa1973-d643-4e74-b84d-e54e7cd3db86%3F_%3D231519eng.pdf&updateUrl=updateUrl2920&ark=/ark:/48223/pf0000231519/PDF/231519eng.pdf.multi&fullScreen=true&locale=en#STEM%20Study%20Final%20Draftv3.indd%3A.154648%3A1417.

<u>**Component 3**</u>

Climate Technopreneurship Fund (CTF)

- The Fund’s investment criteria should be gender-mainstreamed. Following investment criteria to be inserted as part of the CTF’s investment/selection criteria:

  - ➢ At least 50% of the CTF-invested JVs are Women-led: (1) SMEs with at least 26% of the total capital owned by women; (2) SMEs with at least one woman participates in the Board of Directors and/or different types of boars or internal committees such as ESG committees, among others (CEO, COO, CTO, etc.); and/or; (3) SMEs with more than 30% of the total workforce are females.

  - ➢ All the CTF investment JVs/activities are gender-mainstreamed with concrete gender performance indicators and tangible results proven.

  To support this:

- At the fund level, the involvement of gender expertise should support JV-level Gender Action Plan (including gender mainstreaming commitment plan) throughout the project period through high-level review, monitoring and capacity building assistance.

<u>**Component 4**</u>

Technical Assistance (TA)

- The project should promote and facilitate effective country TA programs to generate a viable pool for further candidate female-founded/headed/employed local firms for CTF participation.

- The Programme should identify women representatives for the national network /communication platform to foster women technopreneurship.

- The Programme should identify the areas for cooperation with each country for gender-responsive capacity building (for the current and future participants of the CTF).

- The Programme should facilitate the integration of gender indicators to be developed and incorporated into the M&E plan and activities.

5. Gender Action Plan

This GAP shall be implemented as an integrated mechanism of ESMS. Per component of the Programme, agent of the GAP management are as follows:

- Components 1 and 4: through the gender specialists of the Regional SMU (GGGI) and Country SMU of the respective five countries[211]
- Component 2: through the Gender Specialist within the Acceleration Advisory Consortium for Global Readiness Programme Consulting Services[212]
- Component 3 (Pre-investment stage): through the Gender Specialist of the Environmental & Social and Gender Compliance Team (ESGCT), the Expert Advisory Committee (EAC)[213]
- 4) Component 3 (Post-investment stage): through the Gender Specialist of the Environmental & Social and Gender Compliance Team (ESGCT) and the E&S (and gender) Risk Management (Sub) Committee (ESRMC)[214].

For overall supervision and ensuring the compliance of all the Programme-related activities with the GAP and GCF Gender Policy, as appropriate, throughout all the components, KDB (as AE) would be responsible.

<table>
<tr><th>Activities</th><th>Indicators & Targets</th><th>Timeline</th><th>Responsibilities</th><th>Costs</th></tr>
<tr><td colspan="5"><u>Impact Statement:</u>
▪ Girls and women in the ASEAN region become central agents and major beneficiaries of technological leapfrogging through climate tech innovation in self-combating climate change, which steers the region’s paradigm-shifting transition towards low-carbon and climate-resilient economies and societies.
<u>Outcome Statement:</u>
▪ A gender-mainstreamed climate entrepreneurship acceleration ecosystem is established with a surging technologically skilled, capable, and passionate female entrepreneurs and women-led SMEs as key climate technology innovation agencies are established in all the target countries.</td></tr>
<tr><td colspan="5">Component 1. Country-Driven Acceleration Readiness
(Executing Entity: GGGI, KDB)</td></tr>
<tr><td colspan="5">Output Statement:
Local SME selection shall prioritize candidate SMEs in each of the five target countries which meet the “women-owned SMEs” conditions, as follows:
■ SMEs with at least 26% of the total capital owned by women, and/ or;</td></tr>
</table>

[211] For details, see Section 3.2.2 of the ESMS of this Programme (Annex 6 of the Funding Proposal).
[212] For details, see Section 3.2.3 of the ESMS of this Programme (Annex 6 of the Funding Proposal).
[213] For details, see Section 3.2.3 of the ESMS of this Programme (Annex 6 of the Funding Proposal).
[214] For details, see Section 3.2.3 of the ESMS of this Programme (Annex 6 of the Funding Proposal).

■ SMEs at least one woman participates on the Board of Directors and/or different types of boards or internal committees, such as ESG committees, among others (CEO, COO, CTO), and/ or; ■ SMEs with more than 30% of the workforce are females. All the selected SMEs shall produce the Country SME Gender Profiling Form (CGPF)[215].				
A1-1. Set up a gender specialist/gender expertise within a Regional SMU and National SMU in each of the five target countries (Activity 1.1[216])	1. 1 gender specialist is locally hired in the National SMU through partnership	By the date of completion of a National SMU in each country	KDB/GGGI	USD 50,000 Gender expertise secured through at least one direct recruitment or through partnership (i.e. outsourcing).
A1-2. Identify qualified local JV candidates: Prepare communication content including "women-owned SMEs" status and N-CEAP Investment Criteria for Local Entrepreneurs #6. Gender Mainstreaming	1. Minimum of 30% (committed) and up to 50% (target) of "women-owned SMEs" in the five countries are identified and eager to take part in the Programme through effective communication and awareness-raising[217]. 2. All the candidate country SMEs are convinced of adopting its own	All through the Program operation period	KDB/GGGI	

[215] Annex 2 of this document.
[216] Activity in bracket indicates the matching activity of the proposed Programme (as shown in the <Programme Summary at a Glance> in Section A.21. Executive Summary of the Funding Proposal Document (p.7) respectively.
[217] CTF Investment Criteria #9 (Gender Mainstreaming) targets at least 50% of the CTF-invested JVs are "women-led" or commit gender mainstreaming by establishing and implementing a gender mainstreaming commitment plan (GMCP). At this point, it is not possible to designate a specific target percentage of "women-led" local business applicant as it is widely open and uncertain as future eventualities. The programme will strongly promote "women-led" local firms' participation and to fulfil the Programme-level 50% minimum target, the Programme will strive to promote GMCP development and executing to the extent possible to all the willing beneficiaries of the Programme.

	gender mainstreaming commitment plan.			
A1-3. Integrate gender component to tailor-made acceleration services(N-CEAPs): Support the candidate country SMEs prepare Country SME Gender Profiling Form (CSGP)[218] (Activity 1.3)	1 Guidelines to fill in Country SME Gender Profiling Form (CSGP) developed. 2 N-CEAP program includes a module to assist country SMEs to fill in CSGP, led by gender trainer/expert through partnership 3 CSGP produced by each of the candidate country SMEs based on the guidelines and the country-specific support program of N-CEAP.		KDB/GGGI	
Component 2. Global Acceleration for Collaborative R&DB (Executing Entity: NHIS)				
Output Statement: Every selected JV will submit its own gender mainstreaming strategy (at the company level), detailed gender assessment and action plan for their CTF Proposal, fit-for-purpose.				
A 2-1. Set up Fund-level Gender expertise (through procurement) (Activity 2.1)	1. At least one gender expert is procured, whose function will	By the date, the Global Acceleration Advisory Secretariat is set up.	NHIS/Secretariat	USD 200,000 (5,000 per project/technology, on the

[218] Annex 2 of this document.

	be an integral part of the function of the Global Acceleration Advisory Secretariat			basis of 40 investments approx.)
A2-3. Integrate gender component to collaborative R&DB services: Fund-level Gender expertise assists participating JV candidates to prepare a simplified gender analysis (GA) and gender action plan (GAP)[219] in accordance with the GCF's Gender Policy (B.24/12 adopted in Nov 2019)[220]. (ACTIVITY 2.3.) *For sector/technology-specific guidelines for filling A1-3, see Appendix 5 below.	1. GA prepared for each of the participating JV candidates. 2. GAP established for each of the participating JV candidates. Duly prepared by the JV candidates & reviewed & quality validated by Fund-level Gender expertise for submission to the Fund as part of the CTF application document packages.		NHIS/Secretariat	
Component 3. Climate Technopreneurship (CTF) (Executing Entity: NH ARP/GP)				
Output Statement: ☐ At least 50% of the CTF-invested JVs are Women-led (① with at least 51% of the total capital owned by women, and/or② at least 26% of the total capital owned by women and at least one woman participates on the Board of Director, and/or more than 50% of workforces are females)) JVs, or alternatively, have presented the gender mainstreaming commitment plan to the CTF as part of the funding proposal. ☐ All the CTF investment JVs/activities are gender-mainstreamed with concrete gender performance indicators and tangible results proven.				

[219] Considering the lack of capacity of the applicant JVs (being SMEs in developing countries), the GA and GAP will be simplified to cover the most relevant key aspects of the analysis and action plans specific to the climate technology/sector/business types. Sample GA and GAP shall be provided with guidance to the JV candidates as part of the Acceleration program. See the standard template for the GA & GAP in Appendix 3 below.

[220] https://www.greenclimate.fund/sites/default/files/document/gcf-gender-policy.pdf

A3-2-1-1. Execute Gender Criteria within the CTF Investment Criteria* (ACTIVITY 3.2.1) *Prioritization of "women-led SMEs/JVs" & JVs with gender mainstreaming commitment plan (GMCP)	At least 50% of the CTF-invested JVs meet these priority selection conditions. Fund-level Gender expertise review and validate an individual JV candidate's fulfilment of the prioritization conditions & as necessary, suggest complementary measures to the submitted GMCP.		NH ARP	
A3-2-1-2. Fund-level Gender expertise supports JV-level Gender Action Plan execution (including gender mainstreaming commitment plan) throughout the project period through high-level review, monitoring and capacity-building assistance. (ACTIVITY 3.2.1)	All the funded projects are gender-mainstreamed as planned through gender mainstreaming commitment plan, its process and implementation is supported and monitored by the Fund-level Gender expertise and reflected in the CTF Gender Report. Validated by: Regular gender monitoring & evaluation reports (at JV- and the fund-level).		NH ARP	
A3-3-2. (At JV-level) Country SMU supports JV-level gender action plan (including gender mainstreaming commitment plan) implementation by the	Monitoring Report on GAP implementation is submitted, reviewed and actively monitored by the CTF system throughout the		NH ARP	

funded JVS throughout the project period.	entire implementation period of a JV activities.			
Component 4. Technical Assistance (TA) (Executing Entity: GGGI, KDB)				
Outcome Statement: Effective country TA programs generate a viable pool for further candidate female-founded/headed/employed local firms for CTF participation.				
A4-1. Include the gender component in knowledge-sharing activities (ACTIVITY 4.1.2)	Knowledge-sharing program include gendered climate technology entrepreneurship component in each of the five countries respectively.		KDB/GGGI	USD 50,000
A4-2. Identify the areas for cooperation with each country for gender-responsive capacity building (of current and future participants of the CTF) (ACTIVITY 4.2)	At least one gender-responsive TA Programme rolled out per country.		KDB/GGGI	
A 4-3. Integrate gender component to the national strategy on climate technopreneurship promotion	At least one strategy with concrete action plan(s) are integrated into the national strategies on climate technopreneurship promotion of each of the five countries respectively	Is this linked with the timelines indicated in the M&E plan of the program?	KDB/GGGI	

Appendix 1. Preliminary Stakeholder Mapping by Country

Country	**CAMBODIA**			
Stakeholders Category	Ministerial Actors	Education Initiatives NGOs	Business Initiatives	Ministerial Actors & international organizations
Stakeholders	• Ministry of Education, Youth and Sports • Ministry of Labour and Vocational Training • Ministry of Women's Affairs • Gender and Climate Change Committee • (Ministry of Women's Affairs)	• STEM Sisters Cambodia • Education First Cambodia - Royal University of Phnom Penh • Kampuchean Action for Primary Education (KAPE)	• The Cambodia Women Entrepreneurs Association (CWEA) • SHE Investments	• GGGI Country Office • WB Country Office etc.

Country	**INDONESIA**			
Stakeholders Category	Ministerial Actors	Education Initiatives NGOs	Business Initiatives	Ministerial Actors & international organizations
Stakeholders	• Ministry of Education, Culture, Research, and Technology - General Directorate of Higher Education, Research, and Technology - General Directorate of Vocational Education • Ministry of Women Empowerment and Child Protection	• Universitas Prasetiya Mulya (STEMPreneur Centre - STEM DEPARTMENTS) • Del Polytechnic of Informatics • YCAB Foundation	• SheMeansBusiness Indonesia • Indonesian Women Entrepreneurs Network - IWEN • Wanita Wirausaha Femina	• GGGI Country Office • WB Country Office etc.

	• (Ministry of Finance) Ministry of Cooperatives and SMEs • The Directorate General of Climate Change Control (DJPPI)			

Country	**LAO PDR**			
Stakeholders Category	Ministerial Actors	Education Initiatives NGOs	Business Initiatives	International organizations
Stakeholders	• Ministry of Agriculture and Forestry - • Department for the Advancement of Women • Ministry of Labour and Social Welfare • Ministry of Education and Sports • Lao PDR Women's Union (LWU) • National Environment-Climate Change Committee • Department of • Disaster Management and Climate Change • (DDMCC), Ministry of Natural Resources • and Environment	• Faculty of Environment Science, National University of Laos • Gender Development Association - Laos	• The Business Women's Association • Young Entrepreneurs Association of Laos (YEAL)	• GGGI Country Office • WB Country Office etc.

Country	PHILIPPINES			
Stakeholders Category	Ministerial Actors	Education Initiatives NGOs	Business Initiatives	International Organizations
Stakeholders	• Philippine Commission on Women • Department of Labour and Employment • Ministry of Science Research and Technology • Philippine Department of Environment and Natural Resources (DENR) • Philippine Department of Education • Technical Education and Skills Development Authority • Climate Change Commission	• Stem Leadership Alliance • Institute of Women's Studies • Women's Legal and Human Rights Bureau (WLB) • Women's Rights Movement of the Philippines	• Women's Business Council Philippines • Philippine Women's Economic Network	• GGGI Country Office • WB Country Office etc.

Country	VIETNAM			
Stakeholders Category	Ministerial Actors	Education Initiatives NGOs	Business Initiatives	International Organizations
Stakeholders	• National Committee for the Advancement of Women in Vietnam (NCFAW) • Vietnam Women's Union	• University of Da Nang • BUILD-IT public-private ecosystem • WISE Vietnam (Women's Initiative for	• SWOF SEAF Women's Opportunity Fund • WISE Women's Initiative for Start-	• GGGI Country Office • WB Country Office etc.

	• Ministry of Labour – Invalids and Social Affairs (MOLISA) • Vietnam Women Entrepreneurs Council (VWEC) • Ministry of Science and Technology (MOST) • Ministry of Education and Training (MOET) • National Climate Change Committee	Start-ups and Entrepreneurship)	up and Entrepreneurship • Vietnam Climate Innovation Centre • BUILD-IT public-private ecosystem	

Appendix 2. Gender Profiling Form (GPF)

Each of the country SMEs shall be screened to assess whether the enterprise meets this Programme's "women-owned SME" criteria. This Company Gender Profiling Form (CGPF) shall be filled and submitted to the Programme at Component 1 or Component 2 stage.

CTF Gender Advisory Team shall review the CGPF of the applicant JVs to the CTF as part of the application package and validate if the applicant meets the Gender Mainstreaming Condition 1 ("The women-owned SME" status), one of the CTF Investment Criteria.

Does your company fulfil the following conditions?	YES/NO	If YES, pls provide details.	If YES, can you submit the evidence?
[CONDITION 1-1] **SMEs with at least 26% of the total capital owned by women**	YES	(Pls specify.)	YES: (Pls list up & submit) ■ Document #1: ■ Document #2: ■ Document #3:
			No (Pls explain the reason)
	No	N/A	N/A
[CONDITION 1-2] **SMEs with** **At least one woman participates on the Board of Directors Director and/or different types of boards or internal committees such as ESG Committees, among others (or CEO, COO, CTO)**	YES	(Pls specify.)	YES: (Pls list up & submit) ■ Document #1: ■ Document #2: ■ Document #3:
			No (Pls explain the reason)
	No	N/A	N/A
[CONDITION 1-3]	YES	(Pls specify.)	YES: (Pls list up & submit) ■ Document #1: ■ Document #2:

SMEs with more than 30% of the workforce are females			■ Document #3:
			No (Pls explain the reason)
	No	N/A	N/A
[OTHERS] **Are there any other aspects of your company that could be considered to regard your company as "women-owned"?** **Pls specify, if there are**.	YES	(Pls specify.)	YES: (Pls list up & submit) ■ Document #1: ■ Document #2: ■ Document #3:
			No (Pls explain the reason)
	No	N/A	N/A

The Table above has been filled by (Name of the person/Title or position/Organization/Contact No./ Email Address)

With signature

Date of submission

Appendix 3. Simplified Gender Analysis (GA) & Gender Action Plan (GAP) (General Template)

Participating JV candidates in the Acceleration Program (Component 2), or those JV candidates who are not participating in the Programme but preparing for the CTF application by themselves, need to prepare Simplified Gender Analysis (GA) and Gender Action Plan (GAP) and submit together with the other CTF application documents.

GA and GAP for JV candidates to prepare consists of two levels, namely:

1) LEVEL I: ORGANIZATIONAL (APPLICANT ENTITY, JV or country SME) LEVEL, and;

2) LEVEL II: PROPOSED ENTERPRISE LEVEL

Together with Gender Profiling Form (GPF, See Annex 2 above), GA & GAP documents are part of the CTF Application document packages to submit for the CTF application in Component 3.

This GA & GAP document needs to be drafted by the JV candidate. However, upon request, Fund-level Gender expertise can provide support to prepare it, as part of the Acceleration Program (Component 2).

Over time, the CTF shall disclose good example of the GA&GAP for the other applicants' references.

Following are the key components to be included in the GA+GAP document that needs to be prepared and submitted by the JV candidates for CTF.

Contents	Instruction	Answer (filled by the JV candidates)	Reference, explanation & Reference sources	(Note)
LEVEL I: ORGANIZATIONAL (APPLICANT ENTITY) LEVEL				
Section I-1. Overview of the Applicant entity (JV or Country SME) [221]**.** A Finalized Gender Profiling Form (GPF) (See Appendix 2 above) Shall be provided here.				

[221] Section I-1 of this Format is identical with the Gender Profiling Form (GPF).

Does your company fulfil the following conditions?	YES/NO	If YES, pls provide details.	If YES, can you submit the evidence?
[CONDITION 1-1] **SMEs with at least 26% of the total capital owned by women**	YES	(Pls specify.)	YES: (Pls list up & submit) ■ Document #1: ■ Document #2: ■ Document #3:
			No (Pls explain the reason)
	No	N/A	N/A
[CONDITION 1-2] **SMEs with** **At least one woman participates on the Board of Directors and/or different types of boards or internal committees such as ESG Committees, among others (or CEO, COO, CTO)**	YES	(Pls specify.)	YES: (Pls list up & submit) ■ Document #1: ■ Document #2: ■ Document #3:
			No (Pls explain the reason)
	No	N/A	N/A
[CONDITION 1-3] **SMEs with more than 30% of the workforce are females**	YES	(Pls specify.)	YES: (Pls list up & submit) ■ Document #1: ■ Document #2: ■ Document #3:
			No (Pls explain the reason)

	No	N/A	N/A
[OTHERS] **Are there any other aspects of your company that could be considered to regard your company as "women-owned"?** **Pls specify if there are.**	YES	(Pls specify.)	YES: (Pls list up & submit) ■ Document #1: ■ Document #2: ■ Document #3:
			No (Pls explain the reason)
	No	N/A	N/A

Section I- 2. Entity's Gender Mainstreaming Commitment Plan (GCMP)				
I-2-1. Willingness to commit to Gender Mainstreaming of the entity	I-2-1-1. Is your Entity willing to adopt Gender Mainstreaming measures internally? ☐ Yes ☐ No	(Pls elaborate the reason here.)	*Rationale and advantage of the organizational gender mainstreaming commitment shall be clearly communicated during Acceleration Program (Component 2). Individual JV applicant may approach Fund-level Gender expertise for further explanation before making a decision. (By written communication/ email or virtual meeting etc.)*	*Standardized guidance (manual) on the organizational gender mainstreaming commitment will be developed and provided in written for reference to all the JV applicants.*
I-2-2. Commitment Type (1): Employment	I-2-2-1. Is the Enterprise Team led/managed by a woman/women? ☐ Yes ☐ No	(If yes, pls elaborate here.)	*Individual JV applicant may approach Fund-level Gender expertise for further explanation before making a decision. (By written*	

	I-2-2-2. Will the Enterprise employ girls/women? ☐ Yes ☐ No	(If yes, pls elaborate here.) ■ Total _______ employment of girls or women / (___% out of total ___ number of employments to be created for girls/women.)	*communication/ email or virtual meeting etc.)*	
	I-2-2-3. What level of employment for girls and women will the Enterprise employ? ☐ CEO ☐ COO ☐ CTO ☐ Board member ☐ Director & Senior Manager ☐ Mid-level Manager/Administrator ☐ Line workers ☐ (Outsourced) Service workers (Pls specify:) ☐ Others: Pls specify:	(Pls tick all that applicable with the respective number of employments of girls and women)		
I-2-3. Commitment Type (2): Budget allocation	I-2-3-1. Is there allocation of budget specifically for the organizational gender mainstreaming within your entity? ☐ Yes ☐ No ☐ Not yet. But willing to introduce.	(Pls specify) ☐ Yes: Pls specify. ☐ No: Pls explain the reason for not allocating.		

		◻ Not yet. But willing to introduce: Pls elaborate your plan here.		
I-2-4. Commitment Type (3): Improvement of welfare policy for women employees	I-2-4-1. Are you willing to introduce/improve welfare policy for women employees? ☐ Yes ☐ No	(Pls specify) ☐ Yes: Pls elaborate your plan here. ☐ No: Pls explain the reason for not allocating.		
I-2-5. Other Commitments	**I-2-5-1. Pls specify any other gender mainstreaming commitment your entity is planning to introduce.**	(Pls specify here.)	*Individualized commitment plan shall be reviewed by the fund level Gender expertise and assessed on its individual merits.* *If necessary, guidance will be provided to concrete and substantiate the proposed commitment plan.*	
I-2-6. Gender Mainstreaming Commitment Action Plan	■ **<u>Organizational (Entity-level) Gender Action Plan [Typical Template]</u>**			

Gender Commitment	Indicators & Targets	Timeline	Responsibilities (Department)	Costs
Gender Commitment #1				
Gender Commitment #2				

	Gender Commitment #3				
	[…]				
LEVEL II: PROPOSED ENTERPRISE LEVEL					
Section II-1. Overview of the Proposed Enterprise					
II-1-1. Description of the Enterprise for CTF application	Describe the rationale /components of the proposed enterprises/ budget/ implementing parties (& delivery parties)/Period etc.	(Specify here.)			
II-1-2. Location & Host communities	Any location for the facilities (Office, manufacturing factories etc.) & locations of the supply chains of the proposed enterprises	(Specify here.)			
II-1-3. Key stakeholders	List up the key central/local/community stakeholders of all types (government/non-governmental/civil/ business etc.)	(List up here.)			
II-1-4. Key Result Areas	Indicate the relevant key result areas out of the GCF's 8 result areas.	Pls tick all applicable. [Mitigation] ☐ Energy Generation and access ☐ Low-emission transport ☐ Buildings, cities, industries, and appliances			

		[Adaptation] ☐ Most vulnerable people and communities ☐ Health and well-being, and food and water security		
II-1-5. Climate technology	Specify what climate technology it aims to adopt and transfer	Pls tick all applicable ☐ Renewable energy generation ☐ Transmission, distribution, and electricity storage ☐ Low-emission transport ☐ Residential and industrial energy efficiency technologies ☐ Agricultural technologies and practices ☐ Water management and treatment ☐ Waste management and treatment		

		☐ Ocean energy ☐ Bioenergy (renewable biomass) ☐ Small-scale hydropower ☐ Solar energy ☐ Wind energy ☐ Modern renewable off-grid energy systems ☐ Connection of isolated grid ☐ Battery storage ☐ Low-carbon/energy efficient mass transport ☐ EVs (e.g., e-cars, e-motorbikes, e-bicycles) ☐ EV infrastructure ☐ Alternative transport fuels (next generation advanced zero-emission biofuels, green hydrogen ☐ Building energy modelling		

		☐ Lighting ☐ HVAC ☐ Appliances ☐ Sustainable agriculture ☐ Climate-resilient plant/animal genetics/breeding ☐ Water saving and efficient irrigation methods ☐ Flood and drought management technology ☐ RE-powered equipment ☐ Other low carbon and climate-resilient farming methods and climate-linked insurance products ☐ Nature based solutions ☐ Management of water resources ☐ Small-scale, decentralised wastewater plants		

		☐ Wastewater treatment ☐ Water treatment processing ☐ Water desalination ☐ Mechanical-biological treatment ☐ Aerobic/semi-aerobic digestion ☐ Waste to energy ☐ Composting ☐ Others:		
II-1-6. Climate technology diffusion and commercialization mechanisms	Briefly describe the diffusion and commercialization mechanism of the adopted technology	(Specify here.)		
Section II- 2. Gender Analysis				
II-2-1. Girls & women in host communities	Based on the answer in Section 1.2. above, summarise: 2.1.1. Key gender dimension of the host communities/locations/potential labour force/customers/project partners/participants (with key socio-economic indicators (e.g. population/income level/literacy & education (esp.	(Specify here.) (Or: attach a separate report.)	*Individual JV applicant may approach Fund-level Gender expertise for guidance. (By written communication/ email or virtual meeting etc.)*	

	STEM & ICT) etc. with references.			
	2.1.2. Key gender-related stakeholder entities (e.g. women's unions, women's rights & empowerment advocacy & support groups, etc.)	(Specify here.) (Or: attach a separate report.)		
II-2-2. Vulnerability Assessment	Based on the answer in Section 1.5 & 1.6 above, summarise: 2.1.3. Are girls and women more vulnerable than boys and men in this sector in the current situation? ☐ Yes ☐ No ☐ Do not know.	(Pls elaborate.)	For country-specific gender vulnerability and potential for gender equality and economic empowerment, see Section 3 of this Gender Assessment Report.	
	2.2.2. Would girls and women benefit more than boys and men once this enterprise is properly implemented? ☐ Yes ☐ No	(Pls elaborate: What kind of benefits to what specific groups?) E.g.)		

	☐ Do not know.	Solar (and other renewable) facility manufacturing allows more women with employment and skill training opportunities, economically empowering girls and women.		
II-2-3. Barrier Analysis	2.3.1. Would the proposed enterprise potentially worsen the existent gender inequality between men and women in the country? □ Yes □ No □ Do not know.	(Pls elaborate: What aspect of gender inequality may worsen due to what component of the proposed enterprises?) E.g.) • Employment and technical education/training opportunities are given only to men, worsening economic equality between men and women. • Current green finance in local banks favours male entrepreneurs by requiring additional conditions for women SME owners (legally or	(*Based on answers given in the left column, Fund-level Gender expertise may deepen the barrier identification by reducing it to the typical barrier typology (e.g. institutional/policy/cultural-customary, societal/economic & financial, etc.)	

		customarily) (such as marriage certificate & co-signing by male spouse etc.)		
Section II- 3. Enterprise-level Gender Action Plan				
II-3-1. Identification of Gender Actions *(*A separate set of guidelines shall be provided as an attachment to this Format for guidance to fill this section.)*	3.1.1. How to strengthen the positive impacts of the proposed Enterprise on gender equality and women's empowerment?	Based on the answer to Section 2.2 above, pls identify measures to strengthen the benefits to girls and women & reduce their vulnerabilities to climate change. (Identify here) ■ Gender Action #1: ■ Gender Action #2: ■ Gender Action #3: […]	Fund-level Gender expertise may support the GAP establishment process by providing guidance to answer to these items. (Customized review and consulting as part of the Acceleration Program (Component 2), upon the JV candidate's request.)	
	3.1.2. How to reduce the negative impacts of the proposed enterprise on gender equality and women's empowerment?	Based on the answer to Section 2.3 above, pls identify measures to lift the barriers and reverse the worsening existent gender equality between men and women through the proposed enterprise.		

		(Identify here) ■ Gender Action #4: ■ Gender Action #5: ■ Gender Action #6: […]		
II-3-2. Implementation Plan of the Gender Action	3.1.3. Per each of the Gender Actions identified (#1~) identify agents/ required budget/monitoring and evaluation agent/frequency	Pls fill the GAP table below to the extent possible.		
II-3-3. Gender Action Plan	■ **JV Enterprise-specific Gender Action Plan [Typical Template]** (see table below)			

Activities	**Indicators & Targets**	**Timeline**	**Responsibilities (Agent)**	**Costs**
Gender Action #1				
Gender Action #2				
Gender Action #3				
Gender Action #4				
Gender Action #5				
Gender Action #6				
[…]				

Appendix 4. Stakeholder Consultation Activities

Meeting Type		Date (Venue)	Participants*	Key Discussion Points
Regular Progress Meeting_ Global Level	Kick-off	02 Sep 2021 (@Online)	GP team, GAIA, PwC, GGGI, KDB	- Overall ESMS policies & implementation frameworks - Feedback from individual stakeholders (incl. eligibility and CTF investment criteria etc.) - Alignment of ESMS and GAP with overall Programme management system etc.
	Interim	09 Nov 2021 (@KDB)		
	Progress	03 Nov 2021 (@Online)		
		06 Jan 2022 (@Online)		
		18 Mar 2022 (@Online)		
		05 Aug 2022 (@KDB)		
Country Level	NDA Consultation for No Objection Letter	Aug 2021 – March 2022 (@Online)	KDB, NDA, line ministries, GGGI, key local ecosystem stakeholders	- Inter-ministerial consultation arranged by the NDAs and relevant line ministries invited by the NDAs across the five NOL countries - Main discussion on country ownership
Topical Meeting	ESMS Development	29 Oct 2021 (@Online)	GAIA, GGGI, PwC, KDB	- Validation of AE & EEs' ESMS - Coordination of implementation mechanism across Component 1, 2, 3 & 4
		08 Jun 2022 (@KDB)		
		13 Dec 2022 (@PwC)		
		29 Mar 2023 (@KDB)	GCF, GAIA, PwC, KDB	- GCF Review results and Feedbacking on the draft ESMS
	Gender Assessment & Gender Action Plan (GA & GAP)	12 Nov 2021 (@Online)	GAIA, GGGI, KDB	- Overall Gender assessment process and action plan establishment - Consolidation of the GGGI's and KDB's review & feedback to the draft GA and GAP etc.
		14 Jan 2022 (@Online)	GAIA, GGGI	
		21 Jan 2022 (@Online)		
		27 Jan 2022 (@Online)		
		11 Feb 2022 (@Online)		
		21 Feb 2022 (@Online)	GAIA, GGGI, KDB	

** GAIA: Gaia Consult Inc./PwC: Price Waterhouse Coopers/GGGI: Global Green Growth Institute/KDB: Korean Development Bank*

Appendix 5. Sector/Technology-specific Guidelines for Gender Mainstreaming for CTF Enterprises/Projects

Renewable energy, agriculture, water and waste management are sectors identified with higher potential benefits for women through the CTF support. With the addition of the Early Warning System (EWS) and Disaster Management, the following sections provide some of the typical gender mainstream actions that all the relevant JV applicants are recommended to consider and reflect in their respective GA/GAP (Appendix 3 of this GA/GAP of the Programme (Annex 8 of the Funding Proposal of the R&DB Programme)) for submission as part of the CTF Application.

Appendix 5-1. Renewable Energy

- **Renewable Energy (Solar PV) Sector in the Five Countries**

[Cambodia]

Situated in the 'global sun belt'[222], Cambodia is well-positioned to utilize solar energy. The government has provided clarity in the solar sector by allowing electricity generation for self-consumption through off-grid and grid-connected systems that meet specific criteria. In recent years, start-ups focusing on connecting rural farms and fisheries to affordable off-grid solar solutions have increased, and key incubators such as the Switch to Solar Incubation Programme and Impact Hub's DakDam Incubator programme are supporting the sector. Cambodia's updated Nationally Determined Contribution (NDC) to the UNFCCC includes conditional targets for the energy, industry, transport, building, agriculture, and waste sectors. One of the NDC's mitigation actions for the energy sector is to increase energy access to rural areas, which requires **solar home systems, DC and AC microgrids, and solar battery charging stations**. Cambodia's NDC also includes adaptation actions such as scaled-up climate-resilient agricultural production through increased access to solar irrigation systems, using solar water pumps.

[Indonesia]

The RD&B Program prioritizes the use of solar photovoltaic (PV) technology as a key climate technology. In Indonesia, solar PV technology ranks as the top priority among renewable energy technologies in terms of cost, environmental benefit, economic impact, technology maturity, country ownership, and social impact. (See Table 1.B.38 of the Pre-Feasibility Study of the Program (Annex 2 of the Funding Proposal))

The potential of solar energy in Indonesia is vast, and the government has made significant efforts in the last five years to maximize the use of clean and renewable energy, particularly solar energy. Solar PV technology is considered the most mature commodity in terms of regulation, market, and technology, with various energy start-ups emerging to cater to the solar market.

Solar PV technology has the potential to be utilized in **remote and off-grid areas** across Indonesia, with direct applications in households and industries. The government has also issued regulations to promote the use of clean and renewable energy, including the National Energy Plan and the Provision of Energy-Efficient Solar Lamps for People Who Have Not Accessed Electricity.

Solar Photovoltaic (PV) to have high potential to be implemented in Indonesia (See Table 1.B.42 of the Pre-Feasibility Study of the Program (Annex 2 of the Funding Proposal)), in the form of the following

[222] Within 35 degrees of the Equator.

technologies**: Low-cost smart grid system solar PV**, **local industry to build components of the solar modules, user-friendly apps to increase solar PV investment/installation, micro-inverter solar panel**.

[Lao PDR]

The renewable energy sector in Lao PDR is experiencing significant growth, with solar technology being the dominant sub-sector. The country's prospective technology needs for climate mitigation include **solar technology for power generation and supply, and solar home systems for energy access** (See Table 1.C.13 of the Pre-Feasibility Study of the Program (Annex 2 of the Funding Proposal)). Priority technologies for the RD&B Program in Lao PDR include **isolated home systems (solar home systems) and clean cookstoves**.[223]

[Philippines]

The agricultural sector in the Philippines has potential for growth through the implementation of green technologies, particularly in **climate-smart agriculture (CSA)** initiatives. CSA can be achieved through **solar-powered irrigation systems, drones, and smart farms**, which are cost-efficient compared to traditional methods. The Philippine government has also prioritized renewable energy technologies in its efforts to mitigate climate change, particularly in the energy, transportation, and waste sectors. Renewable energy systems can also be deployed as climate adaptation actions following the National Climate Change Action Plan (NCCAP), particularly in agriculture and fisheries, where climate-related disasters have hampered development. The establishment of Competitive Renewable Energy Zones (CREZ) in the country can guide potential investments from project developers.

[Vietnam] [224]

Vietnam has made significant strides in renewable energy, particularly in wind and solar, over the past three years. By the end of 2021, Vietnam had over 9.5 GW of rooftop solar capacity connected to the national power grid system, which is a massive increase compared to 2019, placing Vietnam among the top 10 countries with the highest capacity of installed solar energy. Solar technology is predicted to account for USD 280 billion of the renewable energy sector's market value, which is expected to reach USD 714 billion. However, renewable energy development in Vietnam remains limited compared to its potential, with only 100 MW of wind power, 15 MW of solar power, and 10 MW of biomass power. As part of mitigation technologies, **solar photovoltaic power plants and solar water heaters** have been included as priority technologies in the energy sector.

- **Gender Perspective in Renewable Energy Sector (Solar PV)**

Among the diverse renewable energy sectors, solar PV is a globally leading sector of gender-balanced employment. The solar sector has about doubled women's employment rate, compared to wind and oil and gas sectors, in 2021[225].

[223] Table 1.C.16. Lao PDR energy priority technologies and Table 1.C.22. Summary of sectors and priority technologies
[224] 1.2.4 list of sectors with growing climate technology
[225] https://www.irena.org/News/pressreleases/2022/Sep/Solar-PV-Employs-More-Women-Than-Any-Renewables

Globally, women's job share in the solar PV sector can be determined as follows: administration (58%), non-STEM technical (38%), other non-technical (35%), and STEM (32%)[226]. This shows that more than half of female workforces are serving administrative work, while a few of them are involved in STEM-based technical jobs. However, this still takes a higher portion compared to the other renewable energy sectors. With the lower level of attaining STEM education of females, expect Philippines, which has about a similar share of STEM education participation, women are marginalized to be involved in the mainstreaming skills of the industry.

International Renewable Energy Agency (IRENA) analysed[227] the barriers that women are facing in the solar PV workplace. The barriers can be divided into three categories which are entry, retention, and advancement. For entry, gender role perception, limited mobility, and hiring practices of that sector became the obstacles; due to fair and transparent policies, maternity leaves, and flex-time, women were hard to be retained in the workplace; the issues that are hindering women's advancement are glass ceiling, lack of mentors, and social norms. These cultural and social perceptions make it women hard to be skilled in the renewable energy sector, even though the solar PV sector shows a relatively higher level of women workforce participation. Compared with other renewable energy sectors, solar PV still hires significant portion of women workers in technical areas (mainly related to the manufacturing of panels). Once installed the panels and the solar PV system need to be regularly cleaned. Often cleaning is conducted manually and this may prefer female labours.

At the very first stage of solar PV development in Southeast Asia, Indonesia was one of the leading countries that take the majority of solar PV capacity[228]. After then, Philippines started to increase solar energy in their energy generation shares. More recently, Vietnam became one of the largest countries in the world, in terms of solar PV growth. Meanwhile, there is steady growth in Cambodia and Laos. These show the variations in solar PV development across the countries. Although the five countries have different stages in solar PV deployment, considering the general gender-biased issues in the solar PV sector (including renewable energy), gender-mainstreamed perspectives are required for all stages of each project. In this context, the occupational environment, health, and safety (EHS) need to reflect gender aspects. Although all the five countries have ratified key labour-related international conventions, in practice, there are limitations, especially for women workers.

Recommended Gender-mainstreaming approaches

<1> General Requirements

All the requirements under the CTF Investment Criteria: Gender Component (See Table 1 of the Gender Assessment above.) will apply to all the EWS and Natural Disaster Technology application projects/enterprises subject to the CTF approval. Thus "women-led enterprises" on the EWS and Natural Disaster Management projects shall be prioritized. The rest of the applicant JVs shall also be encouraged to hire more female employees (with equal working conditions and remuneration as male employees) and improve women employees' welfare policies.

All the CTF Applicant JVs shall need to submit a simplified gender assessment and gender action plan ([Appendix 3. Simplified Gender Analysis (GA) & Gender Action Plan (GAP) (General Template)] of the Gender Assessment/Gender Action Plan (GA/GAP), Annex 8 of the Funding Proposal of the Programme). [Section II- 2. Gender Analysis] requires you to assess the gender perspective of a specific

[226] https://www.irena.org/Digital-content/Digital-Story/2022/Sep/A-Gender-Perspective-on-Solar-Employment/detail
[227] https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2022/Sep/IRENA_Solar_PV_Gender_perspective_2022.pdf?rev=61477241eb9e4db2932757698c554dc2
[228] Sreenath et al. (2002 May). A decade of solar PV deployment in ASEAN: Policy landscape and recommendations. [Conference Paper] Energy Reports 8. Retrieved from: https://reader.elsevier.com/reader/sd/pii/S2352484722010691?token=B4D69AFBBD87BF3AB17F2D40DAE09500B4E6D11F6EEC299563683A291CEC61D4F5D733E5504F20D97D40AC81DCC5535B&originRegion=us-east-1&originCreation=20230424094303

proposed JV enterprises/projects, based on which [Section II-3. Enterprise-level Gender Action Plan] need to be established.

<2> Sector-specific Gender Mainstreaming Considerations

Gender Action 1. Consider gender-specific occupational EHS plan

Female workers in manufacturing panels and assembling solar PV systems, and installation may face differentiated types and levels of dangers and hazards. (The higher sensitivity of women to the exposure of particular toxic/hazardous materials such as the use of conductive pastes containing Cd, Pb, and/or phthalates.)

Gender Action 2. Consider biological differences of women

Ergonomics: Women may have different ergonomic needs than men, which can impact their comfort and safety in the workplace. For example, solar panel installation often requires heavy lifting and working at heights, which may be challenging for women who are smaller or have less upper body strength than their male counterparts. Heights are also one of the significant issues in occupational health and safety concerns. With the 2020 Fatalities for US Roofers having increased 15% as demands for Solar Roof Installations increase, this issue is deemed to have resulted in higher fatality rates for Solar installation projects outside the United States because of more lax safety regulations[229]. With construction being a male-dominated field, there is concern that female workers will be more susceptible to injuries due to infrastructure built without consideration of female-body ergonomics.

Possible solutions include:

- ***Conduct a gender-sensitive risk assessment: Conducting a gender-sensitive risk assessment can help identify potential hazards and risks for female workers and provide recommendations for addressing them[230]. (eg. Flexibility of work hours of female workers under the sun (due to higher proneness of heat stress of female workers[231]))***

- ***Promote gendered ergonomics in design: Ergonomics should be considered in the design phase of infrastructure projects to ensure that they are inclusive of all workers, including women. This can include designing equipment and tools that are better suited to the female body, such as smaller and lighter tools[232].***

- ***Produce health and safety products specifically designed for female workers (e.g. PPE fit for average Asian female bodies, etc.)***

- ***Ensure "zero-tolerance of SEAH and GBV" in all workplaces.***

[229] 2020 Fatalities for US Roofers Increased 15% as Solar Roof Installations Increase | NextBigFuture.com
[230] https://www.ilo.org/wcmsp5/groups/public/---ed_protect/---protrav/---safework/documents/publication/wcms_324653.pdf
[231] https://www.sciencedirect.com/science/article/abs/pii/S0168159112002687
[232] Designing for Gender Equality in the Developing Context: Developing a Gender-Integrated Design Process to Support Designers' Seeing, Process, and Space Making, Ginger Daniel (2013)

Appendix 5.2. Agriculture

- **Agriculture Sector in the Five Countries**

[Cambodia]

Agriculture is a key driver of Cambodia's economy, making up an estimated 21.1% of the country's GDP in 2019.[1] The sector is highlighted as a critical focus for adaptation actions in national climate strategies, including the NDC, as it is highly vulnerable to the effects of climate change, such as flooding and drought. Moreover, farmers are highly vulnerable to the rising temperatures caused by climate change, which can lead to lower productivity and heat stress.

Climate-resilient agriculture technology is the top adaptation technology priority in Cambodia, marking full co-benefits in terms of adaptation impacts, economic, social, environmental and gender co-benefits as well as country ownership (See Table 1.A.13 of Pre-Feasibility Study, Annex 2 of the Funding Proposal of the R&DB Programme).

Several programmes and initiatives are taking place including **smart agriculture initiatives**, supported by both the Cambodian government and international and local entities. Increasingly technology innovation in agriculture sector in Cambodia is focusing on technology and apps for farmers, assuming a high level of digital literacy. Given the level of digital literacy of (esp. female) farmers, training would need to be strengthened. So far, however, smart agriculture has not been extensively reached out to the Cambodia's rural communities and has significant potential for upscaling with proper incentives.

[Indonesia]

Three out of Five Indonesians live in rural areas with agriculture as their predominant occupation. Food sovereignty, self-sufficiency and increasing agricultural productivity are the main foci of government agriculture policy. While Indonesia's 2022-2024 National Medium-Term Development Plan (RPJMN) selected agriculture as one of the priority areas for adaptation, the climate technology is still in its nascent stage in Indonesia, and the agritech is attracting least amount of funding thus far. (See Figure 1.B.4 of the Pre-Feasibility Study, Annex 2 of the Funding Proposal of the R&DB Programme).

The country's agriculture sector is highly vulnerable to climate change-induced extreme weather conditions, such as extreme heating and flooding. Due to decreasing water availability, the country's rice production is projected to decline by more than 25% in several provinces. The priority agri technology in Indonesia are (in order of the ranks): (1) **crop(rice) tolerance to drought and flooding**; (2) **Technology for mariculture development**; (3) **Technology of efficient irrigation (intermittent, SRI, PTT**); (4) **cattle meat technology development among others**. (See Table 1.B.35 of the Pre-Feasibility Study, Annex 2 of the Funding Proposal of the R&DB Programme).

[Lao PDR]

In Lao PDR, more than two-thirds of the workforce is employed in agriculture, mostly in small-scale farming. The country is highly exposed to climate-related hazards and is expected to get worse over time. Temperatures in Lao PDR are predicted to increase by 3.9˚C against baseline conditions before the end of the century, while agriculture yields reduce consistently, driven by the incidence of extreme heat during growing seasons of stable crops such as rice. The country is also heavily exposed to flooding.

To reduce sector GHG emission, the Lao government aims to: (1) enhance low-emission and environmentally friendly practices, such as **smart and precise farming, feeds and feeding** and

livestock manure management; (2) adjust **water management practices** in 50,000 hectares in lowland rice cultivation, and; (3) Increase animal feed by 1.4 million tonnes by 2024, including 1% of **feed and concentrates that reduce emissions.** (See Table 1.C.6. of the Pre-Feasibility Study, Annex 2 of the Funding Proposal of the R&DB Programme). For enhanced adaptive capacity, the country also aims to: (1) promote **climate resilience in farming systems and agricultural infrastructure**, and; (2) promote appropriate technologies for climate change adaptation, including **nature-based and circular economy solutions**. The country is also leading trends in **organic agriculture**. The country has a great potential for soil carbon sequestration. To achieve this, the country needs to shift from widely-used burning practices to soil rehabilitation through conservation agriculture. For this awareness raising and training is required for the farming population to recognize the advance of **soil carbon sequestration techniques**.

Through extensive consultations and prioritization exercises (incl. AHP), the following four technologies in agriculture were identified as top priority for CTF support: (1) **Early warning system and insurance;** (2) **Flood and drought management technology**; (3) **Livestock disease control technology**, and; (4) **Pest management/control technology**.

[Philippines]

The Philippines Innovation Act of 2019 (RA 11293) fosters agriculture as one of the priority areas for fostering innovation and internationalization activities of MSMEs. The country has **also climate-smart agriculture (CSA)** initiatives. The country highlights **climate-responsive agriculture (With Renewable energy-powered post-harvesting and processing equipment technology** to address prohibitive level of cost of farm equipment and facilities with increasing energy (mostly diesel) price they use, as examples) as its NDC adaptation actions. Cropland management and precision agriculture are also prioritized for CTF support.

[Vietnam]

Viet Nam is an agriculture-based country, where cultivated land for farming makes up nearly 40% of the total land mass, and about 20 million people work in the agriculture sector (about 40% of the total employment), contributing 17% of the national GDP. However, in general, the agriculture sector has been receiving comparatively less attention for investment until recently.

The Vietnamese government is driving initiatives for smart rural, smart village and smart agriculture with an aim to create more efficient and modern rural services through inclusive digital transformation. Priority agricultural technologies for Adaptation for CTF support in Vietnam include, among others: (1) **Agricultural ecosystem-based adaptation (AEbA**); (2**) Climate-smart agriculture (CSA);** (3) **Local poultry and waterfowl species with a lower critical temperature (LCT**); (4) **Convert from discrete and small-scale breeding into concentrated breeding**; (5**) Integrated rice farming models: rice-shrimp/fish/duck**, and; (5**) VietGAP-based breeding**. (See Table 1.E.25 of the Pre-Feasibility Study, Annex 2 of the Funding Proposal of the R&DB Programme).

For mitigation actions, (1) **ICM for annual upland crop cultivation** has topped in priority, followed by: (2) **Shifting from double/triple rice cropping system to rice-aquaculture farming system**; (3) **Improved technologies to recycle livestock dung as organic fertilizer**, and; (4**) Reuse of agricultural/crop residues.** (See Table 1.E.30. of the Pre-Feasibility Study, Annex 2 of the Funding Proposal of the R&DB Programme).

- **Gender Perspective in Agriculture Sector**

In many developing countries, the agriculture sector has women's faces. In Cambodia and Indonesia, the agricultural sector is predominantly occupied by women (about 80% in Cambodia & 70-80% in Indonesia respectively). More than 60% of Vietnamese women work in the agriculture sector and more than 50% of the farmers in Lao PDR is also women. In contrast, only less than a quarter of the agricultural population are women in the Philippines. The minority female farmers in the Philippines are concentrated in subsistence farming and food for family consumption, whereas men primarily focus on cash crops.

With varying degrees, across the countries, climate hazards threaten the livelihoods and food security of millions of smallholder farmers and agribusiness who depend on agriculture in our selected countries, especially women". Regardless of the level of participation, women farmers in all five countries are in disadvantageous positions compared with their male counterparts: Women's activities are often under-valued, the farming scale is smaller, and innovation and scale-up are often hampered by uneven grounds generated by socio-cultural, traditional bias and gender roles, which favours males for land ownership, access to formal credit services as well as inheritances. In general, located in rural settings, the agriculture sector is known to be a sector where male-dominated old customs are traditions are perpetuated more strongly than other industries.

Given the limited education and resources available to the rural farming women in the five countries, it is of utmost importance to proactively the women farmers and agricultural entrepreneurs to the Programme and ensure their equal benefiting from it.

- **Recommended Gender-mainstreaming approaches**

<1> General Requirements

All the requirements under the CTF Investment Criteria: Gender Component (See Table 1 of the Gender Assessment above.) will apply to all the EWS and Natural Disaster Technology application projects/enterprises subject to the CTF approval. Thus "women-led enterprises" on the EWS and Natural Disaster Management projects shall be prioritized. The rest of the applicant JVs shall also be encouraged to hire more female employees (with equal working conditions and remuneration as male employees) and improve women employees' welfare policies.

All the CTF Applicant JVs shall need to submit a simplified gender assessment and gender action plan ([Appendix 3. Simplified Gender Analysis (GA) & Gender Action Plan (GAP) (General Template)] of the Gender Assessment/Gender Action Plan (GA/GAP), Annex 8 of the Funding Proposal of the Programme). [Section II- 2. Gender Analysis] requires you to assess the gender perspective of a specific proposed JV enterprises/projects, based on which [Section II-3. Enterprise-level Gender Action Plan] need to be established.

<2> Sector-specific Gender Mainstreaming Considerations

Gender Action 1: To the extent possible, aim to set the women's participation targets as proportionate to the women's share of sector participation.

In countries with predominantly higher level of participation of women in agriculture such as Cambodia (about 74%), Indonesia (70~80%), and Lao PDR (more than 50 %) the Programme shall aim to increase the country-specific targets for women's participation (in terms of: 1) the number of selected local applicant agricultural startups and firms which are "women-owned" (in Component 1), and/or 2) number of (local) female employments as part of the Gender Mainstreaming Commitment Plan (GMCP).

- ***In Component 1, the Programme shall set a more ambitious target for country (local) "women-led" startups/firms' participation in the agriculture sector.***

- ***In Component 2, the Programme shall set more ambitious targets for a JV applicant's gender mainstreaming action plan (GMCP). (e.g., by setting hire women employment/participant rates. & overall outcome targets of women beneficiaries etc.)***

Gender Action 2: Proactively tap into the women farmers' insights for innovation and identification of the appropriate farming technologies.

By sheer size of women occupying their hands-on experiences in the farming sector, women are a reservoir of the traditional and indigenous agricultural technologies and practices: While male farmers are more dominant in machine-based, large-scale commercial scale monoculture (e.g., simple crop/fruit plantations), informal and small-scale, subsistence farming has been similar with mixed farming, agroforestry and aquaculture-rice production combination styles. Thus, women farmers should be the leading participants from the very planning and design stage of the project preparation.

- ***In Component 2, in the JV enterprise preparation stage, the Programme shall ensure the consultation activities have more or equal number of female participants than males at community-level. Also in future stakeholder engagement plan, the JV enterprise for CTF application shall ensure more or equal representation of women in consultation groups.***

Gender Action 3: Consciously posit women as the prototype participants and beneficiaries of the funded enterprises/activities.

Given the majority of farmers are women in most countries (except the Philippines), it is advisable that whatever business plans and designs of concerned enterprises/businesses under CTF support, should assume women's group as a prototype users and beneficiaries. Sounds truism though, this assumption needs to be clearly and consciously articulated due to our deeply entrenched perception positing (often while adult) males as a standard and model for the general humankind from the very proposal and conceptualization state. This will provide gendered technology innovation.

- ***In Components 2 and 3, the JV enterprises in the agriculture sector shall explicitly posit women in the target country(ies) as a prototype target group for technology applications and beneficiaries. As appropriate the JV enterprises shall consider "gendered" designing and planning. Such approaches shall be explicitly reflected in the plan of the JV enterprises (in the proposal document) during CTF application.***

Appendix 5.3. Water

- **Water Sector in the Five Countries**

[Cambodia]

Cambodia remains a largely agrarian country with its high population dependent on subsistence farming for their livelihoods which leaves them highly vulnerable to climate shocks. ADB (2021) mentions in its report on the great potential for the application of **new water and agricultural technologies,**

including drip irrigation, water harvesting, biomass and rural energy.[233] In this sense, **water resource management technology** remains highly relevant as the prioritized adaptation technologies, critical for maintaining 'the fertility of productive agricultural lands for food production, fisheries, and other biodiversity resources'[234].

[Indonesia]

Indonesia identifies water resource management as one of the priority sectors for the adaptation technology for NDC achievements[235]. In addition, **water harvesting (well & infiltration pond)** is selected as the highest in its prioritized adaptation technologies in the water sector, which results in improving health and well-being, and food and water security.

[Lao PDR]

Lao PDR is a country, rich in water resources. Flooding is a major cause of climate change-related disasters in Lao PDR, and it affects the quality of water, the loss of lives, and the livelihoods. The most vulnerable areas to flooding are the plain along the Mekong River in the central and southern parts[236]. 75% of Lao PDR's population relies on agriculture for their livelihood, and food security could be endangered if water demands for the agriculture sector that cannot be met. The food security-water nexus is also broadly recognised in the country's climate strategies in its latest NDC[237], Lao PDR has set a conditional target for the agriculture sector to **develop 50,000 hectares of adjusted water management practice**. Resilience of water infrastructure to climate change can be approached both from a **climate-resilient infrastructure** angle and a water resource/water security angle.[238]

The R&DB Programme highlights investment priority on **watershed management technology, hydropower dam technology, and climate-resilient water supply technology** to enhance the resilience of the water resources sector in Lao PDR.

[Philippines]

In the Philippines, the gap between the available water supply and demand continues to widen. About 12.40 million people are limited to getting water from unsafe sources. Some areas do not have water service providers (WSPs) and most WSPs are neither financially nor technically capable of delivering the required services to a growing population. In addition, only 515 or 68.9 percent of the country's 748 water districts (WDs) are operational at varying levels of efficiency and coverage.[239] As a consequence, the effects of climate change exacerbate the country's water challenges. In this context, **small-scale RE-powered water desalination and mobile RE-powered water treatment** as referred in the R&DB Programme are highly relevant for the supply of fresh water, particularly in water-scarce regions[240] as one of the prioritized adaptation technologies.

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[233] "Sector Assessment, Strategy, and Road Map," ADB.
[234] "Sector Assessment, Strategy, and Road Map," ADB.
[235] See Table 1.B.11. of Demand-Driven Pre-Feasibility Study (Annex 2) of the Programme by KDB and GGGI.
[236] https://climateknowledgeportal.worldbank.org/country/lao-pdr/vulnerability
[237] See Table 1.C.9. of Demand-Driven Pre-Feasibility Study (Annex 2) of the Programme by KDB and GGGI.
[238] See 'Climate Technology Needs Assessment' in Demand-Driven Pre-Feasibility Study (Annex 2) of the Programme by KDB and GGGI, pg.221.
[239] National Economic and Development Authority, Philippine Water Supply and Sanitation Master Plan 2019-2030, 15 September 2021, https://neda.gov.ph/wp-content/uploads/2021/09/120921_PWSSMP_Main-Report.pdf
[240] "Is Small Scale Desalination Coupled with Renewable Energy a Cost-Effective Solution," Applied Sciences, 2021, https://www.mdpi.com/2076-3417/11/12/5419/pdf.

[Vietnam]

In the water sector, there is a high demand for adaptation technologies in Viet Nam. R&DB Programme highlights that further actions must be taken to address the impact of climate change on water resources, such as **implementing integrated water resource management in river basins, monitoring and protecting water resources, ensuring reservoir safety, improving water usage efficiency, enhancing adaptive capacity and resistance to climate change, managing the coastal strip and coastal protection zones in an integrated manner, and spatial planning for the sea and the islands**. In view of this, **water resource management** has been identified as prioritized adaptation technologies for the R&DB Programme.

- **Gender Perspective in the Water Sector**

In most developing countries, women and girls are the primary providers, managers and users of water at household levels but their roles seldom go beyond.

Traditionally, the grid-connected, large-scale water supply, distribution and wastewater treatment infrastructure development was heavily male-dominated areas due to its engineering/technical requirements and (perceived or real) intensity of the physical labour. Having said, that, in urban setting, women (of low-income households) comprised bottom-most parts of the physical labourers (with more precarious working conditions often without formal labour contracts, in informal sectors) of the construction and associated business activities (food stalls, street peddlers selling drinks and snack nearby the construction sites, etc.)

In the context of developing countries, a significant share of a country's population still remains without access to clean and sufficient processed water through modernized water supply and treatment system, neither they are in the protection of their health and hygiene through proper wastewater treatment systems. Lack of sufficient and clean and safe water provision is more acute in remote rural and isolated locations.

In these settings, it is often girls and women who spend more time for water access at household levels: Women and girls spend a substantial amount of time collecting and transporting water for household activities, contributing to a situation of 'time poverty'[241], deepening gender inequality in development. Moreover, women and girls face an increase in vulnerability and Gender-Based Violence (GBV) at water points and while walking to collect water[242] due to the long distances to and isolation at water sources, etc. In this regard, a meaningful engagement of women as water service beneficiaries such as in gender-mainstreaming/sensitive approach will not only promote women's rights but also improve the access to safe water and improved sanitation.

- **Recommended Gender-mainstreaming approaches**

<1> General Requirements

All the requirements under the CTF Investment Criteria: Gender Component (See Table 1 of the Gender Assessment above.) will apply to Water Management Technology application projects/enterprises subject to the CTF approval. Thus "women-led enterprises" on Water Management projects shall be prioritized. The rest of the applicant JVs shall also be encouraged to hire more female employees (with

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[241] https://eugender.itcilo.org/toolkit/online/story_content/external_filea2qs/TA_Water&sanitation.pdf
[242] World Bank, Toolkit for Mainstreaming Gender in Water Operations, March 2016, https://documents1.worldbank.org/curated/en/922021536852796350/pdf/Toolkit-for-Mainstreaming-Gender-in-Water-Operations.pdf

equal working conditions and remuneration as male employees) and improve women employees' welfare policies.

All the CTF Applicant JVs shall need to submit a simplified gender assessment and gender action plan **([Appendix 3. Simplified Gender Analysis (GA) & Gender Action Plan (GAP) (General Template)]** of the Gender Assessment/Gender Action Plan (GA/GAP), Annex 8 of the Funding Proposal of the Programme). [Section II- 2. Gender Analysis] requires you to assess the gender perspective of a specific proposed JV enterprises/projects, based on which **[Section II-3. Enterprise-level Gender Action Plan]** need to be established.

<2> Sector-specific Gender Mainstreaming Considerations

Gender Action 1: Involve women in the development and maintenance of community-level water system

Water projects designed and implemented with full women's participation in all phases of planning, decision-making, and management of water and sanitation are more sustainable and effective[243]. As women continue to bear the primary responsibility for the food, care, and health of the household, women have consequently accumulated considerable knowledge about current water sources and needs. As such, placing women as the key players in designing, operating and maintaining the water systems especially in community levels (in non-technical or non-engineering sector) will not only change the cultural concept and social norms, but also help to transform into women-friendly water systems that reflects their needs and preferences as the sector's primary customers.

With built-in interest, women (and girls) who are often the caretakers and guardians of their families, have great potential to become capable and responsible managers and operators for sustainable water management at the community level. In many cases, externally financed water supply and treatment systems and facilities such as water purifiers, septic tanks, etc. often short-lived due to the failure of institutionalization of a community-based sustainable water management governance on the ground. In this regard, involving women in the maintenance of water system will not only foster women's self-esteem but also economic empowerment by providing technical guidance and financial support which will overall contribute to women's empowerment.

- ***Contract women as national and provincial level staff in the field of water and sanitation management; Provide equal opportunity to women staff along with men, for upgradation of skills through training and opportunity to both males and females in management of water and sanitation system.***

Gender Action 2: Engage women, especially the vulnerable and low-income, for monitoring and Surveillance of drinking water sources and treatment facilities.

Water-borne illnesses such as diarrhea, cholera and typhoid are common in developing countries. As women typically are the ones that ensure the quality and quantity of water for the entire families, proper water treatment through monitoring and surveillance of drinking water sources by women would improve the quality of water for consumption, improved access, and economic and environmental

[243] Women for Water Partnership, With Women Better Results In Water Management, 2022, https://www.womenforwater.org/uploads/7/7/5/1/77516286/20220722_women_for_water_partnership-with_women_better_results_in_water_management.pdf

benefits as well. Involving women in water quality management and monitoring such as water quality testing will contribute to improving the quality of water.

Unlike external companies that conduct regular or quarterly visits, regular/frequent/permanent physical presence of the women residents at the community water facilities would ensure constant monitoring of the performance of the water facilities while nimbly addressing any malfunctions.

- ***Mobilize and institutionalize women's groups (or female students) and provide training and education programs for women to gain the necessary technical and managerial skills for their active water management and monitoring.***

Gender Action 3: Improve safe and convenient access to water and sanitation facilities for the vulnerable

In many cases in developing countries, location of the water supply facilities does not ensure equal access to all community members, especially socio-economically vulnerable and politically marginalized groups. In (increasingly) acute water shortage situation, the water facilities may be subject to illegal appropriation by the elite groups, excluding the vulnerable.

- ***When designing and locating the water and sanitation facilities, give special consideration to the vulnerable group, particularly women and girls who are at higher risk of gender-based violence as well as people with specific needs such as the elderly and persons with disabilities.***

- ***Consider offering collective asset ownership of water utilities and facilities at community-level to avoid the risks of corruption and elite capture; Locate water and sanitation facilities close to dwellings (e.g., through social mapping), in a secure environment and along safe access paths.***

Appendix 5.4. Waste Management

- **Waste Management Sector in the Five Countries**

[Cambodia]

Waste Management Technologies are recognized as priority climate mitigation and adaptation technologies in Cambodia. (See Table 1.A.12 and Table 1.A.13 of the Demand-Driven Pre-Feasibility Study (Annex 2 of the Funding Proposal) of the RD & B Programme)

The waste sector is responsible for approximately 2.1% of GHG emissions in Cambodia, as of 2016. Solid waste and wastewater management is an ongoing challenge in the country with increasing levels of waste generation. Waste composition is also changing with an increasing share of non-degradable waste such as plastics, metals, and electronic wastes. There is a need for additional waste segregation, treatment and reuse infrastructure and management systems. Rapid economic and urban development, combined with a growing middle class, has had a substantial impact on waste generation in Cambodia's major cities and waste collection and management is increasingly recognized as a major issue in the country. Currently, households, local companies and factories frequently dispose of their waste by burning it or dumping it in public places, streams, and deserted locations, posing public health threats and soil, water and other environmental pollution risks.

While there is a general lack of data on the actual amount of recycled waste in Cambodia, **recycling activities of limited scale** are observed in Phnom Penh. Several private companies are exploring **the**

business potential targeting both organic and inorganic waste. Both men and women have a low level of awareness about the waste sector.[244]

[Indonesia]

Every year, Indonesia generates 65.8 million tons of solid waste from households and industries. The rate of trash production is expected to rise as the population expands, reaching 70.8 tons per capita by 2025.

According to the Ministry of Environment and Forestry (2017), only 2.2% of the waste generated is being recycled or processed into other resources, while 66.39% of trash produced in Indonesia is landfilled and 19.62% is unmanaged. The flow of recyclable materials in Indonesia is heavily dependent on the informal sector from street recycling pickers which collect recyclable materials from the streets.

Among the various waste management technologies related to climate change mitigation, mechanical and biological waste treatment (MBT) and in-vessel composting (ICV) technologies rank the highest priorities for CTF investment. Followed are Low-solids anaerobic digestion (LSAD) and high-solids anaerobic digestion. (See Table 1.B. 41 of the Demand-Driven Pre-Feasibility Study (Annex 2 of the Funding Proposal) of the RD & B Programme).

[Lao PDR]

To achieve the national GHG emission reduction target, the Lao government aims to implement 500-tonne-per-day sustainable municipal solid waste management project (Lao PDR NDC 2020).

There is an absence of waste treatment systems that meet the required environmental standards in many of the factories that are located in larger cities. In general, suitable technologies and sustainable waste management are lacking. In addition, regulatory approval procedures for private sector investment are often lacking predictability and gate fees are rather low that often the operations are not commercially viable. In Lao PDR, Municipal solid waste management has the potential to deliver various sustainable development co-benefits, including environmental, climate, public health and sustainable cities benefits.

Priority technologies in waste treatment areas are: **mechanical and biological waste treatment (MBT)**, **Biogas and biofuel technology**, **Gasification treatment**, **Methane recovery from sanitation landfills**. In the waste disposal area, top priorities are: **Landfilling technology** and **semi-aerobic landfilling system**.

[Philippines]

Waste is one of the three climate change mitigation sectors identified in the Technology Needs Assessment for Climate Change Mitigation (CCC, 2018), together with energy and transportation. In terms of **waste heat recovery (WHR), power generation from landfill methane recovery** is included as technology option in the country's NDC.

[Vietnam]

[244] GIZ. (2019). Partnership Ready Cambodia: Waste management. Retrieved from https://www.giz.de/en/downloads/GBN_Sector%20Brief_Kambodscha_Waste_E_WEB.pdf.

In recent years, Viet Nam has installed a number of solid waste treatment facilities that utilise cutting-edge technology and procedures. Innovative techniques, such as **combustion and composting technologies**, are used in conjunction with landfilling to reduce the need for landfill treatment technologies while increasing organic treatment. Power generation from waste incineration, Anaerobic waste treatment with biogas recovery for power generation and compositing are three technologies prioritized for the RD&B Programme in Vietnam. (See Table 1.E.35 of the Pre-Feasibility Study of the Program (Annex 2 of the Funding Proposal))

- **Gender Perspective in the Waste Management Sector**

In the wider informal waste management sector, women are primarily found in the lower tier, working in waste picking and separating at landfill sites. Traditionally, men dominate the higher-income and decision-making roles, whether as truck drivers, scrap dealers, repair shop workers, or in buying and reselling recyclables. This, again, reflects the gendered division of labour in society but also means that women are often excluded when waste management activities are formalized, missing out on protections and benefits, such as social security or higher wages.

[Cambodia]

In Cambodia, informal female workers as garbage collectors and dismantlers in urban areas, such as Phnom Penh, are the key actors in waste recycling in a sector largely lacking central organization. While this provides previous income generation opportunities to low-income Cambodia women, lack of segregation of toxic and hazardous materials, and unregulated unhygienic recycling practices pose serious threats to occupational safety and health.

[Indonesia]

In Indonesia, In the informal waste collection, segregation and recycling sector, women's roles are limited: Informal garbage collection work is often dangerous and invites gender-based violence and culturally women should avoid contact with strangers. If any, women's participation in this sector is concentrated in small-scale, localized scavenging activities of lower-value recyclables, leaving heavier and more valuable materials to their male counterparts, resulting in low remuneration than males. At the household level, however, Indonesian women play an active role in the disposal, sorting, and segregation of residual waste and recycling materials and they are responsible for the household's trash management.[245] Women who are directly involved in the activities of their community organization can reduce the amount of food waste and improve the overall community waste management. However, waste management at the community level is still influenced by the community authorities' stereotypes on women and major decision-making and managerial roles are given to men.[246]

[Lao PDR]

[245] The role of gender in waste management- GA Circular. (2021). Retrieved from GA Circular website: https://www.gacircular.com/publications/ .

[246] Gumilar Hadiningrat. (2020). Women's Role in Food Waste Management in Indonesia (Study Case in Bandung). Retrieved from ResearchGate website: https://www.researchgate.net/publication/347432275_Women's_Role_in_Food_Waste_Management_in_Indonesia_Study_Case_in_Bandung.

Like in Cambodia, Lao women also take part in dangerous and unhygienic waste picking and also segregation activities to sell them to the collection centres. They pick waste both at the streets and at the end-points (at dump sites). A recent report indicates that the livelihoods of the waste pickers are worsening with increasing competition and the structural change in processing (from direct selling to the collection factory in the country to, having to sell to the middlemen, as recyclables are being imported to Thailand and Vietnam.)[247] Nearby Vientiane, about 150 households are reported to reside in informal settlement (Ban Naphasuk) whose livelihoods are associated with the garbage collection work at the dumping sites[248]. Inadequate waste management and sanitation have consequences on the health and well-being of communities and the local environment.

Currently waste management in Vientiane are both being implemented by private and public entities. With the Thailand partnership etc. private waste collection and recyclable segregations are taking roots in a collective and formalized manner with the intervention of private capitals[249]. Still however the upstream garbage collections (from door to door, on the streets) remain informal and individualized, largely taken care of by low-income poor individuals.

Part of the waste management modernization projects are taking place where involved entities and players are taking actions to improve the conditions of the upstream waste pickers for instance,

In collaboration with UNDP and bilateral development cooperations, Lao government (MONRE and MPWT) is carrying out Lao Environmental and Waste Management Project (P175996). GAP has been established as part of the implementation plan, including the following, among others:

(1) With Partnership with Lao Women's Union (LWU) conduct gender-specific education and training Programme including, GBV components and gender-sensitive IEC material development, and'
(2) Promotion of female employment by the Contractor and strengthened Labour conditions, Code of Conduct, OHS measures

[Philippines]

There are gender differences with respect to both the management of, and attitudes towards, waste management. **Women and men play different roles in the management of household waste, recycling, and the education of children in environmental protection.** Given women's primary responsibility for cleaning, food preparation, family health, laundry, and domestic maintenance, women and men may view domestic waste and its disposal differently. In some urban centres, solid waste management has evolved into an organized system of collection, trade and recycling.

Women are less likely to be employed in official rubbish collection by truck or comparable vehicles, mostly due to the fact that the task requires them to travel fairly away from home and are **underrepresented in the category of recycling collectors (8.3%).** Moreover, women included in the

[247] https://laos.oxfam.org/latest/stories/relying-one-another-saving-groups-informal-workers-informal-workers
[248] https://www.nicholasbosoni.com/vientiane-waste-pickers?pgid=k8pe0mqp-055bf8b3-1e36-4388-9cbc-2347c93ca29a & https://think-asia.org/bitstream/handle/11540/1733/Volume%202_No%201_Oct%202005_05.pdf?sequence=1
249 "Currently, UCSC, the Lao Garbage Company, and Chanthabouly Cleansing Pvt. Company (also private) are jointly handling city waste in Vientiane. One nongovernment organization (NGO), the Lao Women's Union, is conducting educational and training programs for waste pickers in the city with support from the SWM for Vientiane Poor Project. The programs focus on topics such as how to handle hazardous waste in the waste stream and how to add value or generate income from collected wastes." (Source: https://laoepf.org.la/wp-content/uploads/2022/11/20221114_EWMP-P175996_SEP_Draft-Rev4-Clean.pdf)

waste and recycling sector are mostly **informal workers,** and it is quite common for Filipinas to operate in junk shops (36%).[250]

Introducing waste management technologies, could also be considered a priority sector for women in the Philippines. Lack of waste management and segregation of industrial, toxic, and hazardous materials endangers mostly low-income households and economically and socially vulnerable women and men, exposing them to major public health risks and pollution. **Women are found to be key agents in the household's waste management while underrepresented in the category of recycling collectors.** The Programme should target Filipino women to promote waste recycling and management inside and outside the household. Education and technical capacity building activities on "how to recycle" targeting women should also be implemented aside from clean waste technologies and infrastructure.

[Vietnam]

Waste management could also be considered a priority sector for Vietnamese women. Women represent 35-40% of formal waste collectors while they dominate the informal sector with a 65% participation rate, often living and working under dangerous and unhealthy conditions. In this sector, men have higher paid professions in the informal waste economy, being dealers and owners of recycling activities, but also in the formal waste economy, such as truck drivers and managers.251

Women usually are the decision-makers concerning domestic food purchases, and thus have a great influence on household waste management and separation but given the weak participation in community decision-making about waste disposal, men are more likely to take part in the community consultation processes and therefore take decisions about the community waste management.252

Investment in recycling technologies and infrastructure in Viet Nam could contribute to the female labour migration from the informal to formal sector, improving their income generation and work conditions. The Programme should target female garbage collectors and recyclers who have been engaging in the previously informal waste industry by providing them with the required skills in a clean and modernized system.

- **Recommended Gender-mainstreaming approaches**

In Cambodia, Lao PDR and the Philippines, women constitute significant parts of the information and highly dangerous and unhygienic occupation of garbage collects and waste pickers at door-to-door level

[250] "While less likely to be employed in formal waste collection by the use of truck or similar vehicles (5%), women are often engaged as street sweepers (37%) within the category of formal employment. Also challenged by the requirements of domestic work, the women in recycling either support their family members or operate on a very small scale and as a result are not adequately represented within the category of recycling collectors (8.3%). The owning and operating of junk shops by women is quite common (36%) and representative of the general trend of this form of employment, i.e. the operation of small businesses. "/ Global Platform for Sustainable Cities. The Role of Gender in Waste Management. (2021, October 13). Retrieved from https://www.thegpsc.org/knowledge-products/gender-and-cities/role-gender-waste-management.

[251] The role of gender in waste management- GA Circular. (2021). Retrieved from GA Circular website: https://www.gacircular.com/publications/. & "Between 10,000 and 16,000 waste workers work every day in Hanoi and Ho Chi Minh City. Known as *dong nat*, *nhom nhua* or *ve chai*, they buy and collect waste from households to resell to recyclers. The majority of them, women, earn less than their male counterparts, and 9 per cent of waste pickers are said to be children." & "UNDP's Wiesen emphasised, "Here in Vietnam, over 60 per cent of the workforce in the informal waste management sector are women. They often work in precarious situations and are exposed to harmful substances and chemicals along the value chain. Disproportionally impacted by plastic pollution, they also have direct exposure to toxic gases and emissions from waste burning and cooking fuels and can suffer from heat-related diseases and skyrocketing air pollution levels." (https://vir.com.vn/fostering-a-just-and-sustainable-future-for-vietnams-female-waste-workers-92838.html)

[252] The State of Gender Equality and Climate Change in Viet Nam (EmPower – Women for Climate Resilient Societies). Retrieved from https://asiapacific.unwomen.org/-/media/field%20office%20eseasia/docs/publications/2021/05/empowerpolicy%20brief%20for%20stakeholdersstate%20of%20gender%20equality%20%20climate%20change%20viet%20nam.pdf?la=en&vs=119.

(upstream of the value chain of the urban solid waste management sector) and at the endpoint dumping site. Their jobs are collecting the primary waste (unprocessed and non-segregated) from the source points and sell them to the "middlemen" (to export the recyclables and some parts to neighbouring countries, as in the case of Lao PDR exporting to Thailand and Vietnam), or to garbage collection /segregation and recycling centres. As in the case of

Often, they constitute the lowest-income level segments of the urban population whose alternative livelihood and alternative lifestyles simply cannot be a viable option. Without an effective gender action plan, recognizing the informal waste pickers groups as legitimate partners for the improvement of the urban solid waste management system, substantial social costs (including immediate loss of jobs and uprooting of livelihoods, leading to deepening impoverishment of the already poor societal groups.

[Best Practice Example]

In the following project (See table below), UNIDO's technical expertise aims to substantially advance gender equality in waste management. The project aims to help overcome gender inequality, empower women and break down the gender division of labour in the waste management sector by offering women secure and stable jobs.

Table 5-3-1: Example of Best Practice for Gender Mainstreaming Assistance Program in the W aste Management Sector

Project Name	Supporting The Socio-Economic Inclusiveness of Women and Youth by Waste Management
Project Management	United Nations Industrial Development Organization (UNIDO)
Country	Guinea
Project Period	Year 2011 - 2015 - Phase 1 (2011 – 2013) - Phase 2 (2014 – 2015)
Purpose	• Supported the socio-economic inclusiveness of women and youth through waste management activities
Main Project Activities	(a) waste collection from households in different locations in Conakry and 03 regional localities; (b) establishment of community-based waste recycling centres on plastics and organic waste; (c) information dissemination and awareness raising on solid waste management.
Outputs	**[Phase 1]** • the capacity-building on solid waste collection and recycling of more than 3,200 youth and women organized in 84 associations; • the establishment of community-based solid waste recycling centres in Conakry, Kindia and Labé **[Phase 2]** • the capacity of more than 5,000 women and youth on waste collection and recycling and to enable more than 500 women and youth to generate revenues through economic initiatives related to waste management

<1> General Requirements

All the requirements under the CTF Investment Criteria: Gender Component (See Table 1 of the Gender Assessment above.) will apply to all the waste management technology-related projects/enterprises subject to the CTF approval. Thus "women-led enterprises" on the waste management projects shall be prioritized. The rest of the applicant JVs shall also be encouraged to hire more female employees (with equal working conditions and remuneration as male employees) and improve women employees' welfare policies.

All the CTF Applicant JVs shall need to submit a simplified gender assessment and gender action plan **([Appendix 3. Simplified Gender Analysis (GA) & Gender Action Plan (GAP) (General Template)]** of the Gender Assessment/Gender Action Plan (GA/GAP), Annex 8 of the Funding Proposal of the Programme). [Section II- 2. Gender Analysis] requires you to assess the gender perspective of a specific proposed JV enterprises/projects, based on which **[Section II-3. Enterprise-level Gender Action Plan]** need to be established.

<2> Sector-specific Gender Mainstreaming Considerations

Women and men play different roles in waste management and recycling. often living and working under dangerous and unhealthy conditions. In this sector, men have higher paid professions in the informal waste economy, being dealers and owners of recycling activities, but also in the formal waste economy, such as truck drivers and managers.

In terms of the providers of services, employment opportunities tend to be different for women and men, with women occupying lower-grade positions than men.

As the waste sector modernizes and applies new technologies that may require greater education and training, it is essential that women are included in this process. It can lead to the dissolution of the informal sector related to the employed workforce, which can lead to job loss, accelerated poverty, and threat to livelihood, and can cause various social conflicts in the transition process.

Women waste pickers/garbage collectors need to be recognized as legitimate and valuable partner for the improvement of the overall urban waste management sector by following recommended actions:

- **Gender Action 1: Directly/Indirectly contract/employ the female waste pickers groups as delivery partners, provide formal and secure employment contract** (with standard HSE and legal labour contracts) to them to secure them with decent jobs and safe and standard work conditions, preventing them from any types of abuses and often (gender-based) violences.

- **Gender Action 2:** (Related to Gender Action 1) **Capacitate them to organize and modernize their practices (by establishing properly registered enterprise/for-profit entities etc.) through various business and garbage collection-related skill trainings.**

- **Gender Action 3:** When JV enterprises involve construction and operation of a sizable waste treatment (or Waste to Energy) facility development, ensure the contractor includes **local employment component prioritizing the local female employees with the relevant experiences (including informal waste collection works)**. If possible, consider setting a certain target (xx%) of the non- and semi-skilled jobs as well as working level management positions reserved for women waste pickers/garbage collectors or those with associated works (waste segregation centre work, etc.) for their smooth transition to formal waste management sector.

Appendix 5.5. Early Warning System (EWSs) and Disaster Management

- **EWS & Disaster Management Sector in the Five Countries**

Disaster management entails both structural (such as dykes, embankment and ground elevation for sea level rise & tsunami prevention) and non-structural (such as climate change and hydrometeorological monitoring and forecasting, early warning, planning, mangrove and breakwater tree planting, and urban planning.).

As part of a non-structural solution for climate change-related natural disaster management technologies, early Warning System (EWSs) are a crucial component of many countries' climate change adaptation strategies and objectives. The aim of an EWS is to avoid or reduce damages inflicted by natural disasters. As such, EWS is designed to disseminate messages and warnings effectively to ensure that communities are in a permanent stage of readiness and to enable early action when disaster strikes. In this sense, an EWS can only be effective if consensus is reached on its benefits by local communities[253].

Early warning systems, which use integrated communication networks to assist communities in getting ready for dangerous climate-related occurrences, are an adaptive response to climate change. A successful EWS fosters long-term sustainability by preserving people, employment, land, and infrastructure[254].

Following are the key priority climate technology areas in each of the countries, assessed by the KDB and GGGI for the R&DB Programme (Annex 2 of the Funding Proposal of the Programme):

[Cambodia]

Cambodia includes a "National end-to-end early warning system with a focus on effective dissemination to population at risk" as its National Adaptation Action within its NDC[255].

[Indonesia]

Indonesia's public health is significantly affected by climate change-related extreme weather and natural disasters. In this context, the following two human health sector technology areas were identified to have the highest relevance considering various factors (costs, environmental benefits, economic impacts, technology maturity, country ownership, social impacts): 1) Early warning & information systems to prevent outbreaks (daily reporting system), and; 2) health bio metering of climate change disease technologies. In particular, the R&DB Programme highlights investment priority on the technologies on **upgrading the hydro-meteorological information & information system/map for climate-related disease (dengue, malaria and pneumonia) to early warning to local action system.**

[Lao PDR]

Being a predominantly agrarian country, Lao PDR's EWS is related to the overall climate resilience building and adaptation strategies of the rural communities and agricultural sector. Early warning system and insurance in livestock industry is one example, as such. The communities would need to be

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[253] The most content and the following analyses are directly quoted or indirectly referred from: Demand-Driven Pre-Feasibility Study (Annex 2) of the Programme by KDB and GGGI.
[254] https://www.un.org/en/climatechange/climate-solutions/early-warning-systems
[255] See Table 1.A.10. of Demand-Driven Pre-Feasibility Study (Annex 2) of the Programme by KDB and GGGI.

mobilized at a village level and in this context, establishment of multipurpose community centers could function as EWS hub and evacuation areas.

Also rich in water resources, Lao PDR has the greatest need for EWS in flood prevention areas. **Developing EWS in flood and drought management** is highly relevant. The deployment of effective EWS can help to reduce the impacts of flash flooding by preventing loss of life and damage to properties of affected areas of the country. In this context, the Lao government aims to develop **end-to-end EWSs** to provide timely, accurate and effective warnings and to enhance the response capacity of all communities that are at risk of rain-induced natural disasters.

[Philippines]

It appears the Philippines has relatively advanced level of EWS in place: The country's Integrated Disease Surveillance and Response (IDSR) system includes the development of early warning and response systems for climate-sensitive health risks. The country also has an approved national health adaptation strategy and has conducted a national assessment of climate change impacts and vulnerability and adaptation for health.

[Vietnam]

Natural disaster prevention and management through strengthened EWS is one of the priorities in Vietnam. Consultations with various stakeholders in the country identified that the EWS technology application in the following sectors were found to be in high relevance and demand: (1) EWS in Forestry for fire prevention and management; (2) Early warning and decision-making support system (DSS) including AI-enabled DSS.

- **Gender Perspective in EWS and Disaster Management**

In general, girls and women worldwide are reported to be more vulnerable to the damages of climate change-related natural disasters and extreme weather conditions such as Tsunami, flood and severe droughts. Higher casualties and incident rates of the girls and women are partly explained by their physical frailties (compared with boys and men, partly biological attributes but also due to girls/women's poorer food intakes) but also partly due to their limited access to the disaster alert and critical information.

Women typically do not receive disaster notifications since men are the only ones who inform them of potential threats. In many cases, due to societal norms and barriers, women were still hesitant to take part in the planning for disaster response and capacity building.[256]Warning information dissemination impacts differ based on gender disparities, due to the different levels of educational attainment, reading levels, and digital literacy, in terms of recognizing, understanding, and acting. People of different genders might have varying degrees of access to formal and unofficial channels of information transmission, differing communication preferences (influenced by gender norms), and unique difficulties in accessing and acting on an early warning[257].

[256] Shah et al. (2022) Gender perspective of flood early warning systems: People-cantered approach. Water 14(14). Retrieved from: https://doi.org/10.3390/w14142261

[257] https://wrd.unwomen.org/sites/default/files/2021-11/GENDER~1_1.PDF

Disaster vulnerability is exacerbated by social marginalization and gender inequity. Women and gender minorities suffer more during and after a disaster as they have had less economic, political, and cultural power. The vulnerability of women and gender minorities during a disaster can be increased by traditional gender norms (such as the idea that males make decisions), stereotypical gender roles, and gender-based violence[258].

However, there are glimmers of changes: Shared experiences of severe disasters can often lead to a wake-up call for a community's self-mobilization and women's bottom-up leadership as an active agent for disaster management, as shown in the Thua Thien Hue Province after a tragic havoc by a tropical typhoon Even in 1999[259].

Except for the Philippines (with relatively equal share of women taking engineering career), the remaining four countries, has put the (hard infrastructure-based) structural solution to Disaster management not an area for high potential for women's participation in the short-term. While the overall STEM education and engineering training for girls and women should continue to be promoted steadily, this Programme would need to focus on more immediate and direct intervention points for women's empowerment and equal benefit sharing of the Programme.

- **Recommended Gender-mainstreaming approaches**

<1> General Requirements

All the requirements under the CTF Investment Criteria: Gender Component (See Table 1 of the Gender Assessment above.) will apply to all the EWS and Natural Disaster Technology application projects/enterprises subject to the CTF approval. Thus "women-led enterprises" on the EWS and Natural Disaster Management projects shall be prioritized. The rest of the applicant JVs shall also be encouraged to hire more female employees (with equal working conditions and remuneration as male employees) and improve women employees' welfare policies.

All the CTF Applicant JVs shall need to submit a simplified gender assessment and gender action plan **([Appendix 3. Simplified Gender Analysis (GA) & Gender Action Plan (GAP) (General Template)]** of the Gender Assessment/Gender Action Plan (GA/GAP), Annex 8 of the Funding Proposal of the Programme). [Section II- 2. Gender Analysis] requires you to assess the gender perspective of a specific proposed JV enterprises/projects, based on which **[Section II-3. Enterprise-level Gender Action Plan]** need to be established.

<2> Sector-specific Gender Mainstreaming Considerations

[258] https://public.wmo.int/en/resources/bulletin/gender-equality-context-of-multi-hazard-early-warning-systems-and-disaster-risk

[259] This is a good example of climate disaster response action of a women-led community in Vietnam. In 1999, a tropical typhoon Even hit the coastal line of Vietnam, the region pounded due to rainfall more than a monthly average of its. The flooding continued for a couple of days, and the villages in Thua Thien Hue province were almost destroyed. A villager reached the local branches of the Vietnam Women's Union to overcome the dangers of rising sea levels. They started to plant Mangroves because they can survive in the saltwater and generate breeding grounds for fish and help to prevent floods from the sea. For the entire process of this project, women, especially the Vietnam Women's Union, made plans, implemented, and extended the activities with community resources. (https://qz.com/1758547/how-vietnamese-women-are-fighting-back-against-climate-change)

Gender Action 1: Diversify the communication modalities to ensure inclusiveness and accessibility by various groups including minority segments of the population.

Setting up an App or an Internet platform based EWS is important but may become urban/high incomer/educated/male-biased. Effective EWS and disaster management practices requires going beyond the system, as the public needs to be able to take early actions once they receive early warning. Linking the system to the locally actionable mechanism remains a challenge in Indonesia and other countries. The public's understanding and awareness on the EWS and its technologies and applications remain to be limited.

To reach out to the entire population of the country that are subject to such disasters and risks often requires special consideration of identifying and deploying viable communication methods and modalities for specific segments of the population whose access are largely challenged (due to their remoteness, as in the case of remote or isolated (as in the case of islets), or due to physical disabilities, such as blind people and the elderly, etc.) communities, often without electricity/Internet/Mobile phone penetrations. In urgent disaster outbreak situations, targeted language (local dialects for ethnic minorities, or visual signs for illiterate/low level of literacy group) and communication methods (e.g., physical disaster communication network set in the community level for those not connected through Internet, or without mobile phone network, etc.)

As there is a difference in educational attainment across gender, mostly women have lower digital literacy. To implement complementary actions at the community level, traditional and offline communication methods must be used in a parallel manner[260]. In addition, considering the literacy levels of the population, voice messages also need to be utilized when delivering warnings. As women's voices are more effective in delivering urgent information, for instance, women can take the role of providing voice messages under an early warning system[261].

- ***Formulate a communications plan, develop knowledge products and information, education and communication (IEC) materials, in easy-to-understand, gender-neutral language(s). Ensure specific communication strategies for marginalized populations (inc. different communication modes (for the blind, with those with hearing impairments, illiterate or low-educated people). Ensure both digital/non-digital methods of communications.***

Gender Action 2: Strengthen community-based climate disaster preparation and management System with equal/heightened roles of the women's groups.

Community-driven climate risk resilience projects are crucial for disaster preparedness and responses. As in the case of Thua Thien Hue province, Vietnam, shown above the Community's self-mobilization and its facilitation need to be strongly promoted and supported. This is one of the types of ecosystem-based adaptation (EbA), which is appropriate for bottom-up approaches. Local communities have ample knowledge (including know-how) on their local environments, so they are proper actors in utilizing community-based resources. Community-driven bottom-up approach, once carefully and appropriately supported, will identify the optimum solution to the problems.

- In case a proposed JV enterprises target specific locations/communities as beneficiaries and/or delivery partners, ***understand the community governance dynamic and esp. role of community girls and women with their potential as active agents. Assess their needs and willingness to participate (as well as their concerns associated). Assess the gaps in their***

[260] https://public.wmo.int/en/our-mandate/focus-areas/natural-hazards-and-disaster-risk-reduction/mhews-checklist/warning-dissemination-and-communication
[261] Edworthy et al. (2003). The use of male of female vices in warning systems: A question of acoustics. Noise & Health 6(21). pp. 39-50. Retrieved from: https://www.noiseandhealth.org/text.asp?2003/6/21/39/31683

capacity and identify measures to fill such gaps. As necessary, ***include the culturally appropriate and feasible capacity-building programme for the girls and women as sole/leading or equal participants of the proposed enterprise/activities.***

- ***If necessary, assist mobilization and organization of girls' and women's groups.*** Structural engagement of women (esp. of the host communities) as an institutional partner for project preparation and execution would invite women as an equal partner for climate actions by empowering them. Gradually, this will shift the cultural perception that women are passive recipients of the decisions and actions made by men in society, generating ripple effects of socio-cultural changes in the country[262].

Gender Action 3: Strengthen the consultation of girls and women throughout the project cycle and empower them.

- Early warning systems and climate monitoring businesses can be made of diverse forms of activities. ***Maintain a gender-sensitive perspective at all stages of the execution of the JV enterprises/business, from the planning to the operation stage to the extent possible.*** ■

[262] https://www.oecd.org/derec/adb/tool-kit-gender-equality-results-indicators.pdf